Measurement of the Drell-Yan Absolute Cross-Section in pp and pd Collisions with a 120 GeV Proton Beam at Fermilab

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Abstract

This analysis note details the extraction of absolute Drell-Yan cross sections for pp and pd interactions using data collected with the runs 2 and 3. We present these findings for preliminary collaboration approval ahead of upcoming conference presentations. This study advances prior analyses by integrating both Liquid Hydrogen (LH_2) and Liquid Deuterium (LD_2) target datasets, alongside implementing significant methodological upgrades to the efficiency correction framework. To mitigate statistical limitations, the reconstruction efficiency is now derived from a global, high-statistics parameterization of the Drift Chamber 1 ($D1$) occupancy, integrated across all kinematic bins (DocDB 11427[1]). Furthermore, the hodoscope efficiency correction has transitioned from RS67 based paddle efficiency corrections to averaged hodoscope paddle efficiencies over all the roadsets 57-70 to a rigorous, event-by-event calculation utilizing dimuon track kinematics and paddle-specific response mappings (DocDB 11467[2]). Building upon these improvements, we report the single-differential cross-sections as a function of transverse momentum, $\mathrm{d}\sigma/\mathrm{d}p_T$, as well as the comprehensive double-differential Drell-Yan cross-sections, $\mathrm{d}^2\sigma/\mathrm{d}x_F\mathrm{d}M$ and $\mathrm{d}^2\sigma/\mathrm{d}x_F\mathrm{d}p_T$, for both target configurations. Finally, we provide comparisons between these extracted distributions and Next-to-Leading-Order (NLO) Quantum Chromodynamics (QCD) predictions.

This work was supported in part by US DOE grant DE-FG02-94ER40847.

[1] `https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11427`
[2] `https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11467`

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1 Introduction

In October 2025, a preliminary evaluation of the *pp* Drell-Yan absolute cross section utilizing a 120 GeV proton beam was documented in DocDB 11383[3]. The present analysis standardizes the extraction framework for both *pp* and *pd* datasets, a necessary harmonization given that the *pp* cross section is requisite for correcting the hydrogen contamination within the liquid deuterium target. To ensure a robust extraction of the physics observables, this analysis incorporates several critical algorithmic and statistical improvements implemented since the preliminary report:

- **Global Reconstruction Efficiency Parameterization:** Prior methodologies parameterized reconstruction efficiency as a function of Station-2 occupancy ($D2$), utilizing distinct curves for each kinematic bin. This approach yielded kinematically segregated samples with poor statistical power and consequently large uncertainties. As established in DocDB 11427[4], the reconstruction efficiency exhibits negligible kinematic dependence. Consequently, we now employ a unified, high-statistics efficiency curve parameterized against Station-1 occupancy ($D1$). This shift to $D1$ also eliminates combinatorial ambiguities when computing efficiencies for mixed events.

- **Dimuon-Level Hodoscope Efficiencies:** Earlier studies applied a global hodoscope and trigger efficiency correction (DocDB 10809[5]) inherently optimized for J/ψ production. To accurately reflect the Drell-Yan continuum, we have adopted an event-by-event calculation. Utilizing paddle-specific efficiencies derived from single-track data (DocDB 11467[6]), the hodoscope efficiency is now computed for each kinematic bin based directly on the constituent tracks of the accepted dimuon pairs.

- **Mixed Event Contamination Correction:** Previously, the reconstruction efficiency correction relied exclusively on the yield of dimuon events in the physical data, neglecting the differing occupancy profiles of mixed events. Because the $D1$ distribution of mixed events diverges from that of regular events, the combinatorial background inherently skewed the efficiency correction. We have addressed this systematic bias by employing a discrete contamination correction method detailed in DocDB 11448[7].

2 Analysis Methodology

The determination of the absolute Drell-Yan cross-section demands a multi-stage process: rigorous selection of dimuon candidates, quantitative background subtraction, precise normalization via integrated luminosity, and detailed correction for geometric acceptance and detector inefficiencies.

2.1 Data and Monte Carlo Samples

This analysis is predicated upon the Roadsets 57, 59, 62, 67, and 70 trigger configuration datasets, encompassing data recorded during Runs 2 and 3 by the SeaQuest spectrometer.

[3] https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11383
[4] https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11427
[5] https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=10809
[6] https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11467
[7] https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11448

The primary physical data arrays for the Liquid Hydrogen (LH_2) target, Liquid Deuterium (LD_2) target, and the requisite empty flask configurations are centralized in two primary directories based on their production source:

Run 2 and 3 Datasets (Roadsets 57, 59, 62, 70):
Located at:`/seaquest/users/harshaka/e906_project/e906-root-ana/work_gpvm/scripts/ /results_runFinalTree/merged_results_e906_mixing/`

- **RS57:**
  - **LH2:**`merged_RS57_LH2_1_1138.root`
  - **LD2:**`merged_RS57_LD2_3_1138.root`
  - **Flask:**`merged_RS57_Empty_2_1138.root`
- **RS59:**
  - **LH2:** `merged_RS59_LH2_1_465.root`
  - **LD2:** `merged_RS59_LD2_3_466.root`
  - **Flask:** `merged_RS59_Empty_2_466.root`
- **RS62:**
  - **LH2:** `merged_RS62_LH2_1_1234.root`
  - **LD2:** `merged_RS62_LD2_3_1237.root`
  - **Flask:** `merged_RS62_Empty_2_1234.root`
- **RS70:**
  - **LH2:** `merged_RS70_LH2_1_264.root`
  - **LD2:** `merged_RS70_LD2_3_266.root`
  - **Flask:** `merged_RS70_Empty_2_267.root`

Run 3 Dataset (Roadset 67):
Located at: `/seaquest/users/apun/e906_projects/rs67_merged_files/`

- **RS67:**
  - **LH2:** `merged_RS67_3089LH2.root`
  - **LD2:** `merged_RS67_3089LD2.root`
  - **Flask:** `merged_RS67_3089flask.root`

To facilitate event-level detector performance corrections, all base ROOT files across the five roadsets were augmented with variables explicitly tracking the reconstruction efficiency (`recoeff`), the hodoscope efficiency (`hodoeff`), and their respective propagated uncertainties (`recoeff_error`, `hodoeff_error`). The updated ntuples, preserving the base naming conventions with an appended `_recoeff_hodoeff.root` suffix, are archived systematically by roadset at:

`/seaquest/users/ckuruppu/rootfiles/`

The inclusion of empty flask data is critical for isolating target-specific interactions by subtracting invariant background generated by the beam interacting with the containment vessel and upstream materials.

Extensive Geant4-based Monte Carlo (MC) simulations were deployed to map the geometric acceptance and reconstruction efficiencies of the spectrometer. The simulated events were propagated through the complete SeaQuest detector geometry and reconstruction chain:

- **Acceptance Studies:** Drell-Yan topologies were generated across a 4π solid angle ("thrown") and filtered through the standard event selection protocol ("accepted"). The baseline files are located in `/seaquest/users/chleung/pT_ReWeight/`, including following files:
  - `mc_drellyan_LH2_M027_S001_4pi_pTxFweight_v2.root`
  - `mc_drellyan_LD2_M027_S001_4pi_pTxFweight_v2.root`
  - `mc_drellyan_LH2_M027_S001_clean_occ_pTxFweight_v2.root`
  - `mc_drellyan_LD2_M027_S001_clean_occ_pTxFweight_v2.root`
- **Efficiency Studies:** To quantify the impact of intense ambient hit multiplicities on track finding, identical kinematic sets were processed both organically ("clean") and with an overlay of random experimental background hits ("messy"). Following files were used to calculate reconstruction efficiency:
  - `mc_drellyan_LH2_M027_S001_clean_occ_pTxFweight_v2.root`
  - `mc_drellyan_LD2_M027_S001_messy_occ_pTxFweight_v2.root`

  Crucially, all simulated events are intrinsically weighted to emulate the transverse momentum (p_T) spectrum mapped in the physical data.

2.2 Event Selection

A stringent, multi-tiered selection protocol is applied to isolate statistically pure Drell-Yan events from the underlying continuum of beam-dump interactions, muon combinatorics, and cosmic background.

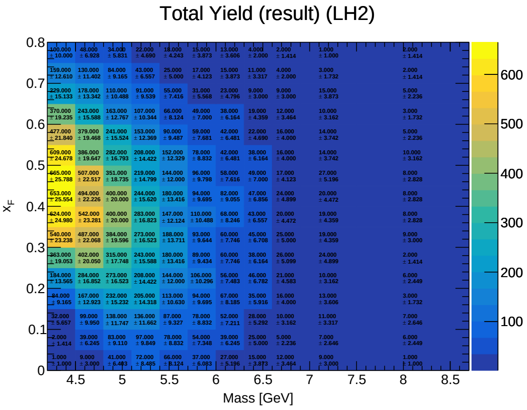
- **Data Quality:** Only spills explicitly validated by the standard SeaQuest offline quality monitoring are retained. An explicit trigger requirement (`MATRIX1 == 1`) enforces top/bottom spectrometer coincidence.
- **Track and Vertex Quality:** We mandate the standard collaboration "Chuck Cuts" (DocDB 2111[8]) to ensure robust spatial reconstruction. This protocol mandates thresholds on track χ^2, hit multiplicity, target-vs-dump vertex probability, and magnet aperture fiducial cuts. The specific boundary conditions are tabulated in tables 1 to 3 of section 10.
- **Kinematic Boundaries:** To isolate the Drell-Yan continuum and suppress charmonium resonant states (J/ψ, ψ'), a lower mass bound of $M_{\mu\mu} > 4.2$ GeV/c^2 is strictly enforced, alongside an integration window of $0 < x_F < 0.8$.

Addition to these Chuck Cuts defined in DocDB: 2111-V42 we defined following cut to ensure target position of dimuons for RS62 dataset only:
$runID > 11500$ We also remove dimuons outside the range of $20 < D1 < 385$ because, the reconstruction efficiency curve is only defined within this range of $D1$ occupancy.

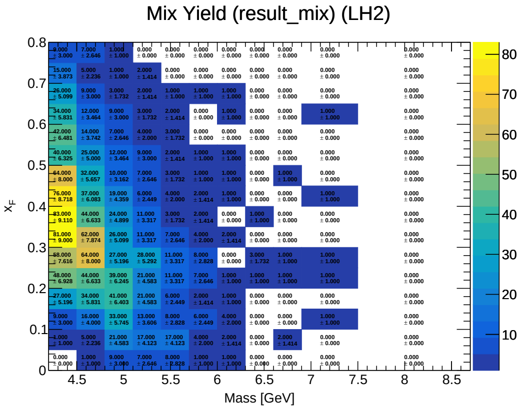
Following this filtering cascade, the primary and mixed-event dimuon yields for all target configurations are isolated into their respective mass and x_F bins, as illustrated in fig. 2.

Addition to above yield distributions made for double differential cross-sections, we newly calculate yield distributions for single differential cross-sections for pT bins.

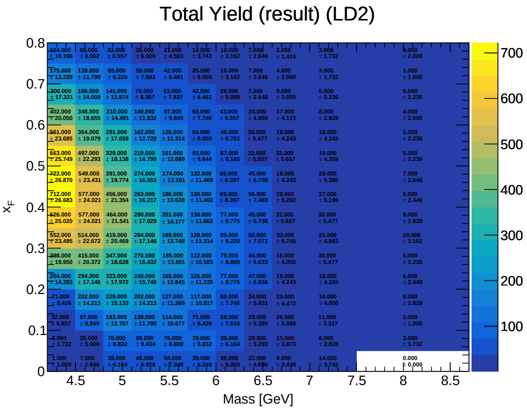
[8] `https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=2111`



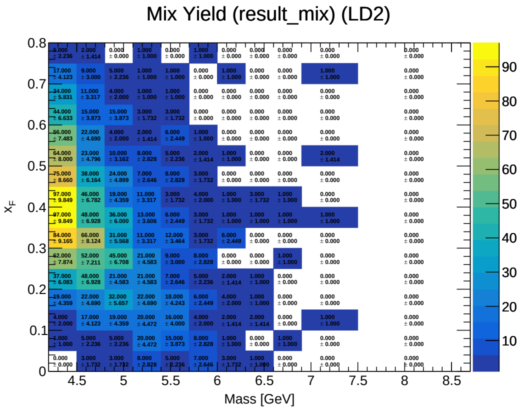
(a) LH2 total dimuon yield



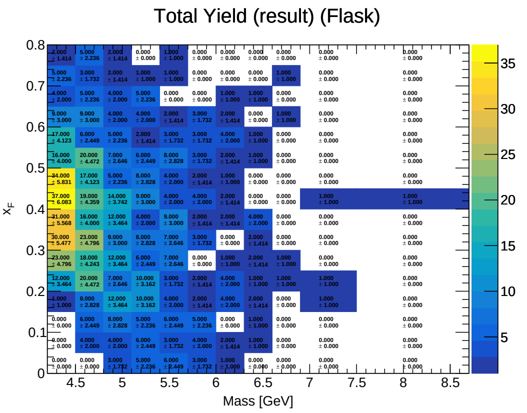
(b) LH2 mix dimuon yield



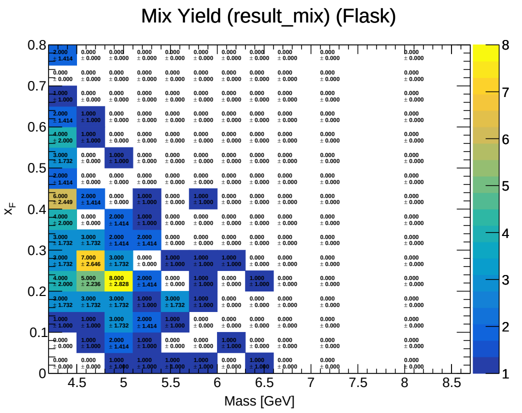
(c) Total LD2 dimuon yield



(d) LD2 mix dimuon yield

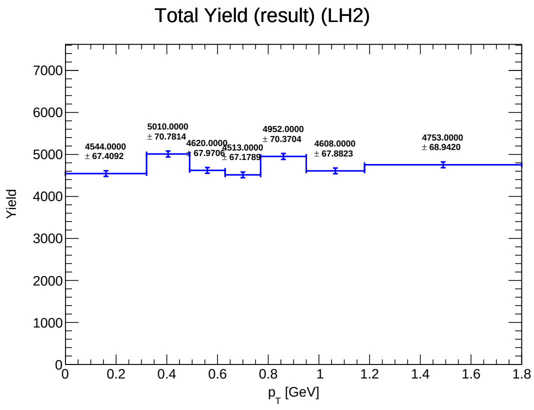


(e) Total Flask dimuon yield

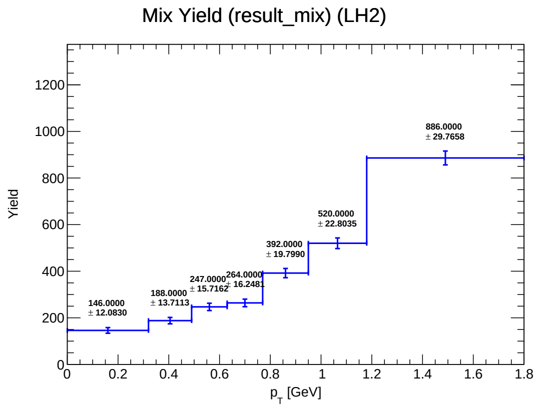


(f) Flask mix dimuon yield

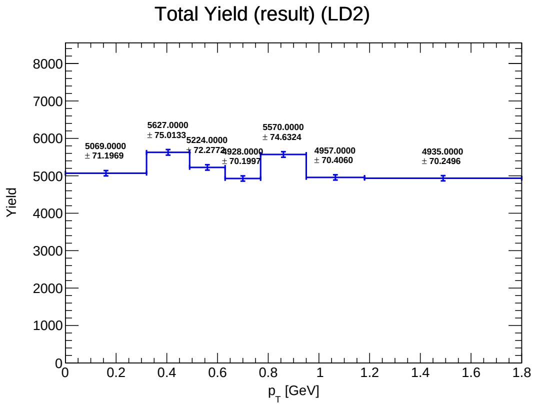
Figure 1: Dimuon distributions after applying event selection criteria.



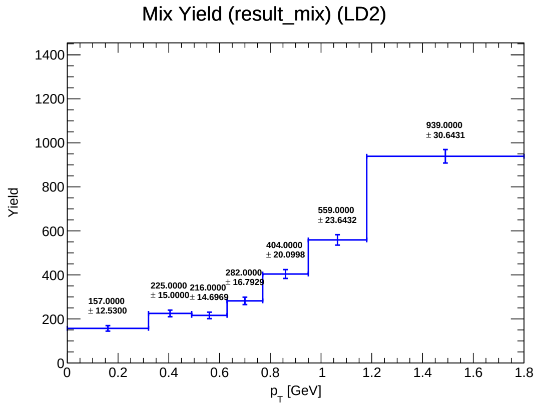
(a) LH2 total dimuon yield



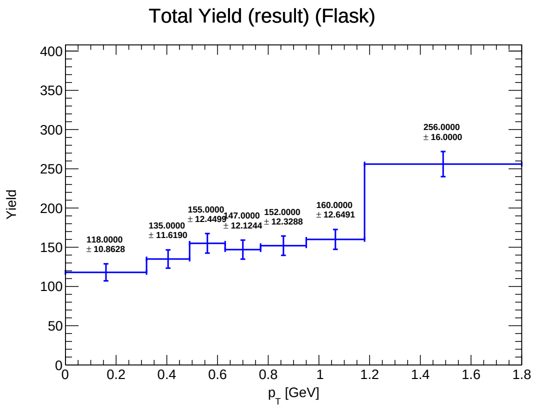
(b) LH2 mix dimuon yield



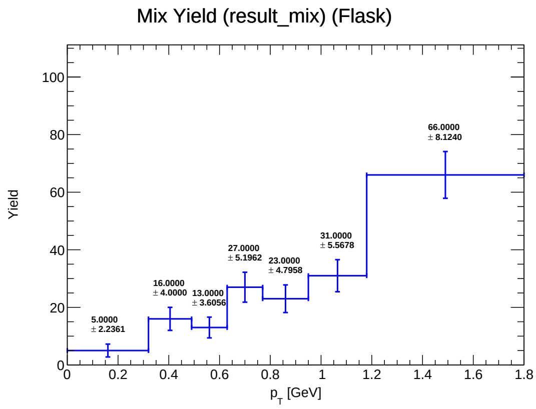
(c) Total LD2 dimuon yield



(d) LD2 mix dimuon yield



(e) Total Flask dimuon yield



(f) Flask mix dimuon yield

Figure 2: Dimuon distributions after applying event selection criteria for pT bins.

2.3 Cross-Section Formalism

The extraction of the double-differential *pp* cross-section within a localized kinematic space $(\Delta M, \Delta x_F)$ follows the formalism outlined in DocDB 11445[9]:

$$\frac{\mathrm{d}^2\sigma_{pp}}{\mathrm{d}M\mathrm{d}x_F} = \frac{1}{\epsilon_{\mathrm{acc}}\mathcal{L}\Delta M\Delta x_F}\left[\frac{Y^{\mathrm{LH2}}_{\mathrm{total}} - Y^{\mathrm{LH2}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{LH2}}_{\mathrm{signal}}\rangle} - \frac{I_{\mathrm{LH2}}}{I_{\mathrm{flask}}}\left(\frac{Y^{\mathrm{flask}}_{\mathrm{total}} - Y^{\mathrm{flask}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{flask}}_{\mathrm{signal}}\rangle}\right)\right] \tag{1}$$

where Y_{total} and Y_{mixed} denote the prompt and combinatorial background yields respectively, and I represents the integrated proton flux (POT) incident on a given target. The true signal efficiency, $\langle\epsilon_{\mathrm{signal}}\rangle$, separates the detector response of real correlated pairs from that of mixed tracks:

$$\langle\epsilon_{\mathrm{signal}}\rangle = \frac{1}{Y_{\mathrm{total}} - Y_{\mathrm{mixed}}}\left[\langle\epsilon_{\mathrm{total}}\rangle Y_{\mathrm{total}} - \langle\epsilon_{\mathrm{mixed}}\rangle Y_{\mathrm{mixed}}\right] \tag{2}$$

Here, $\langle\epsilon_{\mathrm{total}}\rangle$ and $\langle\epsilon_{\mathrm{mixed}}\rangle$ map the composite event efficiencies (DocDB 11448[10]). For an individual accepted dimuon candidate i, the overarching efficiency is the product of the discrete reconstruction and hodoscope survival probabilities:

$$\epsilon^i = \epsilon^i_{\mathrm{reco}} \cdot \epsilon^i_{\mathrm{hodo}} \tag{3}$$

The macroscopic integrated luminosity $\mathcal{L}$ governs the absolute normalization:

$$\mathcal{L} = N_{\mathrm{incident}} \cdot \frac{N_A \rho L}{A} \cdot f_{\mathrm{atten}} \tag{4}$$

where N_A is Avogadro's constant, A is the target molar mass, and f_{atten} defines the hadronic beam attenuation coefficient. For the 50.8 cm LH_2 flask ($\rho_H = 0.0708$ g/cm^3), the target areal density is 3.5966 g/cm^2 with an associated $f_{\mathrm{atten}} = 0.966$ (DocDB 8588[11]). The acceptance parameter ϵ_{acc} is derived purely from Monte Carlo mapping.

Extraction of the *pd* cross section necessitates a subsequent correction for the intrinsic protium (hydrogen) contamination present in the deuterium cryogen:

$$\begin{aligned}\frac{\mathrm{d}^2\sigma_{pd}}{\mathrm{d}M\mathrm{d}x_F} &= \frac{1}{\epsilon_{\mathrm{acc}}\mathcal{L}\Delta M\Delta x_F}\left(\left[\frac{Y^{\mathrm{LD2}}_{\mathrm{total}} - Y^{\mathrm{LD2}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{LD2}}_{\mathrm{signal}}\rangle} - \frac{I_{\mathrm{LD2}}}{I_{\mathrm{flask}}}\left(\frac{Y^{\mathrm{flask}}_{\mathrm{total}} - Y^{\mathrm{flask}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{flask}}_{\mathrm{signal}}\rangle}\right)\right]\right.\\ &\left. - \frac{T_{HD}}{T_{HH}}\frac{I_{\mathrm{LD2}}}{I_{\mathrm{LH2}}}\left[\frac{Y^{\mathrm{LH2}}_{\mathrm{total}} - Y^{\mathrm{LH2}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{LH2}}_{\mathrm{signal}}\rangle} - \frac{I_{\mathrm{LH2}}}{I_{\mathrm{flask}}}\left(\frac{Y^{\mathrm{flask}}_{\mathrm{total}} - Y^{\mathrm{flask}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{flask}}_{\mathrm{signal}}\rangle}\right)\right]\right)\end{aligned} \tag{5}$$

We apply the previously measured fractional target thicknesses $T_{HD} = 0.1084$ and $T_{HH} = 3.5966$ (DocDB 11227[12]), where T_{HD} is explicitly POT-weighted across the varying purity regimes of the LD_2 run periods.

3 Acceptance Correction

The forward topology of the SeaQuest spectrometer necessitates a rigorous unfolding of its geometric acceptance. The survival probability of a generated Drell-Yan pair is highly

[9] https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11445
[10] https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11448
[11] https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=8588
[12] https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11227

dependent on invariant mass (M) and longitudinal momentum fraction (x_F). The binned acceptance fraction is analytically defined via MC as:

$$\text{Acceptance (A)} = \frac{N_{\text{accepted}}}{N_{\text{thrown}}} \tag{6}$$

where N_{accepted} reflects the yield traversing the digital simulated geometry These are events that satisfy the full reconstruction and selection criteria, including the kinematic cuts. They are extracted from the `*clean_occ_pTxFweight*` MC files, which represent the idealized detector response without background overlays. (`*clean_occ_pTxFweight*`) and N_{thrown} denotes the phase space baseline in 4π (`*4pi_pTxFweight*`). To preserve resolution, the array is partitioned into 16 x_F domains (width $\Delta x_F = 0.05$ up to 0.8) and 11 distinct mass bins mapping the high-mass continuum ($4.2 \leq M < 8.7$ GeV/c^2).

Visual representations of the acceptance mappings, demonstrating negligible target-dependent geometric variance between LH_2 and LD_2, are provided in section 11.

4 Reconstruction Efficiency Correction

The algorithmic efficiency of the kTracker spatial reconstruction package degrades under the high hit-multiplicity characteristic of the primary beam. To accurately quantify this effect without fragmenting our statistical confidence across narrow kinematic intervals, we derive the primary efficiency metric ($\epsilon_{\rm reco}$) by comparing "messy" data overlays against idealized "clean" MC tracks.

Following the verification in DocDB 11427[13] that $\epsilon_{\rm reco}$ decouples from the track kinematics, we extract a high-fidelity **global reconstruction efficiency curve** parameterized explicitly against the Drift Chamber 1 ($D1$) hit array, as demonstrated in fig. 3.

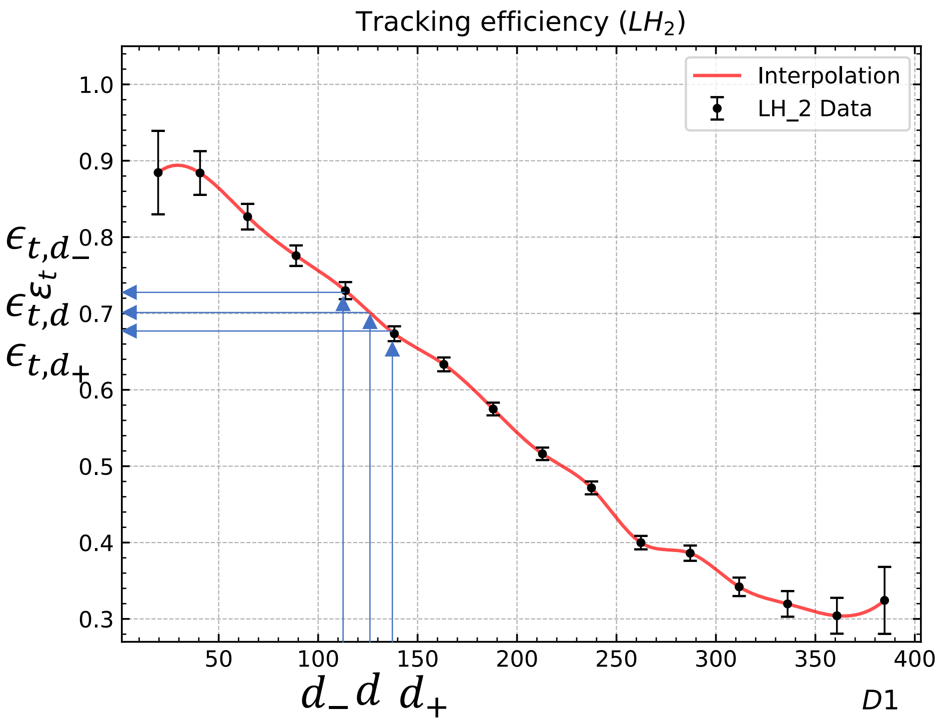


Figure 3: Global Reconstruction Efficiency curve as a function of the $D1$ occupancy variable, integrated over all kinematic bins.

The discrete efficiency weight for any given physical dimuon is mapped dynamically via linear interpolation adjacent to the MC-derived curve edges ($D1^+, D1^-$):

$$\epsilon_i = \epsilon(D1^-) + \left(\frac{\epsilon(D1^+) - \epsilon(D1^-)}{D1^+ - D1^-}\right)(D1^+ - D1_i) \tag{7}$$

For combinatorial **mixed events**, the synthetic occupancy is generalized to the arithmetic mean of the parent tracks:

$$D1_{\rm mixed} = \frac{D1_{\rm pos} + D1_{\rm neg}}{2} \tag{8}$$

4.1 Uncertainty Propagation

Maintaining systematic rigor requires precise propagation of the MC interpolation variance. For a singular data event i, the constituent error scales as:

$$\delta\epsilon_i = \frac{1}{D1^+ - D1^-}\sqrt{(D1^+ - D1_i)^2\,\delta\epsilon(D1^+)^2 + (D1^- - D1_i)^2\,\delta\epsilon(D1^-)^2} \tag{9}$$

[13] `https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11427`

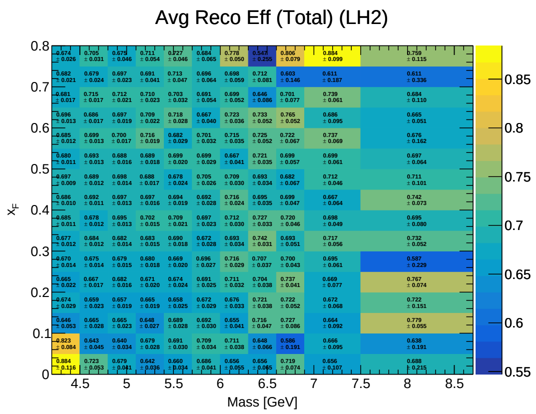
The scalar efficiency $\langle\epsilon\rangle$ for an N-event bin emerges as the simple mean (eq. (10)), and the aggregated statistical uncertainty ($\delta_{\mathrm{prop}}\langle\epsilon\rangle$) propagates quadratically (eq. (11)):

$$\langle\epsilon\rangle = \frac{1}{N}\sum_{i=1}^{N}\epsilon_i \tag{10}$$

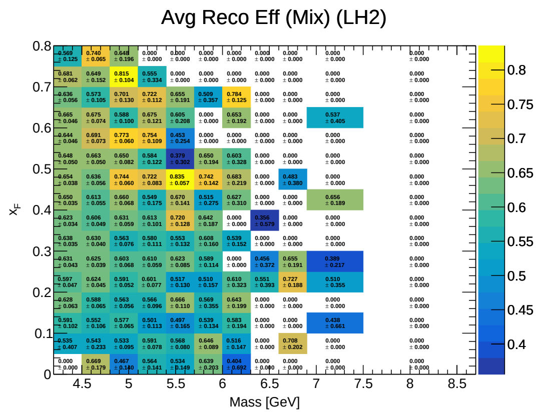
$$\delta_{\mathrm{prop}}\langle\epsilon\rangle = \frac{1}{N}\sqrt{\sum_{i=1}^{N}(\delta\epsilon_i)^2} \tag{11}$$

We generate 2D bin-by-bin topological maps of these corrected efficiencies for both prompt and mixed events, presented in fig. 5. Covariance matrix inclusion is pending for a future iteration of this calculation matrix (per DocDB 11431[14]).

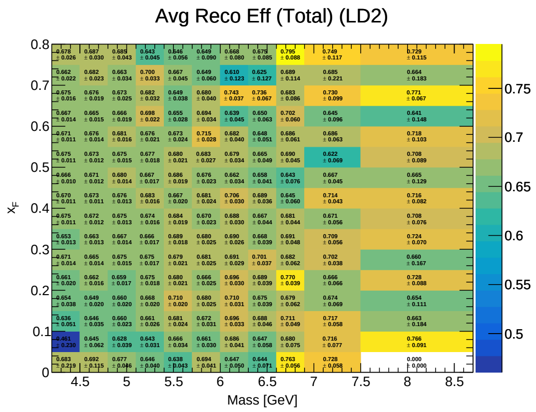
[14] `https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11431`



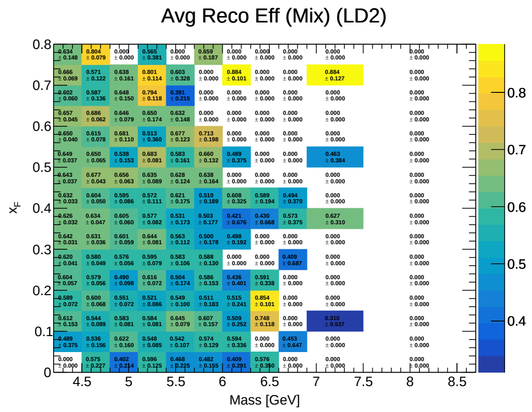
(a) LH2 average reconstruction efficiency for total



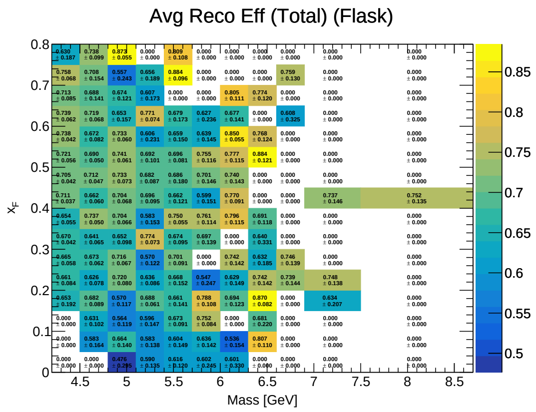
(b) LH2 average reconstruction efficiency for mix



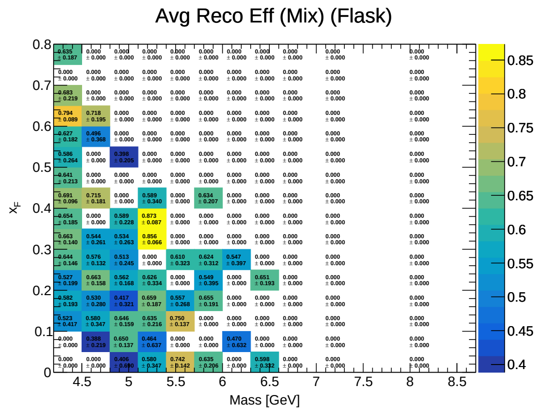
(c) LD2 average reconstruction efficiency for total



(d) LD2 average reconstruction efficiency for mix

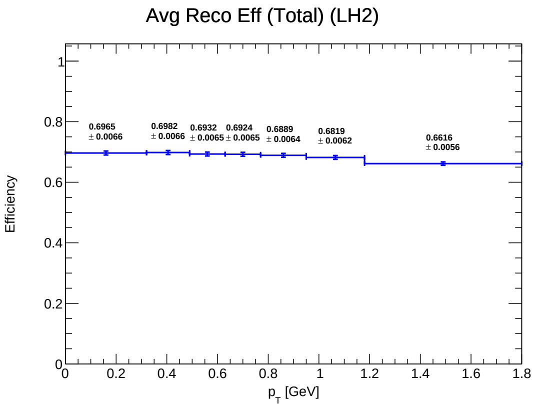


(e) Flask average reconstruction efficiency for total

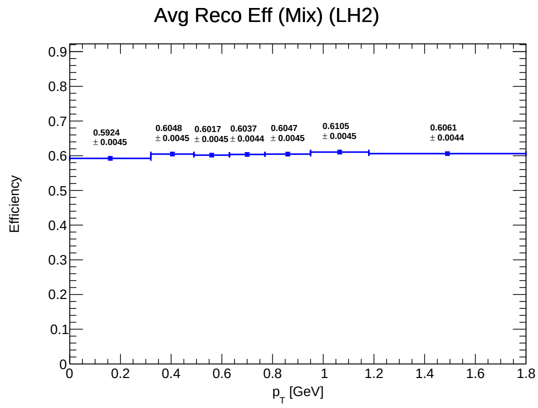


(f) Flask average reconstruction efficiency for mix

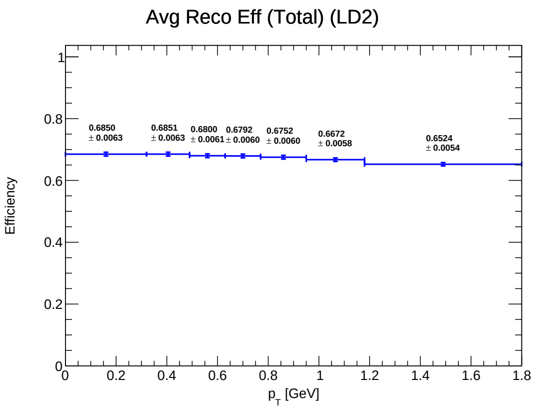
Figure 4: Average Reconstruction Efficiencies calculated each kinematic bin with the propagated uncertainties.



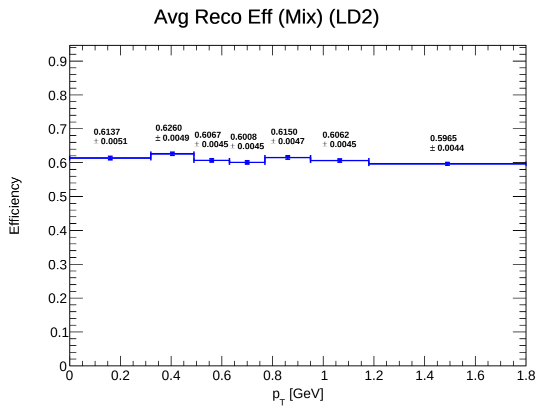
(a) LH2 average reconstruction efficiency for total



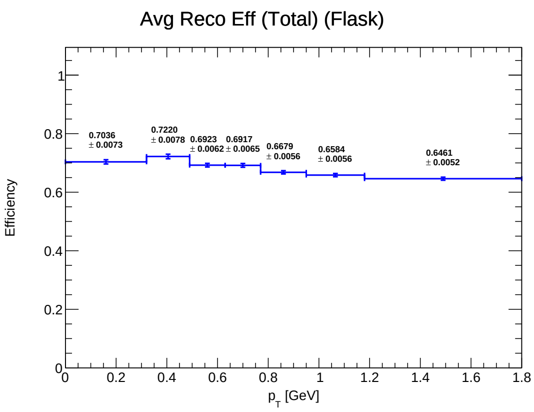
(b) LH2 average reconstruction efficiency for mix



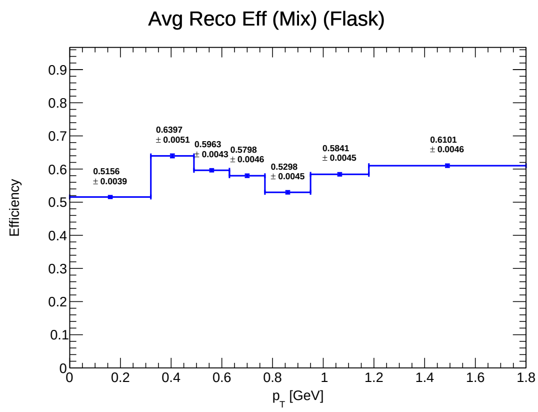
(c) LD2 average reconstruction efficiency for total



(d) LD2 average reconstruction efficiency for mix



(e) Flask average reconstruction efficiency for total

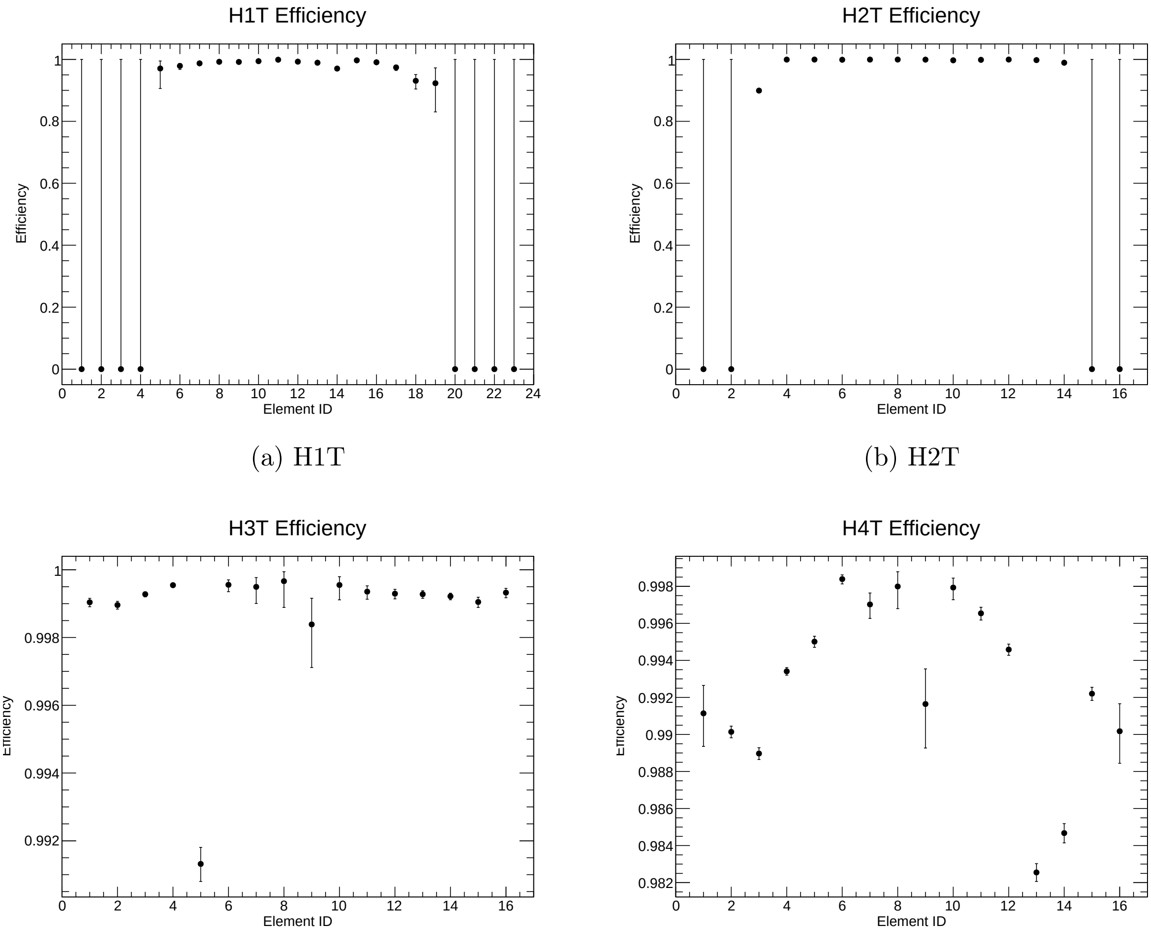


(f) Flask average reconstruction efficiency for mix

Figure 5: Average Reconstruction Efficiencies calculated each pT bin with the propagated uncertainties.

5 Hodoscope Efficiency Correction

To modernize the historical application of a rigid 0.845 ± 0.125 correction factor (DocDB 10809[15]), we have developed an event-by-event dynamic hodoscope resolution metric. Using the extrapolated position vectors at tracking stations 1 and 3, we define the trigger Road ID for every reconstructed dimuon leg. Intersecting these IDs with the dedicated single-track paddle efficiency matrices provided in DocDB 11467[16], we enforce a granular efficiency map (visualized in fig. 6).

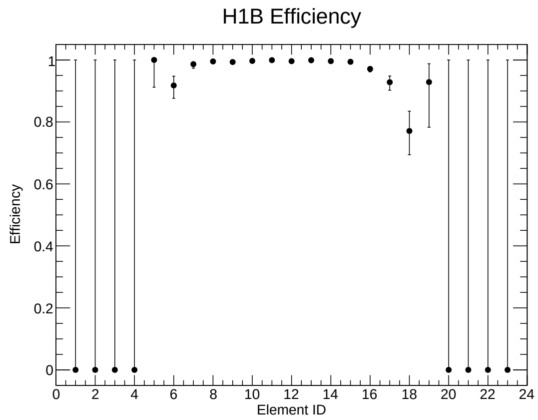


(a) H1T (b) H2T

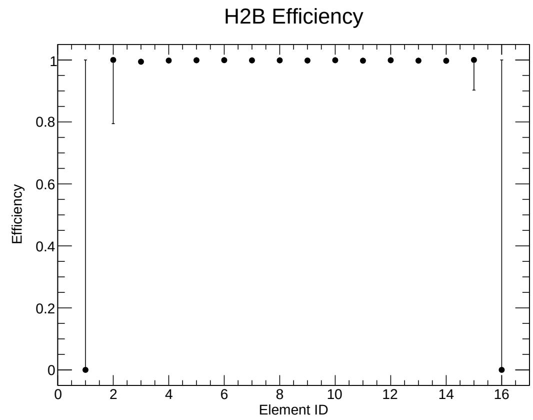
(c) H3T (d) H4T

Figure 6: Hodoscope Paddle Efficiencies (Top Planes).

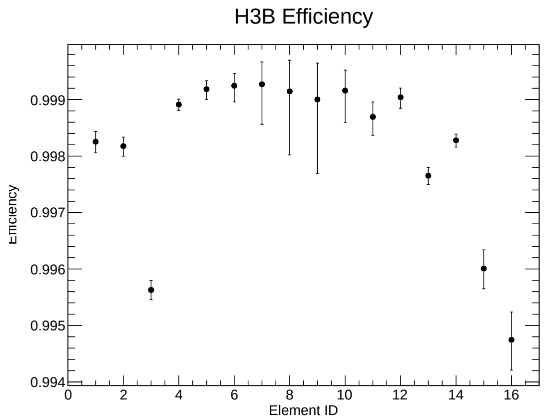
[15] `https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=10809`
[16] `https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11467`



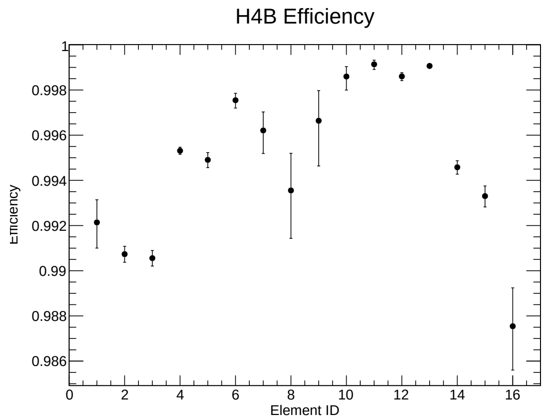
(e) H1B



(f) H2B



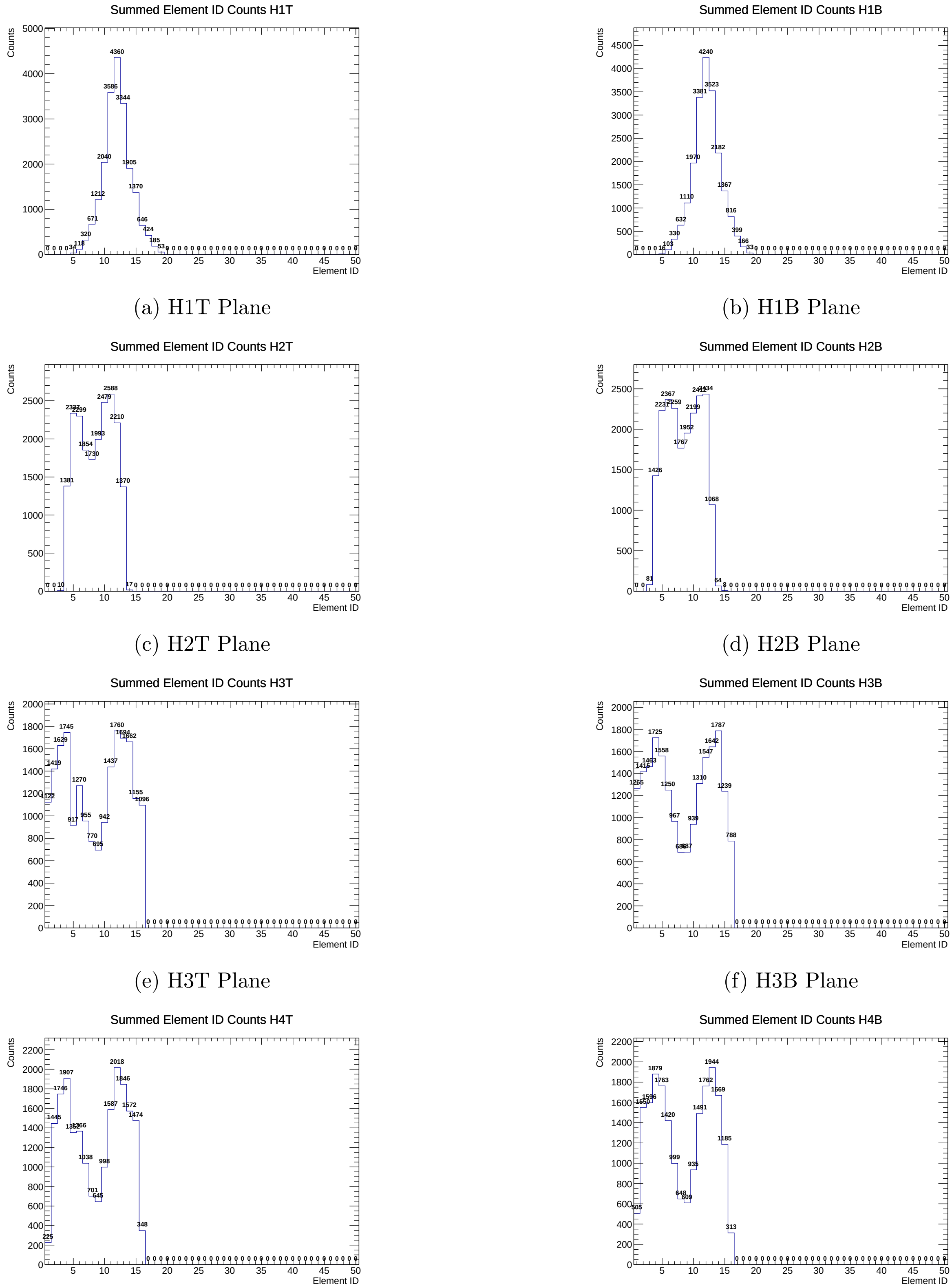
(g) H3B



(h) H4B

Figure 6: Hodoscope Paddle Efficiencies (Bottom Planes) – Continued.

The topological hit multiplicity distributions spanning the paddle planes are characterized in fig. 7, with the ultimate $\langle \epsilon_{\text{hodo}} \rangle$ response metric rendered across the M-x_F phase space in fig. 8.



(a) H1T Plane

(b) H1B Plane

(c) H2T Plane

(d) H2B Plane

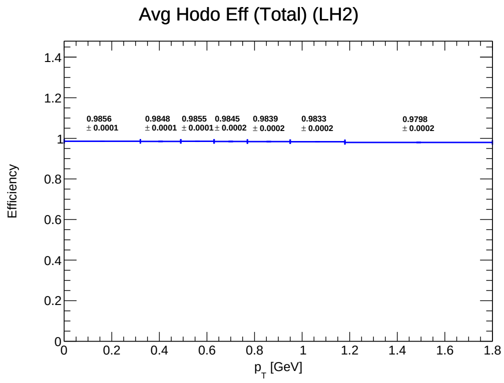
(e) H3T Plane

(f) H3B Plane

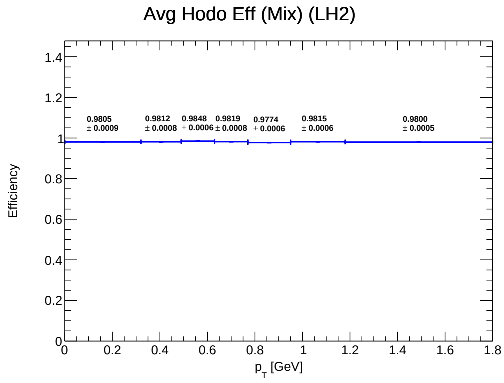
(g) H4T Plane

(h) H4B Plane

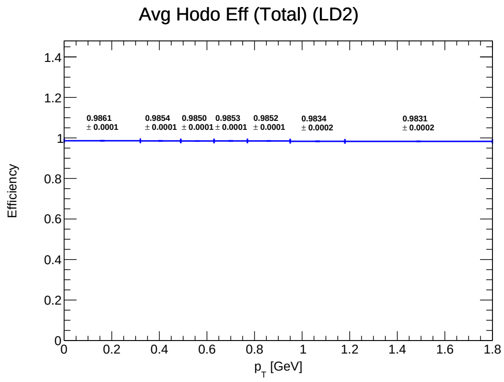
Figure 7: Hodoscope Paddle hit distributions arranged by Plane (Rows 1-4), Top vs Bottom.



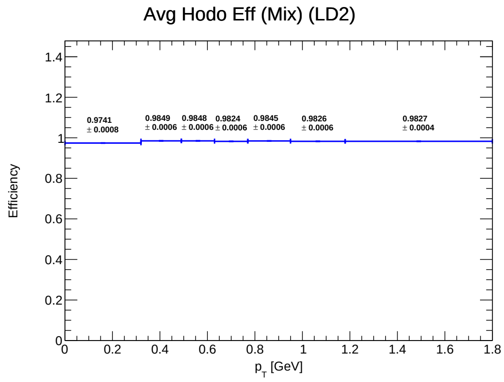
(a) LH2 average hodoscope efficiency for total



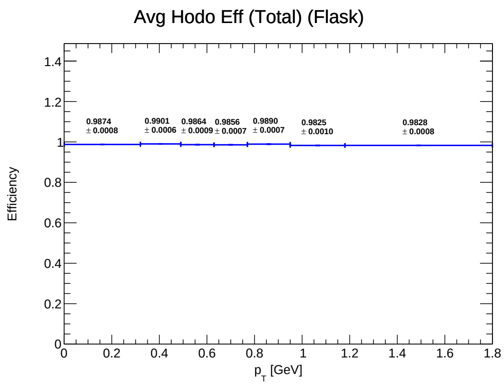
(b) LH2 average hodoscope efficiency for mix



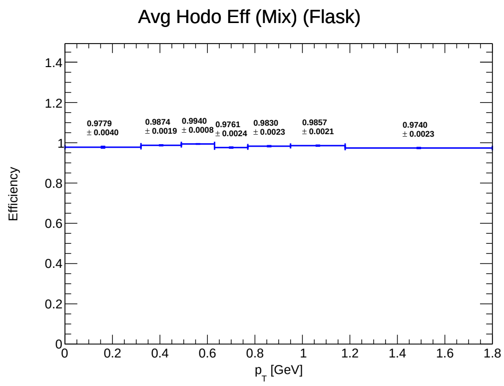
(c) LD2 average hodoscope efficiency for total



(d) LD2 average hodoscope efficiency for mix



(e) Flask average hodoscope efficiency for total



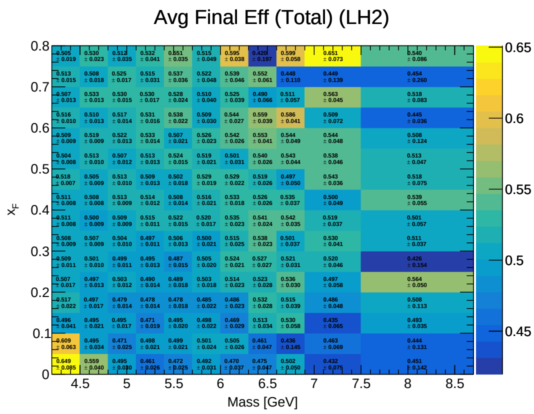
(f) Flask average hodoscope efficiency for mix

Figure 8: Average Hodoscope Efficiencies calculated each kinematic bin with the propagated uncertainties.

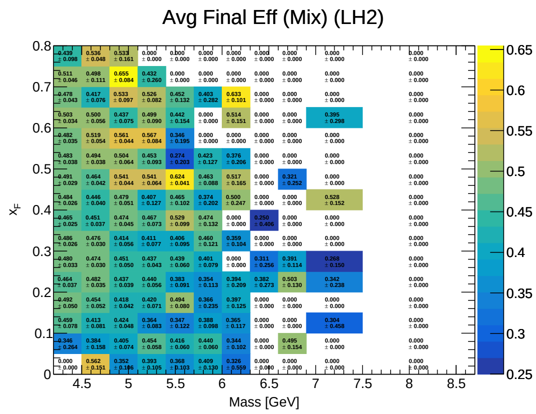
6 Total Efficiency Correction

The total analytical efficiency scalar $\epsilon^{i}_{\text{total}}$ per spatial bin is realized sequentially via the tensor product of the reconstruction and hodoscope efficiencies (eq. (3), aligned with DocDB 11448[17]). The resulting kinematics mappings, overlaid with propagated statistical error envelopes, trace the detector's survival probability gradient as presented in fig. 10.

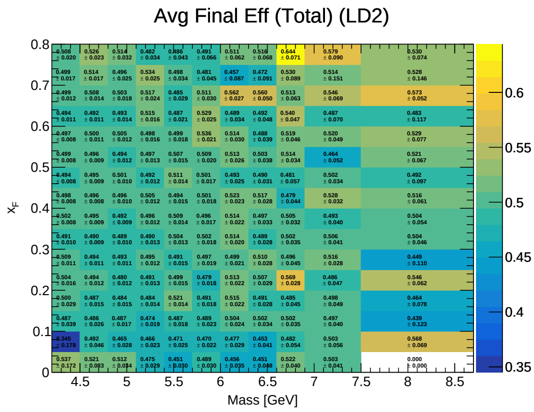
[17] https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11448



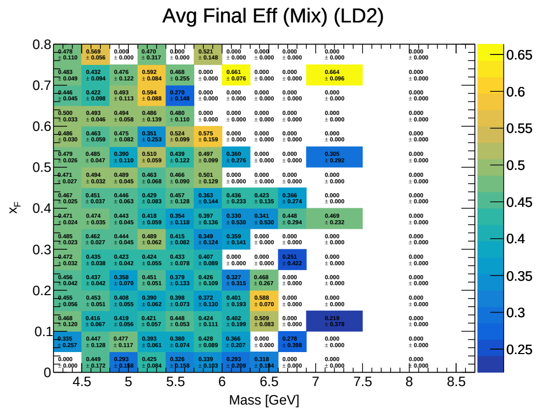
(a) LH2 average total efficiency for total



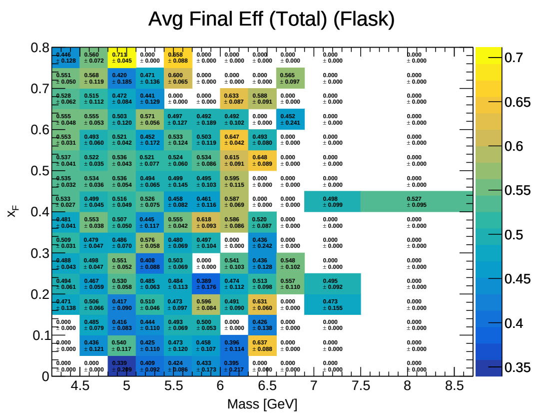
(b) LH2 average total efficiency for mix



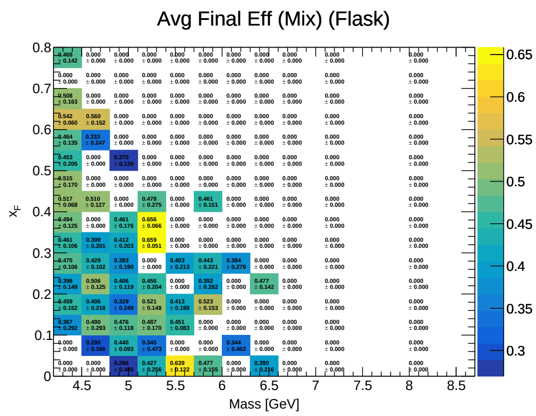
(c) LD2 average total efficiency for total



(d) LD2 average total efficiency for mix

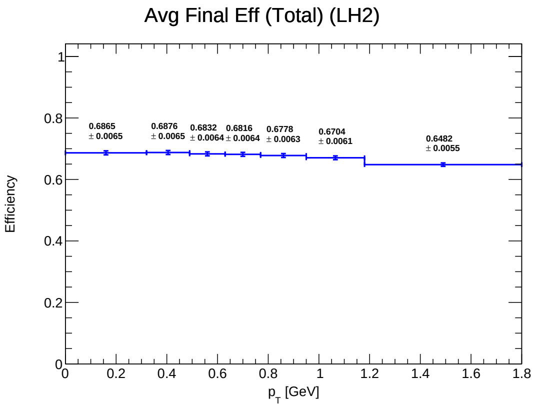


(e) Flask average total efficiency for total

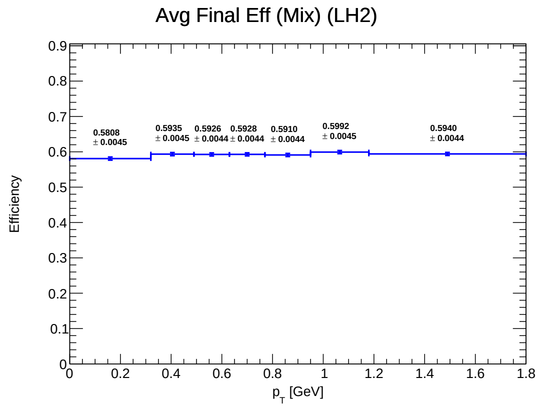


(f) Flask average total efficiency for mix

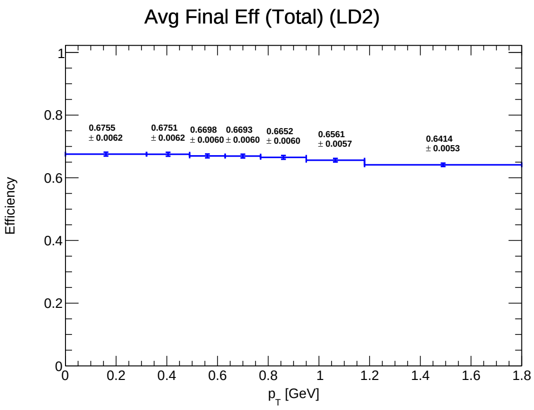
Figure 9: Average total Efficiencies calculated each kinematic bin with the propagated uncertainties.



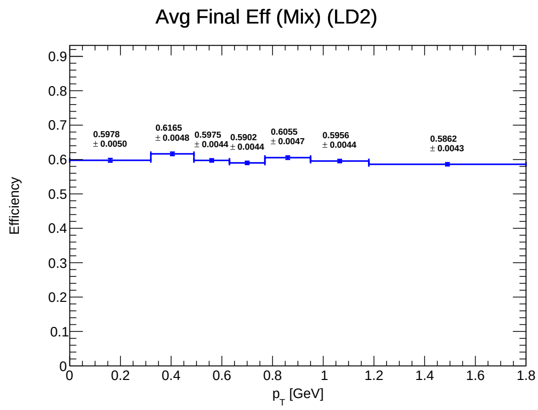
(a) LH2 average total efficiency for total



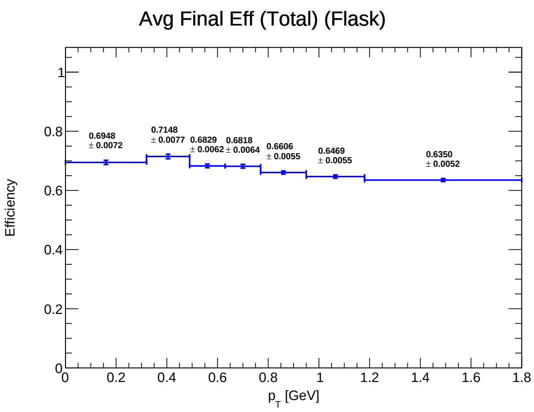
(b) LH2 average total efficiency for mix



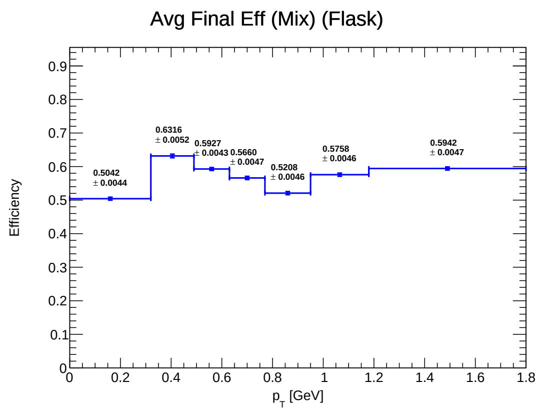
(c) LD2 average total efficiency for total



(d) LD2 average total efficiency for mix



(e) Flask average total efficiency for total



(f) Flask average total efficiency for mix

Figure 10: Average total Efficiencies calculated each kinematic bin with the propagated uncertainties.

7 Determination of Corrected Yields

Isolating the pure physical event rate from raw counting metrics requires scaling by the average signal efficiency correction $\langle \epsilon_{\text{signal}} \rangle$, defined analytically in eq. (2). This unfolding extracts the definitive background-suppressed yield for each distinct experimental geometry (see DocDB 11448[18]):

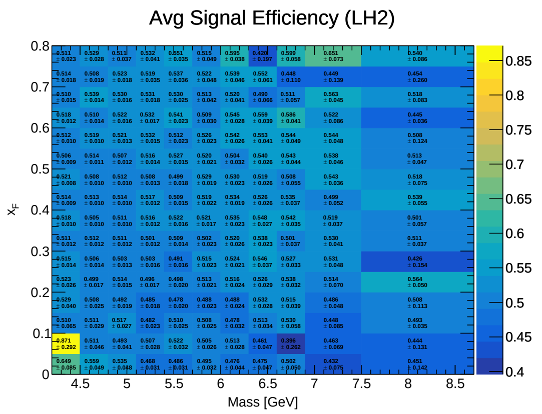
$$Y_{\text{signal}} = \frac{Y_{\text{total}} - Y_{\text{mix}}}{\langle \epsilon_{\text{signal}} \rangle} \tag{12}$$

Pathological boundaries in this relation are bounded explicitly:

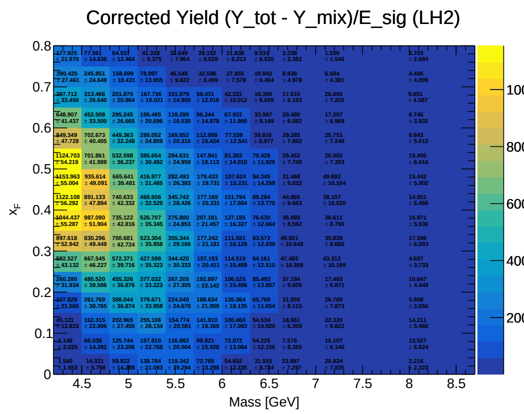
- When $Y_{\text{mix}} = 0$, the array implies null combinatorics, reducing to $\langle \epsilon_{\text{signal}} \rangle = \epsilon_{\text{total}}$.
- In statistically void phase space ($Y_{\text{total}} = 0$ and $Y_{\text{mix}} = 0$), we enforce $\langle \epsilon_{\text{signal}} \rangle = 0$ alongside $Y_{\text{signal}} = 0$.
- A singularity where $Y_{\text{total}} = Y_{\text{mix}} \neq 0$ is structurally averted, as the empirical physical prompt yield Y_{total} uniformly dominates Y_{mix}.

The spatially resolved maps of the final corrected yields are detailed in fig. 12.

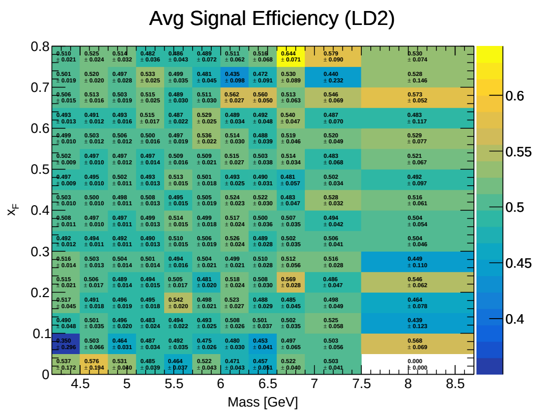
[18] `https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11448`



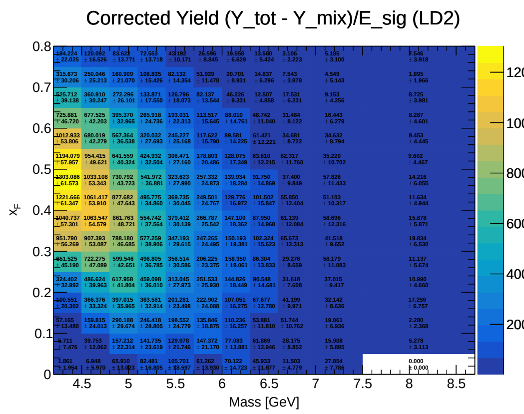
(a) LH2 average signal efficiency correction



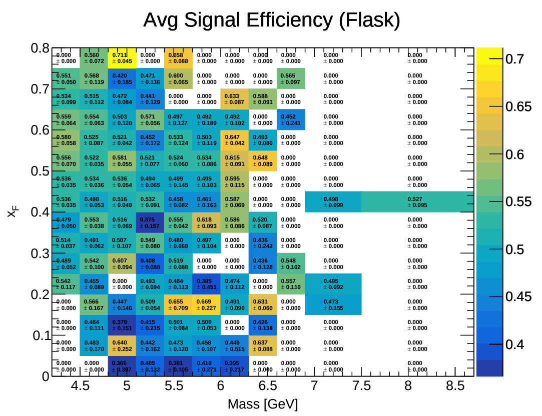
(b) Corrected signal yield for LH2



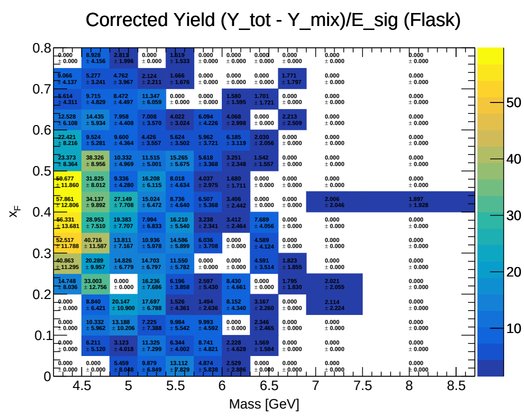
(c) LD2 average signal efficiency correction



(d) Corrected signal yield for LD2

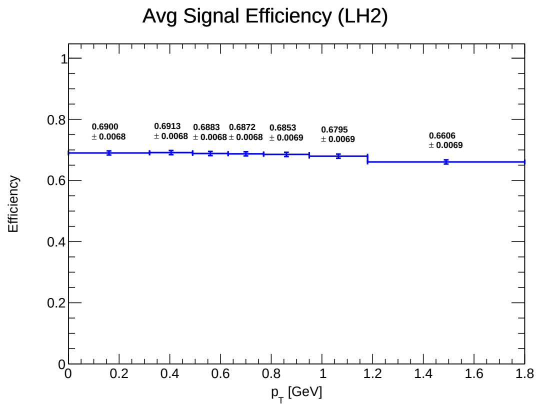


(e) Flask average signal efficiency correction

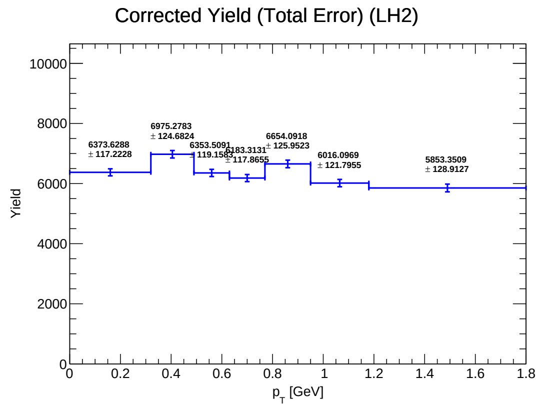


(f) Corrected signal yield for Flask

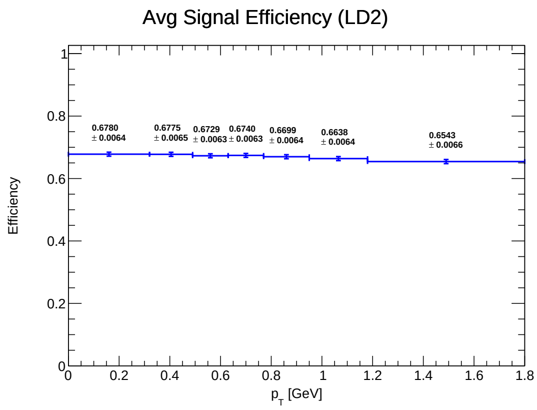
Figure 11: Average signal efficiency $\langle \epsilon_{\text{signal}} \rangle$ (left panels) and final corrected signal yields (right panels).



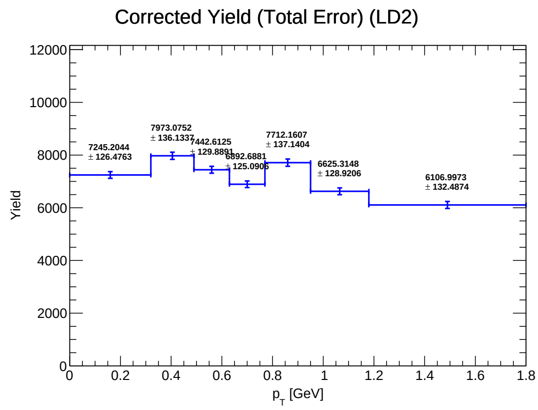
(a) LH2 average signal efficiency correction



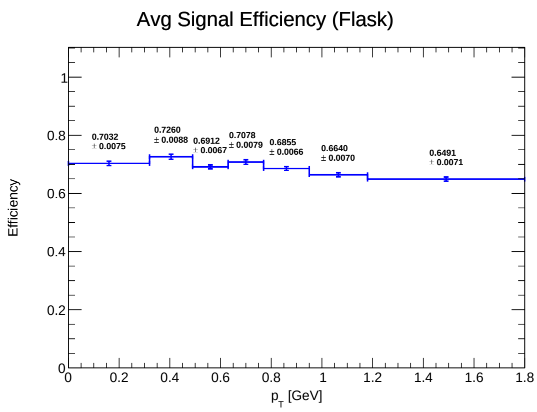
(b) Corrected signal yield for LH2



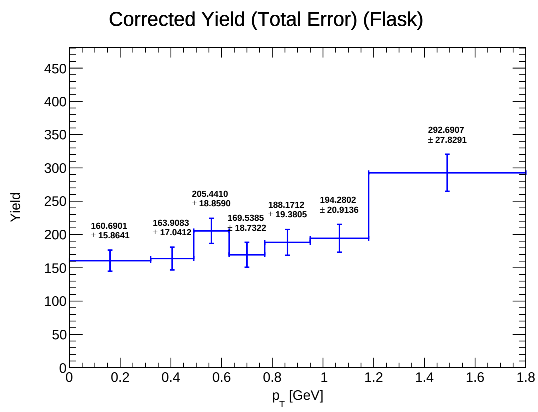
(c) LD2 average signal efficiency correction



(d) Corrected signal yield for LD2



(e) Flask average signal efficiency correction



(f) Corrected signal yield for Flask

Figure 12: Average signal efficiency $\langle\epsilon_{\text{signal}}\rangle$ (left panels) and final corrected signal yields (right panels).

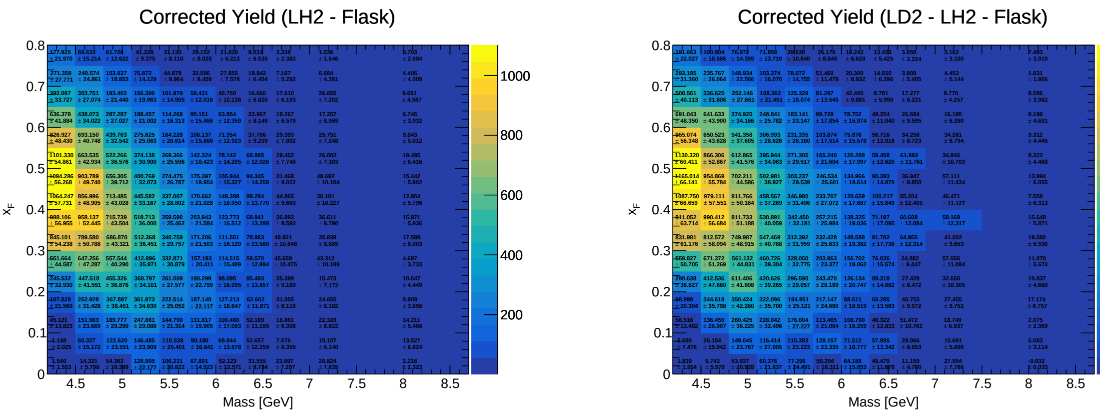
The definitive LH_2 corrected yield arises mathematically following the dual subtraction of uncorrelated mixed pairs and integrated upstream flask background interactions:

$$Y_{\text{corrected}}^{\text{LH2}} = \frac{Y_{\text{total}}^{\text{LH2}} - Y_{\text{mixed}}^{\text{LH2}}}{\langle\epsilon_{\text{signal}}^{\text{LH2}}\rangle} - \frac{I_{\text{LH2}}}{I_{\text{flask}}}\left(\frac{Y_{\text{total}}^{\text{flask}} - Y_{\text{mixed}}^{\text{flask}}}{\langle\epsilon_{\text{signal}}^{\text{flask}}\rangle}\right) \tag{13}$$

The analogous LD_2 formalism incorporates the supplementary protium isotopic compensation:

$$
\begin{aligned}
Y_{\text{corrected}}^{\text{LD2}} \quad = \quad & \frac{Y_{\text{total}}^{\text{LD2}} - Y_{\text{mixed}}^{\text{LD2}}}{\langle \epsilon_{\text{signal}}^{\text{LD2}} \rangle} - \frac{I_{\text{LD2}}}{I_{\text{flask}}} \left(\frac{Y_{\text{total}}^{\text{flask}} - Y_{\text{mixed}}^{\text{flask}}}{\langle \epsilon_{\text{signal}}^{\text{flask}} \rangle} \right) \\
& - \frac{T_{HD}}{T_{HH}} \frac{I_{\text{LD2}}}{I_{\text{LH2}}} \left[\frac{Y_{\text{total}}^{\text{LH2}} - Y_{\text{mixed}}^{\text{LH2}}}{\langle \epsilon_{\text{signal}}^{\text{LH2}} \rangle} - \frac{I_{\text{LH2}}}{I_{\text{flask}}} \left(\frac{Y_{\text{total}}^{\text{flask}} - Y_{\text{mixed}}^{\text{flask}}}{\langle \epsilon_{\text{signal}}^{\text{flask}} \rangle} \right) \right]
\end{aligned} \tag{14}
$$

The phase space evolution of these isolated signals is captured in fig. 13.



(a) LH2 (b) LD2

Figure 13: Corrected yields for LH2 (a) and LD2 (b) after correcting for efficiencies, with propagated uncertainties.

8 Uncertainty Calculation

We bifurcate the error estimation into stochastic bounds and systematic limits. Utilizing standard partial derivative formalisms over the basis vector established in eq. (13), the variance propagates through the measurement topology.

Statistical Uncertainty Components

Because counting variance dominates sparse bin peripheries, the partial derivatives coupling to raw counts take the form:

$$\frac{\partial Y_{\text{corrected}}}{\partial Y_{\text{total}}^{\text{LH2}}} = \frac{1}{\langle \epsilon_{\text{signal}}^{\text{LH2}} \rangle} \tag{15}$$

$$\frac{\partial Y_{\text{corrected}}}{\partial Y_{\text{mixed}}^{\text{LH2}}} = -\frac{1}{\langle \epsilon_{\text{signal}}^{\text{LH2}} \rangle} \tag{16}$$

$$\frac{\partial Y_{\text{corrected}}}{\partial Y_{\text{total}}^{\text{flask}}} = -\frac{I_{\text{LH2}}}{I_{\text{flask}}} \frac{1}{\langle \epsilon_{\text{signal}}^{\text{flask}} \rangle} \tag{17}$$

$$\frac{\partial Y_{\text{corrected}}}{\partial Y_{\text{mixed}}^{\text{flask}}} = \frac{I_{\text{LH2}}}{I_{\text{flask}}} \frac{1}{\langle \epsilon_{\text{signal}}^{\text{flask}} \rangle} \tag{18}$$

Systematic Uncertainty Components from Efficiencies

The covariance of the synthetic signal efficiencies introduces foundational systematic boundaries:

$$\frac{\partial Y_{\text{corrected}}}{\partial \langle \epsilon_{\text{signal}}^{\text{LH2}} \rangle} = -\frac{Y_{\text{total}}^{\text{LH2}} - Y_{\text{mixed}}^{\text{LH2}}}{\langle \epsilon_{\text{signal}}^{\text{LH2}} \rangle^2} \tag{19}$$

$$\frac{\partial Y_{\text{corrected}}}{\partial \langle \epsilon_{\text{signal}}^{\text{flask}} \rangle} = \frac{I_{\text{LH2}}}{I_{\text{flask}}} \frac{Y_{\text{total}}^{\text{flask}} - Y_{\text{mixed}}^{\text{flask}}}{\langle \epsilon_{\text{signal}}^{\text{flask}} \rangle^2} \tag{20}$$

Uncertainty Propagation

Assuming ideal Poissonian variance over the constituent arrays ($\sigma_Y^2 \approx Y$), the absolute statistical error bounds map natively via quadrature addition:

$$\begin{aligned} \delta Y_{\text{corrected}}^{\text{stat}} = \Bigg[& \left(\frac{\partial Y_{\text{corrected}}}{\partial Y_{\text{total}}^{\text{LH2}}} \right)^2 Y_{\text{total}}^{\text{LH2}} + \left(\frac{\partial Y_{\text{corrected}}}{\partial Y_{\text{mix}}^{\text{LH2}}} \right)^2 Y_{\text{mix}}^{\text{LH2}} \\ & + \left(\frac{\partial Y_{\text{corrected}}}{\partial Y_{\text{total}}^{\text{flask}}} \right)^2 Y_{\text{total}}^{\text{flask}} + \left(\frac{\partial Y_{\text{corrected}}}{\partial Y_{\text{mix}}^{\text{flask}}} \right)^2 Y_{\text{mix}}^{\text{flask}} \Bigg]^{1/2} \end{aligned} \tag{21}$$

By construction, the fractional statistical dispersion holds equivalently across the integrated cross section:

$$\frac{\delta \sigma^{\text{stat}}}{\sigma} = \frac{\delta Y_{\text{corrected}}^{\text{stat}}}{Y_{\text{corrected}}} \tag{22}$$

The macroscopic systematic limit, isolated to detector and spatial correction drift, evolves from the efficiency deviations via:

$$\delta Y^{\text{sys,eff}}_{\text{corrected}} = \left[\left(\frac{\partial Y_{\text{corrected}}}{\partial \langle \epsilon^{\text{LH2}}_{\text{signal}} \rangle} \, \delta \langle \epsilon^{\text{LH2}}_{\text{signal}} \rangle \right)^2 + \left(\frac{\partial Y_{\text{corrected}}}{\partial \langle \epsilon^{\text{flask}}_{\text{signal}} \rangle} \, \delta \langle \epsilon^{\text{flask}}_{\text{signal}} \rangle \right)^2 \right]^{1/2} \tag{23}$$

The ultimate systematic constraint $\delta\sigma^{\text{syst.}}$ represents the integration of three primary perturbations:

- Monte Carlo spatial acceptance mappings ($\mathcal{A}$) [section 3]
- Detector and analytical correction yield deviations ($Y_{\text{corrected}}$) [Detailed above]
- Invariant mass contamination via the ψ' radiative tail ($Y_{\psi'}/Y_{DY}$) [section 14]

Evaluating these linearly against the macroscopic continuum yield resolves the absolute error fields:

$$\delta\sigma^{\text{syst.}}_{\text{acc}} = \left(\frac{\delta\mathcal{A}}{\mathcal{A}} \right) \sigma \tag{24}$$

$$\delta\sigma^{\text{syst.}}_{\text{yield}} = \left(\frac{\delta Y^{\text{sys,eff}}_{\text{corrected}}}{Y_{\text{corrected}}} \right) \sigma \tag{25}$$

$$\delta\sigma^{\text{syst.}}_{\psi'} = (Y_{\psi'}/Y_{DY}) \times \sigma \tag{26}$$

To preserve structural coherence, the non-correlated topological components combine into the aggregate bound via Cartesian sum:

$$\delta\sigma^{\text{syst.}} = \sqrt{\left(\delta\sigma^{\text{syst.}}_{\text{acc}}\right)^2 + \left(\delta\sigma^{\text{syst.}}_{\text{yield}}\right)^2 + \left(\delta\sigma^{\text{syst.}}_{\psi'}\right)^2} \tag{27}$$

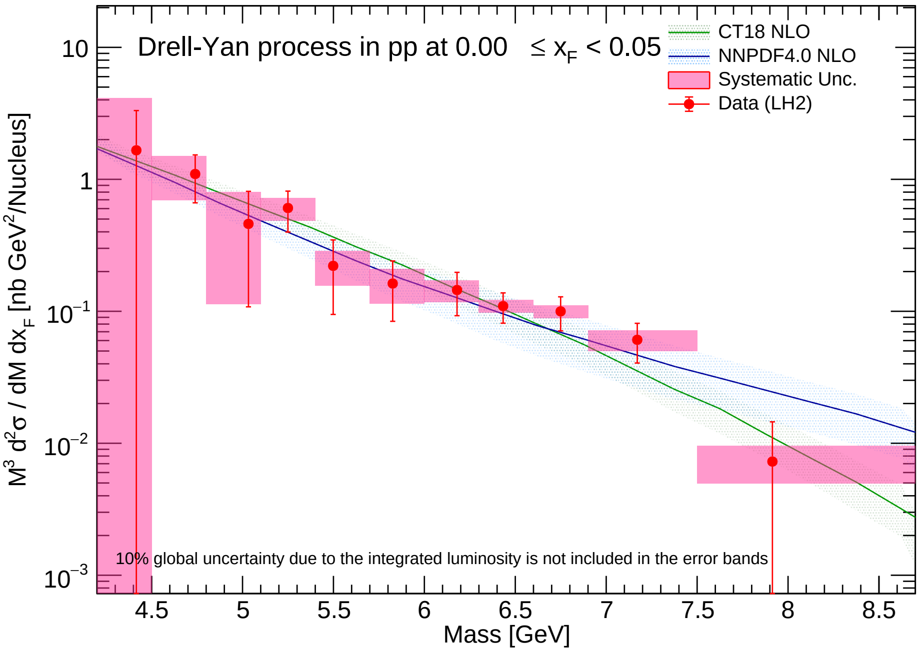
(Note: Error translation into the LD_2 framework follows a parallel methodology modified structurally to accommodate the scalar variance of the internal isotopic subtraction).

9 Cross-Section Results

9.1 LH2 Results (Reported at mass bin average)

We define the primary measurement space plotting the absolute double-differential cross-section of the LH_2 dataset aligned with the internal mass centroid of each kinematic grid. These values are plotted alongside Next-to-Leading-Order calculations using the CT18 and NNPDF4.0 parameterizations to evaluate theoretical consistency.

Statistical voids present in boundary grids currently induce elevated uncertainty spans, which will resolve following the planned merging of adjacent roadsets.



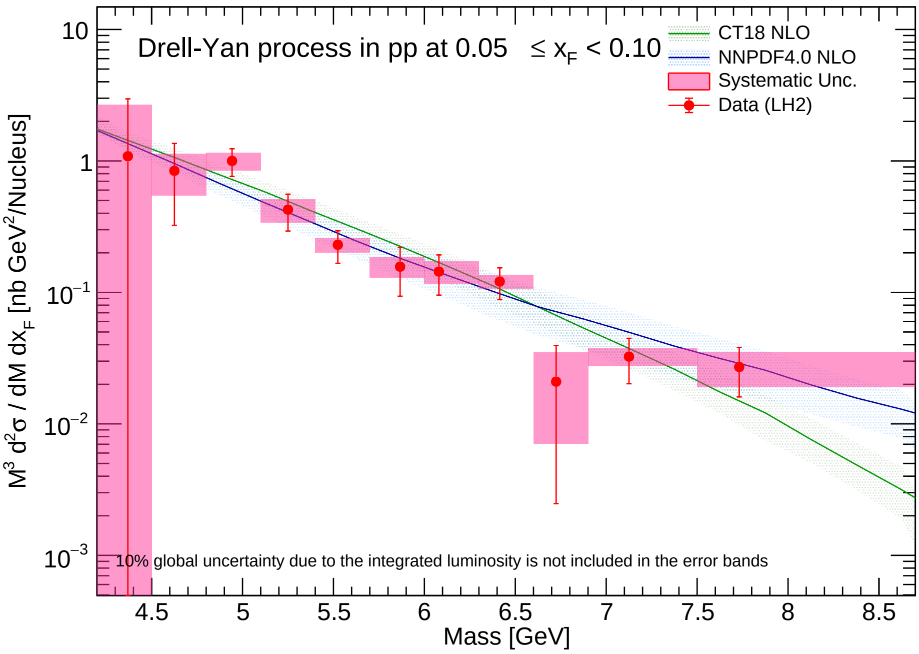
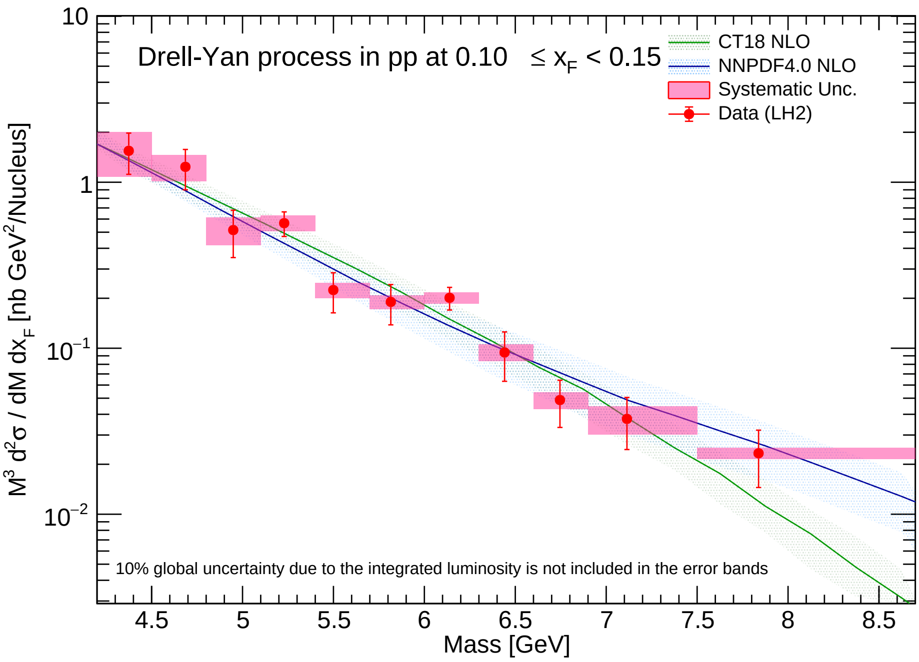


Figure 14: Double-differential cross-section for LH2 (xF bins 0.00-0.10) reported at mass bin average.



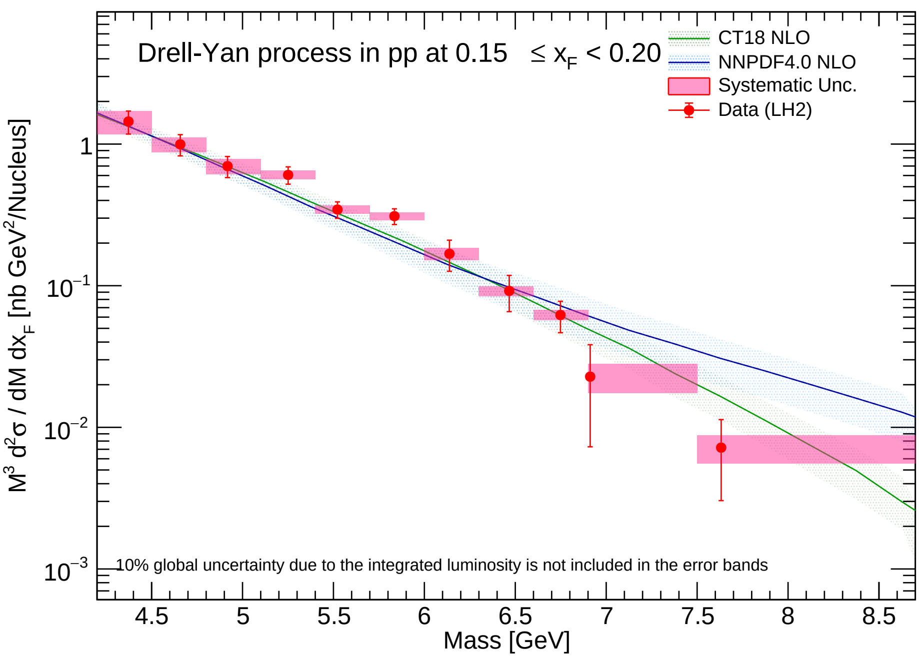


Figure 15: Double-differential cross-section for LH2 (xF bins 0.10-0.20) reported at mass bin average.

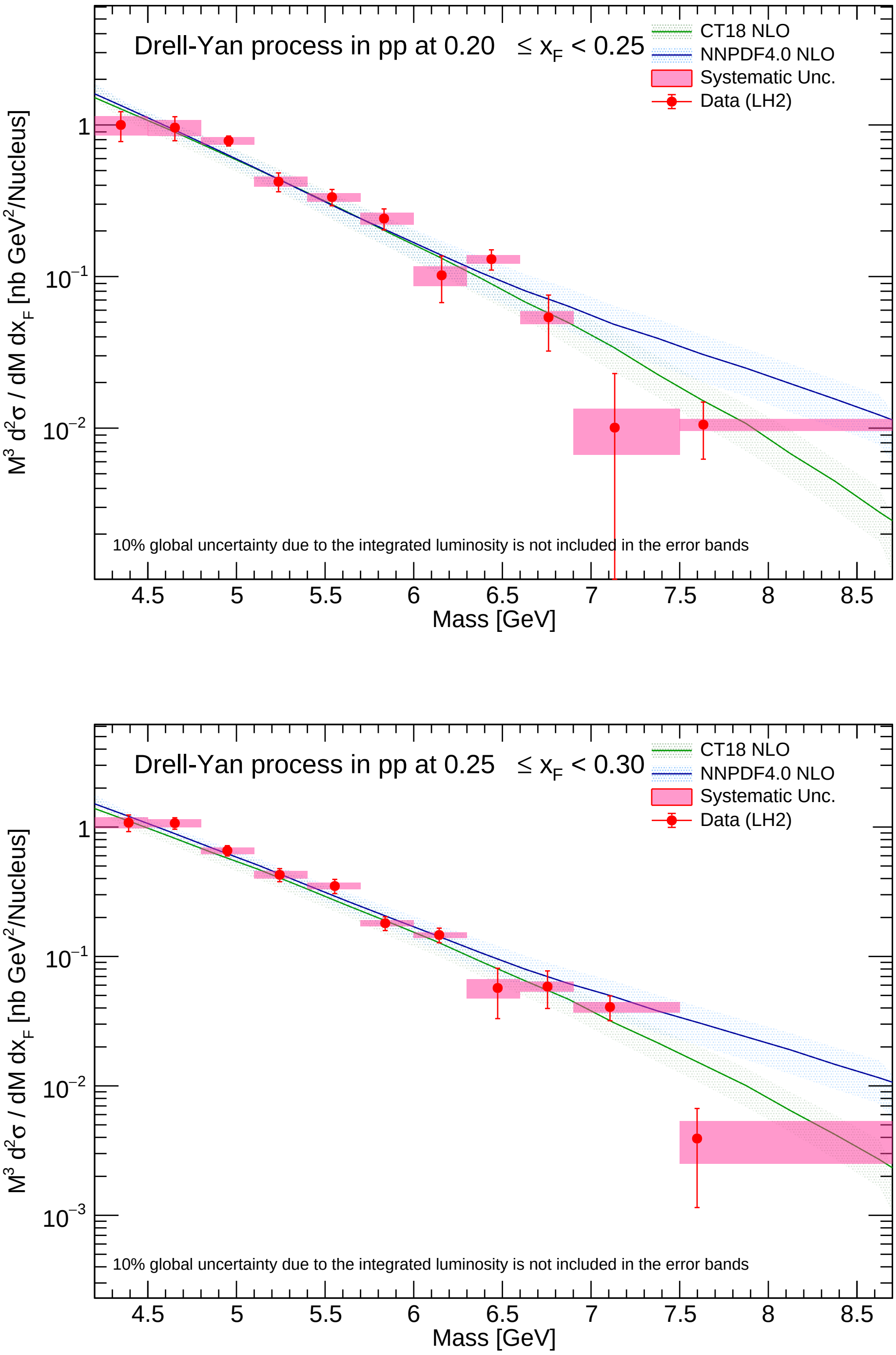


Figure 16: Double-differential cross-section for LH2 (xF bins 0.20-0.30) reported at mass bin average.

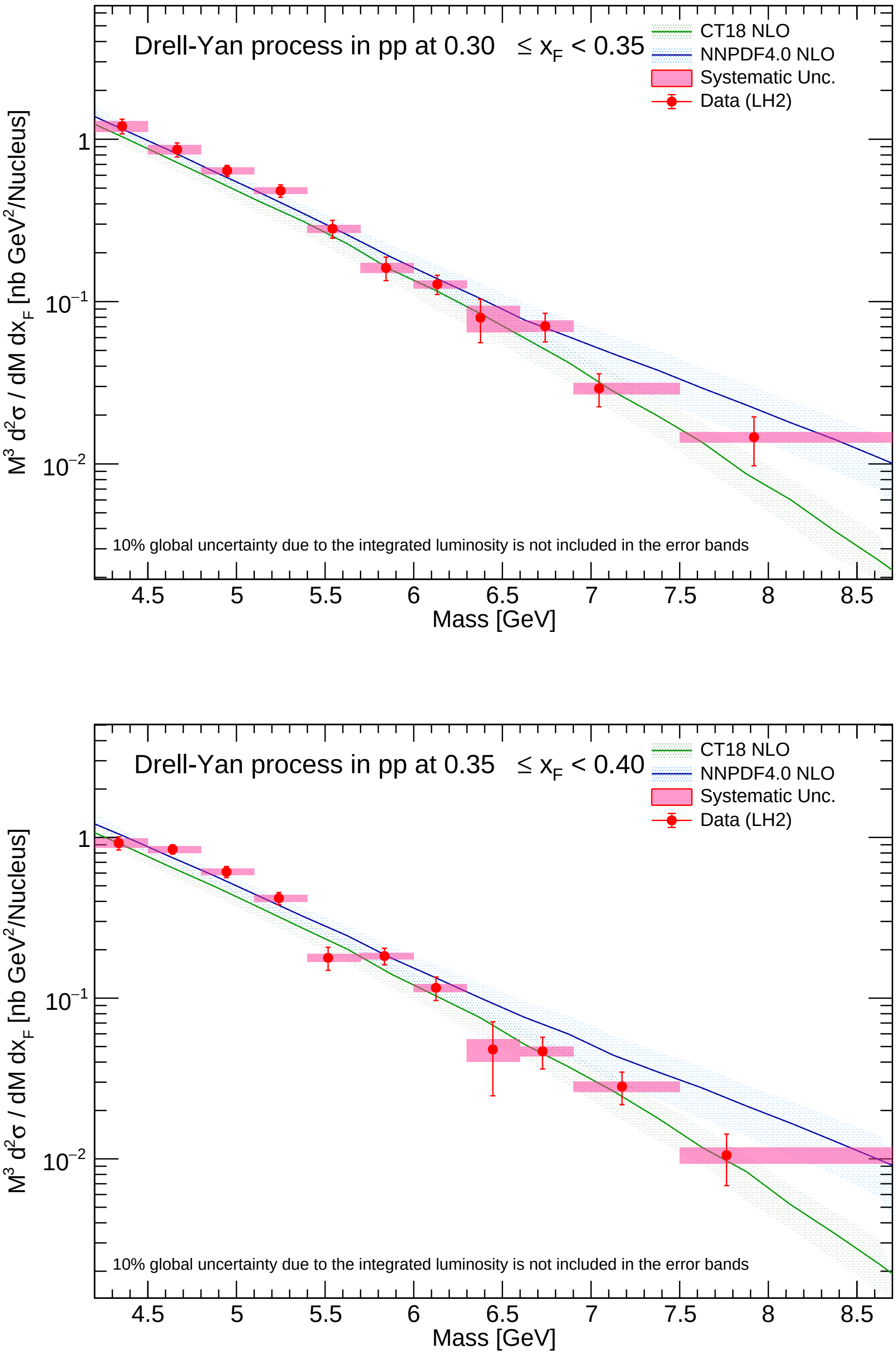
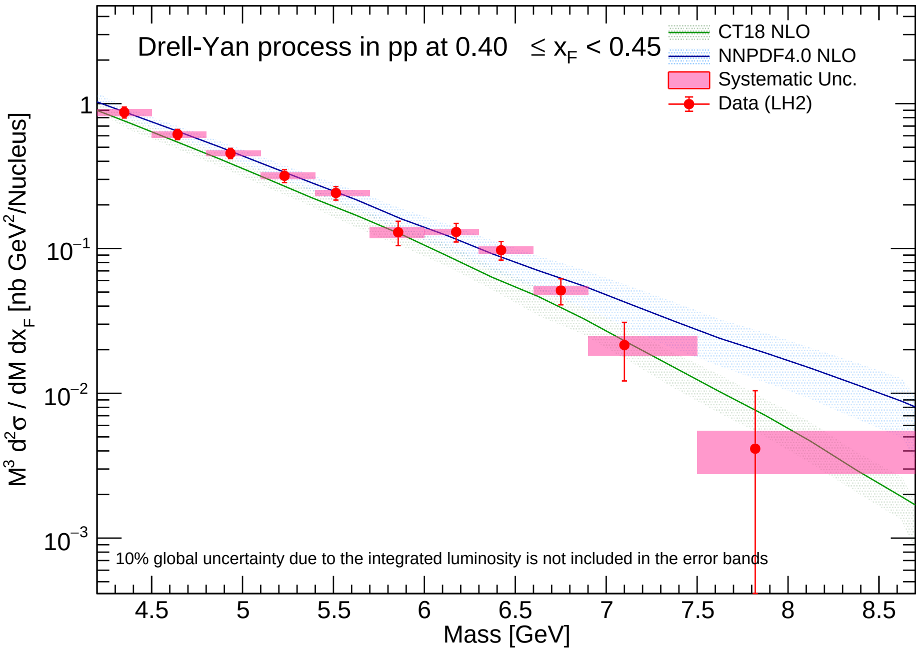


Figure 17: Double-differential cross-section for LH2 (xF bins 0.30-0.40) reported at mass bin average.



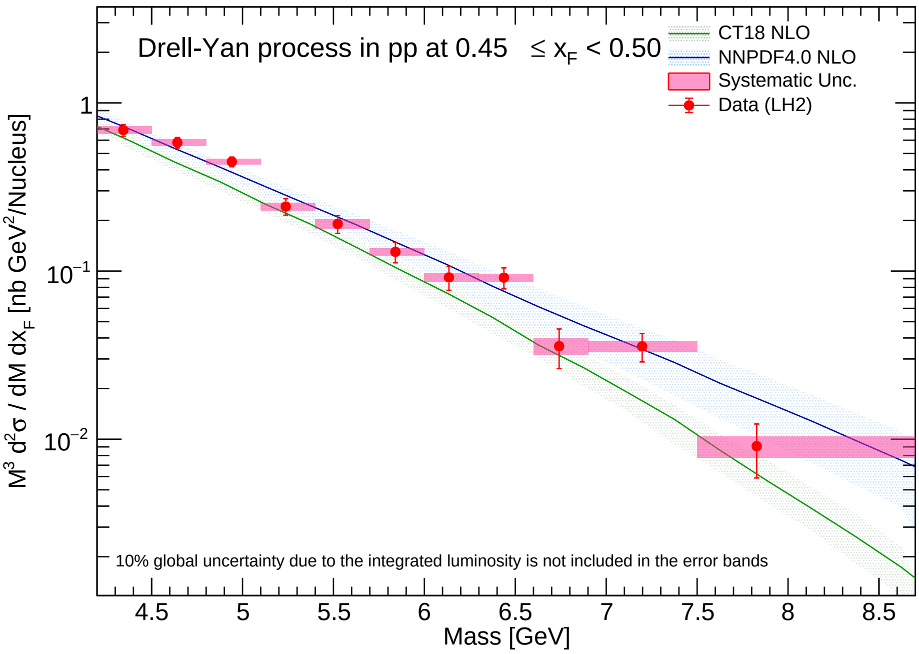
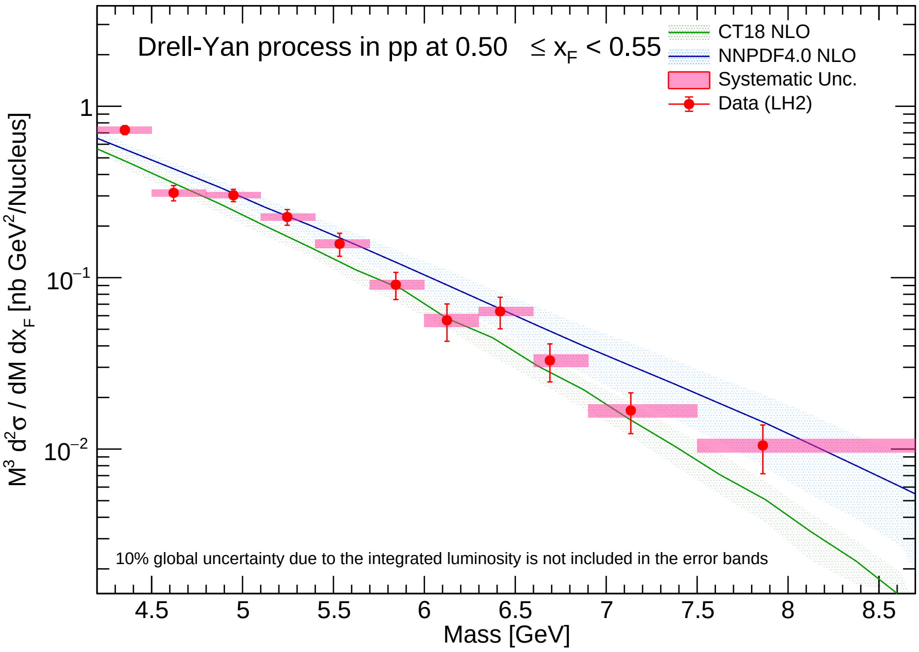


Figure 18: Double-differential cross-section for LH2 (xF bins 0.40-0.50) reported at mass bin average.



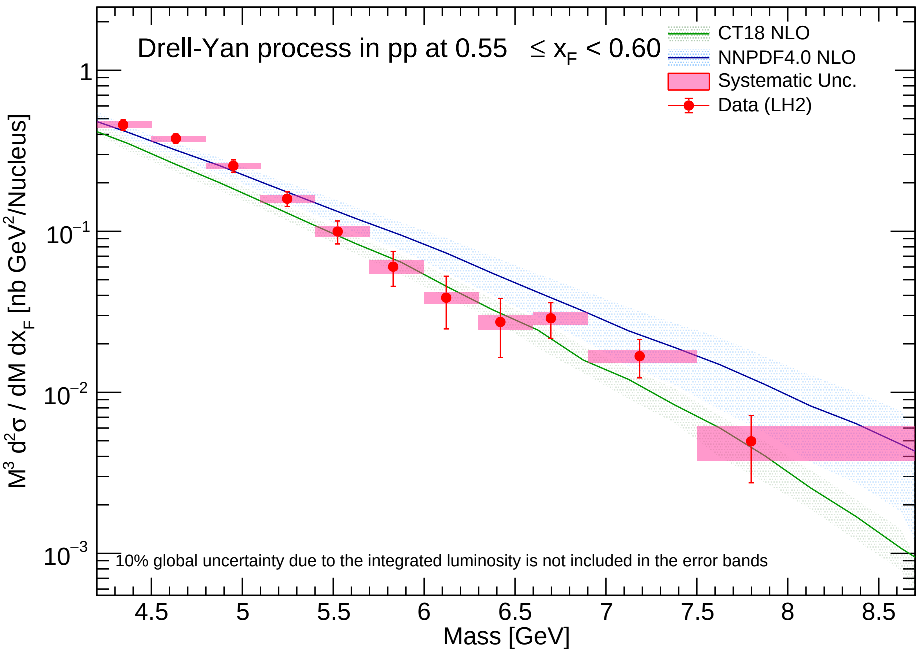


Figure 19: Double-differential cross-section for LH2 (xF bins 0.50-0.60) reported at mass bin average.

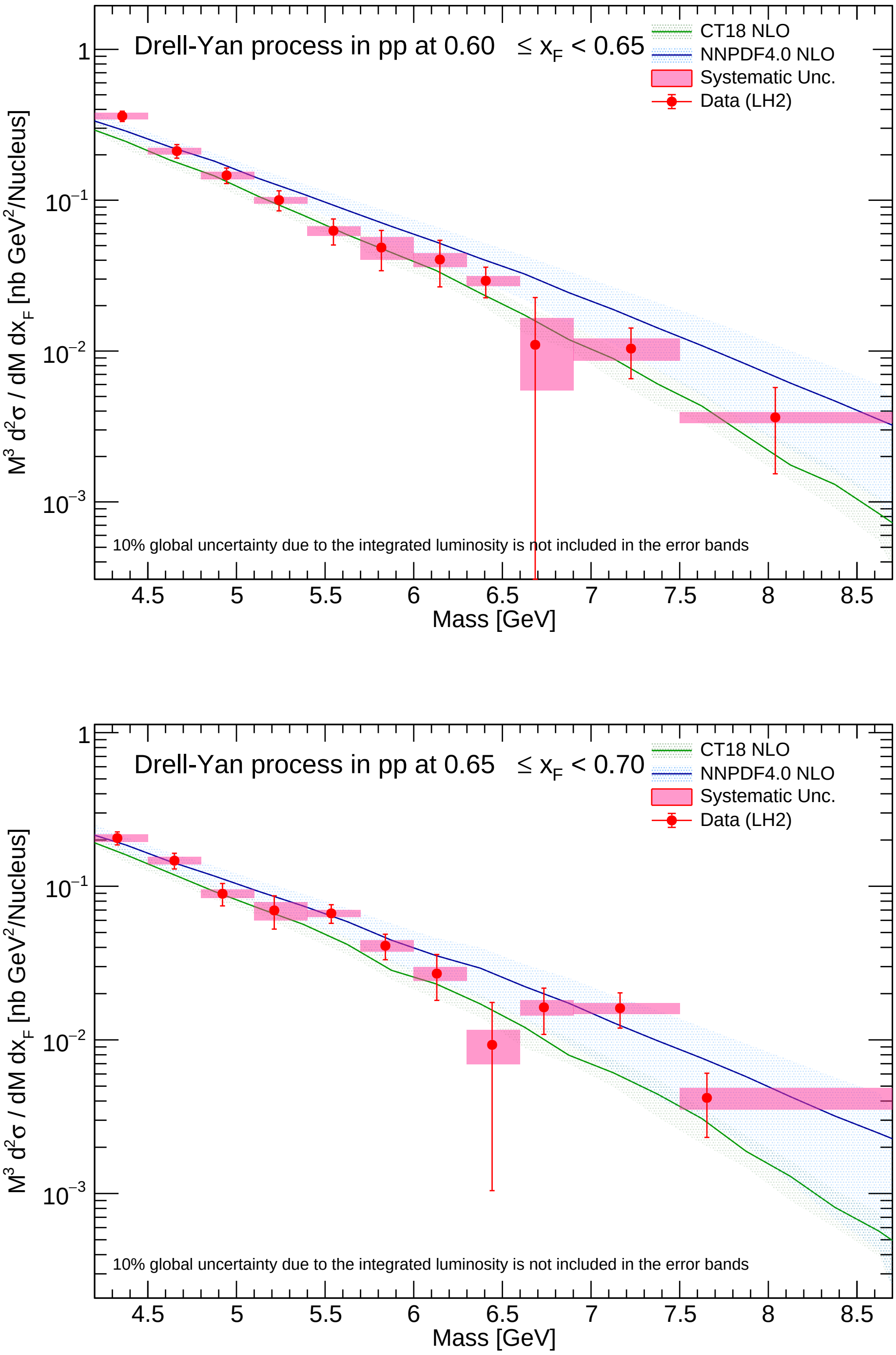
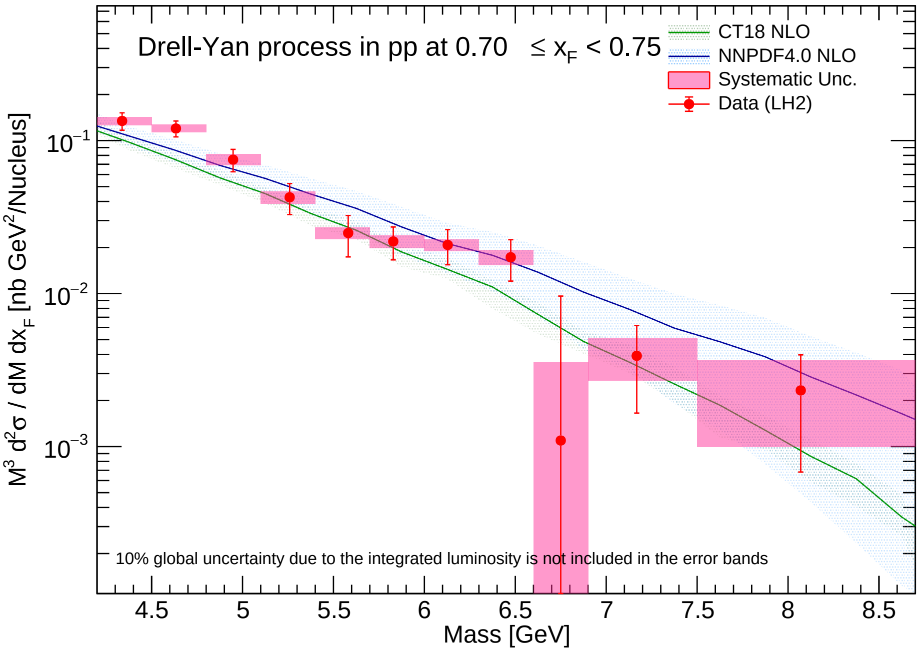


Figure 20: Double-differential cross-section for LH2 (xF bins 0.60-0.70) reported at mass bin average.



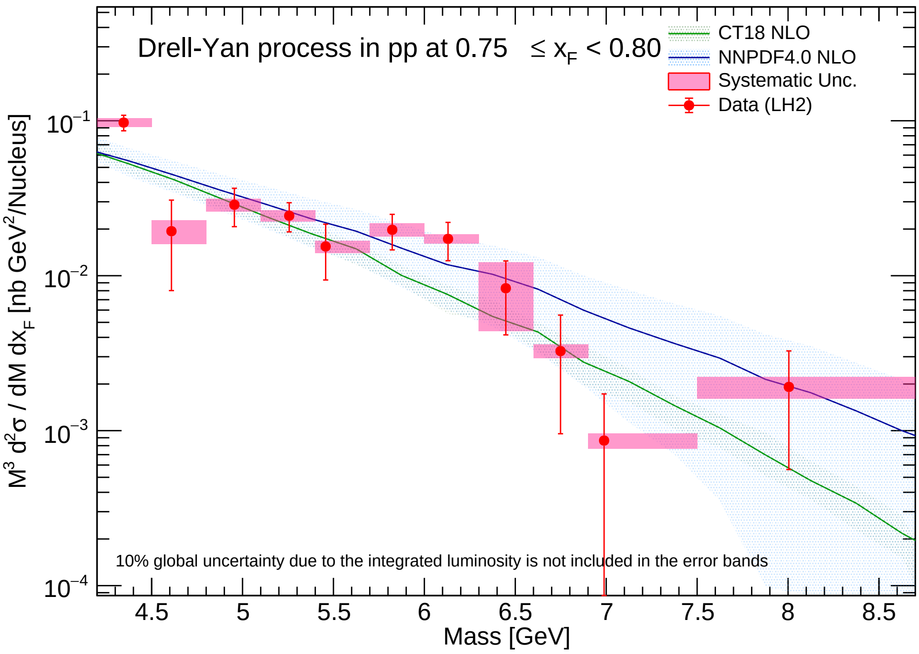
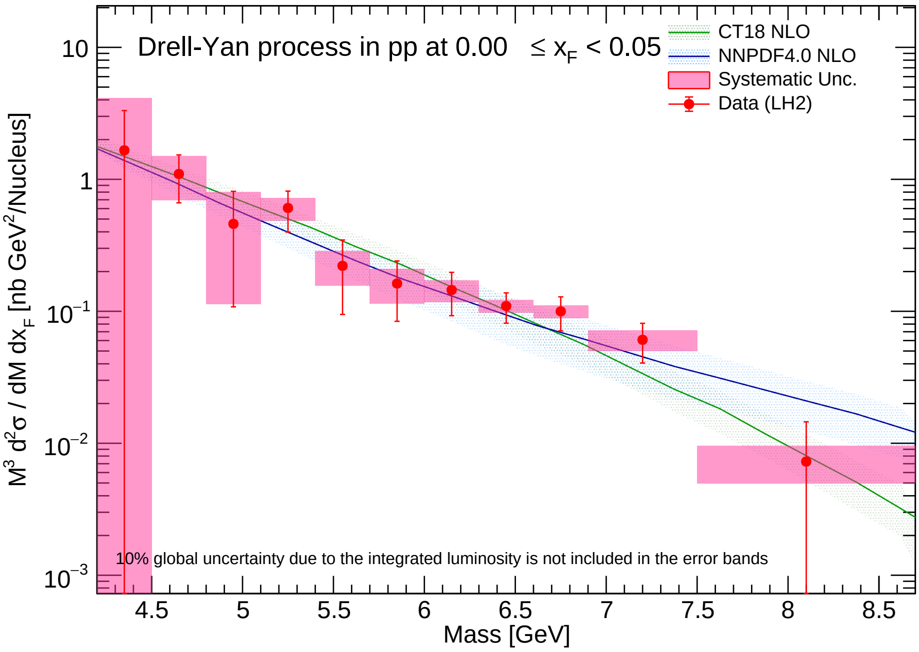


Figure 21: Double-differential cross-section for LH2 (xF bins 0.70-0.80) reported at mass bin average.

9.2 LH2 Results (Reported at mass bin center)

For symmetry against previous analytical frameworks, we additionally project the LH_2 measurements evaluated strictly at the discrete geometric boundary center of the mass bin.



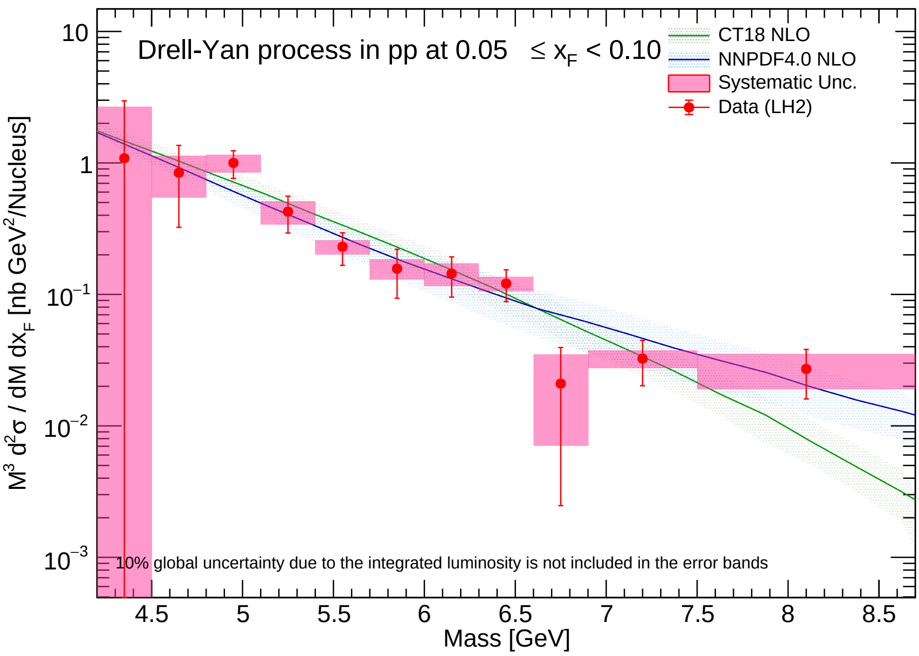


Figure 22: Double-differential cross-section for LH2 (xF bins 0.00-0.10) reported at geometric mass bin center.

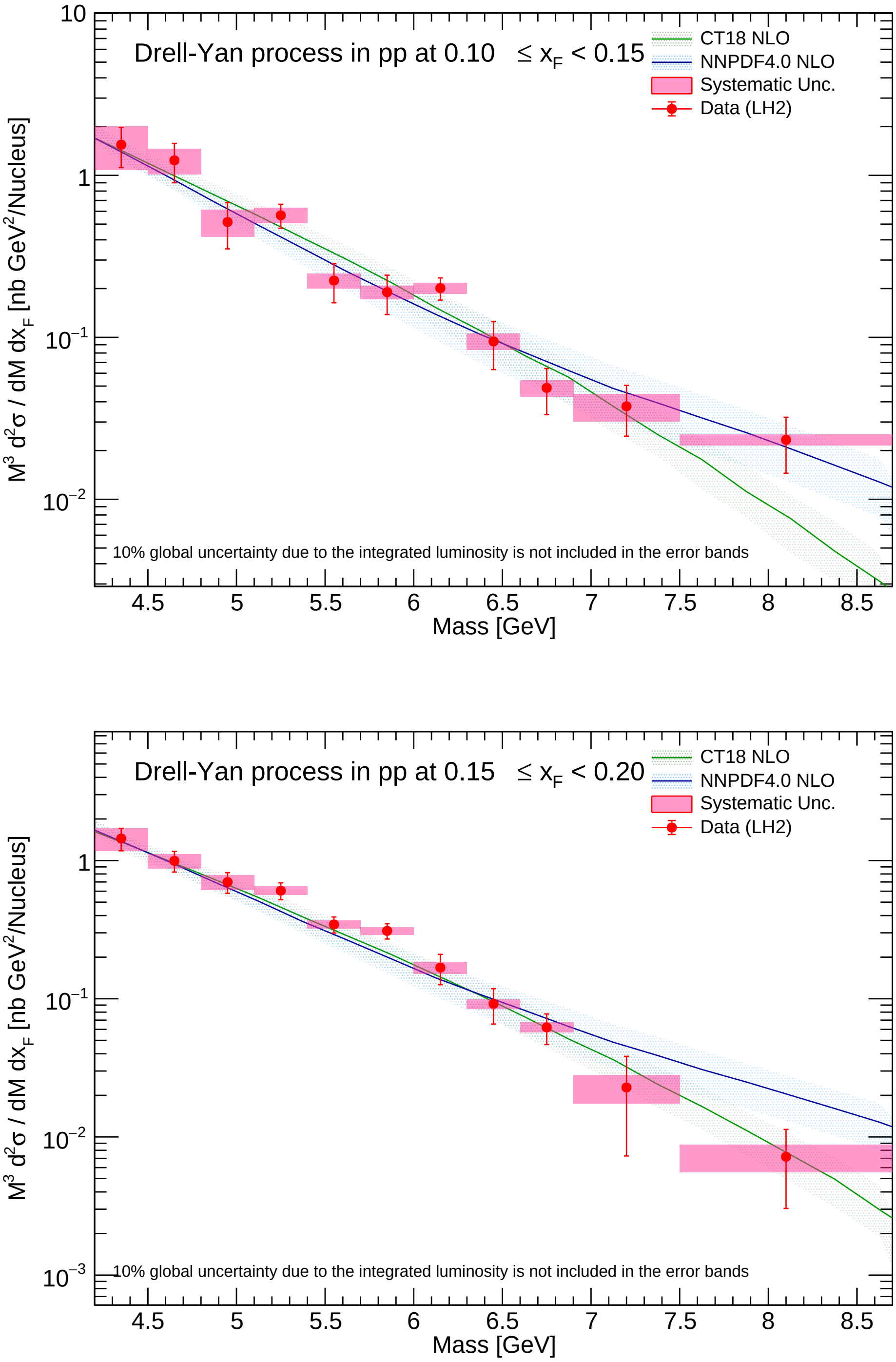
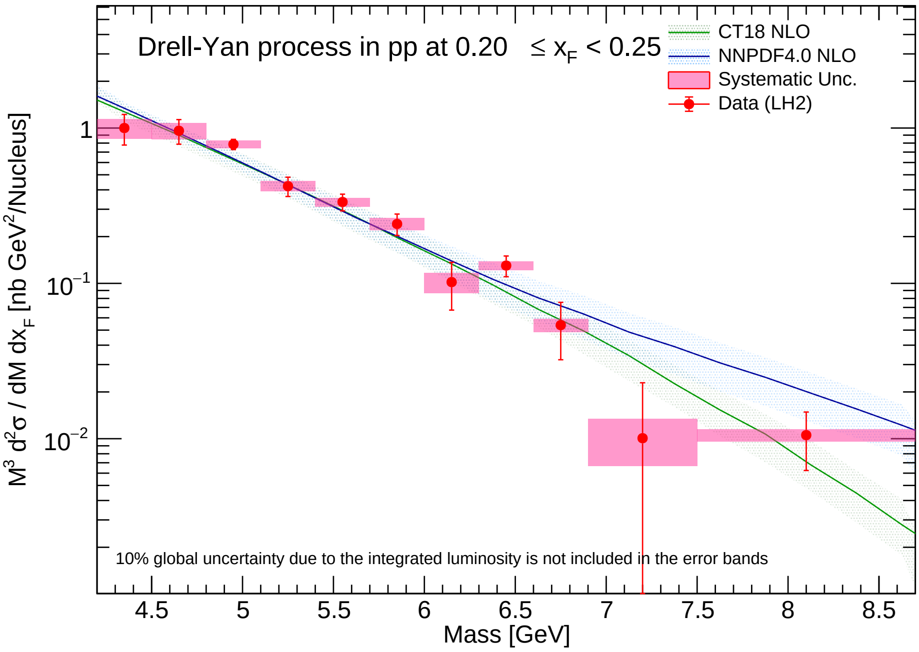


Figure 23: Double-differential cross-section for LH2 (xF bins 0.10-0.20) reported at geometric mass bin center.



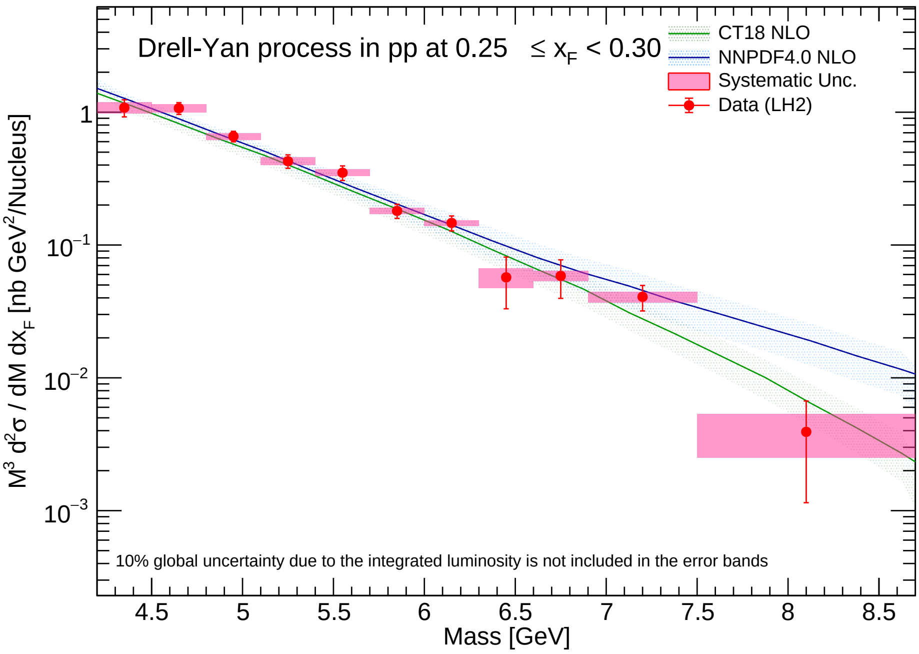


Figure 24: Double-differential cross-section for LH2 (xF bins 0.20-0.30) reported at geometric mass bin center.

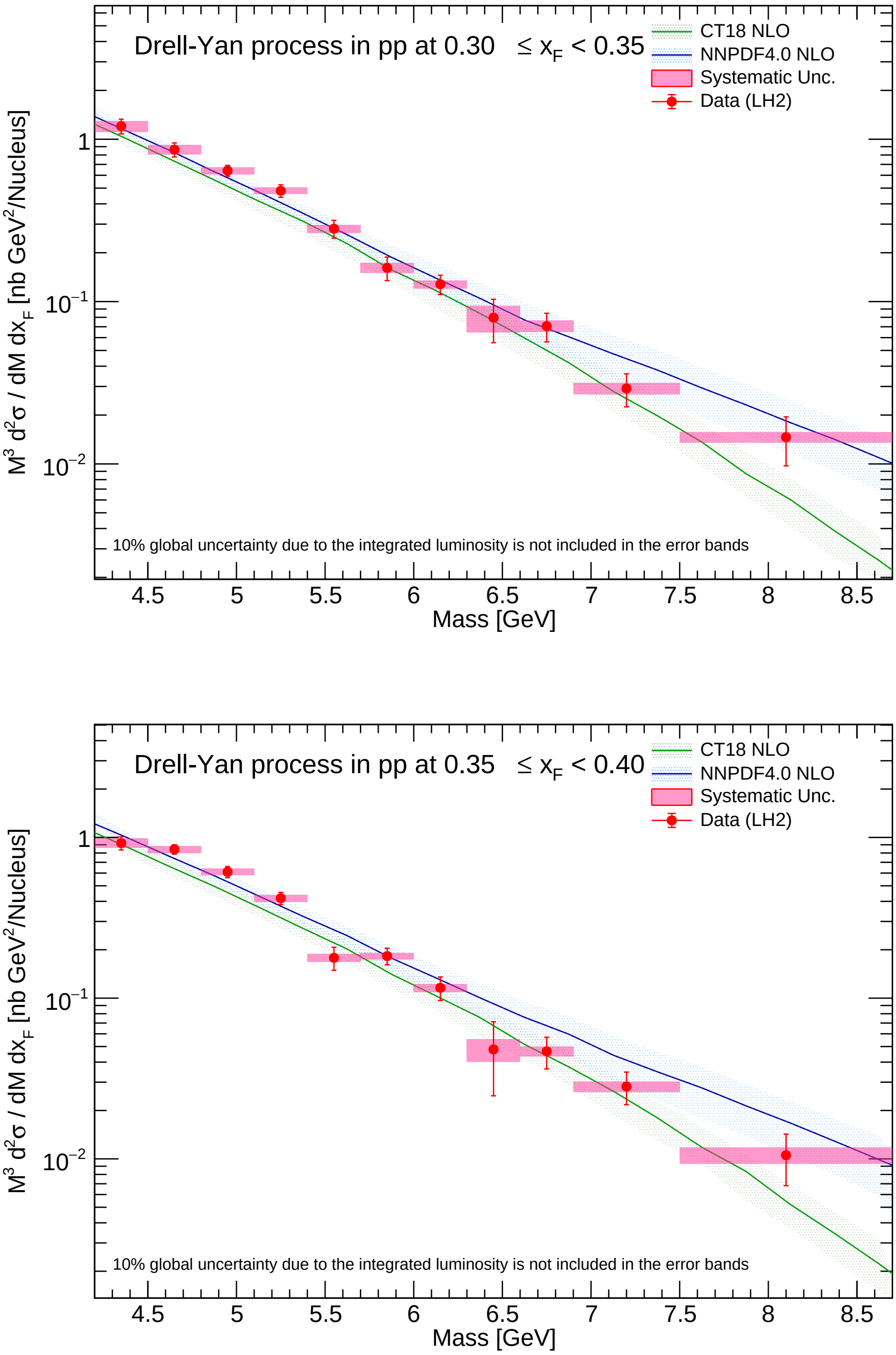
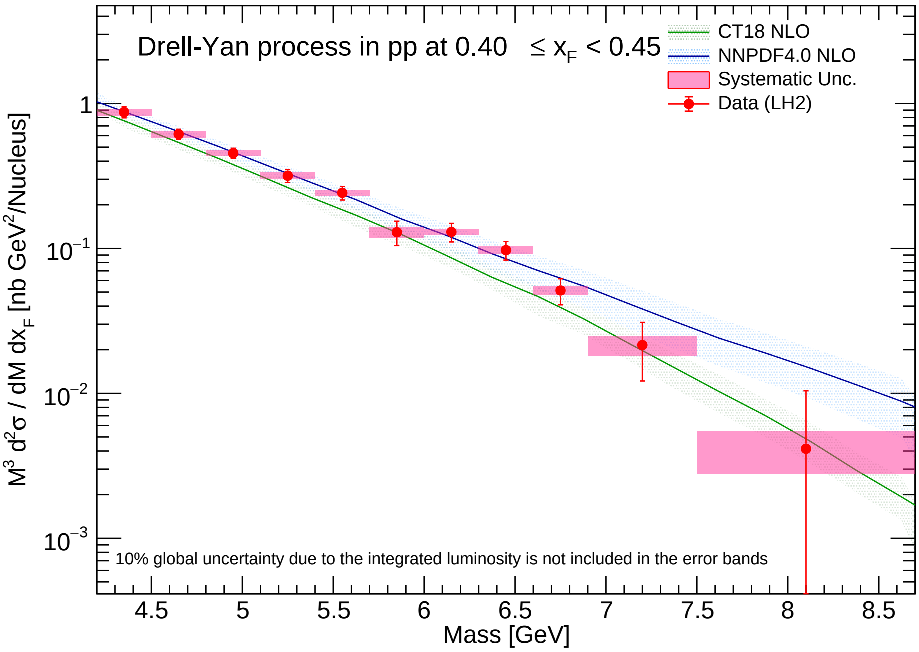


Figure 25: Double-differential cross-section for LH2 (xF bins 0.30-0.40) reported at geometric mass bin center.



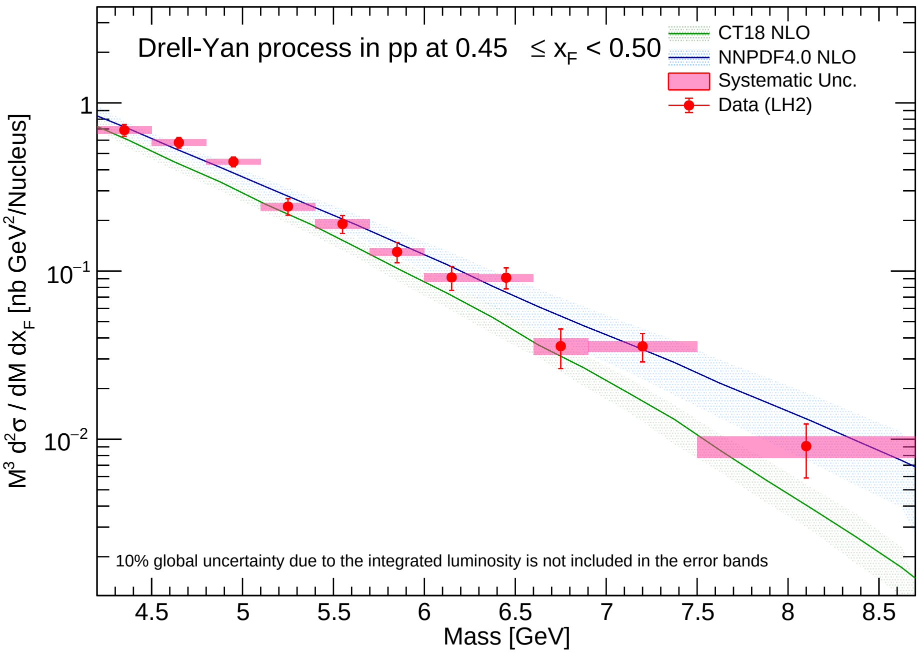


Figure 26: Double-differential cross-section for LH2 (xF bins 0.40-0.50) reported at geometric mass bin center.

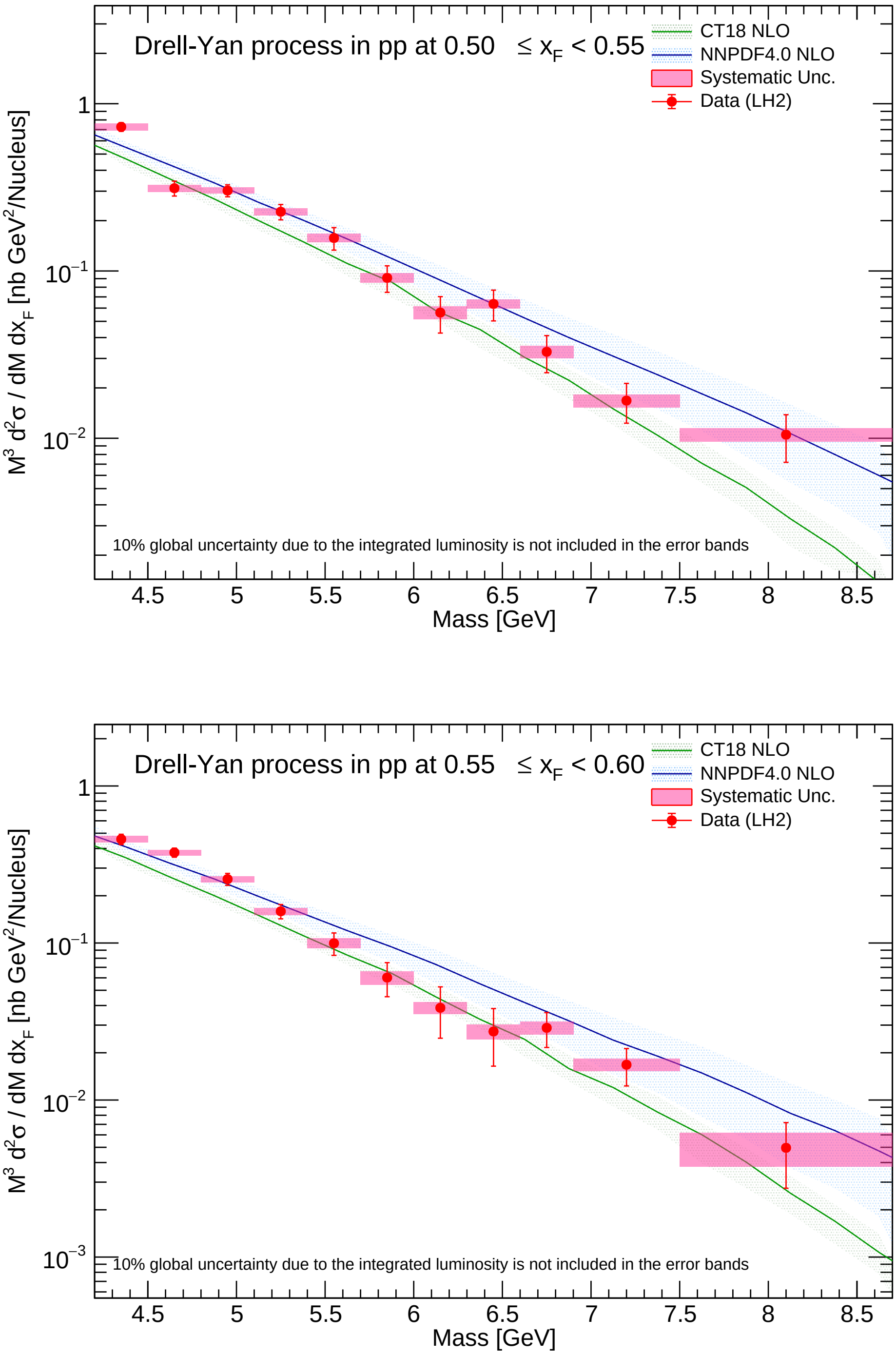
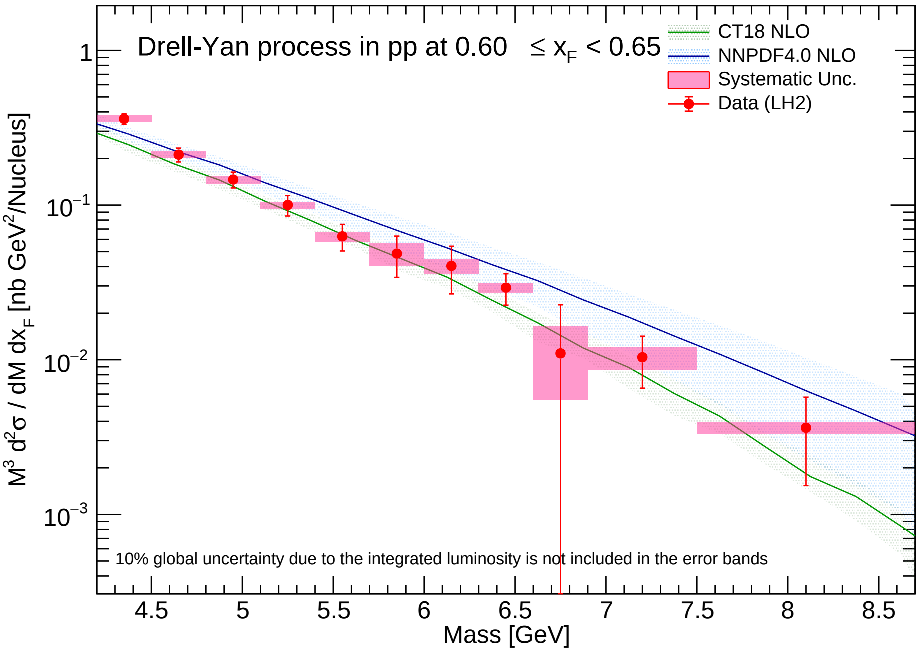


Figure 27: Double-differential cross-section for LH2 (xF bins 0.50-0.60) reported at geometric mass bin center.



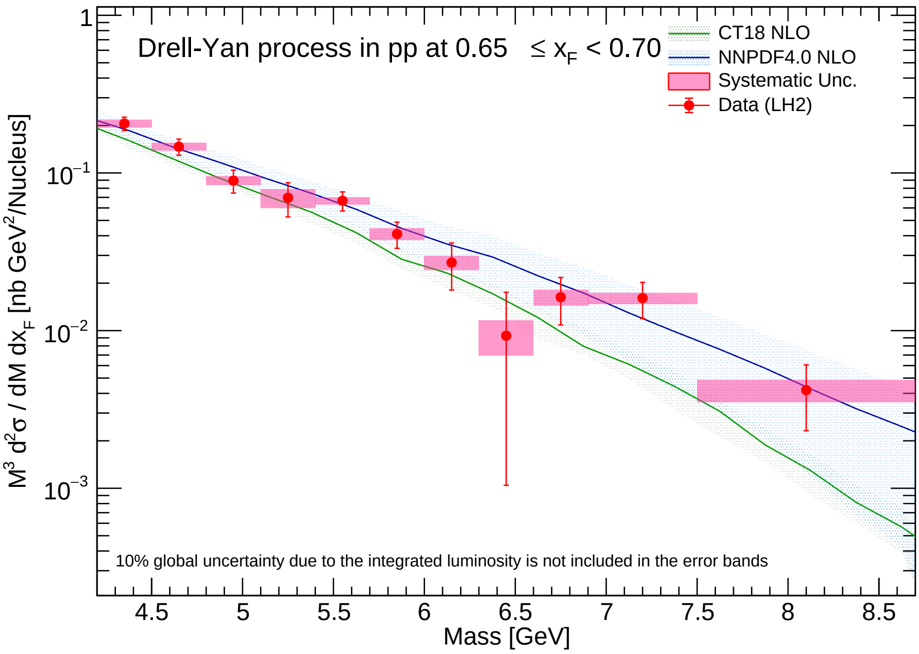
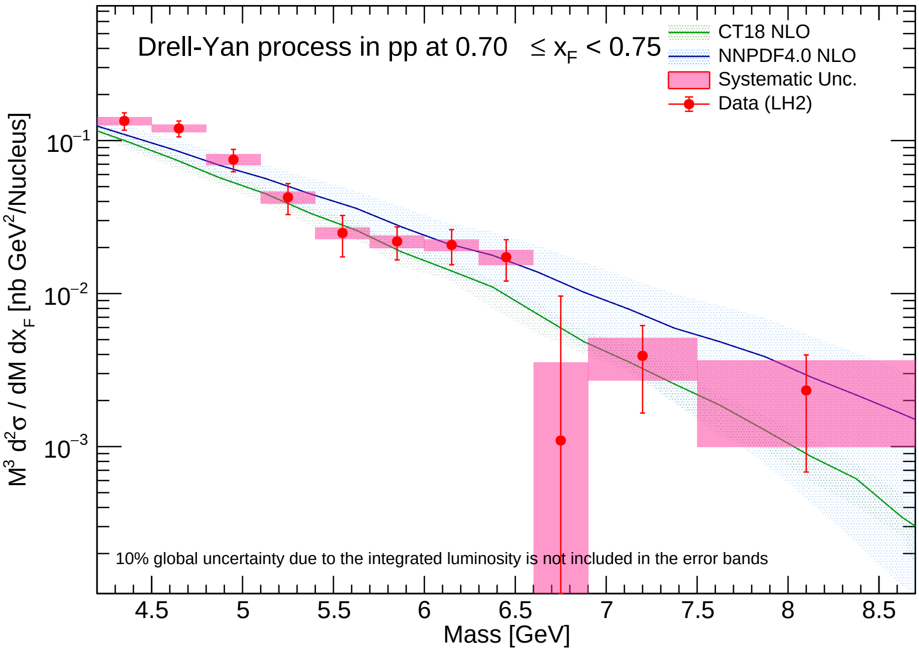


Figure 28: Double-differential cross-section for LH2 (xF bins 0.60-0.70) reported at geometric mass bin center.



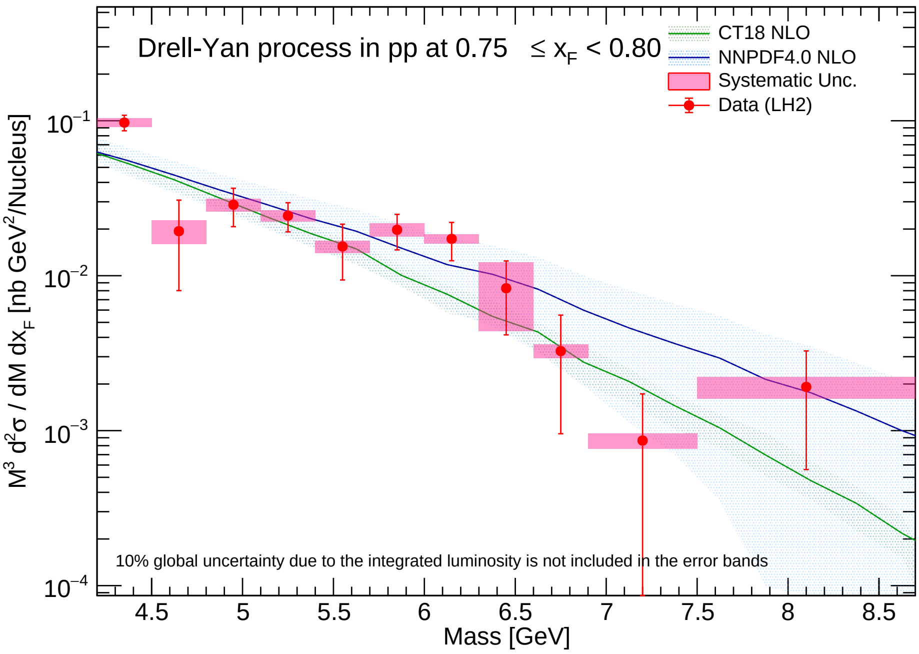
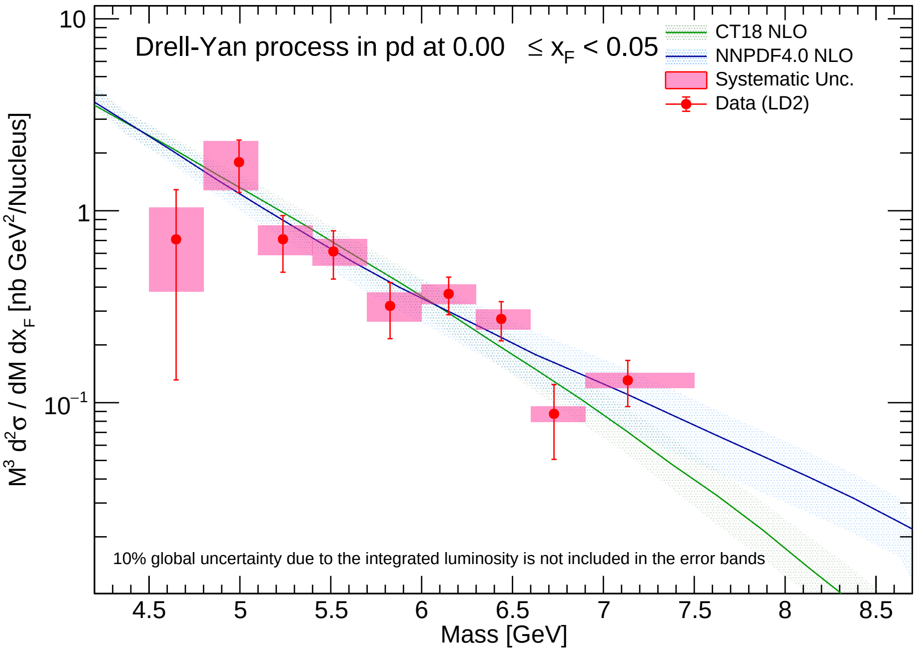


Figure 29: Double-differential cross-section for LH2 (xF bins 0.70-0.80) reported at geometric mass bin center.

9.3 LD2 Results (Reported at mass bin average)

The equivalent double-differential parameterization for the Liquid Deuterium topology is charted against the localized mass bin centroid, tracking identical NLO PDF alignments.



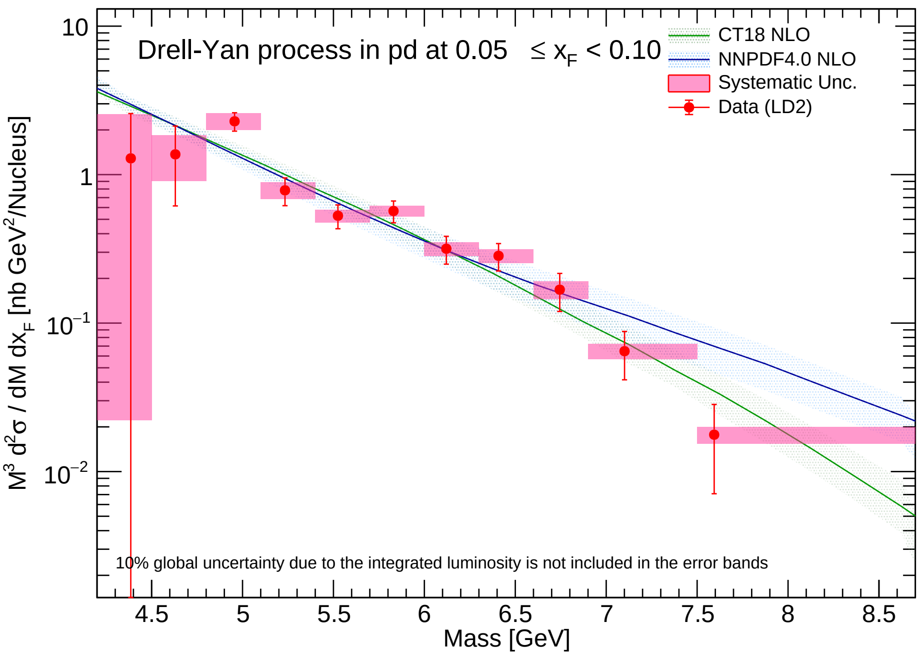
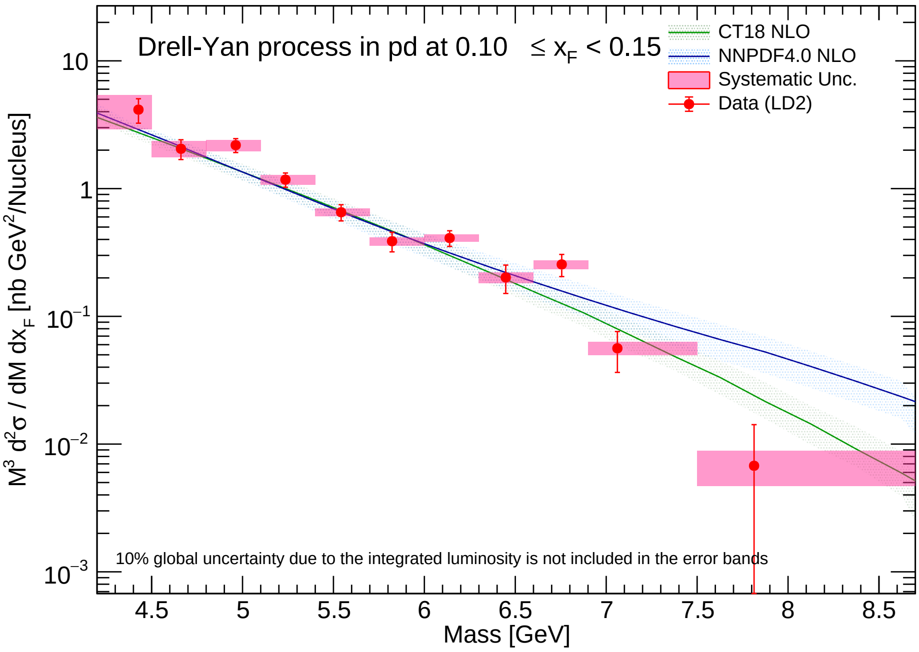


Figure 30: Double-differential cross-section for LD2 (xF bins 0.00-0.10) reported at mass bin average.



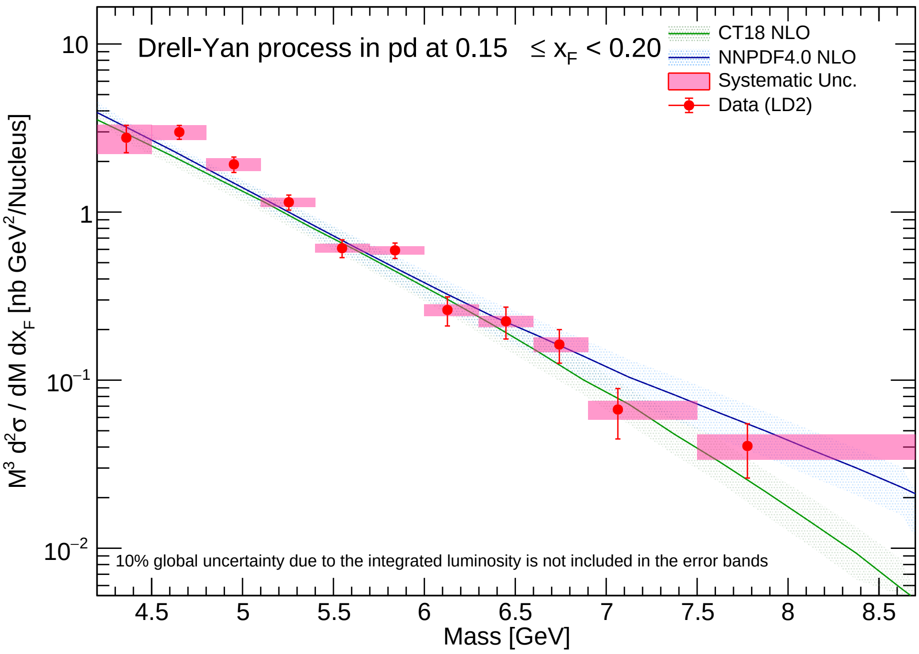


Figure 31: Double-differential cross-section for LD2 (xF bins 0.10-0.20) reported at mass bin average.

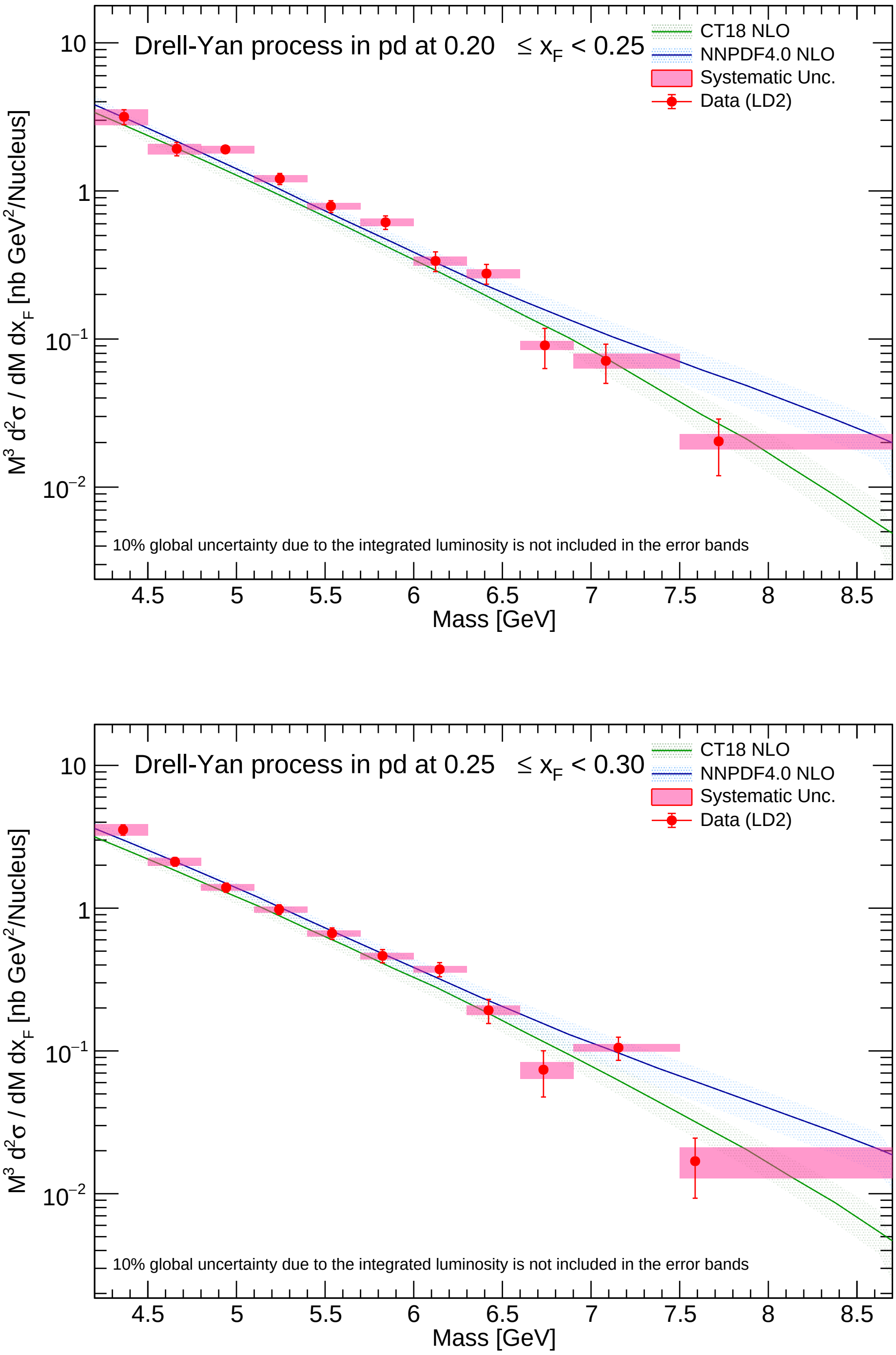
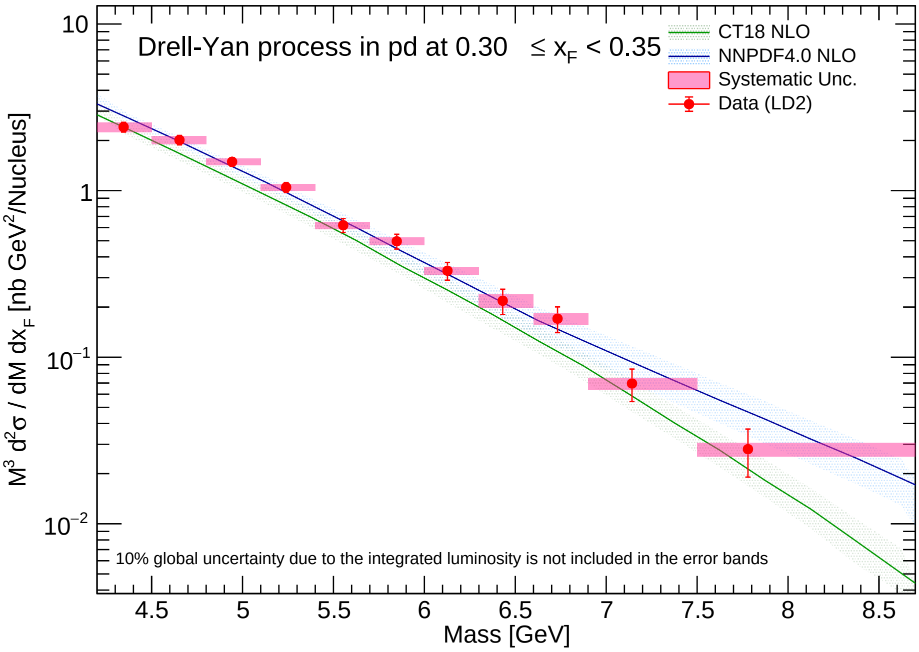


Figure 32: Double-differential cross-section for LD2 (xF bins 0.20-0.30) reported at mass bin average.



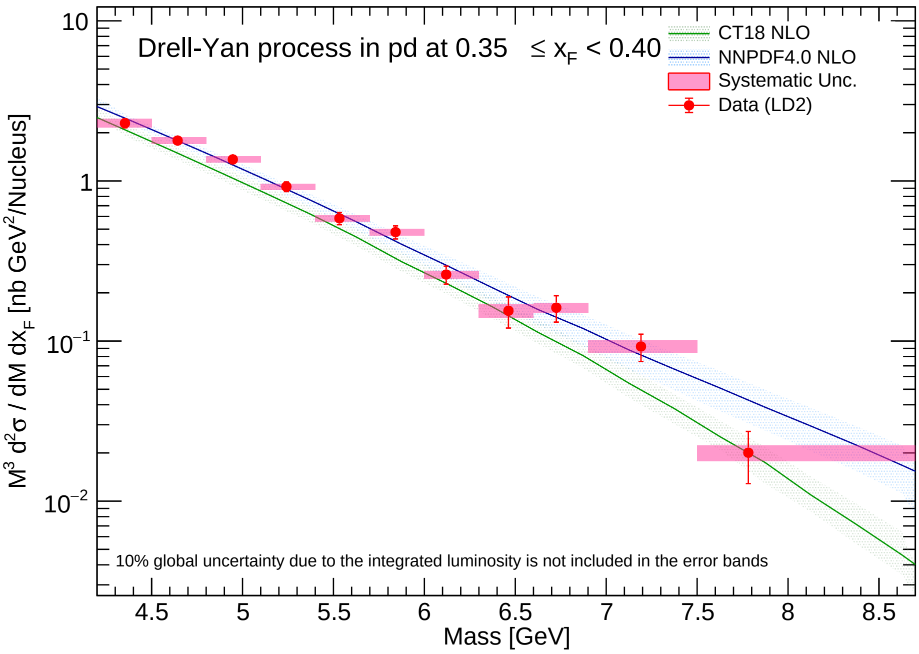


Figure 33: Double-differential cross-section for LD2 (xF bins 0.30-0.40) reported at mass bin average.

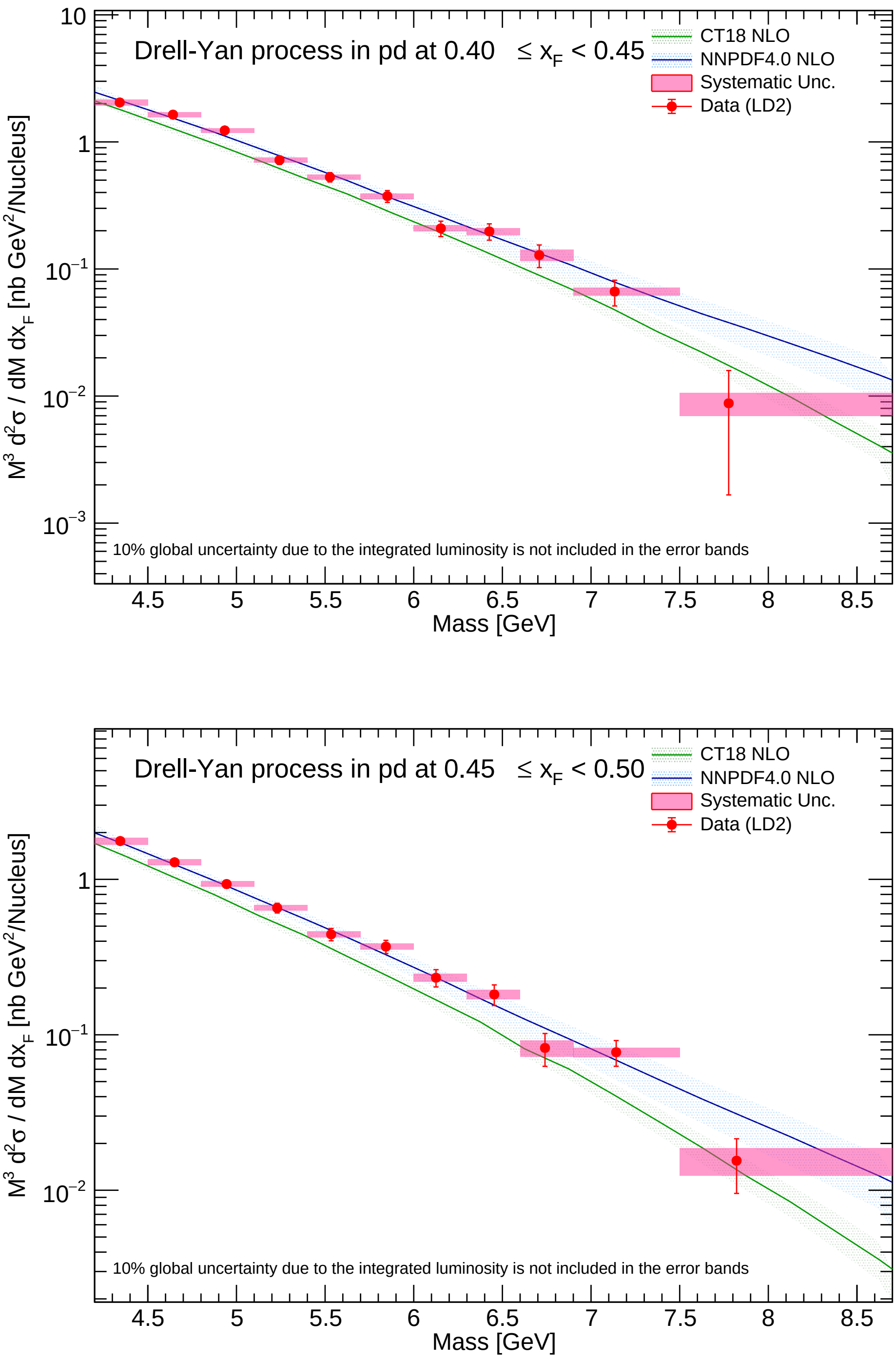
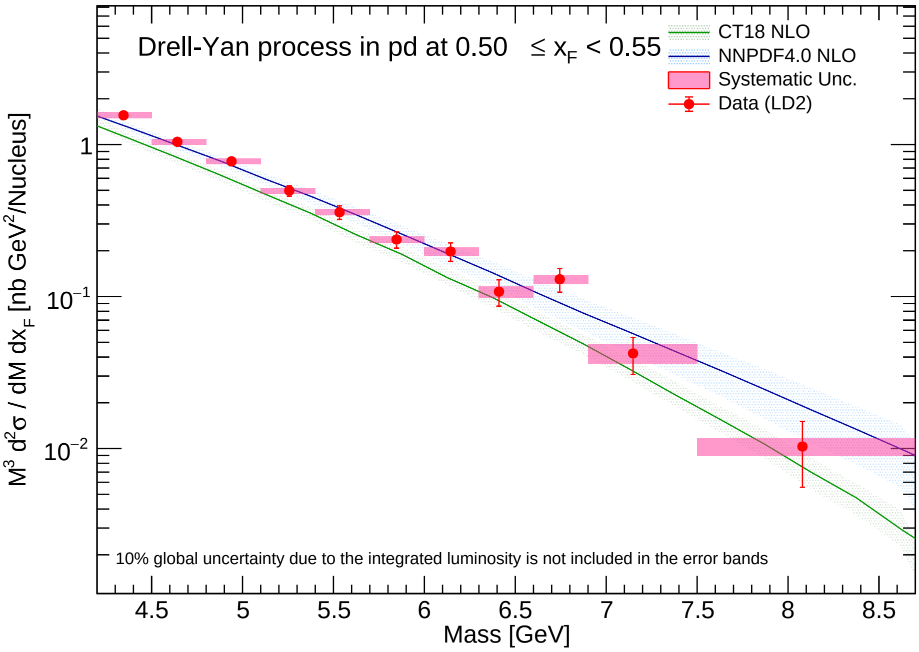


Figure 34: Double-differential cross-section for LD2 (xF bins 0.40-0.50) reported at mass bin average.



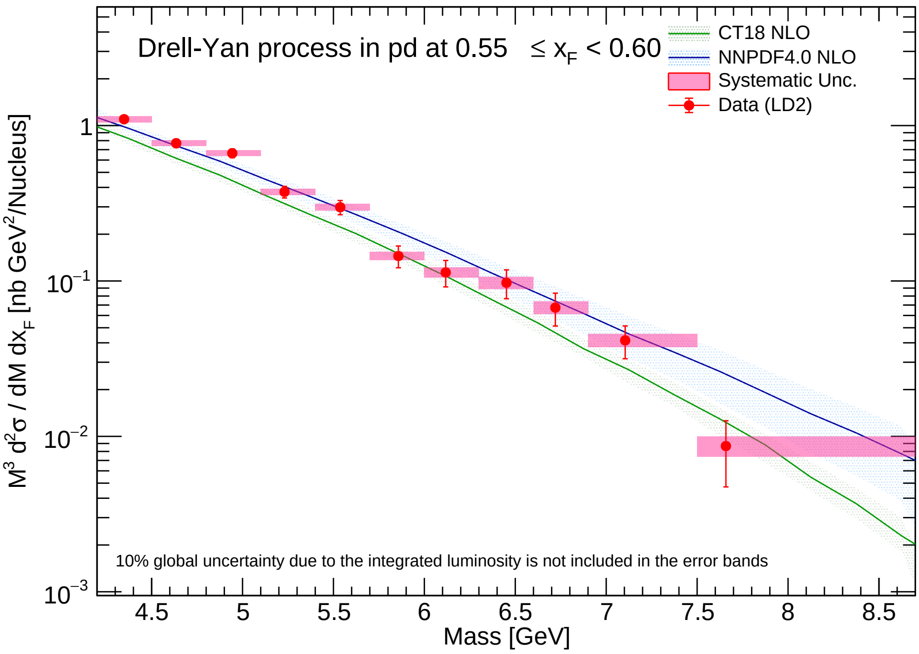


Figure 35: Double-differential cross-section for LD2 (xF bins 0.50-0.60) reported at mass bin average.

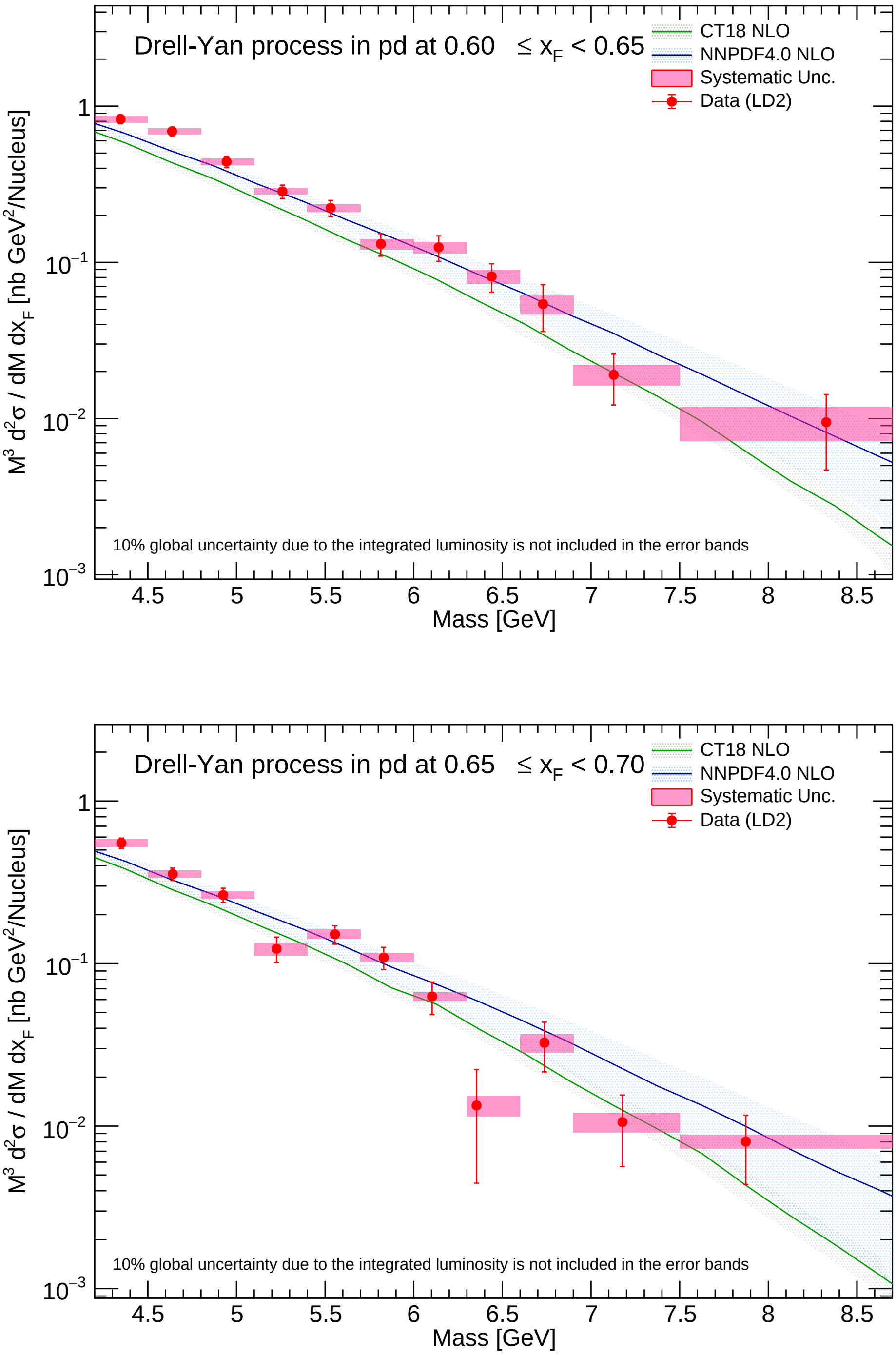


Figure 36: Double-differential cross-section for LD2 (xF bins 0.60-0.70) reported at mass bin average.

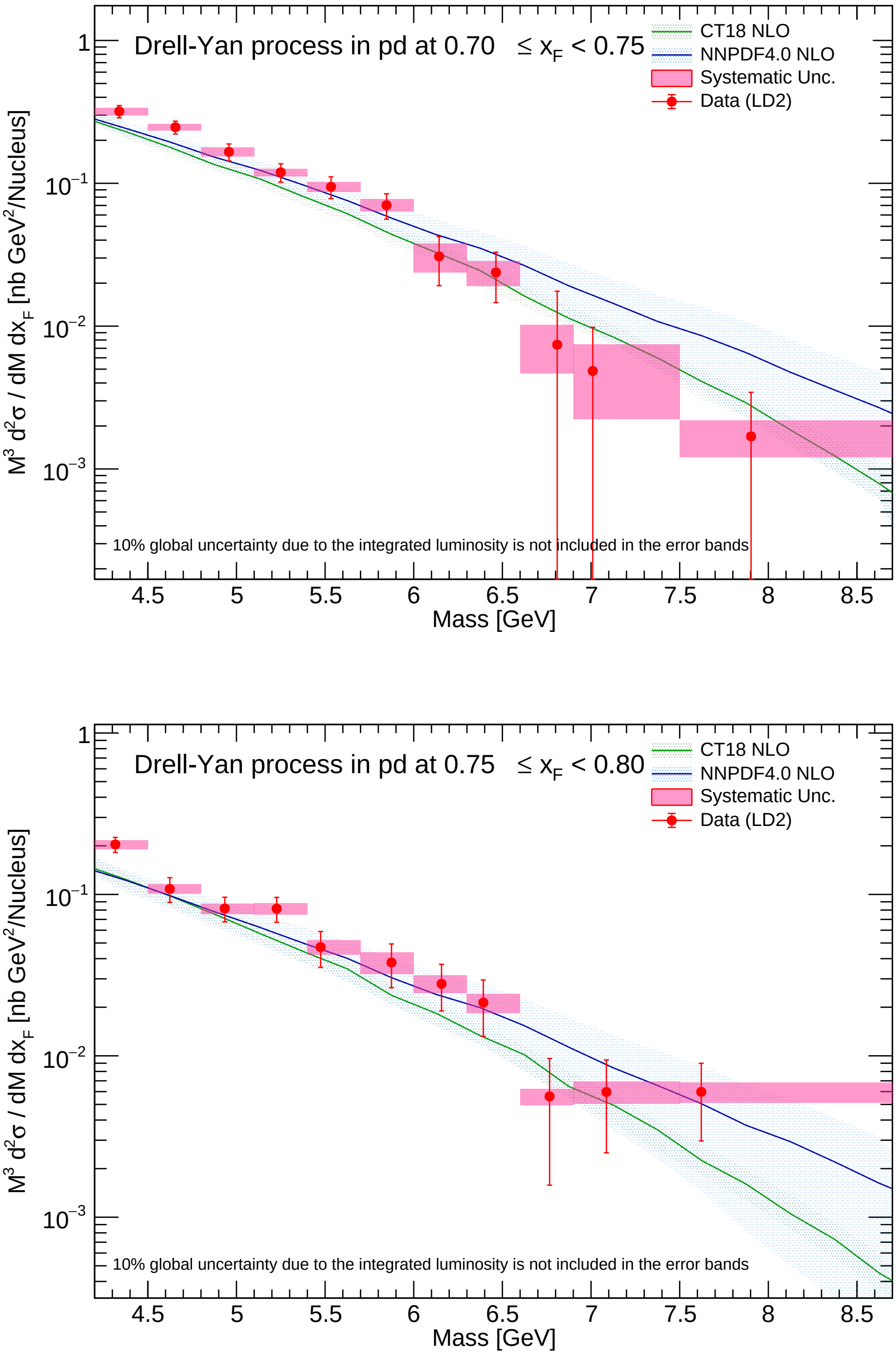
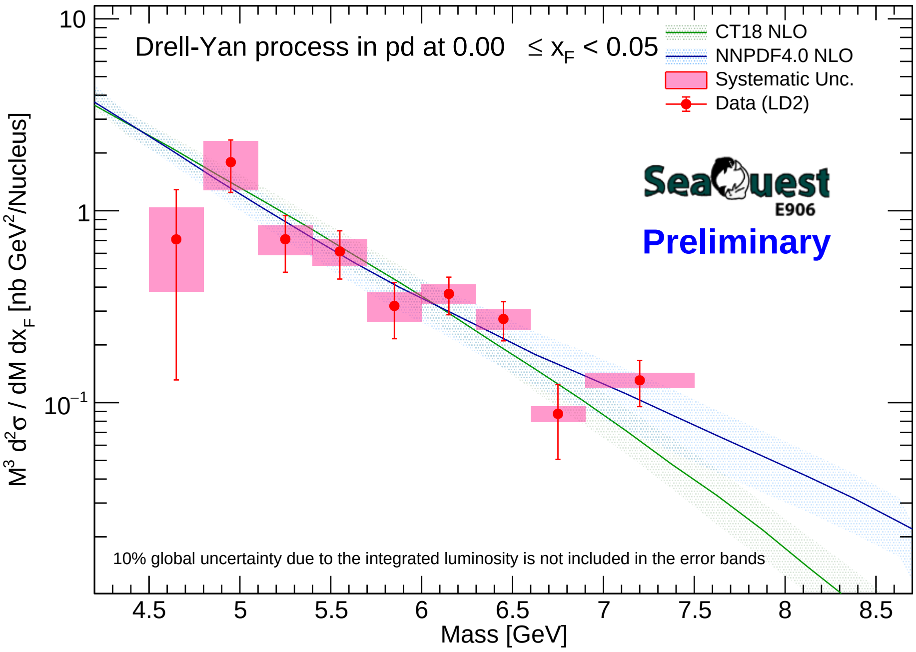


Figure 37: Double-differential cross-section for LD2 (xF bins 0.70-0.80) reported at mass bin average.

9.4 LD2 Results (Reported at mass bin center)

Similarly, the LD_2 cross-section measurements projected strictly onto the geometric bin centers are plotted for rigorous comparison architectures.



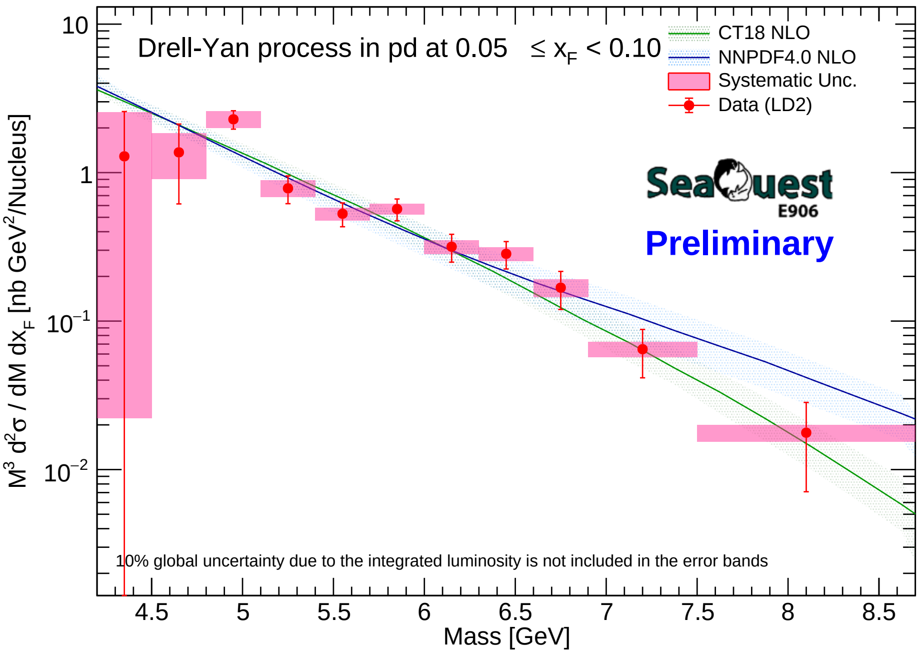
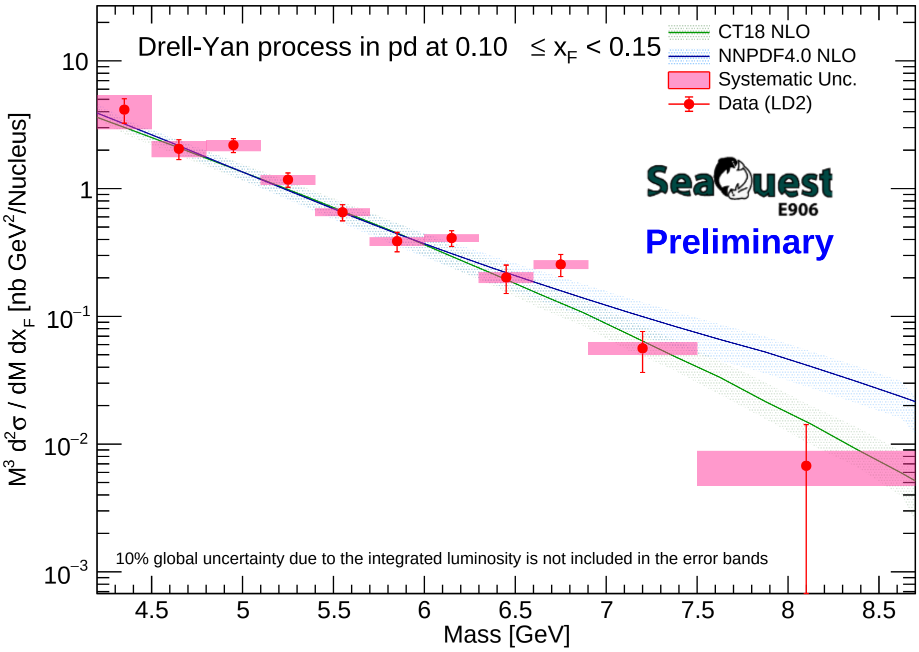


Figure 38: Double-differential cross-section for LD2 (xF bins 0.00-0.10) reported at geometric mass bin center.



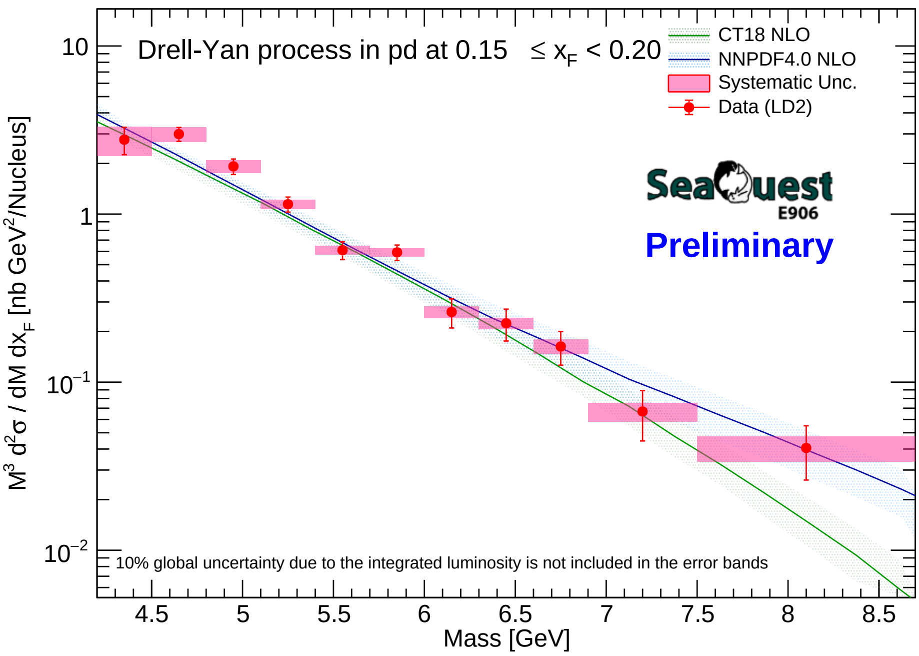


Figure 39: Double-differential cross-section for LD2 (xF bins 0.10-0.20) reported at geometric mass bin center.

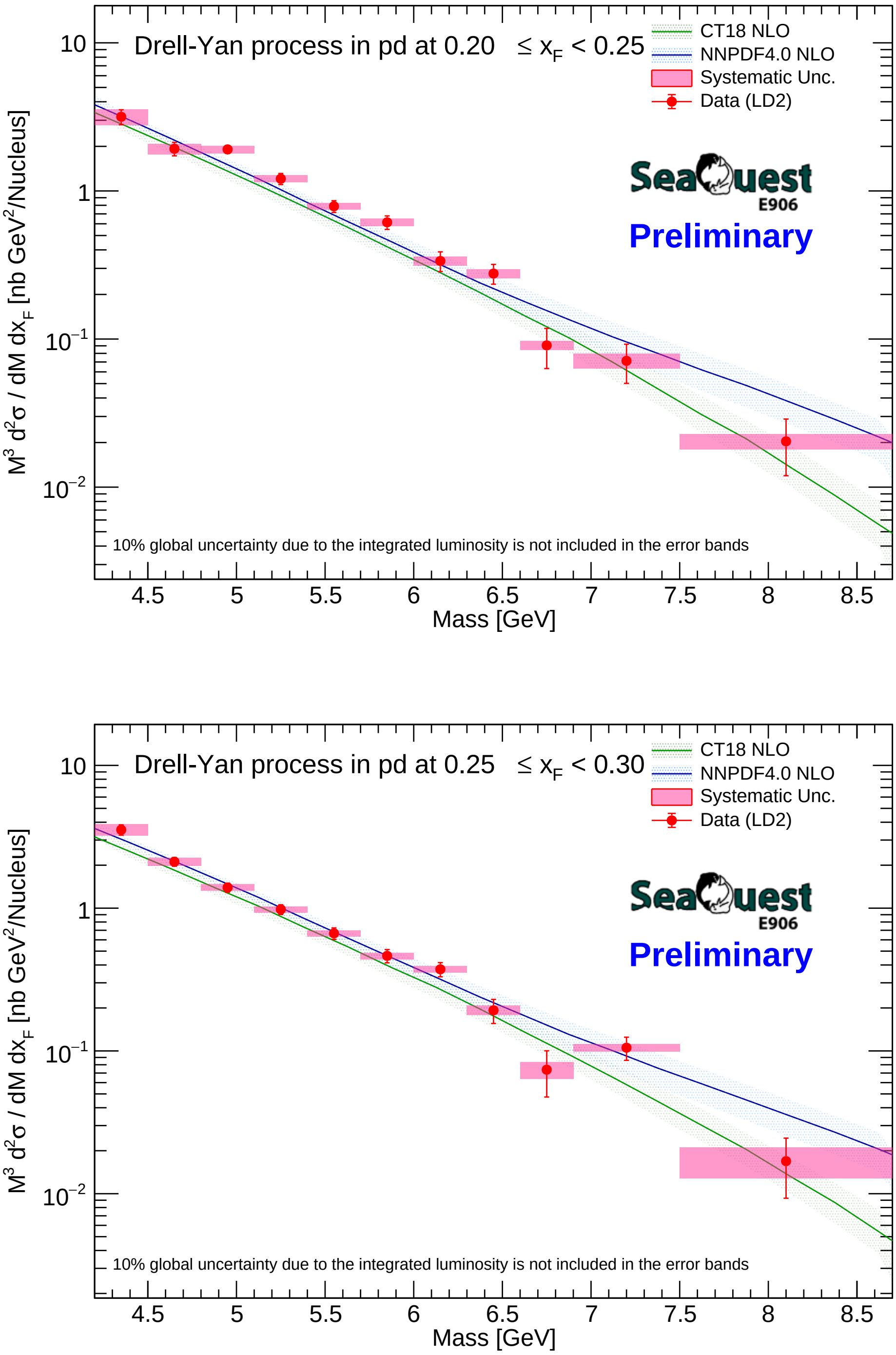
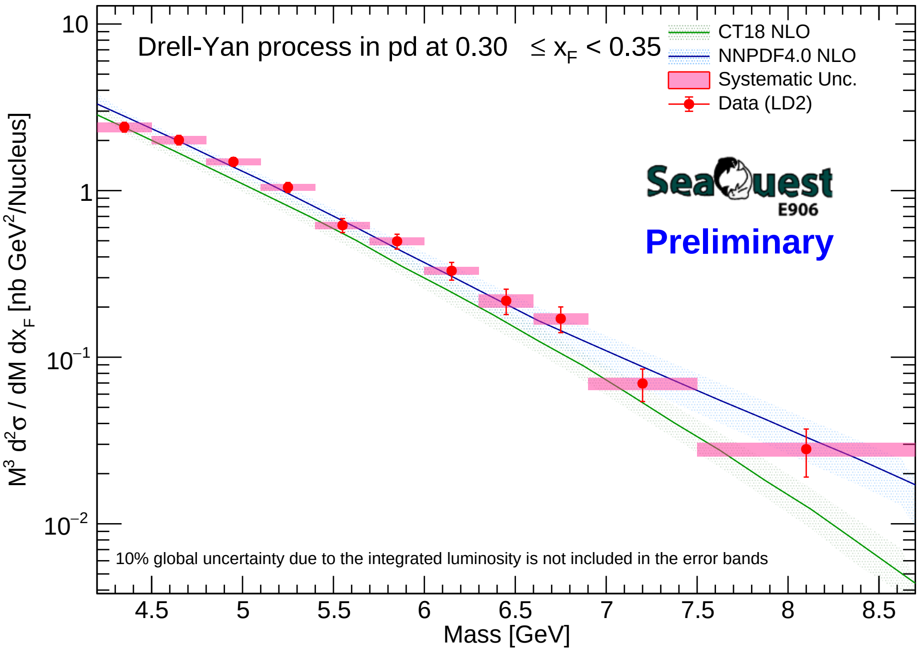


Figure 40: Double-differential cross-section for LD2 (xF bins 0.20-0.30) reported at geometric mass bin center.



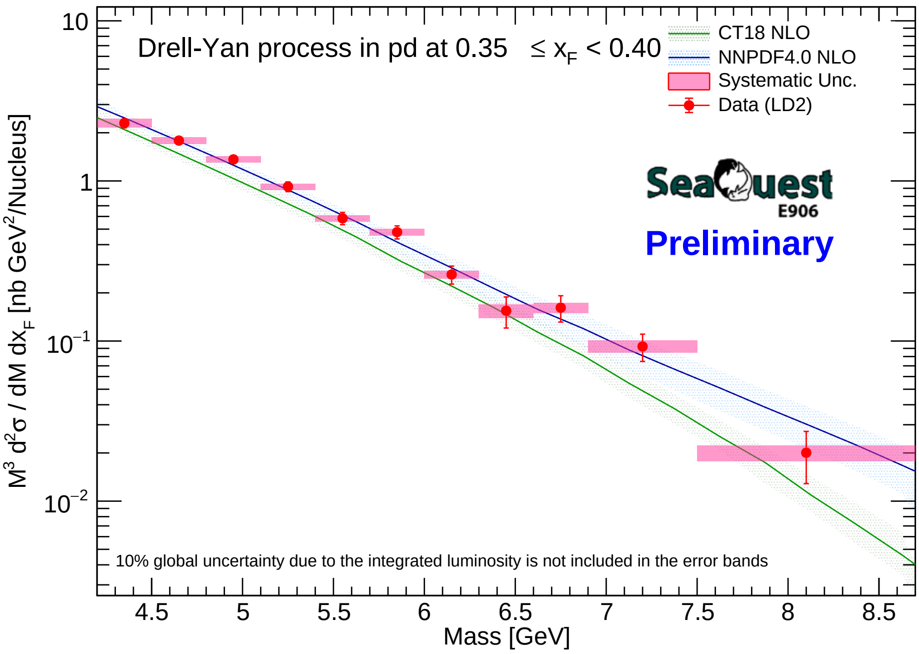
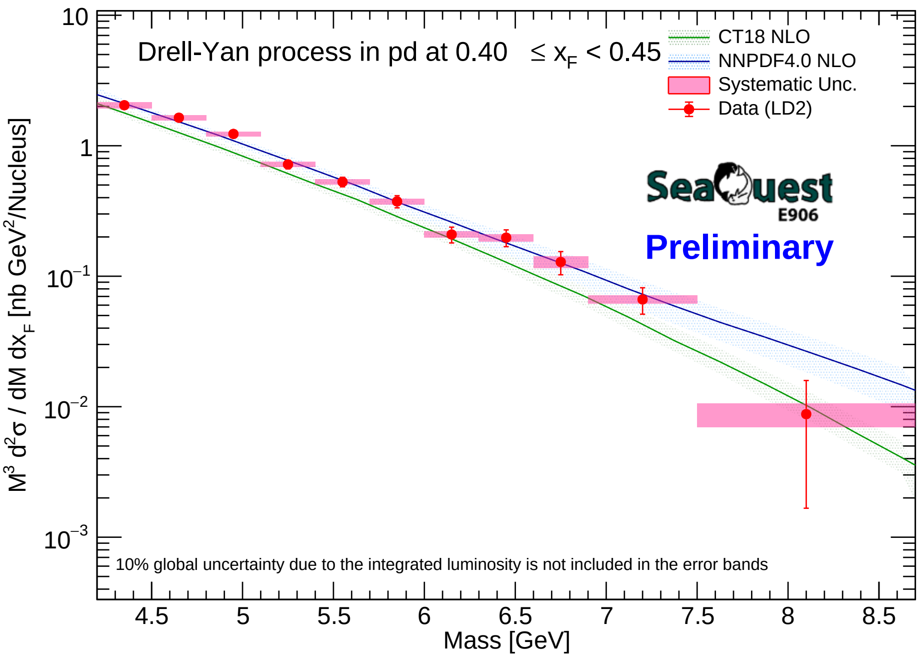


Figure 41: Double-differential cross-section for LD2 (xF bins 0.30-0.40) reported at geometric mass bin center.



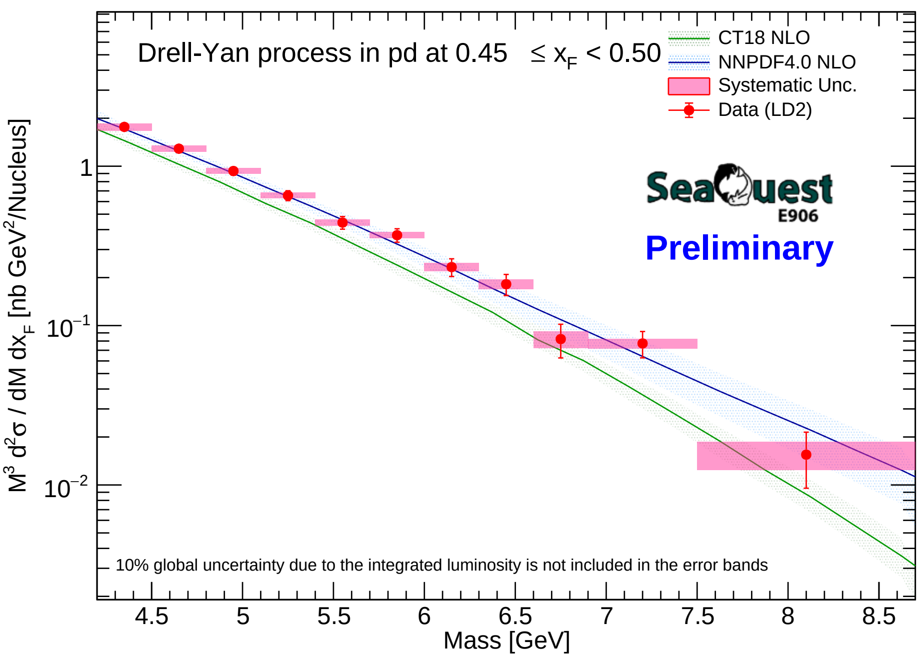


Figure 42: Double-differential cross-section for LD2 (xF bins 0.40-0.50) reported at geometric mass bin center.

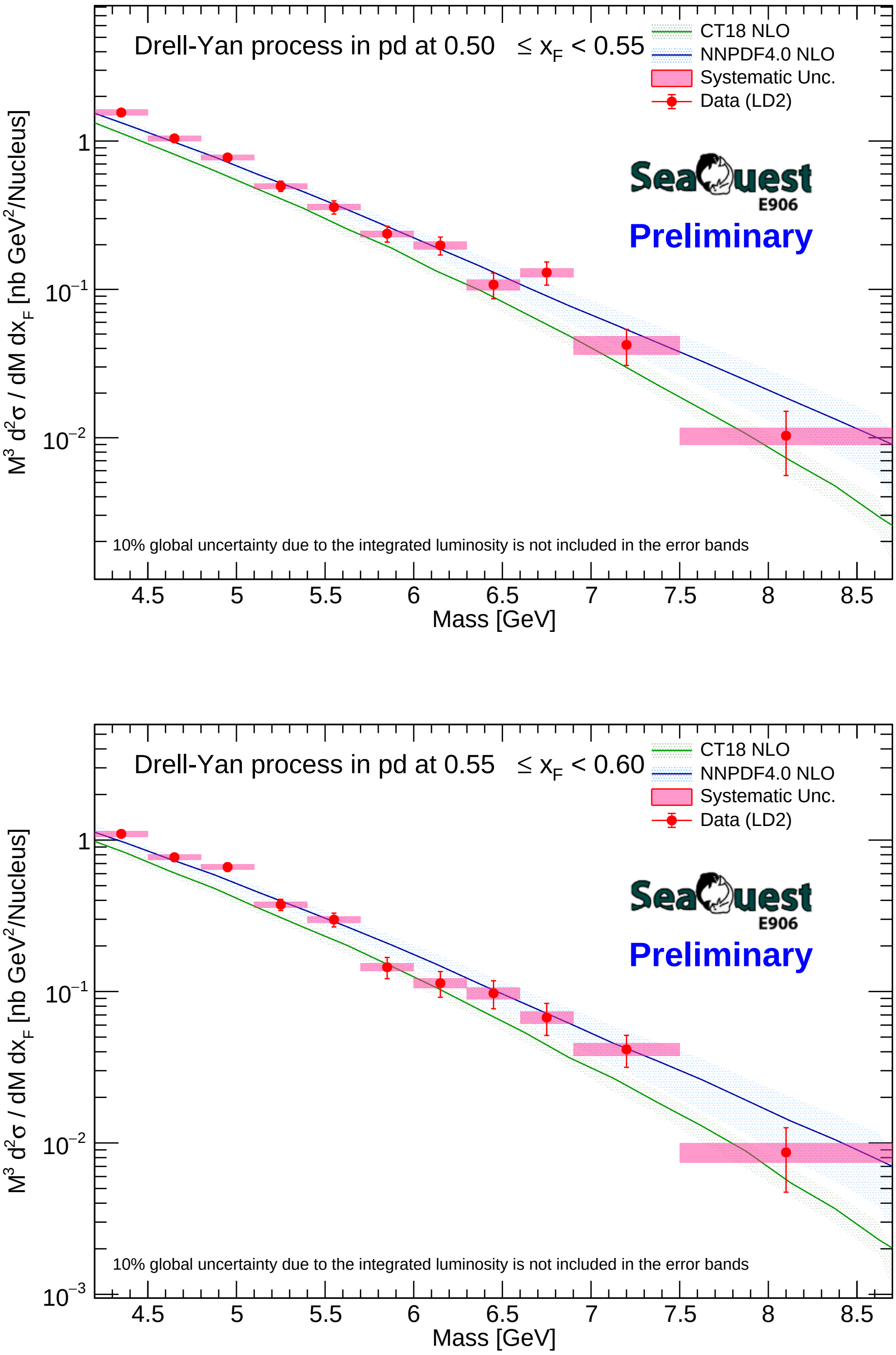
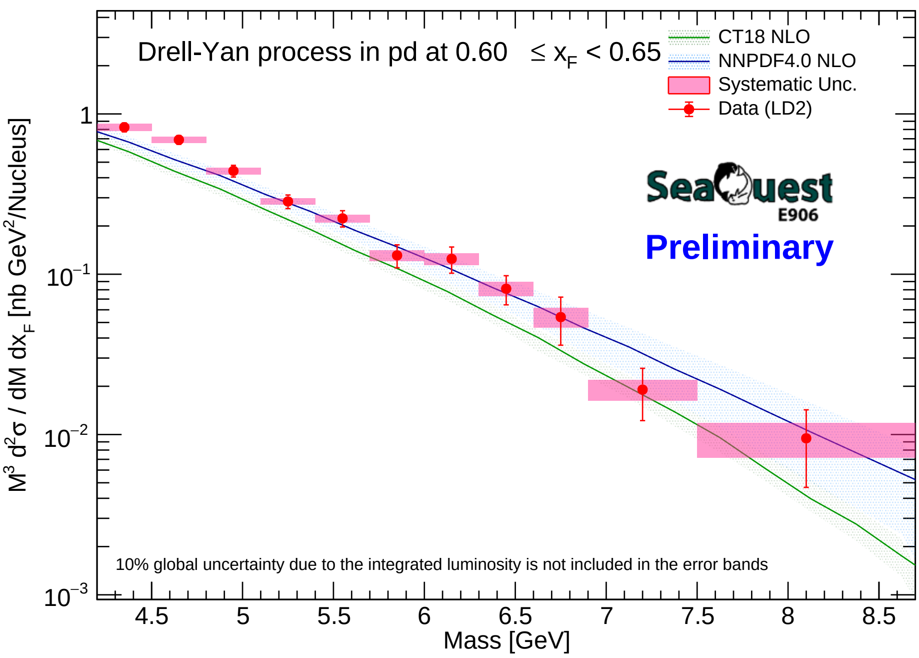


Figure 43: Double-differential cross-section for LD2 (xF bins 0.50-0.60) reported at geometric mass bin center.



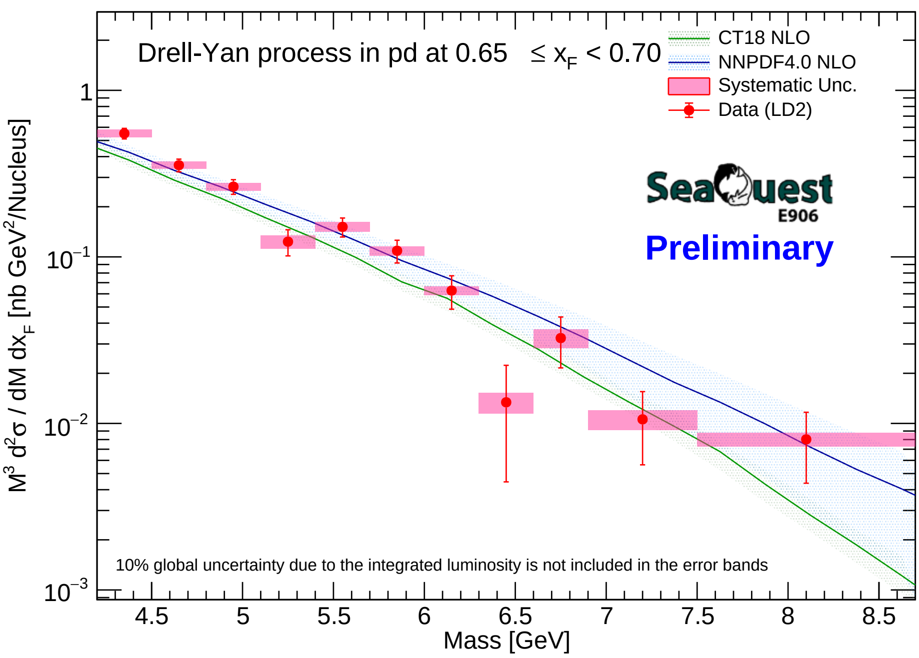


Figure 44: Double-differential cross-section for LD2 (xF bins 0.60-0.70) reported at geometric mass bin center.

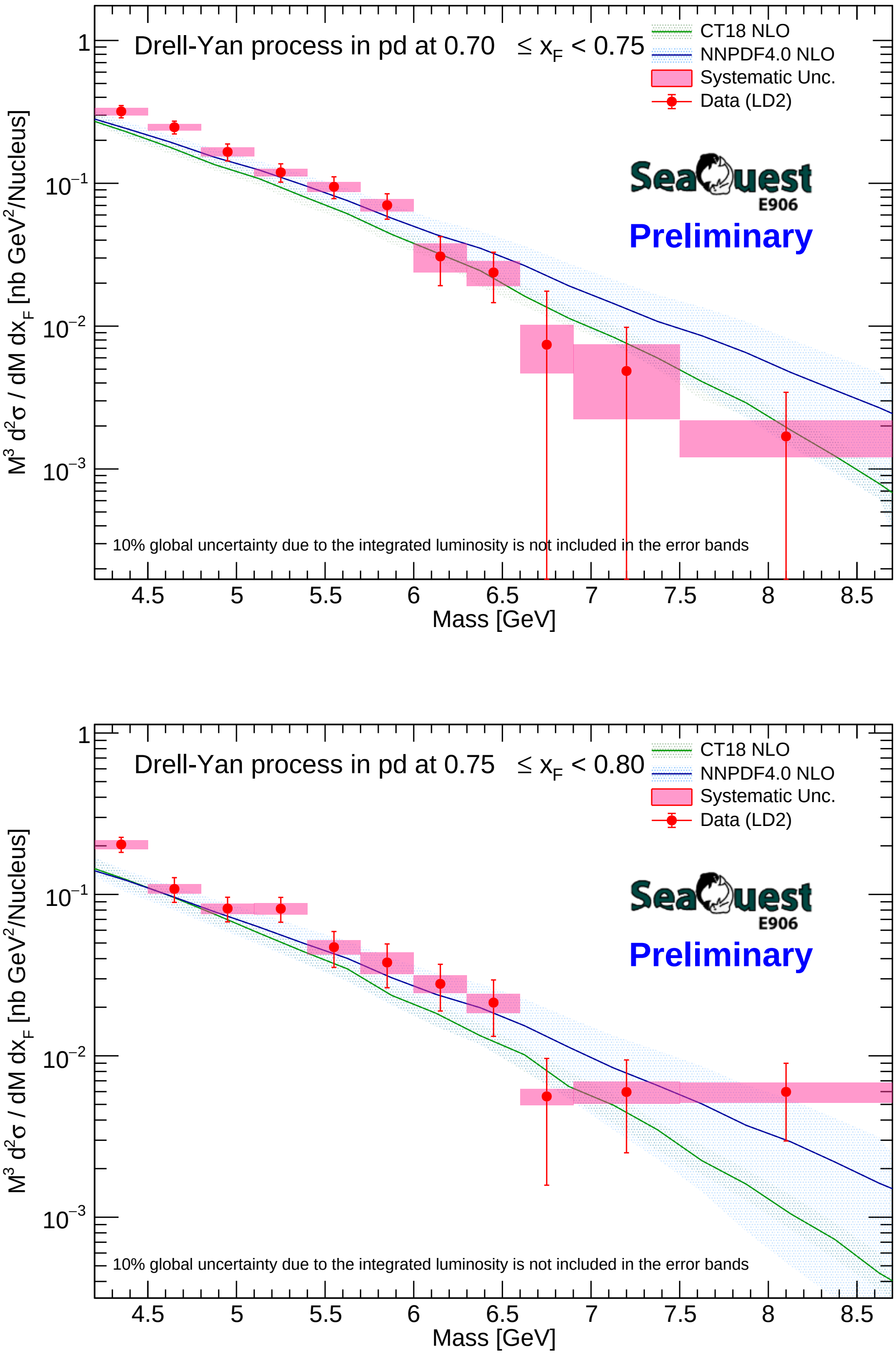


Figure 45: Double-differential cross-section for LD2 (xF bins 0.70-0.80) reported at geometric mass bin center.

9.5 LH2 and LD2 cross-section summary plots

The global measurement landscapes for both configurations are structurally merged into target-specific continuum summary representations.

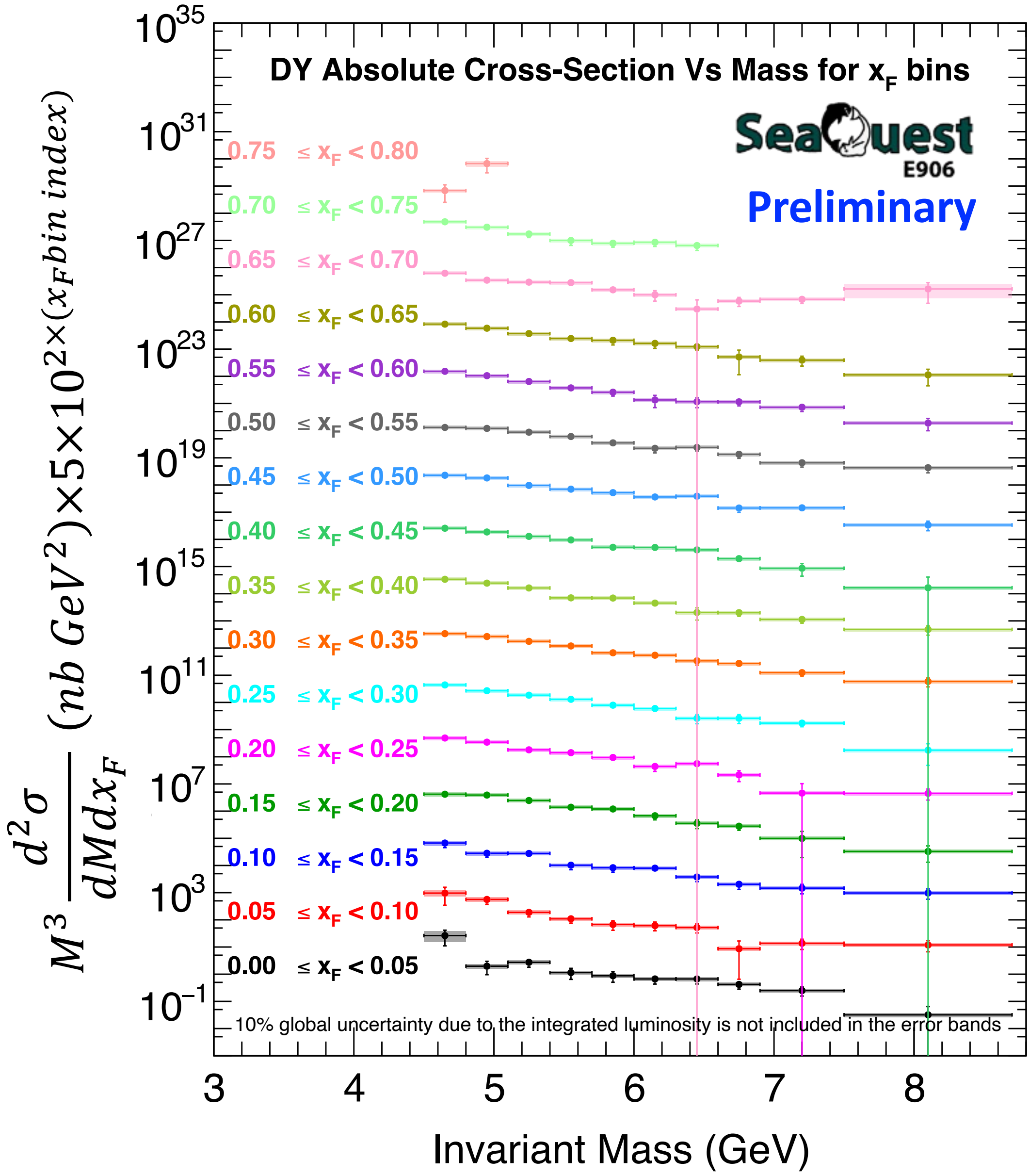


Figure 46: Cross-section summary plots for both LH2 taget.

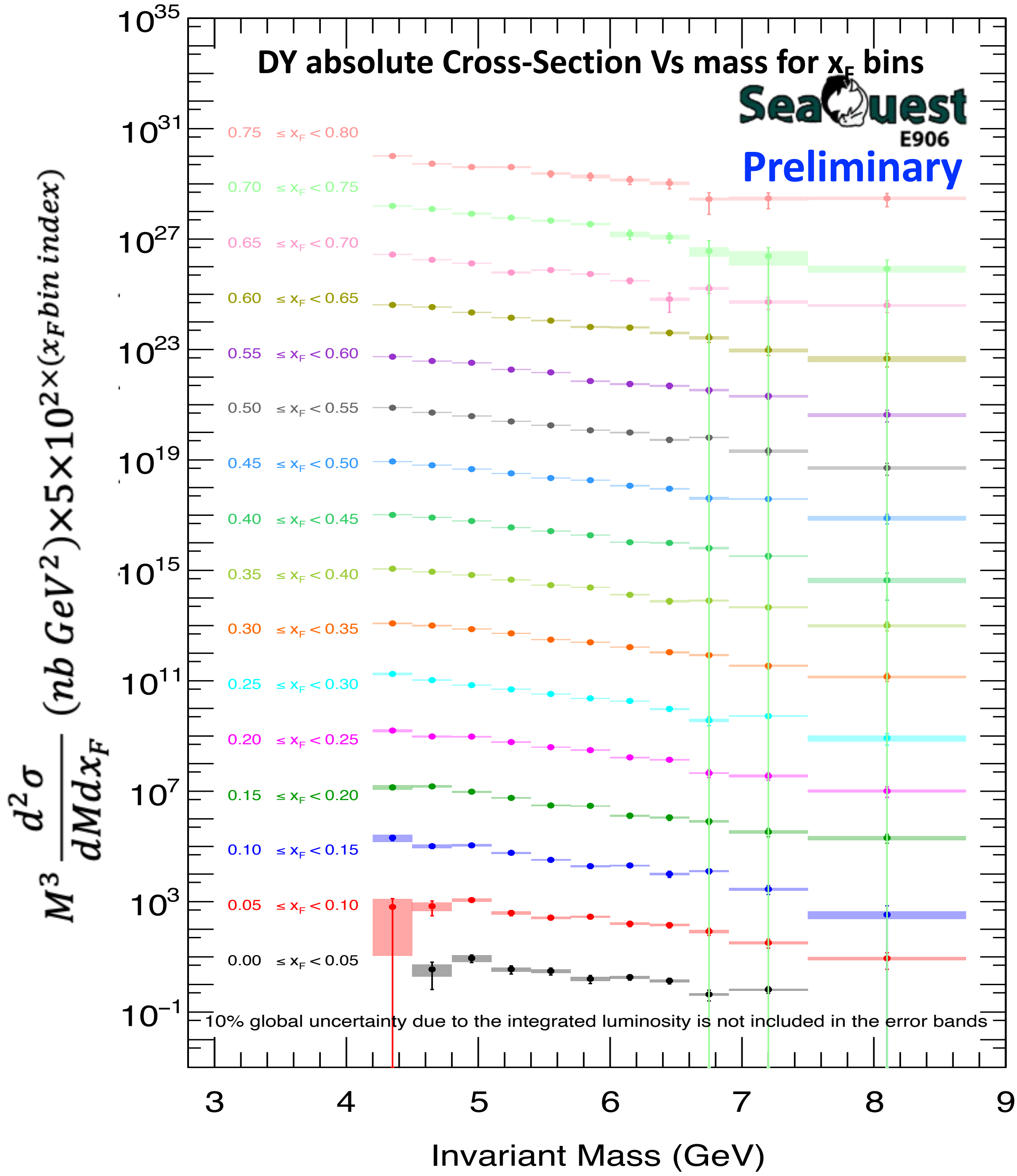


Figure 47: Cross-section summary plots for both LD2 taget.

9.6 LH2 cross section comparison

To benchmark the systematic drift between the current modeling framework and the preliminary iteration detailed in DocDB 11383[19], the differential traces are overlayed at geometric bin centers.

[19] https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11383

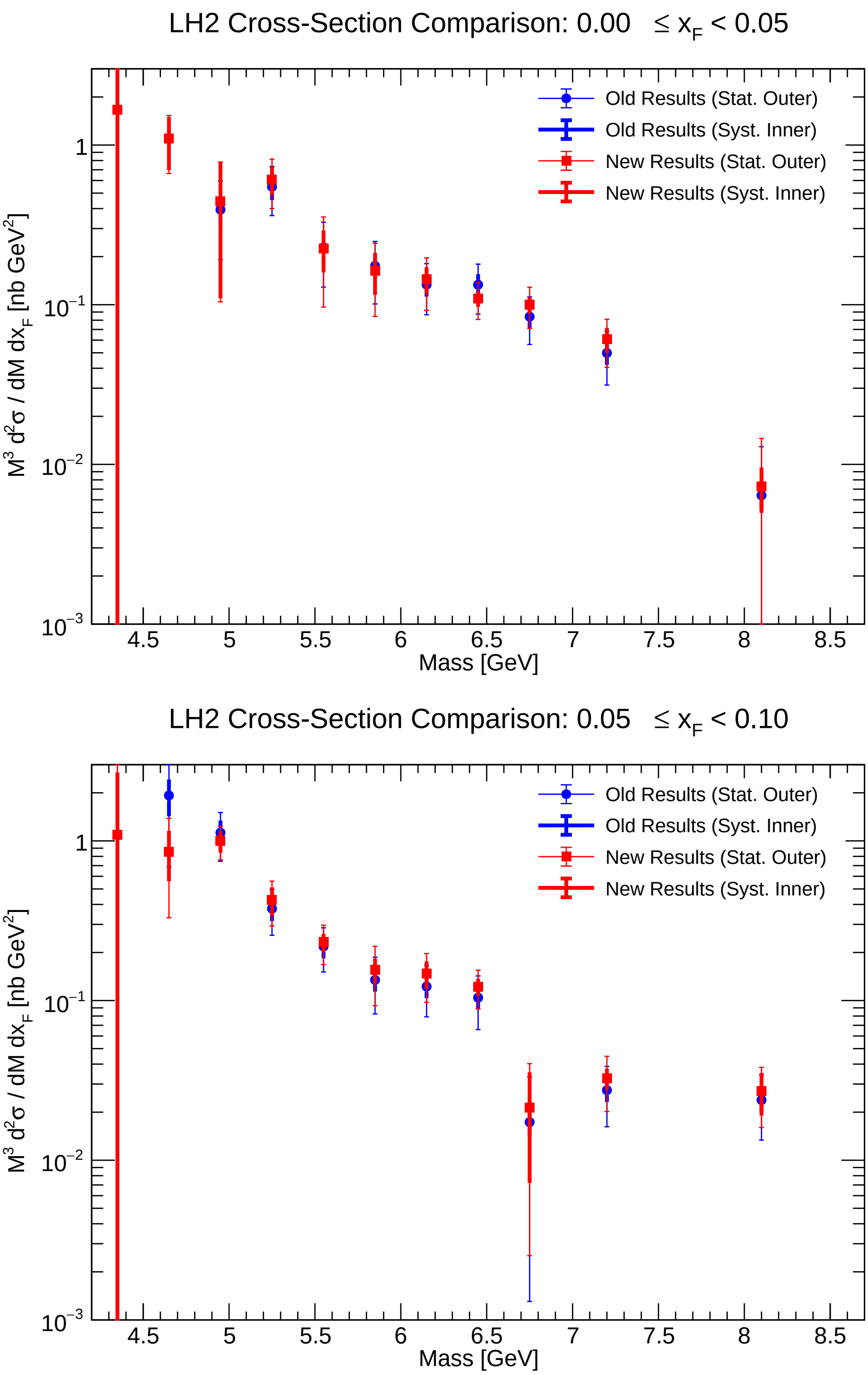


Figure 48: Comparison of double-differential cross-section for LH2 (xF bins 0.00-0.10).

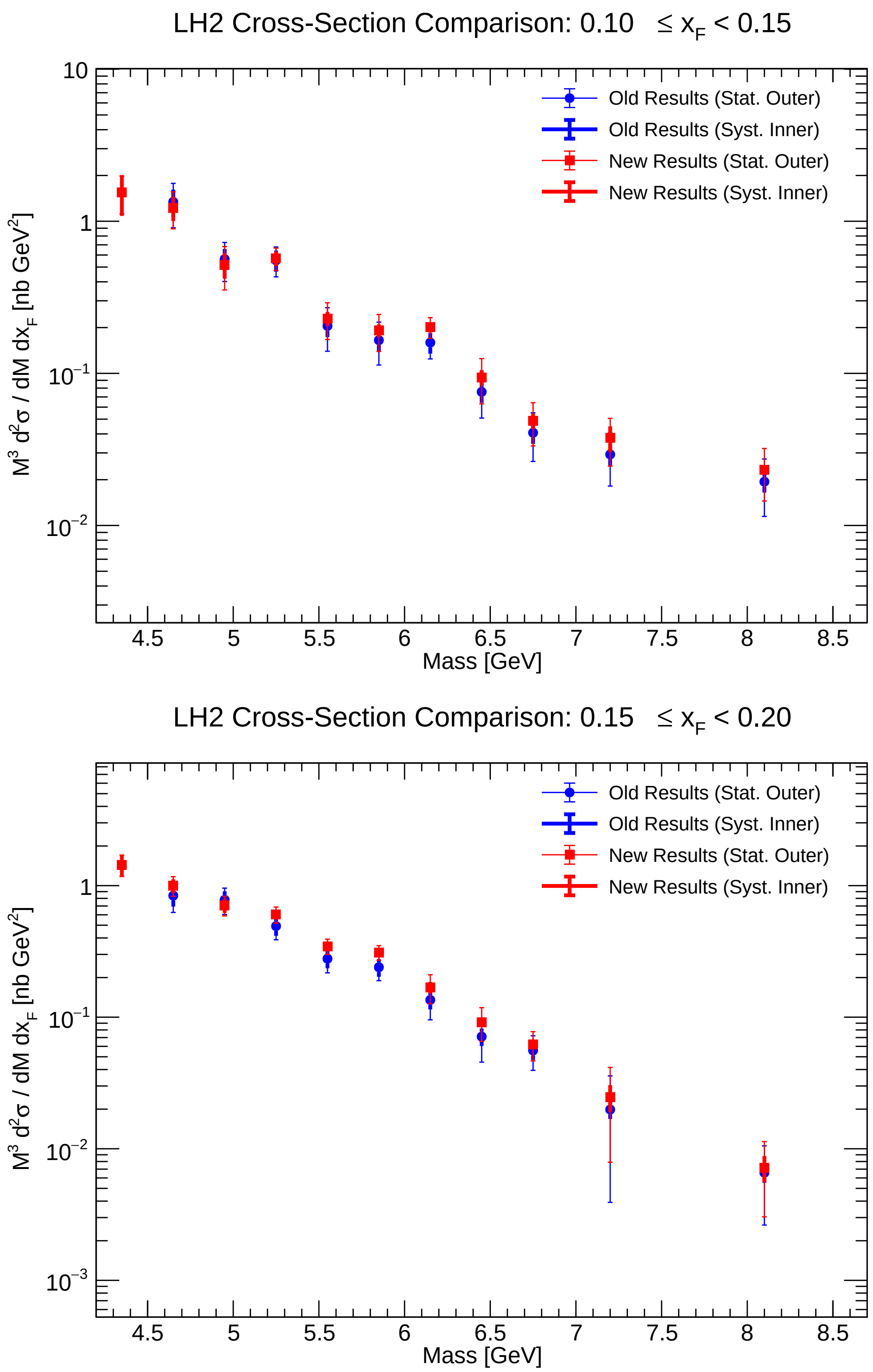
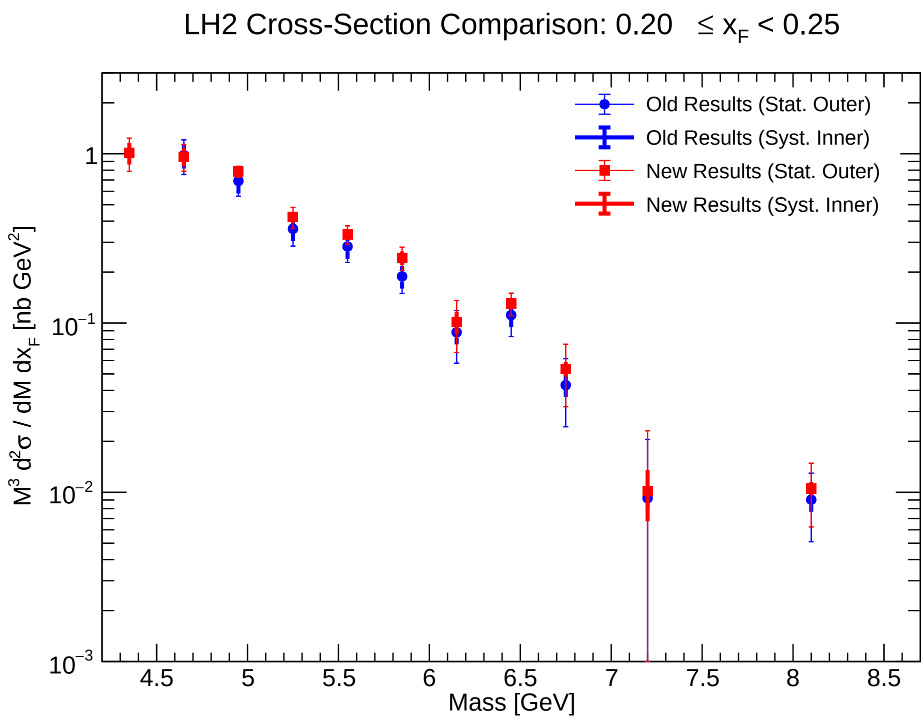


Figure 49: Comparison of double-differential cross-section for LH2 (xF bins 0.10-0.20).



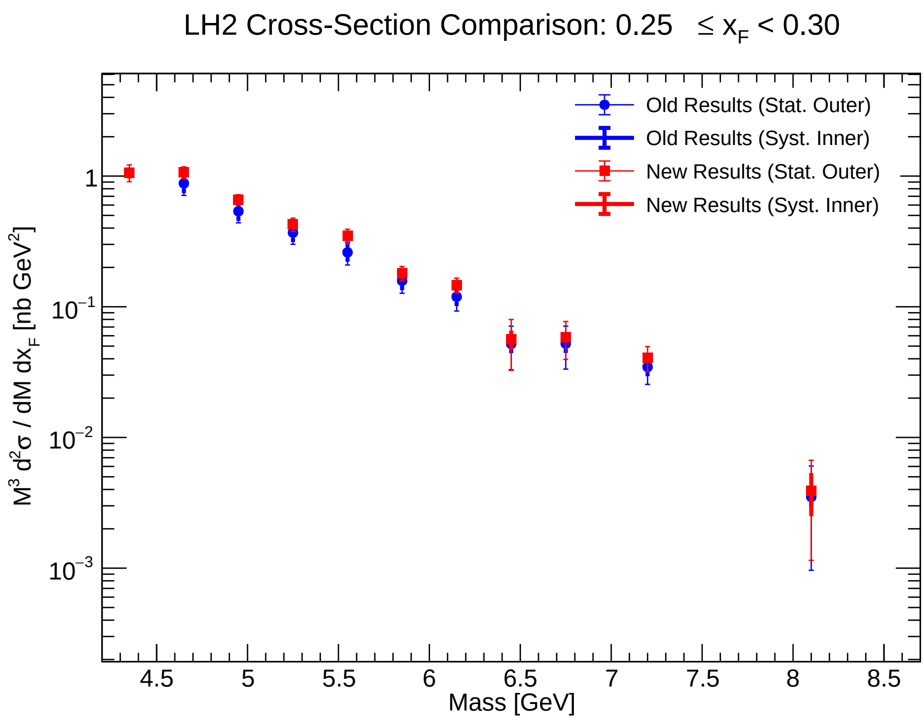
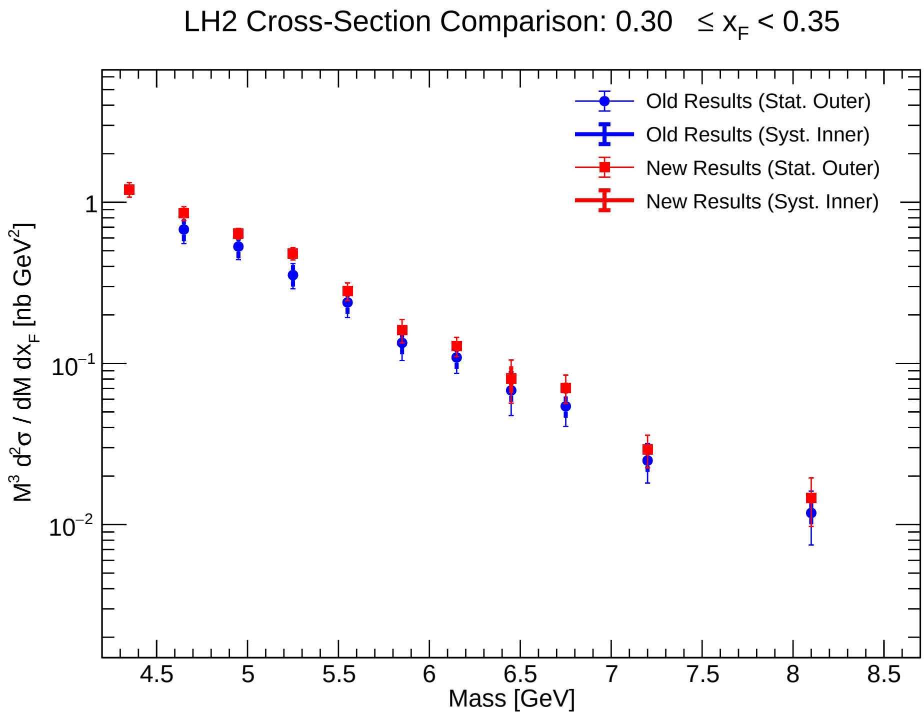


Figure 50: Comparison of double-differential cross-section for LH2 (xF bins 0.20-0.30).



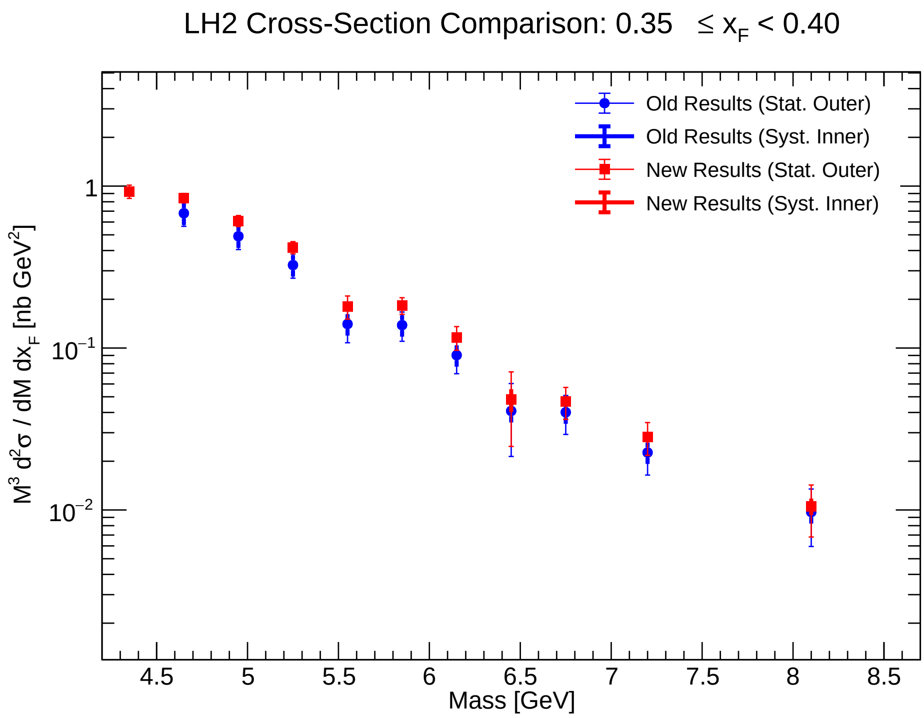


Figure 51: Comparison of double-differential cross-section for LH2 (xF bins 0.30-0.40).

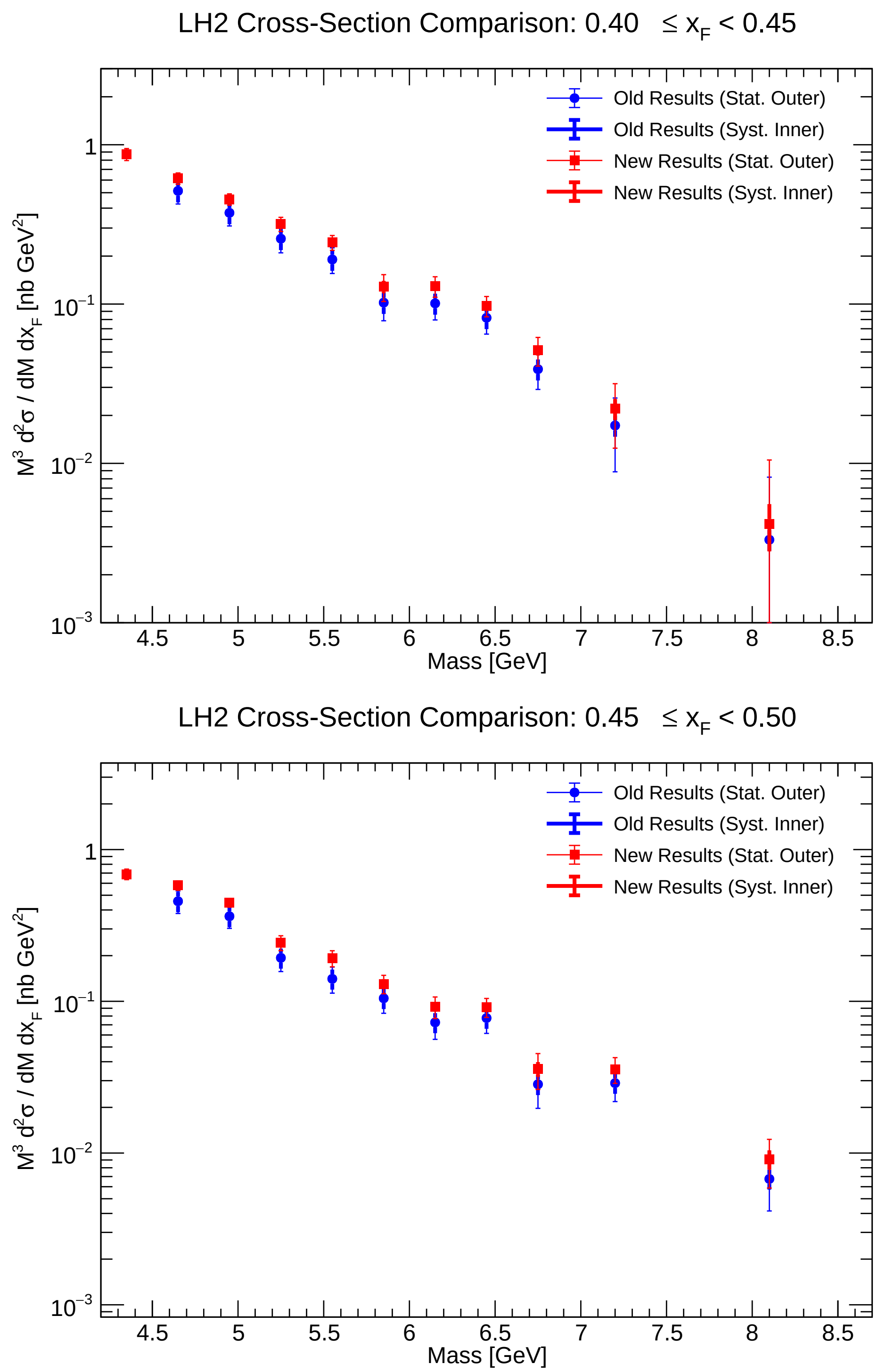


Figure 52: Comparison of double-differential cross-section for LH2 (xF bins 0.40-0.50).

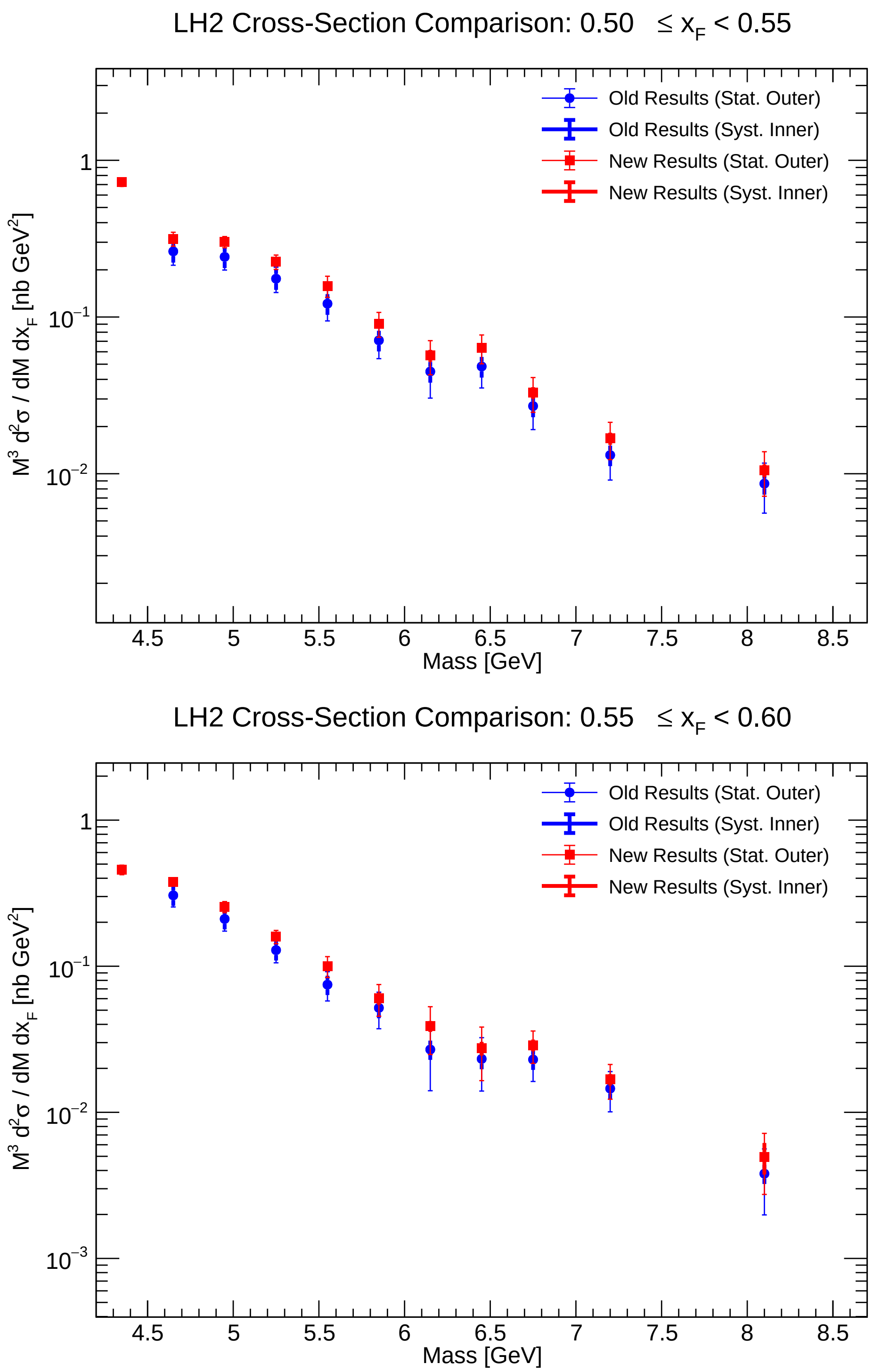


Figure 53: Comparison of double-differential cross-section for LH2 (xF bins 0.50-0.60).

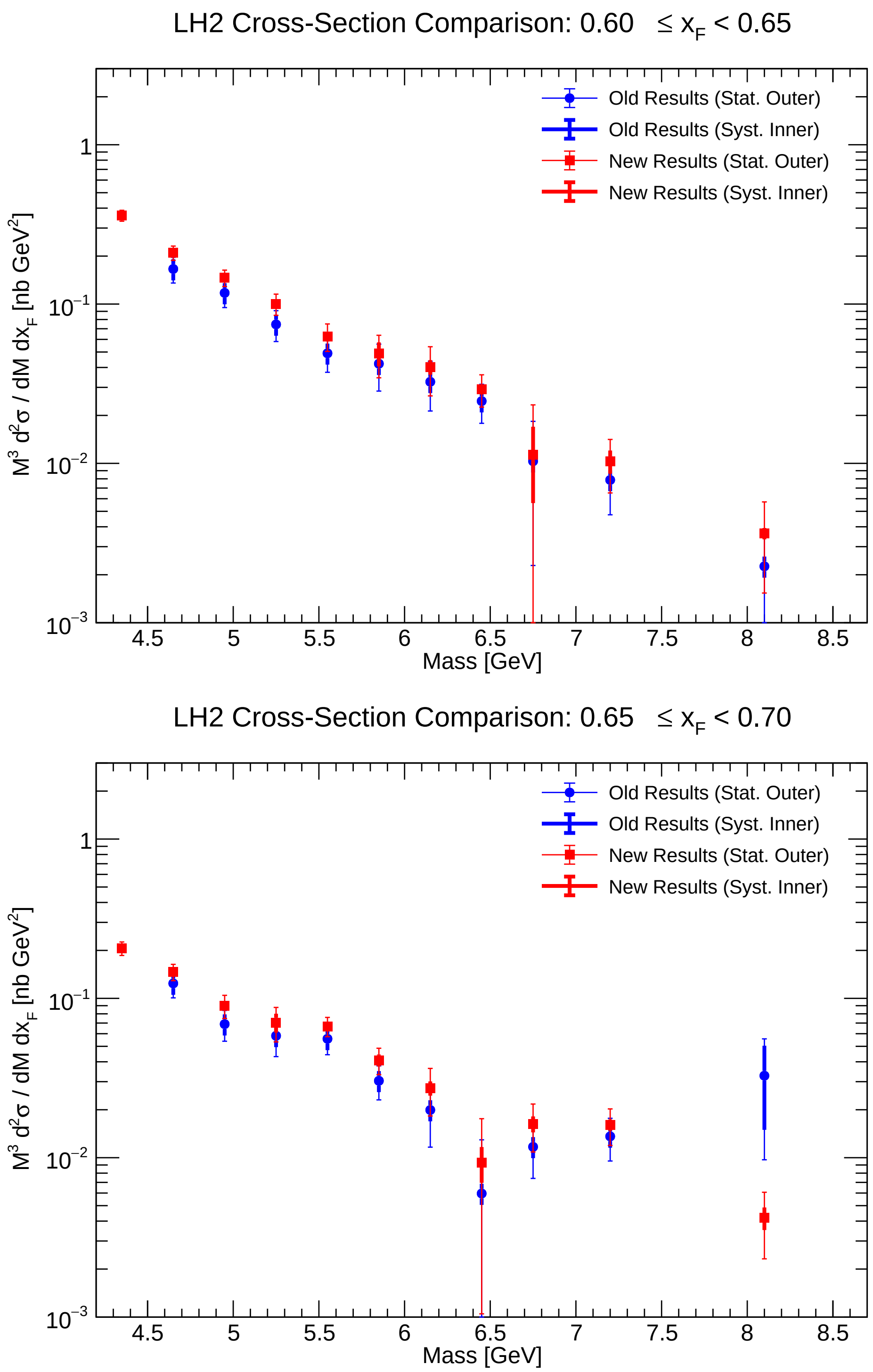
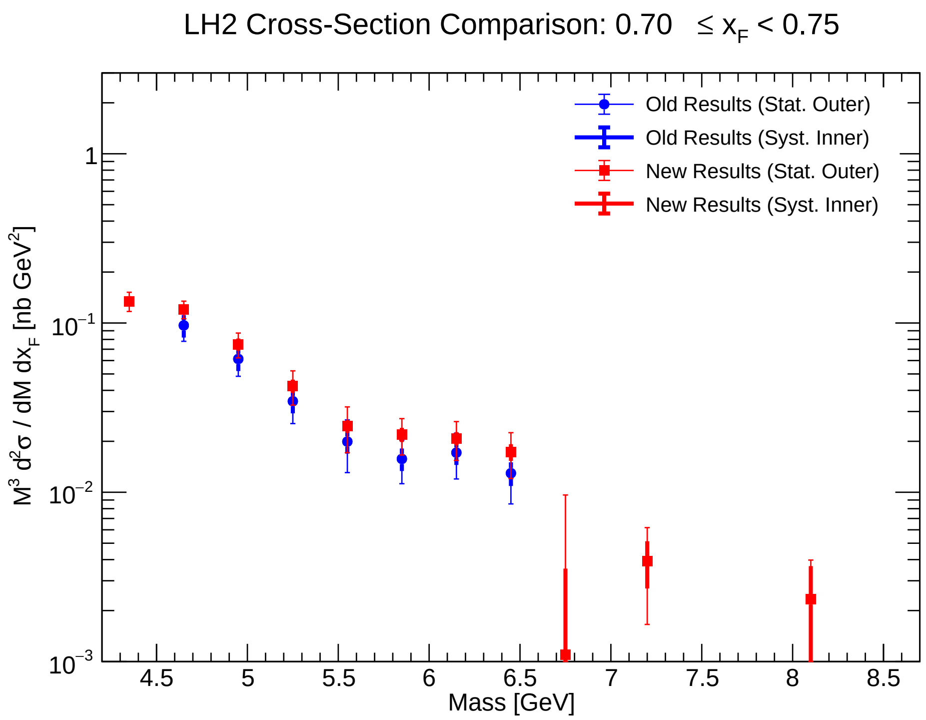


Figure 54: Comparison of double-differential cross-section for LH2 (xF bins 0.60-0.70).



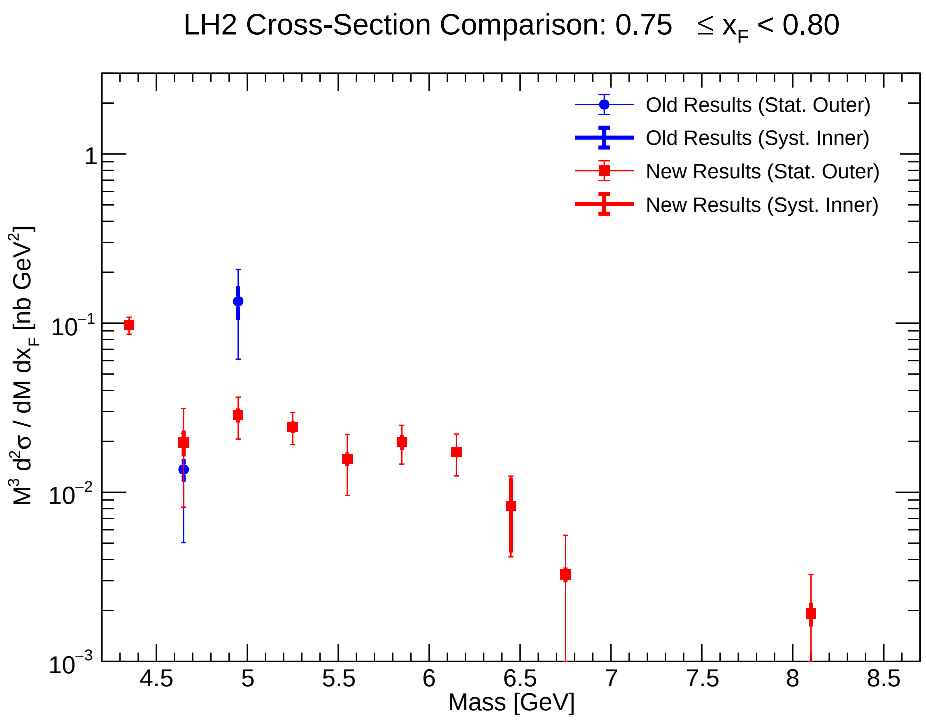


Figure 55: Comparison of double-differential cross-section for LH2 (xF bins 0.70-0.80).

10 Appendix: Event Selection Criteria (Chuck Cuts)

This standardized matrix parameterizes the specific topological limits applied to suppress upstream background interactions, cosmics, and combinatoric tracking vectors inside the spectrometer logic.

In the subsequent tables, the y-axis displacement generated by the beam geometry defines the intrinsic offset $y_{\text{beam}} = 1.6$ cm.

10.1 Single Track Cuts

These discrete geometric and analytic bounds govern individual constituent muon trajectories.

Table 1: Single track selection criteria.

Variable	**Condition**
Track Fit Quality	
Target χ^2	$\chi^2_{\text{target}} < 15$
Reduced χ^2	$\chi^2/(N_{\text{hits}} - 5) < 12$
Vertex Assumption	$\chi^2_{\text{target}} < 1.5 \times \chi^2_{\text{dump}}$ AND $\chi^2_{\text{target}} < 1.5 \times \chi^2_{\text{upstream}}$
Hits & Geometry	
Number of Hits	$N_{\text{hits}} > 13$
Station 1 Z-Momentum	$9 < p_{z,\text{st1}} < 75$ GeV/c
Single Track Vertex z	$-320 < z_v < -5$ cm
Aperture & Trajectory	
Radial pos. at target ($z \approx -130$ cm)	$x^2 + (y - y_{\text{beam}})^2 < 320$ cm^2
Radial pos. at dump ($z \approx 50$ cm)	$16 < x^2 + (y - y_{\text{beam}})^2 < 1100$ cm^2
Vertical Focusing	$y_{\text{st1}}/y_{\text{st3}} < 1$
Vertical Projection	$y_{\text{st1}} \cdot y_{\text{st3}} > 0$
Min Vertical Momentum	$\lvert p_{y,\text{st1}} \rvert > 0.02$ GeV/c
Momentum Conservation	
KMag Momentum Kick	$\lvert \lvert p_{x,\text{st1}} - p_{x,\text{st3}} \rvert - 0.416 \rvert < 0.008$ GeV/c
Vertical Bend (Null)	$\lvert p_{y,\text{st1}} - p_{y,\text{st3}} \rvert < 0.008$ GeV/c
Longitudinal (Null)	$\lvert p_{z,\text{st1}} - p_{z,\text{st3}} \rvert < 0.08$ GeV/c

10.2 Dimuon Cuts

This secondary matrix verifies the topological fidelity of the merged pairs relative to the experimental fiducial volume.

10.3 Occupancy and Topology Cuts

This boundary matrix isolates standard physics triggers explicitly from high-density combinatorial noise loops within the kTracker routine.

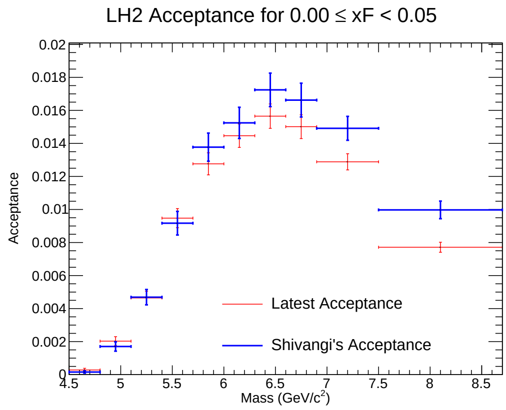
Table 2: Dimuon kinematic and vertex selection criteria.

Variable	Condition
Vertex Position	
Transverse Offset (x)	$\|dx\| < 0.25$ cm
Vertical Offset (y)	$\|dy - y_{\text{beam}}\| < 0.22$ cm
Radial Vertex	$dx^2 + (dy - y_{\text{beam}})^2 < 0.06$ cm^2
Longitudinal Vertex (z)	$-280 < dz < -5$ cm
Vertex Fit Quality	$\chi^2_{\text{dimuon}} < 18$
Vertex Consistency	$\|\chi^2_{\text{trk1}} + \chi^2_{\text{trk2}} - \chi^2_{\text{dimuon}}\| < 2$
Kinematics	
Invariant Mass	$4.2 < M_{\mu\mu} < 8.8$ GeV/c^2
Feynman-x	$-0.1 < x_F < 0.95$
Transverse Scaling x_T	$0.05 < x_T \leq 0.58$
Costh (Collins-Soper)	$\|\cos\theta\| < 0.5$
Longitudinal Momentum	$38 < p_z < 116$ GeV/c
Transverse Momentum limits	$\|dp_x\| < 1.8$ GeV/c, $\|dp_y\| < 2.0$ GeV/c
Total p_T	$dp_x^2 + dp_y^2 < 5.0$ (GeV/c)^2
Track Separation	sep < 270 cm

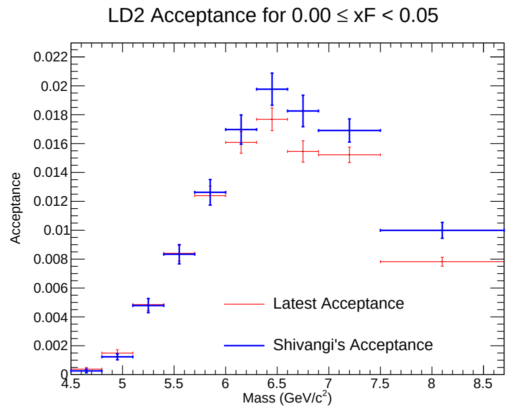
Table 3: Occupancy and topological cuts.

Variable	Condition
Chamber Occupancy	
Drift Chamber 1 Hits	$D1 < 400$
Drift Chamber 2 Hits	$D2 < 400$
Drift Chamber 3 Hits	$D3 < 400$
Total Chamber Hits	$D1 + D2 + D3 < 1000$
Topology	
Opposite Quadrants	$y^{\text{trk1}}_{\text{st3}} \cdot y^{\text{trk2}}_{\text{st3}} < 0$
Total Hits on Tracks	$N^{\text{trk1}}_{\text{hits}} + N^{\text{trk2}}_{\text{hits}} > 29$
Station 1 Hits Sum	$N^{\text{trk1}}_{\text{hits, st1}} + N^{\text{trk2}}_{\text{hits, st1}} > 8$
Station 1 X-Sum	$\|x^{\text{trk1}}_{\text{st1}} + x^{\text{trk2}}_{\text{st1}}\| < 42$ cm

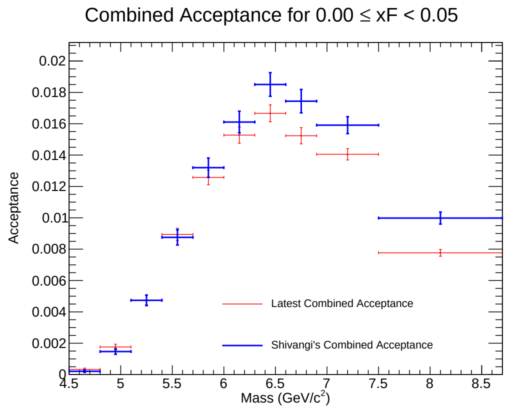
11 Appendix: Acceptance Plots



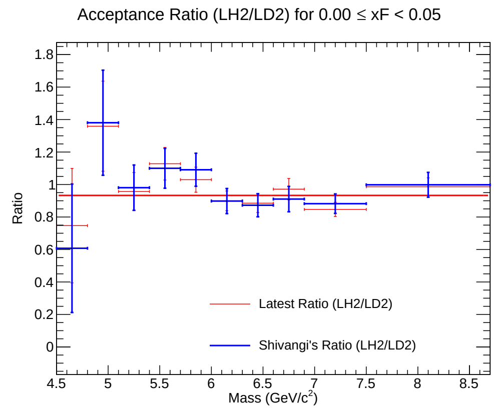
(a) Acceptance for LH2



(b) Acceptance for LD2

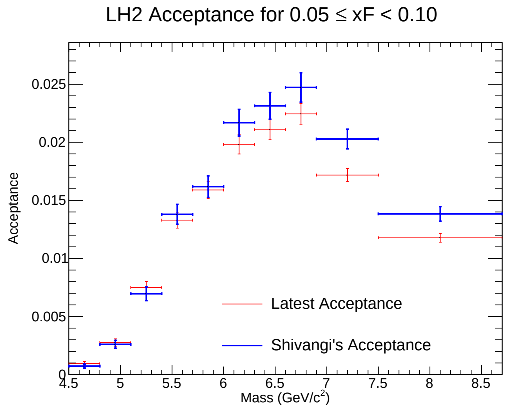


(c) Combined Acceptance

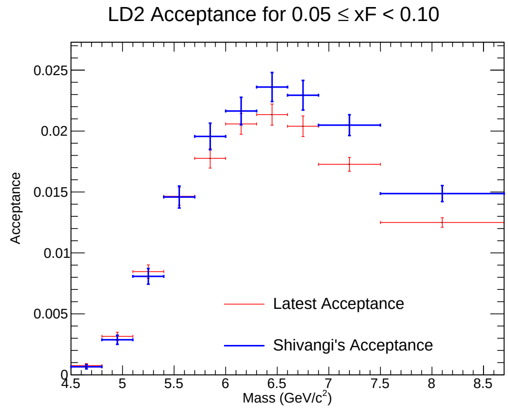


(d) Acceptance Ratio (LH2/LD2)

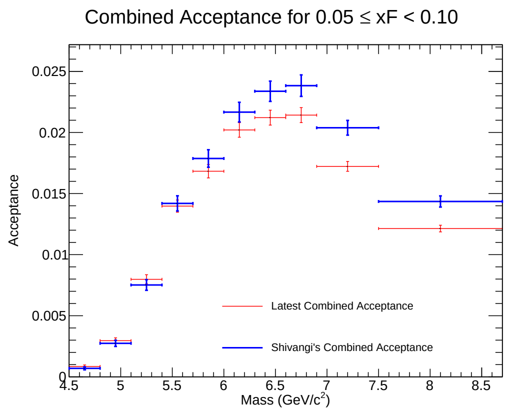
Figure 56: Acceptance plots for $0.00 \leq x_F < 0.05$.



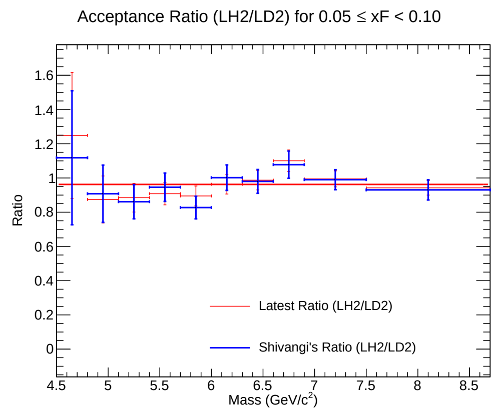
(a) Acceptance for LH2



(b) Acceptance for LD2

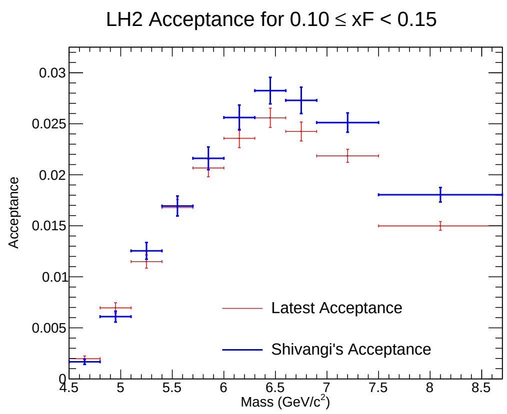


(c) Combined Acceptance

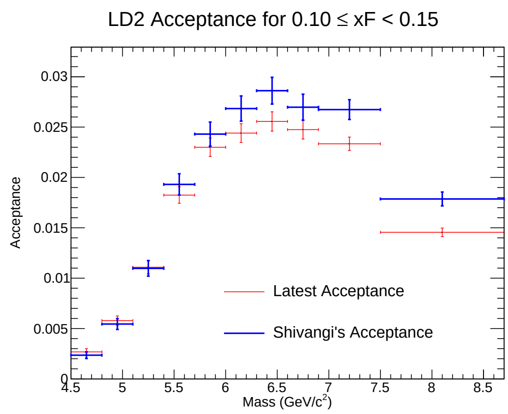


(d) Acceptance Ratio (LH2/LD2)

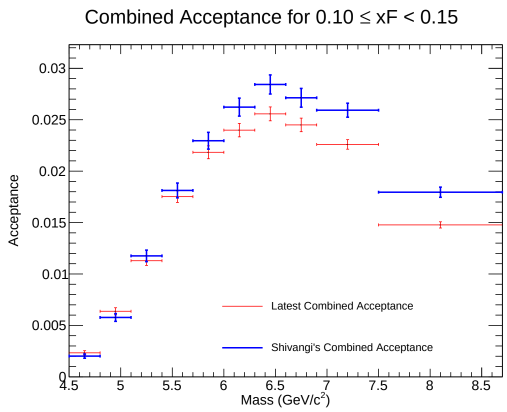
Figure 57: Acceptance plots for $0.05 \leq x_F < 0.10$.



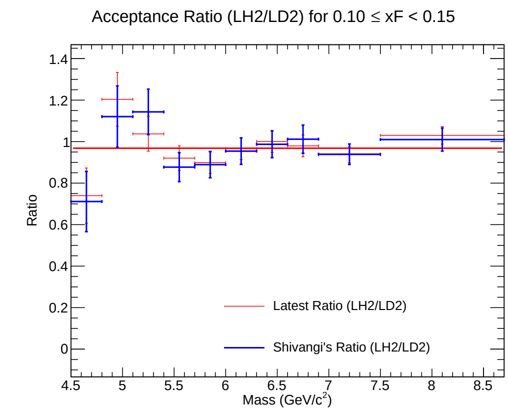
(a) Acceptance for LH2



(b) Acceptance for LD2

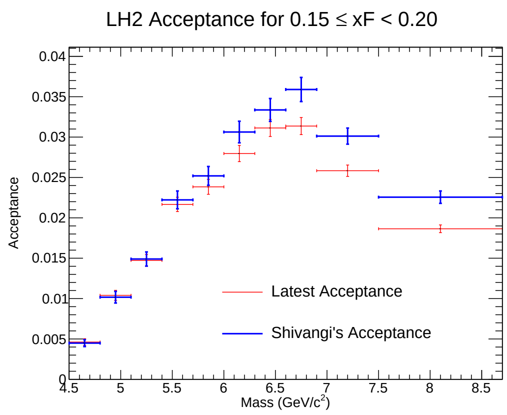


(c) Combined Acceptance

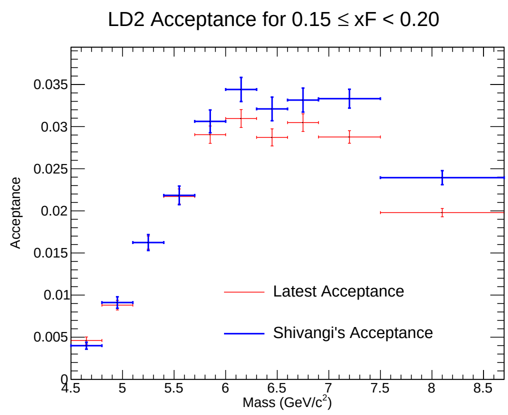


(d) Acceptance Ratio (LH2/LD2)

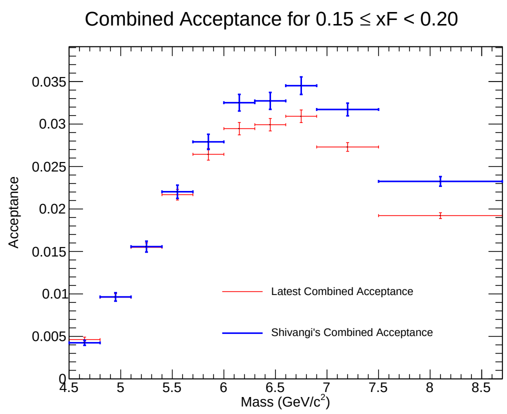
Figure 58: Acceptance plots for $0.10 \leq x_F < 0.15$.



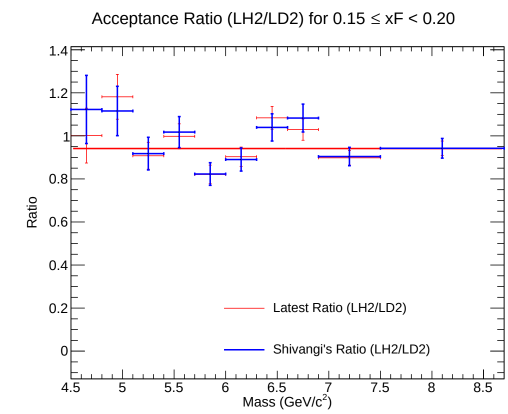
(a) Acceptance for LH2



(b) Acceptance for LD2

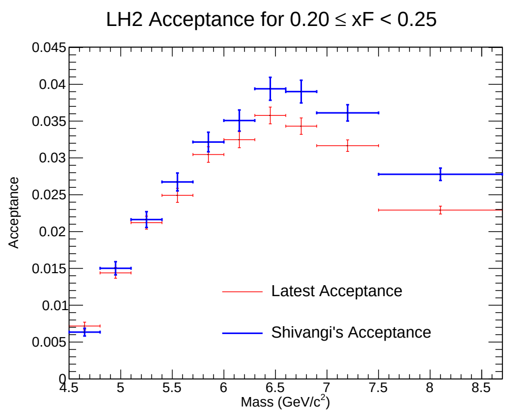


(c) Combined Acceptance

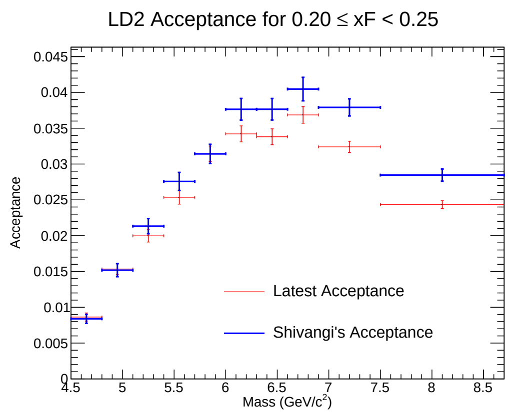


(d) Acceptance Ratio (LH2/LD2)

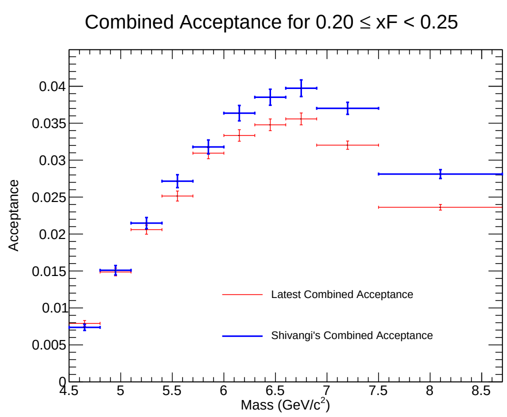
Figure 59: Acceptance plots for $0.15 \le x_F < 0.20$.



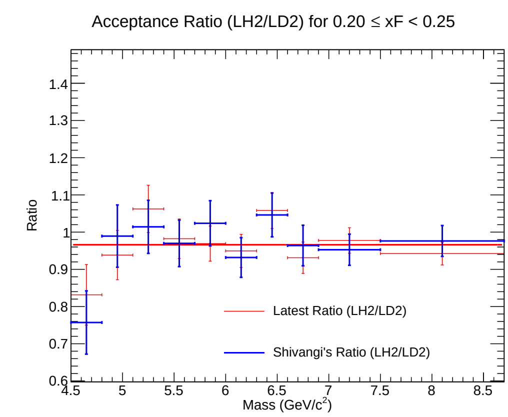
(a) Acceptance for LH2



(b) Acceptance for LD2

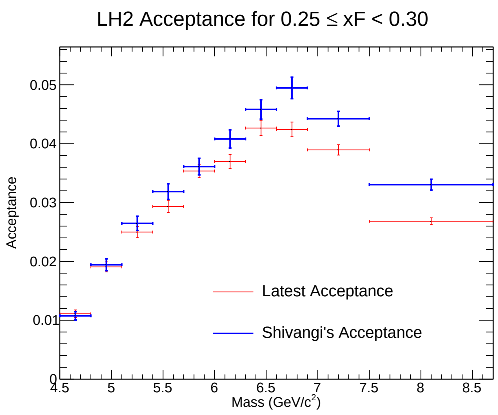


(c) Combined Acceptance

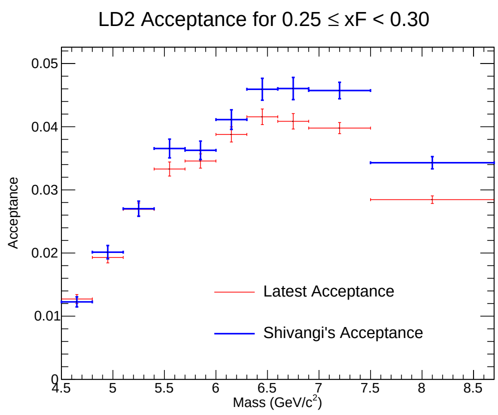


(d) Acceptance Ratio (LH2/LD2)

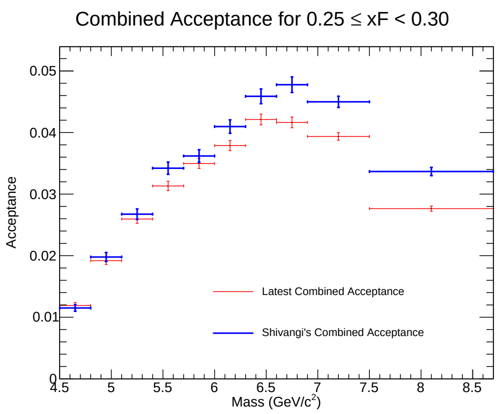
Figure 60: Acceptance plots for $0.20 \leq x_F < 0.25$.



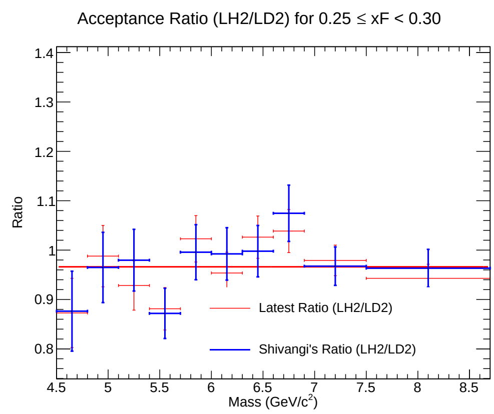
(a) Acceptance for LH2



(b) Acceptance for LD2

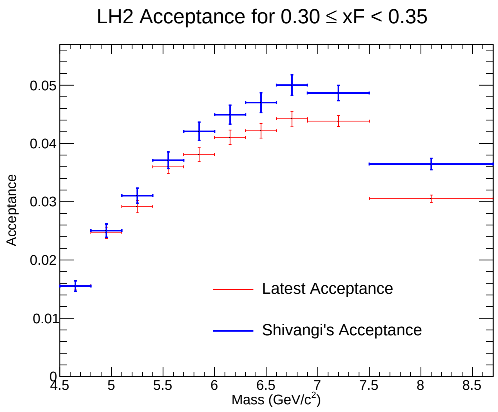


(c) Combined Acceptance

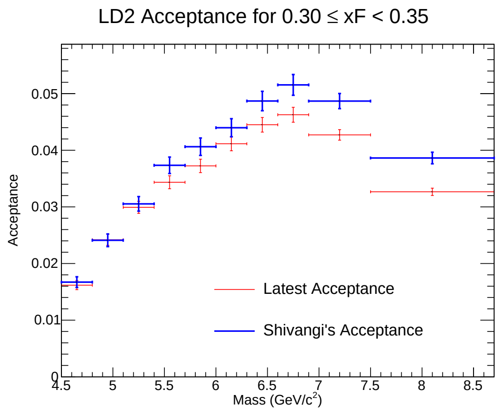


(d) Acceptance Ratio (LH2/LD2)

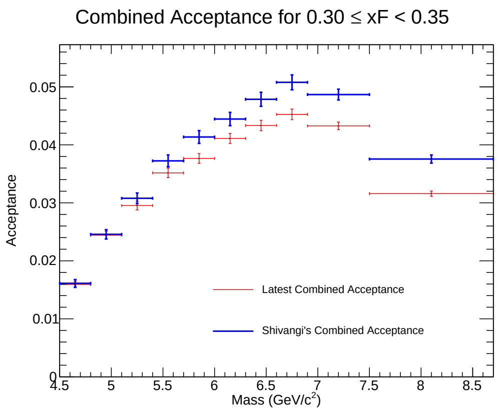
Figure 61: Acceptance plots for $0.25 \leq x_F < 0.30$.



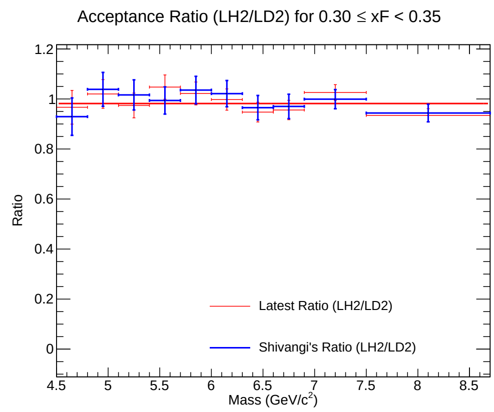
(a) Acceptance for LH2



(b) Acceptance for LD2

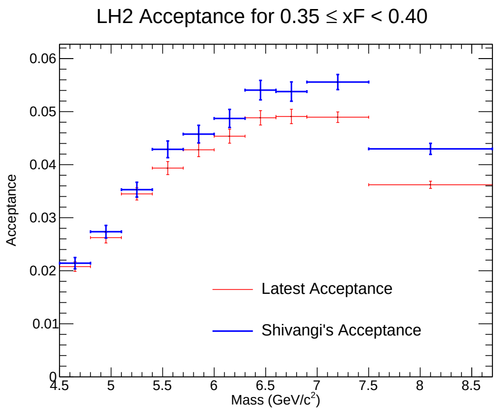


(c) Combined Acceptance

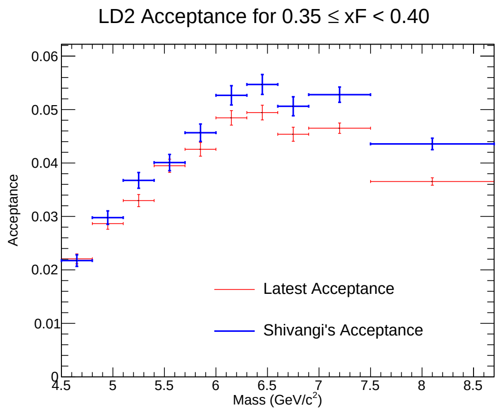


(d) Acceptance Ratio (LH2/LD2)

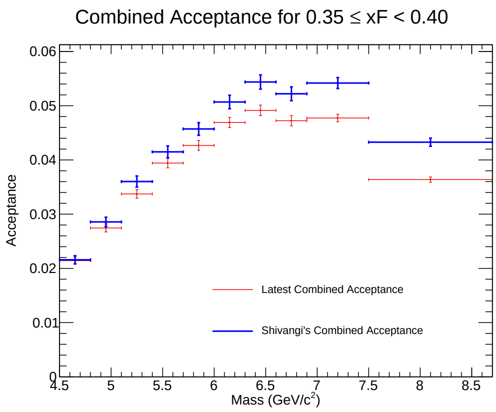
Figure 62: Acceptance plots for $0.30 \leq x_F < 0.35$.



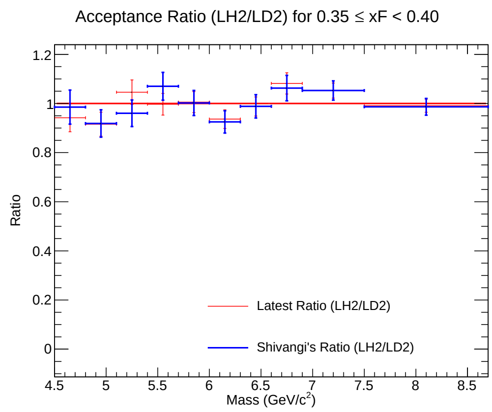
(a) Acceptance for LH2



(b) Acceptance for LD2

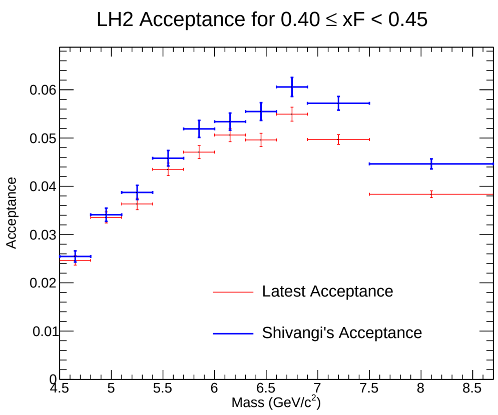


(c) Combined Acceptance

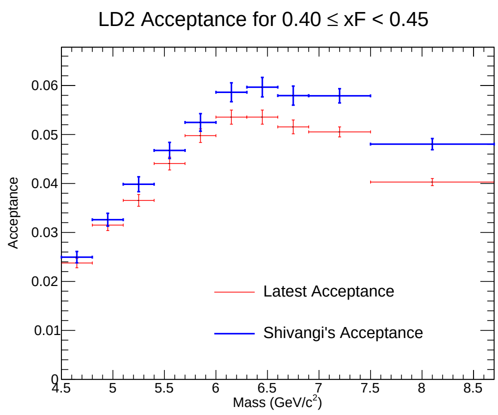


(d) Acceptance Ratio (LH2/LD2)

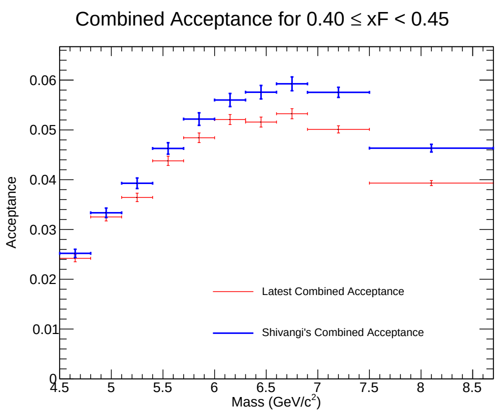
Figure 63: Acceptance plots for $0.35 \leq x_F < 0.40$.



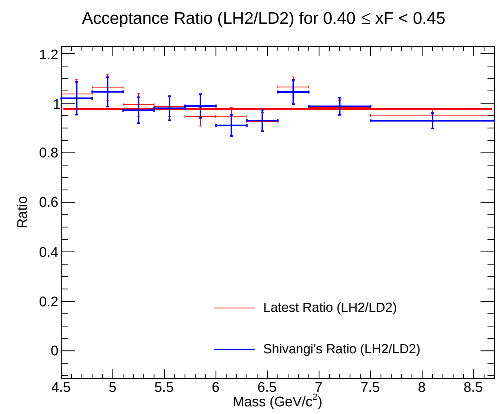
(a) Acceptance for LH2



(b) Acceptance for LD2

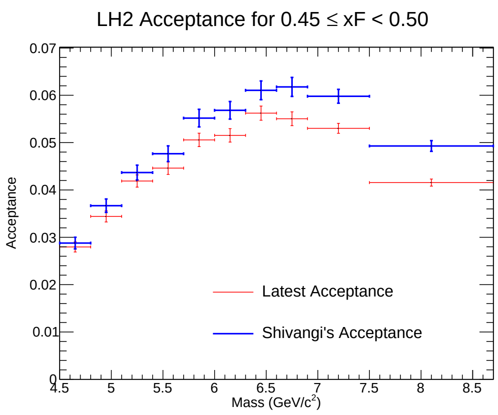


(c) Combined Acceptance

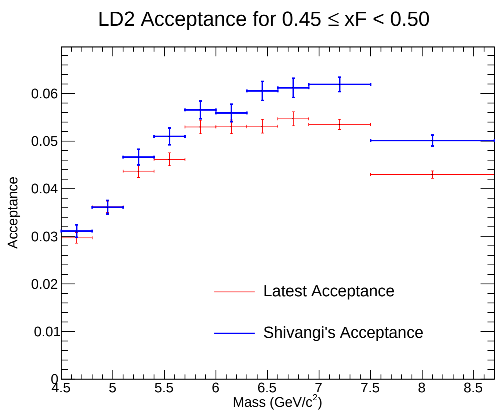


(d) Acceptance Ratio (LH2/LD2)

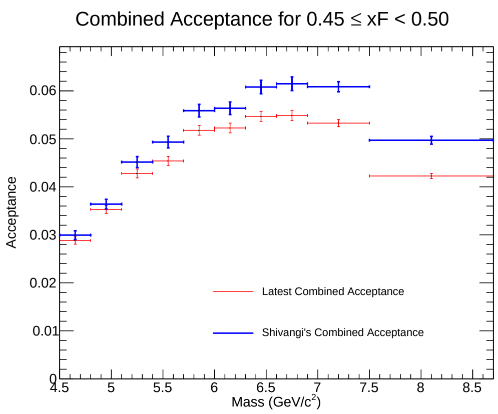
Figure 64: Acceptance plots for $0.40 \leq x_F < 0.45$.



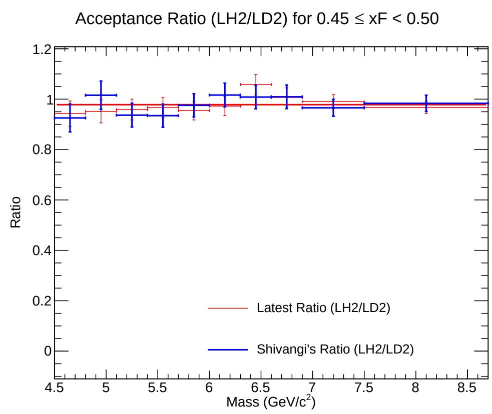
(a) Acceptance for LH2



(b) Acceptance for LD2

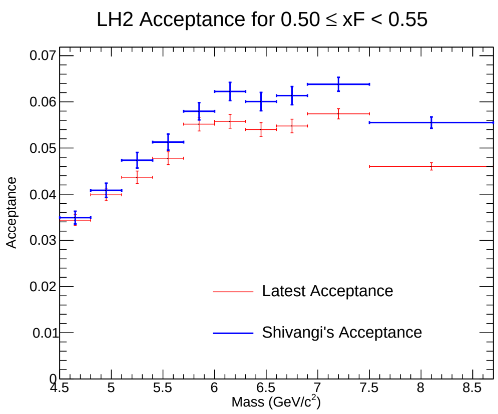


(c) Combined Acceptance

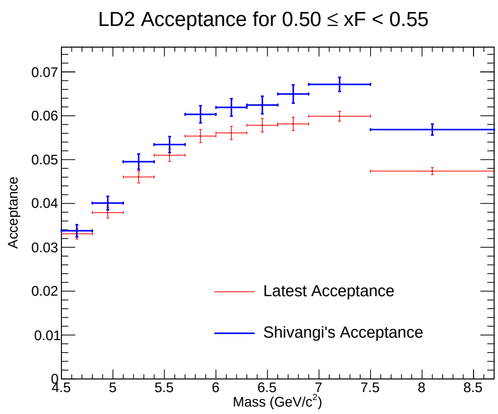


(d) Acceptance Ratio (LH2/LD2)

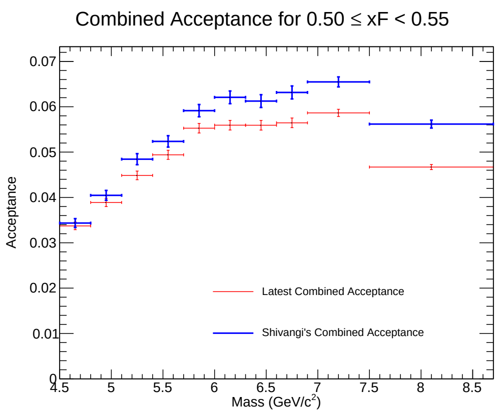
Figure 65: Acceptance plots for $0.45 \leq x_F < 0.50$.



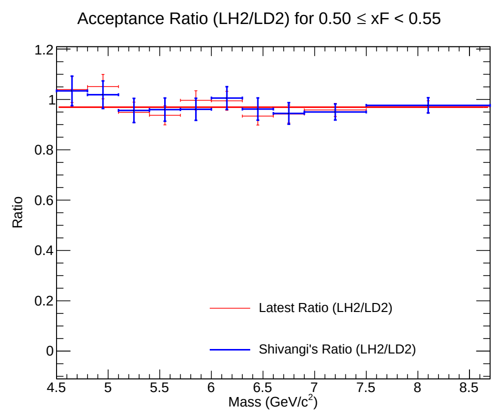
(a) Acceptance for LH2



(b) Acceptance for LD2

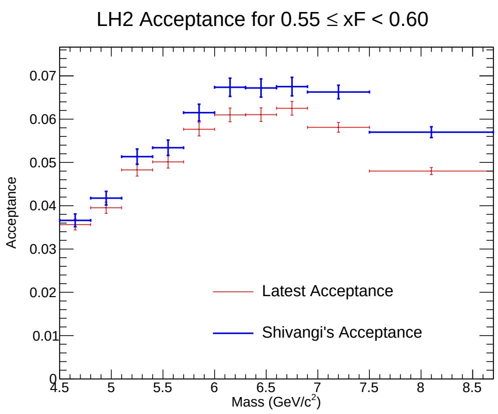


(c) Combined Acceptance

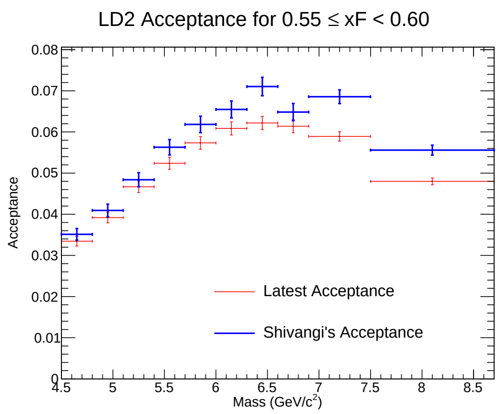


(d) Acceptance Ratio (LH2/LD2)

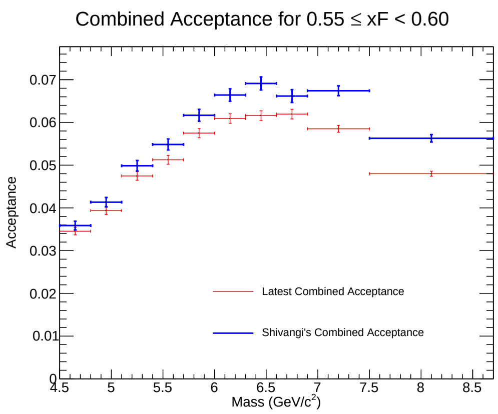
Figure 66: Acceptance plots for $0.50 \leq x_F < 0.55$.



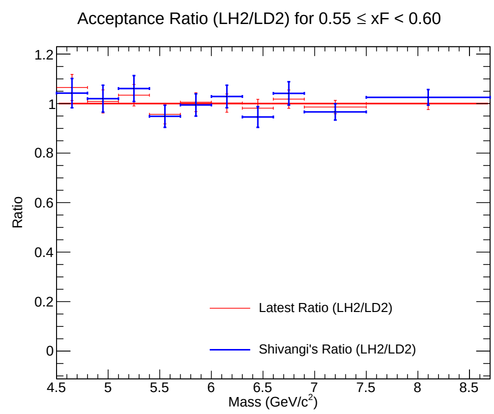
(a) Acceptance for LH2



(b) Acceptance for LD2

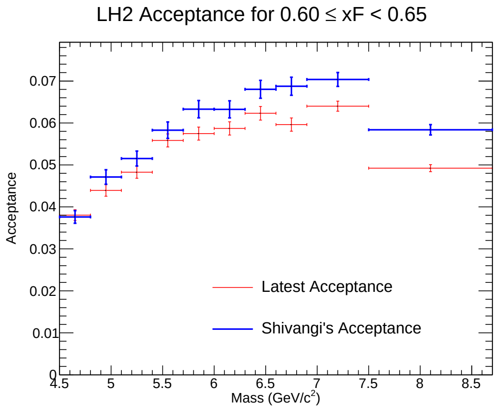


(c) Combined Acceptance

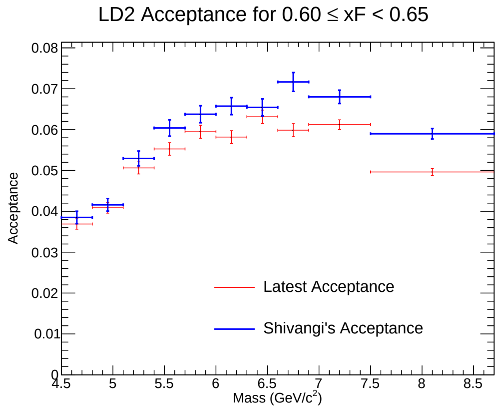


(d) Acceptance Ratio (LH2/LD2)

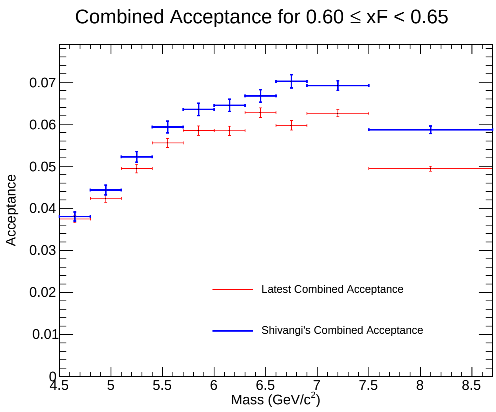
Figure 67: Acceptance plots for $0.55 \leq x_F < 0.60$.



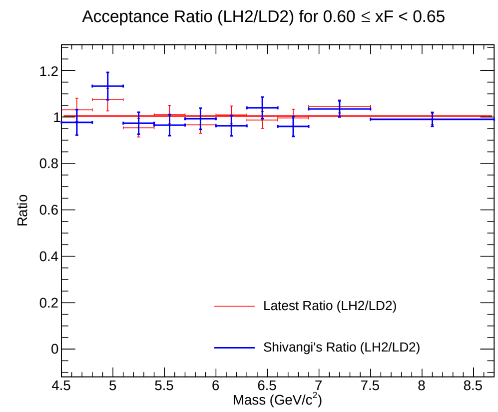
(a) Acceptance for LH2



(b) Acceptance for LD2

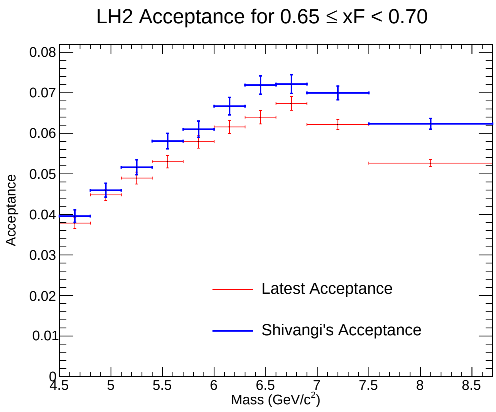


(c) Combined Acceptance

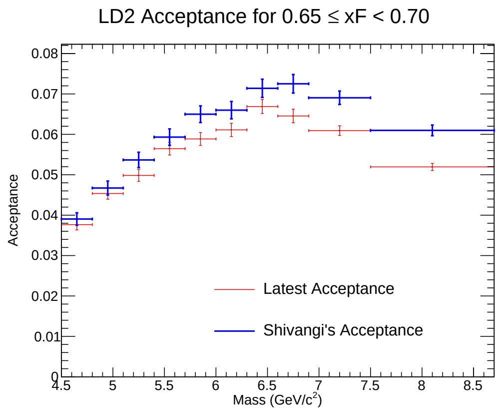


(d) Acceptance Ratio (LH2/LD2)

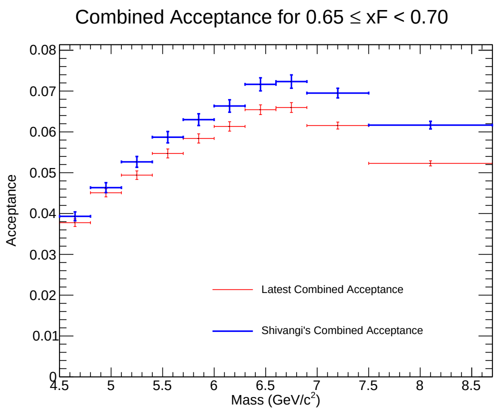
Figure 68: Acceptance plots for $0.60 \leq x_F < 0.65$.



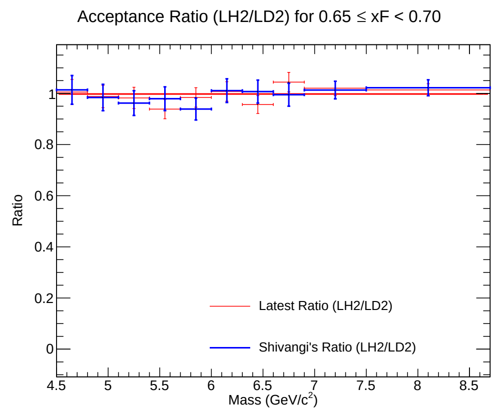
(a) Acceptance for LH2



(b) Acceptance for LD2

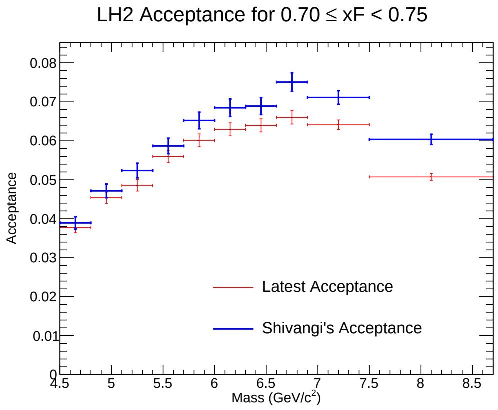


(c) Combined Acceptance

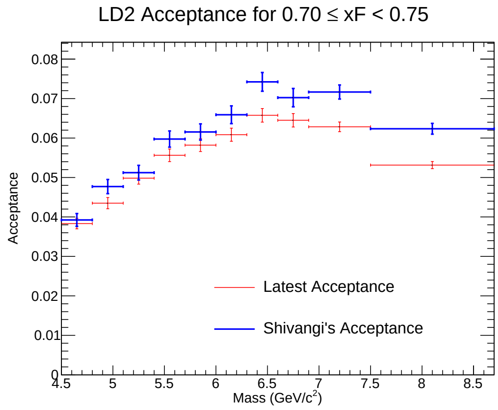


(d) Acceptance Ratio (LH2/LD2)

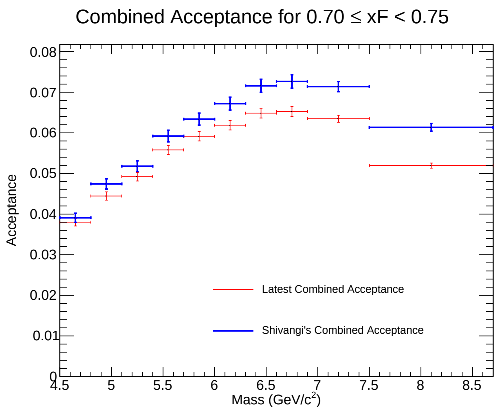
Figure 69: Acceptance plots for $0.65 \leq x_F < 0.70$.



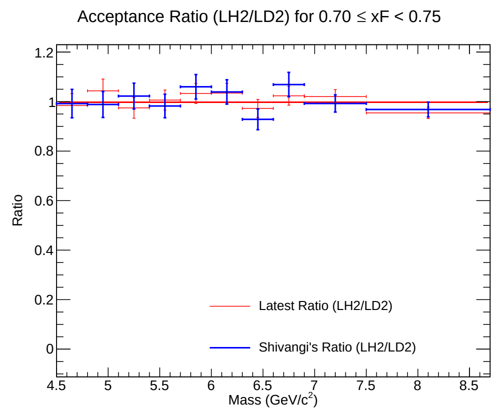
(a) Acceptance for LH2



(b) Acceptance for LD2

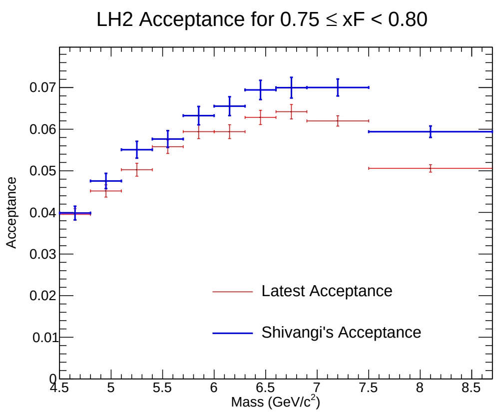


(c) Combined Acceptance

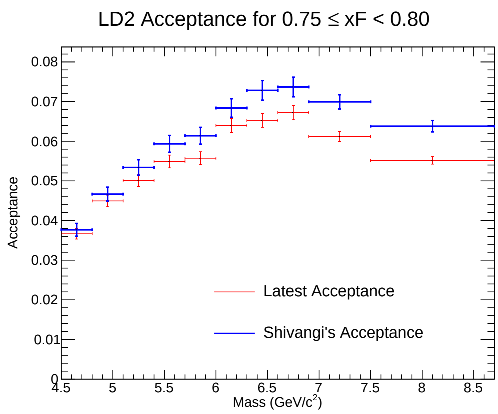


(d) Acceptance Ratio (LH2/LD2)

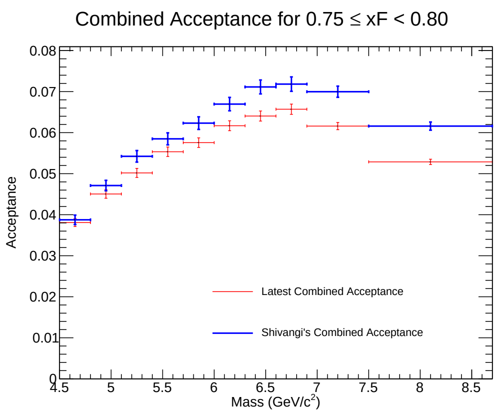
Figure 70: Acceptance plots for $0.70 \leq x_F < 0.75$.



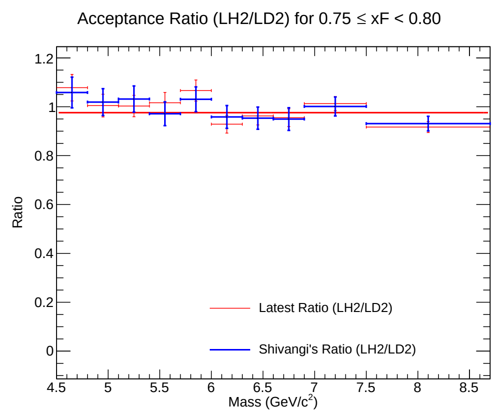
(a) Acceptance for LH2



(b) Acceptance for LD2



(c) Combined Acceptance



(d) Acceptance Ratio (LH2/LD2)

Figure 71: Acceptance plots for $0.75 \leq x_F < 0.80$.

12 Appendix: Data Yield Tables

This section contains the tabulated yield arrays for the three respective target volumes (LH2, LD2, Flask) employed in this analysis cascade. Each table parameterizes the raw prompt hit counts (Y_{total}), the extracted combinatorial matrices (Y_{mix}), the synthetic signal efficiency scalar ($\langle\epsilon_{\text{signal}}\rangle$), and the resultant isolated physical yield ($Y_{\text{corrected}}$).

Table 4: Combined Data Table for LH2

xF bin	**Mass (GeV)**	$Y_{total} \pm \delta Y_{total}$	$Y_{mix} \pm \delta Y_{mix}$	$\langle \epsilon_{signal} \rangle$	$Y_{corr} \pm \delta Y_{corr}$
[0.00, 0.05)	[4.20, 4.50)	1 ± 1.000	0 ± 0.000	0.6494 ± 0.0850	1.540 ± 1.553
[0.00, 0.05)	[4.50, 4.80)	9 ± 3.000	1 ± 1.000	0.5586 ± 0.0490	14.321 ± 5.798
[0.00, 0.05)	[4.80, 5.10)	41 ± 6.403	9 ± 3.000	0.5349 ± 0.0485	59.822 ± 14.288
[0.00, 0.05)	[5.10, 5.40)	72 ± 8.485	7 ± 2.646	0.4684 ± 0.0311	138.784 ± 21.093
[0.00, 0.05)	[5.40, 5.70)	66 ± 8.124	8 ± 2.828	0.4860 ± 0.0313	119.342 ± 19.294
[0.00, 0.05)	[5.70, 6.00)	37 ± 6.083	1 ± 1.000	0.4947 ± 0.0316	72.765 ± 13.298
[0.00, 0.05)	[6.00, 6.30)	27 ± 5.196	1 ± 1.000	0.4757 ± 0.0444	54.652 ± 12.235
[0.00, 0.05)	[6.30, 6.60)	15 ± 3.873	0 ± 0.000	0.4754 ± 0.0474	31.555 ± 8.734
[0.00, 0.05)	[6.60, 6.90)	12 ± 3.464	0 ± 0.000	0.5021 ± 0.0499	23.897 ± 7.297
[0.00, 0.05)	[6.90, 7.50)	9 ± 3.000	0 ± 0.000	0.4322 ± 0.0754	20.824 ± 7.835
[0.00, 0.05)	[7.50, 8.70)	1 ± 1.000	0 ± 0.000	0.4512 ± 0.1417	2.216 ± 2.323
[0.05, 0.10)	[4.20, 4.50)	2 ± 1.414	1 ± 1.000	0.8712 ± 0.2920	1.148 ± 2.025
[0.05, 0.10)	[4.50, 4.80)	39 ± 6.245	5 ± 2.236	0.5110 ± 0.0457	66.538 ± 14.282
[0.05, 0.10)	[4.80, 5.10)	83 ± 9.110	21 ± 4.583	0.4931 ± 0.0413	125.744 ± 23.206
[0.05, 0.10)	[5.10, 5.40)	97 ± 9.849	17 ± 4.123	0.5069 ± 0.0278	157.810 ± 22.768
[0.05, 0.10)	[5.40, 5.70)	78 ± 8.832	17 ± 4.123	0.5219 ± 0.0320	116.882 ± 20.004
[0.05, 0.10)	[5.70, 6.00)	54 ± 7.348	4 ± 2.000	0.5055 ± 0.0264	98.921 ± 15.928
[0.05, 0.10)	[6.00, 6.30)	39 ± 6.245	2 ± 1.414	0.5134 ± 0.0282	72.072 ± 13.084
[0.05, 0.10)	[6.30, 6.60)	25 ± 5.000	0 ± 0.000	0.4610 ± 0.0467	54.226 ± 12.156
[0.05, 0.10)	[6.60, 6.90)	5 ± 2.236	2 ± 1.414	0.3960 ± 0.2622	7.576 ± 8.355
[0.05, 0.10)	[6.90, 7.50)	7 ± 2.646	0 ± 0.000	0.4634 ± 0.0693	15.107 ± 6.140
[0.05, 0.10)	[7.50, 8.70)	6 ± 2.449	0 ± 0.000	0.4436 ± 0.1315	13.527 ± 6.824
[0.10, 0.15)	[4.20, 4.50)	32 ± 5.657	9 ± 3.000	0.5097 ± 0.0652	45.121 ± 13.823
[0.10, 0.15)	[4.50, 4.80)	99 ± 9.950	16 ± 4.000	0.5114 ± 0.0290	162.315 ± 22.906
[0.10, 0.15)	[4.80, 5.10)	138 ± 11.747	33 ± 5.745	0.5173 ± 0.0273	202.965 ± 27.455
[0.10, 0.15)	[5.10, 5.40)	136 ± 11.662	13 ± 3.606	0.4822 ± 0.0232	255.106 ± 28.134
[0.10, 0.15)	[5.40, 5.70)	87 ± 9.327	8 ± 2.828	0.5104 ± 0.0253	154.774 ± 20.581
[0.10, 0.15)	[5.70, 6.00)	78 ± 8.832	6 ± 2.449	0.5077 ± 0.0251	141.810 ± 19.369
[0.10, 0.15)	[6.00, 6.30)	52 ± 7.211	4 ± 2.000	0.4778 ± 0.0325	100.460 ± 17.083
[0.10, 0.15)	[6.30, 6.60)	28 ± 5.292	0 ± 0.000	0.5134 ± 0.0340	54.534 ± 10.920
[0.10, 0.15)	[6.60, 6.90)	10 ± 3.162	0 ± 0.000	0.5302 ± 0.0577	18.861 ± 6.308
[0.10, 0.15)	[6.90, 7.50)	11 ± 3.317	1 ± 1.000	0.4480 ± 0.0853	22.320 ± 8.822
[0.10, 0.15)	[7.50, 8.70)	7 ± 2.646	0 ± 0.000	0.4926 ± 0.0351	14.211 ± 5.466
[0.15, 0.20)	[4.20, 4.50)	84 ± 9.165	27 ± 5.196	0.5286 ± 0.0403	107.828 ± 21.560
[0.15, 0.20)	[4.50, 4.80)	167 ± 12.923	34 ± 5.831	0.5081 ± 0.0251	261.769 ± 30.765
[0.15, 0.20)	[4.80, 5.10)	232 ± 15.232	41 ± 6.403	0.4922 ± 0.0194	388.044 ± 36.874
[0.15, 0.20)	[5.10, 5.40)	205 ± 14.318	21 ± 4.583	0.4846 ± 0.0176	379.671 ± 33.958
[0.15, 0.20)	[5.40, 5.70)	113 ± 10.630	6 ± 2.449	0.4776 ± 0.0199	224.040 ± 24.670
[0.15, 0.20)	[5.70, 6.00)	94 ± 9.695	2 ± 1.414	0.4877 ± 0.0229	188.634 ± 21.959

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Table 4 – continued from previous page

xF bin	Mass (GeV)	$Y_{total} \pm \delta Y_{total}$	$Y_{mix} \pm \delta Y_{mix}$	$\langle \epsilon_{signal} \rangle$	$Y_{corr} \pm \delta Y_{corr}$
[0.15, 0.20)	[6.00, 6.30)	67 ± 8.185	1 ± 1.000	0.4876 ± 0.0236	135.364 ± 18.135
[0.15, 0.20)	[6.30, 6.60)	35 ± 5.916	0 ± 0.000	0.5322 ± 0.0283	65.769 ± 11.654
[0.15, 0.20)	[6.60, 6.90)	16 ± 4.000	0 ± 0.000	0.5152 ± 0.0389	31.055 ± 8.110
[0.15, 0.20)	[6.90, 7.50)	13 ± 3.606	0 ± 0.000	0.4856 ± 0.0476	26.769 ± 7.873
[0.15, 0.20)	[7.50, 8.70)	3 ± 1.732	0 ± 0.000	0.5078 ± 0.1132	5.908 ± 3.656
[0.20, 0.25)	[4.20, 4.50)	184 ± 13.565	48 ± 6.928	0.5225 ± 0.0262	260.280 ± 31.934
[0.20, 0.25)	[4.50, 4.80)	284 ± 16.852	44 ± 6.633	0.4995 ± 0.0165	480.520 ± 39.586
[0.20, 0.25)	[4.80, 5.10)	273 ± 16.523	39 ± 6.245	0.5139 ± 0.0151	455.326 ± 36.876
[0.20, 0.25)	[5.10, 5.40)	208 ± 14.422	21 ± 4.583	0.4960 ± 0.0173	377.032 ± 33.223
[0.20, 0.25)	[5.40, 5.70)	144 ± 12.000	11 ± 3.317	0.4977 ± 0.0204	267.205 ± 27.305
[0.20, 0.25)	[5.70, 6.00)	106 ± 10.296	7 ± 2.646	0.5132 ± 0.0208	192.897 ± 22.142
[0.20, 0.25)	[6.00, 6.30)	56 ± 7.483	1 ± 1.000	0.5163 ± 0.0235	106.525 ± 15.406
[0.20, 0.25)	[6.30, 6.60)	46 ± 6.782	1 ± 1.000	0.5264 ± 0.0291	85.493 ± 13.857
[0.20, 0.25)	[6.60, 6.90)	21 ± 4.583	1 ± 1.000	0.5377 ± 0.0323	37.194 ± 9.005
[0.20, 0.25)	[6.90, 7.50)	10 ± 3.162	1 ± 1.000	0.5145 ± 0.0699	17.493 ± 6.871
[0.20, 0.25)	[7.50, 8.70)	6 ± 2.449	0 ± 0.000	0.5635 ± 0.0501	10.647 ± 4.449
[0.25, 0.30)	[4.20, 4.50)	363 ± 19.053	58 ± 7.616	0.5147 ± 0.0143	592.527 ± 43.132
[0.25, 0.30)	[4.50, 4.80)	402 ± 20.050	64 ± 8.000	0.5063 ± 0.0136	667.545 ± 46.227
[0.25, 0.30)	[4.80, 5.10)	315 ± 17.748	27 ± 5.196	0.5032 ± 0.0132	572.371 ± 39.716
[0.25, 0.30)	[5.10, 5.40)	243 ± 15.588	28 ± 5.292	0.5028 ± 0.0156	427.598 ± 35.323
[0.25, 0.30)	[5.40, 5.70)	180 ± 13.416	11 ± 3.317	0.4907 ± 0.0160	344.420 ± 30.333
[0.25, 0.30)	[5.70, 6.00)	89 ± 9.434	8 ± 2.828	0.5153 ± 0.0235	157.193 ± 20.411
[0.25, 0.30)	[6.00, 6.30)	60 ± 7.746	0 ± 0.000	0.5239 ± 0.0211	114.515 ± 15.489
[0.25, 0.30)	[6.30, 6.60)	38 ± 6.164	3 ± 1.732	0.5455 ± 0.0368	64.161 ± 12.510
[0.25, 0.30)	[6.60, 6.90)	26 ± 5.099	1 ± 1.000	0.5265 ± 0.0331	47.482 ± 10.309
[0.25, 0.30)	[6.90, 7.50)	24 ± 4.899	1 ± 1.000	0.5310 ± 0.0481	43.312 ± 10.199
[0.25, 0.30)	[7.50, 8.70)	2 ± 1.414	0 ± 0.000	0.4258 ± 0.1544	4.697 ± 3.733
[0.30, 0.35)	[4.20, 4.50)	540 ± 23.238	81 ± 9.000	0.5114 ± 0.0118	897.618 ± 52.942
[0.30, 0.35)	[4.50, 4.80)	487 ± 22.068	62 ± 7.874	0.5119 ± 0.0115	830.296 ± 49.449
[0.30, 0.35)	[4.80, 5.10)	384 ± 19.596	26 ± 5.099	0.5109 ± 0.0116	700.681 ± 42.724
[0.30, 0.35)	[5.10, 5.40)	273 ± 16.523	11 ± 3.317	0.5007 ± 0.0121	523.304 ± 35.958
[0.30, 0.35)	[5.40, 5.70)	188 ± 13.711	7 ± 2.646	0.5094 ± 0.0143	355.344 ± 29.166
[0.30, 0.35)	[5.70, 6.00)	93 ± 9.644	4 ± 2.000	0.5021 ± 0.0227	177.242 ± 21.181
[0.30, 0.35)	[6.00, 6.30)	60 ± 7.746	2 ± 1.414	0.5199 ± 0.0259	111.551 ± 16.129
[0.30, 0.35)	[6.30, 6.60)	45 ± 6.708	0 ± 0.000	0.5385 ± 0.0225	83.572 ± 12.939
[0.30, 0.35)	[6.60, 6.90)	25 ± 5.000	0 ± 0.000	0.5008 ± 0.0371	49.921 ± 10.648
[0.30, 0.35)	[6.90, 7.50)	19 ± 4.359	0 ± 0.000	0.5302 ± 0.0414	35.839 ± 8.685
[0.30, 0.35)	[7.50, 8.70)	9 ± 3.000	0 ± 0.000	0.5114 ± 0.0371	17.598 ± 6.003
[0.35, 0.40)	[4.20, 4.50)	624 ± 24.980	83 ± 9.110	0.5180 ± 0.0102	1044.437 ± 55.287
[0.35, 0.40)	[4.50, 4.80)	542 ± 23.281	44 ± 6.633	0.5045 ± 0.0101	987.090 ± 51.904
[0.35, 0.40)	[4.80, 5.10)	400 ± 20.000	24 ± 4.899	0.5115 ± 0.0101	735.122 ± 42.816
[0.35, 0.40)	[5.10, 5.40)	283 ± 16.823	11 ± 3.317	0.5164 ± 0.0119	526.707 ± 35.345
[0.35, 0.40)	[5.40, 5.70)	147 ± 12.124	3 ± 1.732	0.5221 ± 0.0155	275.800 ± 24.853
[0.35, 0.40)	[5.70, 6.00)	110 ± 10.488	2 ± 1.414	0.5213 ± 0.0175	207.181 ± 21.457
[0.35, 0.40)	[6.00, 6.30)	68 ± 8.246	0 ± 0.000	0.5347 ± 0.0225	127.185 ± 16.327
[0.35, 0.40)	[6.30, 6.60)	43 ± 6.557	1 ± 1.000	0.5481 ± 0.0267	76.630 ± 12.664
[0.35, 0.40)	[6.60, 6.90)	20 ± 4.472	0 ± 0.000	0.5421 ± 0.0353	36.893 ± 8.592

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Table 4 – continued from previous page

xF bin	Mass (GeV)	$Y_{total} \pm \delta Y_{total}$	$Y_{mix} \pm \delta Y_{mix}$	$\langle \epsilon_{signal} \rangle$	$Y_{corr} \pm \delta Y_{corr}$
[0.35, 0.40)	[6.90, 7.50)	19 ± 4.359	0 ± 0.000	0.5190 ± 0.0367	36.611 ± 8.790
[0.35, 0.40)	[7.50, 8.70)	8 ± 2.828	0 ± 0.000	0.5009 ± 0.0575	15.971 ± 5.936
[0.40, 0.45)	[4.20, 4.50)	653 ± 25.554	76 ± 8.718	0.5142 ± 0.0093	1122.108 ± 56.292
[0.40, 0.45)	[4.50, 4.80)	494 ± 22.226	37 ± 6.083	0.5128 ± 0.0095	891.133 ± 47.894
[0.40, 0.45)	[4.80, 5.10)	400 ± 20.000	19 ± 4.359	0.5144 ± 0.0100	740.633 ± 42.332
[0.40, 0.45)	[5.10, 5.40)	244 ± 15.620	6 ± 2.449	0.5167 ± 0.0124	460.606 ± 32.529
[0.40, 0.45)	[5.40, 5.70)	180 ± 13.416	4 ± 2.000	0.5090 ± 0.0146	345.742 ± 28.426
[0.40, 0.45)	[5.70, 6.00)	94 ± 9.695	2 ± 1.414	0.5193 ± 0.0222	177.169 ± 20.331
[0.40, 0.45)	[6.00, 6.30)	82 ± 9.055	1 ± 1.000	0.5336 ± 0.0187	151.794 ± 17.884
[0.40, 0.45)	[6.30, 6.60)	47 ± 6.856	0 ± 0.000	0.5264 ± 0.0264	89.284 ± 13.770
[0.40, 0.45)	[6.60, 6.90)	24 ± 4.899	0 ± 0.000	0.5349 ± 0.0368	44.865 ± 9.663
[0.40, 0.45)	[6.90, 7.50)	20 ± 4.472	1 ± 1.000	0.4986 ± 0.0522	38.107 ± 10.020
[0.40, 0.45)	[7.50, 8.70)	8 ± 2.828	0 ± 0.000	0.5387 ± 0.0551	14.851 ± 5.466
[0.45, 0.50)	[4.20, 4.50)	665 ± 25.788	64 ± 8.000	0.5208 ± 0.0083	1153.963 ± 55.004
[0.45, 0.50)	[4.50, 4.80)	507 ± 22.517	32 ± 5.657	0.5077 ± 0.0097	935.614 ± 49.091
[0.45, 0.50)	[4.80, 5.10)	351 ± 18.735	10 ± 3.162	0.5123 ± 0.0104	665.641 ± 39.481
[0.45, 0.50)	[5.10, 5.40)	219 ± 14.799	7 ± 2.646	0.5084 ± 0.0132	416.977 ± 31.485
[0.45, 0.50)	[5.40, 5.70)	144 ± 12.000	3 ± 1.732	0.4991 ± 0.0182	282.493 ± 26.383
[0.45, 0.50)	[5.70, 6.00)	96 ± 9.798	1 ± 1.000	0.5294 ± 0.0194	179.433 ± 19.731
[0.45, 0.50)	[6.00, 6.30)	58 ± 7.616	1 ± 1.000	0.5296 ± 0.0229	107.624 ± 15.231
[0.45, 0.50)	[6.30, 6.60)	49 ± 7.000	0 ± 0.000	0.5194 ± 0.0256	94.345 ± 14.258
[0.45, 0.50)	[6.60, 6.90)	17 ± 4.123	1 ± 1.000	0.5085 ± 0.0555	31.468 ± 9.022
[0.45, 0.50)	[6.90, 7.50)	27 ± 5.196	0 ± 0.000	0.5433 ± 0.0356	49.692 ± 10.104
[0.45, 0.50)	[7.50, 8.70)	8 ± 2.828	0 ± 0.000	0.5181 ± 0.0752	15.442 ± 5.902
[0.50, 0.55)	[4.20, 4.50)	609 ± 24.678	40 ± 6.325	0.5059 ± 0.0090	1124.703 ± 54.219
[0.50, 0.55)	[4.50, 4.80)	386 ± 19.647	25 ± 5.000	0.5143 ± 0.0106	701.861 ± 41.989
[0.50, 0.55)	[4.80, 5.10)	282 ± 16.793	12 ± 3.464	0.5069 ± 0.0124	532.598 ± 36.237
[0.50, 0.55)	[5.10, 5.40)	208 ± 14.422	9 ± 3.000	0.5160 ± 0.0143	385.654 ± 30.492
[0.50, 0.55)	[5.40, 5.70)	152 ± 12.329	2 ± 1.414	0.5270 ± 0.0153	284.631 ± 24.959
[0.50, 0.55)	[5.70, 6.00)	78 ± 8.832	1 ± 1.000	0.5205 ± 0.0212	147.941 ± 18.113
[0.50, 0.55)	[6.00, 6.30)	42 ± 6.481	1 ± 1.000	0.5037 ± 0.0320	81.393 ± 14.010
[0.50, 0.55)	[6.30, 6.60)	38 ± 6.164	0 ± 0.000	0.5396 ± 0.0262	70.428 ± 11.925
[0.50, 0.55)	[6.60, 6.90)	16 ± 4.000	0 ± 0.000	0.5432 ± 0.0440	29.452 ± 7.740
[0.50, 0.55)	[6.90, 7.50)	14 ± 3.742	0 ± 0.000	0.5384 ± 0.0465	26.002 ± 7.303
[0.50, 0.55)	[7.50, 8.70)	10 ± 3.162	0 ± 0.000	0.5129 ± 0.0468	19.495 ± 6.416
[0.55, 0.60)	[4.20, 4.50)	477 ± 21.840	42 ± 6.481	0.5122 ± 0.0104	849.349 ± 47.728
[0.55, 0.60)	[4.50, 4.80)	379 ± 19.468	14 ± 3.742	0.5194 ± 0.0098	702.673 ± 40.405
[0.55, 0.60)	[4.80, 5.10)	241 ± 15.524	7 ± 2.646	0.5207 ± 0.0130	449.363 ± 32.248
[0.55, 0.60)	[5.10, 5.40)	153 ± 12.369	4 ± 2.000	0.5320 ± 0.0148	280.052 ± 24.809
[0.55, 0.60)	[5.40, 5.70)	90 ± 9.487	3 ± 1.732	0.5122 ± 0.0230	169.852 ± 20.315
[0.55, 0.60)	[5.70, 6.00)	59 ± 7.681	0 ± 0.000	0.5263 ± 0.0234	112.098 ± 15.424
[0.55, 0.60)	[6.00, 6.30)	42 ± 6.481	0 ± 0.000	0.5417 ± 0.0263	77.539 ± 12.541
[0.55, 0.60)	[6.30, 6.60)	22 ± 4.690	0 ± 0.000	0.5525 ± 0.0405	39.816 ± 8.977
[0.55, 0.60)	[6.60, 6.90)	16 ± 4.000	0 ± 0.000	0.5444 ± 0.0486	29.393 ± 7.802
[0.55, 0.60)	[6.90, 7.50)	14 ± 3.742	0 ± 0.000	0.5437 ± 0.0480	25.751 ± 7.248
[0.55, 0.60)	[7.50, 8.70)	5 ± 2.236	0 ± 0.000	0.5080 ± 0.1237	9.843 ± 5.012
[0.60, 0.65)	[4.20, 4.50)	370 ± 19.235	34 ± 5.831	0.5178 ± 0.0116	648.907 ± 41.437

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Table 4 – continued from previous page

xF bin	Mass (GeV)	$Y_{total} \pm \delta Y_{total}$	$Y_{mix} \pm \delta Y_{mix}$	$\langle \epsilon_{signal} \rangle$	$Y_{corr} \pm \delta Y_{corr}$
[0.60, 0.65)	[4.50, 4.80)	243 ± 15.588	12 ± 3.464	0.5105 ± 0.0135	452.508 ± 33.500
[0.60, 0.65)	[4.80, 5.10)	163 ± 12.767	9 ± 3.000	0.5216 ± 0.0157	295.245 ± 26.665
[0.60, 0.65)	[5.10, 5.40)	107 ± 10.344	3 ± 1.732	0.5321 ± 0.0172	195.465 ± 20.696
[0.60, 0.65)	[5.40, 5.70)	66 ± 8.124	2 ± 1.414	0.5410 ± 0.0227	118.290 ± 16.030
[0.60, 0.65)	[5.70, 6.00)	49 ± 7.000	0 ± 0.000	0.5091 ± 0.0301	96.244 ± 14.878
[0.60, 0.65)	[6.00, 6.30)	38 ± 6.164	1 ± 1.000	0.5447 ± 0.0282	67.922 ± 11.990
[0.60, 0.65)	[6.30, 6.60)	19 ± 4.359	0 ± 0.000	0.5594 ± 0.0393	33.967 ± 8.149
[0.60, 0.65)	[6.60, 6.90)	12 ± 3.464	0 ± 0.000	0.5859 ± 0.0408	20.480 ± 6.082
[0.60, 0.65)	[6.90, 7.50)	10 ± 3.162	1 ± 1.000	0.5215 ± 0.0861	17.257 ± 6.969
[0.60, 0.65)	[7.50, 8.70)	3 ± 1.732	0 ± 0.000	0.4447 ± 0.0356	6.746 ± 3.932
[0.65, 0.70)	[4.20, 4.50)	229 ± 15.133	26 ± 5.099	0.5104 ± 0.0152	397.712 ± 33.450
[0.65, 0.70)	[4.50, 4.80)	178 ± 13.342	9 ± 3.000	0.5391 ± 0.0140	313.466 ± 26.640
[0.65, 0.70)	[4.80, 5.10)	110 ± 10.488	3 ± 1.732	0.5300 ± 0.0160	201.875 ± 20.964
[0.65, 0.70)	[5.10, 5.40)	91 ± 9.539	2 ± 1.414	0.5306 ± 0.0177	167.736 ± 19.021
[0.65, 0.70)	[5.40, 5.70)	55 ± 7.416	1 ± 1.000	0.5295 ± 0.0246	101.979 ± 14.905
[0.65, 0.70)	[5.70, 6.00)	31 ± 5.568	1 ± 1.000	0.5134 ± 0.0421	58.431 ± 12.016
[0.65, 0.70)	[6.00, 6.30)	23 ± 4.796	1 ± 1.000	0.5197 ± 0.0414	42.331 ± 10.012
[0.65, 0.70)	[6.30, 6.60)	9 ± 3.000	0 ± 0.000	0.4902 ± 0.0663	18.360 ± 6.605
[0.65, 0.70)	[6.60, 6.90)	9 ± 3.000	0 ± 0.000	0.5111 ± 0.0573	17.610 ± 6.193
[0.65, 0.70)	[6.90, 7.50)	15 ± 3.873	0 ± 0.000	0.5628 ± 0.0448	26.655 ± 7.202
[0.65, 0.70)	[7.50, 8.70)	5 ± 2.236	0 ± 0.000	0.5181 ± 0.0834	9.651 ± 4.587
[0.70, 0.75)	[4.20, 4.50)	159 ± 12.610	15 ± 3.873	0.5135 ± 0.0178	280.425 ± 27.461
[0.70, 0.75)	[4.50, 4.80)	130 ± 11.402	5 ± 2.236	0.5084 ± 0.0191	245.851 ± 24.649
[0.70, 0.75)	[4.80, 5.10)	84 ± 9.165	1 ± 1.000	0.5230 ± 0.0177	158.699 ± 18.431
[0.70, 0.75)	[5.10, 5.40)	43 ± 6.557	2 ± 1.414	0.5190 ± 0.0346	78.997 ± 13.955
[0.70, 0.75)	[5.40, 5.70)	25 ± 5.000	0 ± 0.000	0.5371 ± 0.0362	46.545 ± 9.822
[0.70, 0.75)	[5.70, 6.00)	17 ± 4.123	0 ± 0.000	0.5215 ± 0.0482	32.596 ± 8.459
[0.70, 0.75)	[6.00, 6.30)	15 ± 3.873	0 ± 0.000	0.5385 ± 0.0462	27.855 ± 7.578
[0.70, 0.75)	[6.30, 6.60)	11 ± 3.317	0 ± 0.000	0.5516 ± 0.0609	19.942 ± 6.404
[0.70, 0.75)	[6.60, 6.90)	4 ± 2.000	0 ± 0.000	0.4475 ± 0.1098	8.938 ± 4.978
[0.70, 0.75)	[6.90, 7.50)	3 ± 1.732	0 ± 0.000	0.4489 ± 0.1393	6.684 ± 4.381
[0.70, 0.75)	[7.50, 8.70)	2 ± 1.414	0 ± 0.000	0.4539 ± 0.2600	4.406 ± 4.009
[0.75, 0.80)	[4.20, 4.50)	100 ± 10.000	9 ± 3.000	0.5115 ± 0.0233	177.925 ± 21.970
[0.75, 0.80)	[4.50, 4.80)	48 ± 6.928	7 ± 2.646	0.5286 ± 0.0284	77.561 ± 14.636
[0.75, 0.80)	[4.80, 5.10)	34 ± 5.831	1 ± 1.000	0.5113 ± 0.0367	64.537 ± 12.464
[0.75, 0.80)	[5.10, 5.40)	22 ± 4.690	0 ± 0.000	0.5323 ± 0.0413	41.328 ± 9.375
[0.75, 0.80)	[5.40, 5.70)	18 ± 4.243	0 ± 0.000	0.5513 ± 0.0346	32.649 ± 7.964
[0.75, 0.80)	[5.70, 6.00)	15 ± 3.873	0 ± 0.000	0.5146 ± 0.0493	29.152 ± 8.028
[0.75, 0.80)	[6.00, 6.30)	13 ± 3.606	0 ± 0.000	0.5953 ± 0.0378	21.838 ± 6.213
[0.75, 0.80)	[6.30, 6.60)	4 ± 2.000	0 ± 0.000	0.4197 ± 0.1969	9.531 ± 6.535
[0.75, 0.80)	[6.60, 6.90)	2 ± 1.414	0 ± 0.000	0.5992 ± 0.0584	3.338 ± 2.382
[0.75, 0.80)	[6.90, 7.50)	1 ± 1.000	0 ± 0.000	0.6510 ± 0.0726	1.536 ± 1.546
[0.75, 0.80)	[7.50, 8.70)	2 ± 1.414	0 ± 0.000	0.5401 ± 0.0864	3.703 ± 2.684

Table 5: Combined Data Table for LD2

xF bin	Mass (GeV)	$Y_{total} \pm \delta Y_{total}$	$Y_{mix} \pm \delta Y_{mix}$	$\langle \epsilon_{signal} \rangle$	$Y_{corr} \pm \delta Y_{corr}$
[0.00, 0.05)	[4.20, 4.50)	1 ± 1.000	0 ± 0.000	0.5373 ± 0.1718	1.861 ± 1.954
[0.00, 0.05)	[4.50, 4.80)	7 ± 2.646	3 ± 1.732	0.5757 ± 0.1938	6.948 ± 5.970
[0.00, 0.05)	[4.80, 5.10)	38 ± 6.164	3 ± 1.732	0.5310 ± 0.0396	65.910 ± 13.023
[0.00, 0.05)	[5.10, 5.40)	48 ± 6.928	8 ± 2.828	0.4850 ± 0.0391	82.481 ± 16.805
[0.00, 0.05)	[5.40, 5.70)	54 ± 7.348	5 ± 2.236	0.4636 ± 0.0370	105.701 ± 18.597
[0.00, 0.05)	[5.70, 6.00)	39 ± 6.245	7 ± 2.646	0.5223 ± 0.0430	61.262 ± 13.930
[0.00, 0.05)	[6.00, 6.30)	36 ± 6.000	3 ± 1.732	0.4706 ± 0.0428	70.122 ± 14.723
[0.00, 0.05)	[6.30, 6.60)	22 ± 4.690	1 ± 1.000	0.4572 ± 0.0511	45.933 ± 11.677
[0.00, 0.05)	[6.60, 6.90)	6 ± 2.449	0 ± 0.000	0.5216 ± 0.0402	11.503 ± 4.779
[0.00, 0.05)	[6.90, 7.50)	14 ± 3.742	0 ± 0.000	0.5026 ± 0.0411	27.854 ± 7.786
[0.05, 0.10)	[4.20, 4.50)	3 ± 1.732	1 ± 1.000	0.3502 ± 0.2958	5.711 ± 7.476
[0.05, 0.10)	[4.50, 4.80)	25 ± 5.000	5 ± 2.236	0.5031 ± 0.0657	39.753 ± 12.062
[0.05, 0.10)	[4.80, 5.10)	78 ± 8.832	5 ± 2.236	0.4643 ± 0.0314	157.212 ± 22.314
[0.05, 0.10)	[5.10, 5.40)	89 ± 9.434	20 ± 4.472	0.4868 ± 0.0340	141.735 ± 23.619
[0.05, 0.10)	[5.40, 5.70)	79 ± 8.888	15 ± 3.873	0.4924 ± 0.0350	129.978 ± 21.746
[0.05, 0.10)	[5.70, 6.00)	78 ± 8.832	8 ± 2.828	0.4750 ± 0.0264	147.372 ± 21.170
[0.05, 0.10)	[6.00, 6.30)	38 ± 6.164	1 ± 1.000	0.4800 ± 0.0301	77.083 ± 13.881
[0.05, 0.10)	[6.30, 6.60)	28 ± 5.292	0 ± 0.000	0.4526 ± 0.0407	61.869 ± 12.946
[0.05, 0.10)	[6.60, 6.90)	15 ± 3.873	1 ± 1.000	0.4969 ± 0.0649	28.175 ± 8.852
[0.05, 0.10)	[6.90, 7.50)	8 ± 2.828	0 ± 0.000	0.5029 ± 0.0558	15.908 ± 5.895
[0.05, 0.10)	[7.50, 8.70)	3 ± 1.732	0 ± 0.000	0.5684 ± 0.0686	5.278 ± 3.113
[0.10, 0.15)	[4.20, 4.50)	32 ± 5.657	4 ± 2.000	0.4898 ± 0.0482	57.165 ± 13.480
[0.10, 0.15)	[4.50, 4.80)	97 ± 9.849	17 ± 4.123	0.5006 ± 0.0346	159.815 ± 24.013
[0.10, 0.15)	[4.80, 5.10)	163 ± 12.767	19 ± 4.359	0.4962 ± 0.0203	290.188 ± 29.674
[0.10, 0.15)	[5.10, 5.40)	139 ± 11.790	20 ± 4.472	0.4829 ± 0.0238	246.418 ± 28.805
[0.10, 0.15)	[5.40, 5.70)	114 ± 10.677	16 ± 4.000	0.4936 ± 0.0223	198.552 ± 24.779
[0.10, 0.15)	[5.70, 6.00)	71 ± 8.426	4 ± 2.000	0.4932 ± 0.0251	135.846 ± 18.875
[0.10, 0.15)	[6.00, 6.30)	58 ± 7.616	2 ± 1.414	0.5080 ± 0.0260	110.236 ± 16.257
[0.10, 0.15)	[6.30, 6.60)	29 ± 5.385	2 ± 1.414	0.5011 ± 0.0372	53.881 ± 11.810
[0.10, 0.15)	[6.60, 6.90)	26 ± 5.099	0 ± 0.000	0.5025 ± 0.0348	51.744 ± 10.762
[0.10, 0.15)	[6.90, 7.50)	11 ± 3.317	1 ± 1.000	0.5246 ± 0.0584	19.061 ± 6.936
[0.10, 0.15)	[7.50, 8.70)	1 ± 1.000	0 ± 0.000	0.4386 ± 0.1227	2.280 ± 2.368
[0.15, 0.20)	[4.20, 4.50)	71 ± 8.426	19 ± 4.359	0.5172 ± 0.0447	100.551 ± 20.302
[0.15, 0.20)	[4.50, 4.80)	202 ± 14.213	22 ± 4.690	0.4913 ± 0.0181	366.376 ± 33.324
[0.15, 0.20)	[4.80, 5.10)	229 ± 15.133	32 ± 5.657	0.4962 ± 0.0191	397.015 ± 35.965
[0.15, 0.20)	[5.10, 5.40)	202 ± 14.213	22 ± 4.690	0.4951 ± 0.0177	363.581 ± 32.914
[0.15, 0.20)	[5.40, 5.70)	127 ± 11.269	18 ± 4.243	0.5415 ± 0.0204	201.281 ± 23.498
[0.15, 0.20)	[5.70, 6.00)	117 ± 10.817	6 ± 2.449	0.4980 ± 0.0205	222.902 ± 24.088
[0.15, 0.20)	[6.00, 6.30)	60 ± 7.746	4 ± 2.000	0.5231 ± 0.0272	107.051 ± 16.275
[0.15, 0.20)	[6.30, 6.60)	34 ± 5.831	1 ± 1.000	0.4876 ± 0.0289	67.677 ± 12.780
[0.15, 0.20)	[6.60, 6.90)	20 ± 4.472	0 ± 0.000	0.4854 ± 0.0450	41.199 ± 9.971
[0.15, 0.20)	[6.90, 7.50)	16 ± 4.000	0 ± 0.000	0.4978 ± 0.0490	32.142 ± 8.636
[0.15, 0.20)	[7.50, 8.70)	8 ± 2.828	0 ± 0.000	0.4635 ± 0.0779	17.259 ± 6.757
[0.20, 0.25)	[4.20, 4.50)	204 ± 14.283	37 ± 6.083	0.5148 ± 0.0212	324.402 ± 32.992
[0.20, 0.25)	[4.50, 4.80)	294 ± 17.146	48 ± 6.928	0.5055 ± 0.0167	486.624 ± 39.963
[0.20, 0.25)	[4.80, 5.10)	323 ± 17.972	21 ± 4.583	0.4887 ± 0.0139	617.958 ± 41.804

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Table 5 – continued from previous page

xF bin	Mass (GeV)	$Y_{total} \pm \delta Y_{total}$	$Y_{mix} \pm \delta Y_{mix}$	$\langle \epsilon_{signal} \rangle$	$Y_{corr} \pm \delta Y_{corr}$
[0.20, 0.25)	[5.10, 5.40)	248 ± 15.748	21 ± 4.583	0.4944 ± 0.0151	459.098 ± 36.010
[0.20, 0.25)	[5.40, 5.70)	165 ± 12.845	7 ± 2.646	0.5047 ± 0.0167	313.045 ± 27.973
[0.20, 0.25)	[5.70, 6.00)	126 ± 11.225	5 ± 2.236	0.4811 ± 0.0197	251.533 ± 25.930
[0.20, 0.25)	[6.00, 6.30)	77 ± 8.775	2 ± 1.414	0.5179 ± 0.0242	144.826 ± 18.449
[0.20, 0.25)	[6.30, 6.60)	47 ± 6.856	1 ± 1.000	0.5080 ± 0.0305	90.548 ± 14.681
[0.20, 0.25)	[6.60, 6.90)	18 ± 4.243	0 ± 0.000	0.5693 ± 0.0276	31.618 ± 7.608
[0.20, 0.25)	[6.90, 7.50)	18 ± 4.243	0 ± 0.000	0.4863 ± 0.0466	37.015 ± 9.417
[0.20, 0.25)	[7.50, 8.70)	6 ± 2.449	0 ± 0.000	0.5460 ± 0.0625	10.990 ± 4.660
[0.25, 0.30)	[4.20, 4.50)	398 ± 19.950	62 ± 7.874	0.5157 ± 0.0140	651.525 ± 45.190
[0.25, 0.30)	[4.50, 4.80)	415 ± 20.372	52 ± 7.211	0.5026 ± 0.0134	722.275 ± 47.089
[0.25, 0.30)	[4.80, 5.10)	347 ± 18.628	45 ± 6.708	0.5037 ± 0.0139	599.546 ± 42.651
[0.25, 0.30)	[5.10, 5.40)	270 ± 16.432	21 ± 4.583	0.5012 ± 0.0141	496.805 ± 36.785
[0.25, 0.30)	[5.40, 5.70)	185 ± 13.601	9 ± 3.000	0.4937 ± 0.0164	356.514 ± 30.586
[0.25, 0.30)	[5.70, 6.00)	112 ± 10.583	8 ± 2.828	0.5043 ± 0.0211	206.225 ± 23.375
[0.25, 0.30)	[6.00, 6.30)	79 ± 8.888	0 ± 0.000	0.4989 ± 0.0213	158.350 ± 19.061
[0.25, 0.30)	[6.30, 6.60)	44 ± 6.633	0 ± 0.000	0.5098 ± 0.0277	86.304 ± 13.833
[0.25, 0.30)	[6.60, 6.90)	16 ± 4.000	1 ± 1.000	0.5124 ± 0.0559	29.276 ± 8.659
[0.25, 0.30)	[6.90, 7.50)	30 ± 5.477	0 ± 0.000	0.5156 ± 0.0280	58.179 ± 11.083
[0.25, 0.30)	[7.50, 8.70)	5 ± 2.236	0 ± 0.000	0.4489 ± 0.1095	11.137 ± 5.674
[0.30, 0.35)	[4.20, 4.50)	552 ± 23.495	84 ± 9.165	0.4917 ± 0.0120	951.790 ± 56.269
[0.30, 0.35)	[4.50, 4.80)	514 ± 22.672	66 ± 8.124	0.4937 ± 0.0114	907.393 ± 53.087
[0.30, 0.35)	[4.80, 5.10)	419 ± 20.469	31 ± 5.568	0.4923 ± 0.0112	788.180 ± 46.685
[0.30, 0.35)	[5.10, 5.40)	294 ± 17.146	11 ± 3.317	0.4902 ± 0.0133	577.259 ± 38.906
[0.30, 0.35)	[5.40, 5.70)	189 ± 13.748	12 ± 3.464	0.5098 ± 0.0150	347.193 ± 29.615
[0.30, 0.35)	[5.70, 6.00)	128 ± 11.314	3 ± 1.732	0.5055 ± 0.0191	247.265 ± 24.495
[0.30, 0.35)	[6.00, 6.30)	85 ± 9.220	6 ± 2.449	0.5260 ± 0.0239	150.193 ± 19.381
[0.30, 0.35)	[6.30, 6.60)	50 ± 7.071	0 ± 0.000	0.4886 ± 0.0281	102.324 ± 15.623
[0.30, 0.35)	[6.60, 6.90)	33 ± 5.745	0 ± 0.000	0.5025 ± 0.0350	65.673 ± 12.313
[0.30, 0.35)	[6.90, 7.50)	21 ± 4.583	0 ± 0.000	0.5058 ± 0.0405	41.518 ± 9.652
[0.30, 0.35)	[7.50, 8.70)	10 ± 3.162	0 ± 0.000	0.5042 ± 0.0462	19.834 ± 6.530
[0.35, 0.40)	[4.20, 4.50)	626 ± 25.020	97 ± 9.849	0.5083 ± 0.0108	1040.737 ± 57.301
[0.35, 0.40)	[4.50, 4.80)	577 ± 24.021	48 ± 6.928	0.4974 ± 0.0099	1063.547 ± 54.579
[0.35, 0.40)	[4.80, 5.10)	464 ± 21.541	36 ± 6.000	0.4967 ± 0.0107	861.763 ± 48.721
[0.35, 0.40)	[5.10, 5.40)	290 ± 17.029	13 ± 3.606	0.4993 ± 0.0126	554.742 ± 37.564
[0.35, 0.40)	[5.40, 5.70)	201 ± 14.177	6 ± 2.449	0.5140 ± 0.0151	379.412 ± 30.139
[0.35, 0.40)	[5.70, 6.00)	136 ± 11.662	3 ± 1.732	0.4985 ± 0.0180	266.787 ± 25.542
[0.35, 0.40)	[6.00, 6.30)	77 ± 8.775	1 ± 1.000	0.5167 ± 0.0235	147.100 ± 18.362
[0.35, 0.40)	[6.30, 6.60)	45 ± 6.708	1 ± 1.000	0.5003 ± 0.0361	87.950 ± 14.968
[0.35, 0.40)	[6.60, 6.90)	32 ± 5.657	1 ± 1.000	0.5070 ± 0.0349	61.139 ± 12.084
[0.35, 0.40)	[6.90, 7.50)	30 ± 5.477	1 ± 1.000	0.4941 ± 0.0418	58.696 ± 12.316
[0.35, 0.40)	[7.50, 8.70)	8 ± 2.828	0 ± 0.000	0.5038 ± 0.0545	15.878 ± 5.871
[0.40, 0.45)	[4.20, 4.50)	712 ± 26.683	97 ± 9.849	0.5034 ± 0.0098	1221.666 ± 61.347
[0.40, 0.45)	[4.50, 4.80)	577 ± 24.021	46 ± 6.782	0.5003 ± 0.0096	1061.417 ± 53.910
[0.40, 0.45)	[4.80, 5.10)	456 ± 21.354	19 ± 4.359	0.4979 ± 0.0107	877.682 ± 47.643
[0.40, 0.45)	[5.10, 5.40)	263 ± 16.217	11 ± 3.317	0.5083 ± 0.0128	495.775 ± 34.860
[0.40, 0.45)	[5.40, 5.70)	186 ± 13.638	3 ± 1.732	0.4949 ± 0.0153	369.735 ± 30.045
[0.40, 0.45)	[5.70, 6.00)	130 ± 11.402	4 ± 2.000	0.5050 ± 0.0190	249.501 ± 24.767

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Table 5 – continued from previous page

xF bin	Mass (GeV)	$Y_{total} \pm \delta Y_{total}$	$Y_{mix} \pm \delta Y_{mix}$	$\langle \epsilon_{signal} \rangle$	$Y_{corr} \pm \delta Y_{corr}$
[0.40, 0.45)	[6.00, 6.30)	69 ± 8.307	1 ± 1.000	0.5240 ± 0.0232	129.776 ± 16.972
[0.40, 0.45)	[6.30, 6.60)	56 ± 7.483	3 ± 1.732	0.5222 ± 0.0303	101.502 ± 15.847
[0.40, 0.45)	[6.60, 6.90)	28 ± 5.292	1 ± 1.000	0.4834 ± 0.0472	55.850 ± 12.404
[0.40, 0.45)	[6.90, 7.50)	27 ± 5.196	0 ± 0.000	0.5283 ± 0.0322	51.103 ± 10.317
[0.40, 0.45)	[7.50, 8.70)	6 ± 2.449	0 ± 0.000	0.5157 ± 0.0609	11.634 ± 4.944
[0.45, 0.50)	[4.20, 4.50)	722 ± 26.870	75 ± 8.660	0.4965 ± 0.0090	1303.086 ± 61.573
[0.45, 0.50)	[4.50, 4.80)	549 ± 23.431	38 ± 6.164	0.4946 ± 0.0101	1033.108 ± 53.343
[0.45, 0.50)	[4.80, 5.10)	391 ± 19.774	24 ± 4.899	0.5022 ± 0.0112	730.792 ± 43.723
[0.45, 0.50)	[5.10, 5.40)	274 ± 16.553	7 ± 2.646	0.4926 ± 0.0129	541.972 ± 36.881
[0.45, 0.50)	[5.40, 5.70)	174 ± 13.191	8 ± 2.828	0.5129 ± 0.0152	323.623 ± 27.990
[0.45, 0.50)	[5.70, 6.00)	132 ± 11.489	3 ± 1.732	0.5013 ± 0.0176	257.332 ± 24.873
[0.45, 0.50)	[6.00, 6.30)	69 ± 8.307	0 ± 0.000	0.4931 ± 0.0250	139.934 ± 18.284
[0.45, 0.50)	[6.30, 6.60)	45 ± 6.708	0 ± 0.000	0.4905 ± 0.0312	91.750 ± 14.869
[0.45, 0.50)	[6.60, 6.90)	18 ± 4.243	0 ± 0.000	0.4813 ± 0.0565	37.400 ± 9.849
[0.45, 0.50)	[6.90, 7.50)	29 ± 5.385	0 ± 0.000	0.5015 ± 0.0340	57.826 ± 11.433
[0.45, 0.50)	[7.50, 8.70)	7 ± 2.646	0 ± 0.000	0.4924 ± 0.0966	14.216 ± 6.055
[0.50, 0.55)	[4.20, 4.50)	663 ± 25.749	64 ± 8.000	0.5016 ± 0.0091	1194.079 ± 57.957
[0.50, 0.55)	[4.50, 4.80)	497 ± 22.293	23 ± 4.796	0.4966 ± 0.0098	954.415 ± 49.621
[0.50, 0.55)	[4.80, 5.10)	329 ± 18.138	10 ± 3.162	0.4972 ± 0.0124	641.559 ± 40.324
[0.50, 0.55)	[5.10, 5.40)	219 ± 14.799	8 ± 2.828	0.4966 ± 0.0136	424.932 ± 32.504
[0.50, 0.55)	[5.40, 5.70)	161 ± 12.689	5 ± 2.236	0.5090 ± 0.0164	306.471 ± 27.160
[0.50, 0.55)	[5.70, 6.00)	93 ± 9.644	2 ± 1.414	0.5089 ± 0.0207	178.803 ± 20.486
[0.50, 0.55)	[6.00, 6.30)	67 ± 8.185	1 ± 1.000	0.5153 ± 0.0270	128.075 ± 17.349
[0.50, 0.55)	[6.30, 6.60)	32 ± 5.657	0 ± 0.000	0.5031 ± 0.0377	63.610 ± 12.215
[0.50, 0.55)	[6.60, 6.90)	32 ± 5.657	0 ± 0.000	0.5135 ± 0.0339	62.317 ± 11.760
[0.50, 0.55)	[6.90, 7.50)	19 ± 4.359	2 ± 1.414	0.4827 ± 0.0677	35.220 ± 10.702
[0.50, 0.55)	[7.50, 8.70)	5 ± 2.236	0 ± 0.000	0.5207 ± 0.0667	9.602 ± 4.467
[0.55, 0.60)	[4.20, 4.50)	561 ± 23.685	56 ± 7.483	0.4986 ± 0.0100	1012.933 ± 53.806
[0.55, 0.60)	[4.50, 4.80)	364 ± 19.079	22 ± 4.690	0.5029 ± 0.0120	680.019 ± 42.279
[0.55, 0.60)	[4.80, 5.10)	291 ± 17.059	4 ± 2.000	0.5058 ± 0.0120	567.364 ± 36.538
[0.55, 0.60)	[5.10, 5.40)	162 ± 12.728	2 ± 1.414	0.4999 ± 0.0164	320.032 ± 27.693
[0.55, 0.60)	[5.40, 5.70)	128 ± 11.314	6 ± 2.449	0.4975 ± 0.0195	245.227 ± 25.168
[0.55, 0.60)	[5.70, 6.00)	64 ± 8.000	1 ± 1.000	0.5356 ± 0.0217	117.622 ± 15.790
[0.55, 0.60)	[6.00, 6.30)	46 ± 6.782	0 ± 0.000	0.5135 ± 0.0303	89.581 ± 14.225
[0.55, 0.60)	[6.30, 6.60)	30 ± 5.477	0 ± 0.000	0.4884 ± 0.0386	61.421 ± 12.221
[0.55, 0.60)	[6.60, 6.90)	18 ± 4.243	0 ± 0.000	0.5190 ± 0.0455	34.681 ± 8.722
[0.55, 0.60)	[6.90, 7.50)	18 ± 4.243	0 ± 0.000	0.5198 ± 0.0491	34.632 ± 8.794
[0.55, 0.60)	[7.50, 8.70)	5 ± 2.236	0 ± 0.000	0.5289 ± 0.0768	9.453 ± 4.445
[0.60, 0.65)	[4.20, 4.50)	402 ± 20.050	44 ± 6.633	0.4932 ± 0.0127	725.881 ± 46.720
[0.60, 0.65)	[4.50, 4.80)	348 ± 18.655	15 ± 3.873	0.4915 ± 0.0121	677.525 ± 42.203
[0.60, 0.65)	[4.80, 5.10)	210 ± 14.491	15 ± 3.873	0.4932 ± 0.0159	395.370 ± 32.965
[0.60, 0.65)	[5.10, 5.40)	140 ± 11.832	3 ± 1.732	0.5152 ± 0.0166	265.918 ± 24.736
[0.60, 0.65)	[5.40, 5.70)	97 ± 9.849	3 ± 1.732	0.4870 ± 0.0220	193.031 ± 22.313
[0.60, 0.65)	[5.70, 6.00)	60 ± 7.746	0 ± 0.000	0.5286 ± 0.0255	113.517 ± 15.645
[0.60, 0.65)	[6.00, 6.30)	43 ± 6.557	0 ± 0.000	0.4886 ± 0.0341	88.010 ± 14.761
[0.60, 0.65)	[6.30, 6.60)	24 ± 4.899	0 ± 0.000	0.4924 ± 0.0483	48.742 ± 11.040
[0.60, 0.65)	[6.60, 6.90)	17 ± 4.123	0 ± 0.000	0.5400 ± 0.0474	31.484 ± 8.122

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Table 5 – continued from previous page

xF bin	Mass (GeV)	$Y_{total} \pm \delta Y_{total}$	$Y_{mix} \pm \delta Y_{mix}$	$\langle \epsilon_{signal} \rangle$	$Y_{corr} \pm \delta Y_{corr}$
[0.60, 0.65)	[6.90, 7.50)	8 ± 2.828	0 ± 0.000	0.4865 ± 0.0702	16.443 ± 6.279
[0.60, 0.65)	[7.50, 8.70)	4 ± 2.000	0 ± 0.000	0.4827 ± 0.1165	8.287 ± 4.601
[0.65, 0.70)	[4.20, 4.50)	300 ± 17.321	34 ± 5.831	0.5060 ± 0.0145	525.712 ± 39.138
[0.65, 0.70)	[4.50, 4.80)	196 ± 14.000	11 ± 3.317	0.5126 ± 0.0160	360.910 ± 30.247
[0.65, 0.70)	[4.80, 5.10)	141 ± 11.874	4 ± 2.000	0.5031 ± 0.0192	272.296 ± 26.101
[0.65, 0.70)	[5.10, 5.40)	70 ± 8.367	1 ± 1.000	0.5154 ± 0.0246	133.871 ± 17.550
[0.65, 0.70)	[5.40, 5.70)	63 ± 7.937	1 ± 1.000	0.4890 ± 0.0296	126.796 ± 18.073
[0.65, 0.70)	[5.70, 6.00)	42 ± 6.481	0 ± 0.000	0.5113 ± 0.0297	82.137 ± 13.544
[0.65, 0.70)	[6.00, 6.30)	26 ± 5.099	0 ± 0.000	0.5625 ± 0.0269	46.226 ± 9.331
[0.65, 0.70)	[6.30, 6.60)	7 ± 2.646	0 ± 0.000	0.5597 ± 0.0501	12.507 ± 4.858
[0.65, 0.70)	[6.60, 6.90)	9 ± 3.000	0 ± 0.000	0.5134 ± 0.0633	17.531 ± 6.231
[0.65, 0.70)	[6.90, 7.50)	5 ± 2.236	0 ± 0.000	0.5462 ± 0.0694	9.153 ± 4.256
[0.65, 0.70)	[7.50, 8.70)	5 ± 2.236	0 ± 0.000	0.5731 ± 0.0519	8.725 ± 3.981
[0.70, 0.75)	[4.20, 4.50)	175 ± 13.229	17 ± 4.123	0.5005 ± 0.0192	315.673 ± 30.206
[0.70, 0.75)	[4.50, 4.80)	139 ± 11.790	9 ± 3.000	0.5199 ± 0.0195	250.046 ± 25.213
[0.70, 0.75)	[4.80, 5.10)	85 ± 9.220	5 ± 2.236	0.4972 ± 0.0276	160.909 ± 21.070
[0.70, 0.75)	[5.10, 5.40)	59 ± 7.681	1 ± 1.000	0.5329 ± 0.0253	108.835 ± 15.426
[0.70, 0.75)	[5.40, 5.70)	42 ± 6.481	1 ± 1.000	0.4992 ± 0.0352	82.132 ± 14.354
[0.70, 0.75)	[5.70, 6.00)	25 ± 5.000	0 ± 0.000	0.4814 ± 0.0453	51.929 ± 11.478
[0.70, 0.75)	[6.00, 6.30)	10 ± 3.162	1 ± 1.000	0.4348 ± 0.0975	20.701 ± 8.931
[0.70, 0.75)	[6.30, 6.60)	7 ± 2.646	0 ± 0.000	0.4718 ± 0.0910	14.837 ± 6.296
[0.70, 0.75)	[6.60, 6.90)	4 ± 2.000	0 ± 0.000	0.5303 ± 0.0890	7.543 ± 3.978
[0.70, 0.75)	[6.90, 7.50)	3 ± 1.732	1 ± 1.000	0.4396 ± 0.2318	4.549 ± 5.143
[0.70, 0.75)	[7.50, 8.70)	1 ± 1.000	0 ± 0.000	0.5278 ± 0.1456	1.895 ± 1.966
[0.75, 0.80)	[4.20, 4.50)	104 ± 10.198	5 ± 2.236	0.5097 ± 0.0212	194.224 ± 22.025
[0.75, 0.80)	[4.50, 4.80)	65 ± 8.062	2 ± 1.414	0.5246 ± 0.0238	120.092 ± 16.526
[0.75, 0.80)	[4.80, 5.10)	43 ± 6.557	0 ± 0.000	0.5142 ± 0.0320	83.622 ± 13.771
[0.75, 0.80)	[5.10, 5.40)	36 ± 6.000	1 ± 1.000	0.4824 ± 0.0359	72.553 ± 13.718
[0.75, 0.80)	[5.40, 5.70)	21 ± 4.583	0 ± 0.000	0.4862 ± 0.0430	43.192 ± 10.171
[0.75, 0.80)	[5.70, 6.00)	14 ± 3.742	1 ± 1.000	0.4888 ± 0.0723	26.596 ± 8.845
[0.75, 0.80)	[6.00, 6.30)	10 ± 3.162	0 ± 0.000	0.5113 ± 0.0623	19.558 ± 6.629
[0.75, 0.80)	[6.30, 6.60)	7 ± 2.646	0 ± 0.000	0.5162 ± 0.0676	13.560 ± 5.424
[0.75, 0.80)	[6.60, 6.90)	2 ± 1.414	0 ± 0.000	0.6439 ± 0.0714	3.106 ± 2.223
[0.75, 0.80)	[6.90, 7.50)	3 ± 1.732	0 ± 0.000	0.5786 ± 0.0898	5.185 ± 3.100
[0.75, 0.80)	[7.50, 8.70)	4 ± 2.000	0 ± 0.000	0.5301 ± 0.0743	7.546 ± 3.918

Table 6: Combined Data Table for Flask

xF bin	Mass (GeV)	$Y_{total} \pm \delta Y_{total}$	$Y_{mix} \pm \delta Y_{mix}$	$\langle \epsilon_{signal} \rangle$	$Y_{corr} \pm \delta Y_{corr}$
[0.00, 0.05)	[4.80, 5.10)	3 ± 1.732	1 ± 1.000	0.3663 ± 0.3968	5.459 ± 8.048
[0.00, 0.05)	[5.10, 5.40)	5 ± 2.236	1 ± 1.000	0.4049 ± 0.1316	9.879 ± 6.849
[0.00, 0.05)	[5.40, 5.70)	6 ± 2.449	1 ± 1.000	0.3813 ± 0.1055	13.112 ± 7.829
[0.00, 0.05)	[5.70, 6.00)	3 ± 1.732	1 ± 1.000	0.4103 ± 0.2705	4.874 ± 5.838
[0.00, 0.05)	[6.00, 6.30)	1 ± 1.000	0 ± 0.000	0.3954 ± 0.2174	2.529 ± 2.886
[0.05, 0.10)	[4.50, 4.80)	4 ± 2.000	1 ± 1.000	0.4830 ± 0.1702	6.211 ± 5.120
[0.05, 0.10)	[4.80, 5.10)	4 ± 2.000	2 ± 1.414	0.6403 ± 0.2519	3.123 ± 4.018
[0.05, 0.10)	[5.10, 5.40)	6 ± 2.449	1 ± 1.000	0.4415 ± 0.1625	11.325 ± 7.299
[0.05, 0.10)	[5.40, 5.70)	3 ± 1.732	0 ± 0.000	0.4729 ± 0.1202	6.344 ± 4.002
[0.05, 0.10)	[5.70, 6.00)	4 ± 2.000	0 ± 0.000	0.4576 ± 0.1065	8.741 ± 4.821
[0.05, 0.10)	[6.00, 6.30)	2 ± 1.414	1 ± 1.000	0.4489 ± 0.5152	2.228 ± 4.628
[0.05, 0.10)	[6.30, 6.60)	1 ± 1.000	0 ± 0.000	0.6373 ± 0.0878	1.569 ± 1.584
[0.10, 0.15)	[4.50, 4.80)	6 ± 2.449	1 ± 1.000	0.4839 ± 0.1114	10.332 ± 5.962
[0.10, 0.15)	[4.80, 5.10)	8 ± 2.828	3 ± 1.732	0.3791 ± 0.1511	13.188 ± 10.206
[0.10, 0.15)	[5.10, 5.40)	5 ± 2.236	2 ± 1.414	0.4153 ± 0.2150	7.225 ± 7.388
[0.10, 0.15)	[5.40, 5.70)	6 ± 2.449	1 ± 1.000	0.5008 ± 0.0840	9.984 ± 5.542
[0.10, 0.15)	[5.70, 6.00)	5 ± 2.236	0 ± 0.000	0.5004 ± 0.0529	9.993 ± 4.592
[0.10, 0.15)	[6.30, 6.60)	1 ± 1.000	0 ± 0.000	0.4263 ± 0.1378	2.346 ± 2.465
[0.15, 0.20)	[4.20, 4.50)	1 ± 1.000	3 ± 1.732	0.0000 ± 0.0000	0.000 ± 0.000
[0.15, 0.20)	[4.50, 4.80)	8 ± 2.828	3 ± 1.732	0.5656 ± 0.1675	8.840 ± 6.421
[0.15, 0.20)	[4.80, 5.10)	12 ± 3.464	3 ± 1.732	0.4467 ± 0.1465	20.147 ± 10.900
[0.15, 0.20)	[5.10, 5.40)	10 ± 3.162	1 ± 1.000	0.5086 ± 0.0541	17.697 ± 6.788
[0.15, 0.20)	[5.40, 5.70)	4 ± 2.000	3 ± 1.732	0.6553 ± 0.7088	1.526 ± 4.361
[0.15, 0.20)	[5.70, 6.00)	2 ± 1.414	1 ± 1.000	0.6695 ± 0.2271	1.494 ± 2.636
[0.15, 0.20)	[6.00, 6.30)	4 ± 2.000	0 ± 0.000	0.4907 ± 0.0897	8.152 ± 4.340
[0.15, 0.20)	[6.30, 6.60)	2 ± 1.414	0 ± 0.000	0.6315 ± 0.0600	3.167 ± 2.260
[0.15, 0.20)	[6.90, 7.50)	1 ± 1.000	0 ± 0.000	0.4731 ± 0.1551	2.114 ± 2.224
[0.20, 0.25)	[4.20, 4.50)	12 ± 3.464	4 ± 2.000	0.5424 ± 0.1175	14.748 ± 8.036
[0.20, 0.25)	[4.50, 4.80)	20 ± 4.472	5 ± 2.236	0.4545 ± 0.0889	33.003 ± 12.756
[0.20, 0.25)	[4.80, 5.10)	7 ± 2.646	8 ± 2.828	0.0000 ± 0.0000	0.000 ± 0.000
[0.20, 0.25)	[5.10, 5.40)	10 ± 3.162	2 ± 1.414	0.4927 ± 0.0943	16.236 ± 7.686
[0.20, 0.25)	[5.40, 5.70)	3 ± 1.732	0 ± 0.000	0.4842 ± 0.1129	6.196 ± 3.858
[0.20, 0.25)	[5.70, 6.00)	2 ± 1.414	1 ± 1.000	0.3850 ± 0.4507	2.597 ± 5.430
[0.20, 0.25)	[6.00, 6.30)	4 ± 2.000	0 ± 0.000	0.4745 ± 0.1119	8.430 ± 4.661
[0.20, 0.25)	[6.30, 6.60)	1 ± 1.000	1 ± 1.000	0.0000 ± 0.0000	0.000 ± 0.000
[0.20, 0.25)	[6.60, 6.90)	1 ± 1.000	0 ± 0.000	0.5570 ± 0.1101	1.795 ± 1.830
[0.20, 0.25)	[6.90, 7.50)	1 ± 1.000	0 ± 0.000	0.4949 ± 0.0920	2.021 ± 2.055
[0.25, 0.30)	[4.20, 4.50)	23 ± 4.796	3 ± 1.732	0.4894 ± 0.0523	40.863 ± 11.295
[0.25, 0.30)	[4.50, 4.80)	18 ± 4.243	7 ± 2.646	0.5422 ± 0.1003	20.289 ± 9.957
[0.25, 0.30)	[4.80, 5.10)	12 ± 3.464	3 ± 1.732	0.6070 ± 0.0937	14.826 ± 6.779
[0.25, 0.30)	[5.10, 5.40)	6 ± 2.449	0 ± 0.000	0.4081 ± 0.0885	14.703 ± 6.797
[0.25, 0.30)	[5.40, 5.70)	7 ± 2.646	1 ± 1.000	0.5195 ± 0.0876	11.550 ± 5.782
[0.25, 0.30)	[6.00, 6.30)	1 ± 1.000	1 ± 1.000	0.0000 ± 0.0000	0.000 ± 0.000
[0.25, 0.30)	[6.30, 6.60)	2 ± 1.414	0 ± 0.000	0.4356 ± 0.1277	4.591 ± 3.514
[0.25, 0.30)	[6.60, 6.90)	1 ± 1.000	0 ± 0.000	0.5485 ± 0.1024	1.823 ± 1.855
[0.30, 0.35)	[4.20, 4.50)	30 ± 5.477	3 ± 1.732	0.5141 ± 0.0368	52.517 ± 11.788

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Table 6 – continued from previous page

xF bin	Mass (GeV)	$Y_{total} \pm \delta Y_{total}$	$Y_{mix} \pm \delta Y_{mix}$	$\langle \epsilon_{signal} \rangle$	$Y_{corr} \pm \delta Y_{corr}$
[0.30, 0.35)	[4.50, 4.80)	23 ± 4.796	3 ± 1.732	0.4912 ± 0.0621	40.716 ± 11.587
[0.30, 0.35)	[4.80, 5.10)	9 ± 3.000	2 ± 1.414	0.5068 ± 0.1073	13.811 ± 7.167
[0.30, 0.35)	[5.10, 5.40)	8 ± 2.828	2 ± 1.414	0.5486 ± 0.0795	10.936 ± 5.978
[0.30, 0.35)	[5.40, 5.70)	7 ± 2.646	0 ± 0.000	0.4799 ± 0.0690	14.586 ± 5.899
[0.30, 0.35)	[5.70, 6.00)	3 ± 1.732	0 ± 0.000	0.4970 ± 0.1042	6.036 ± 3.708
[0.30, 0.35)	[6.30, 6.60)	2 ± 1.414	0 ± 0.000	0.4358 ± 0.2418	4.589 ± 4.124
[0.35, 0.40)	[4.20, 4.50)	31 ± 5.568	4 ± 2.000	0.4793 ± 0.0502	56.331 ± 13.681
[0.35, 0.40)	[4.50, 4.80)	16 ± 4.000	0 ± 0.000	0.5526 ± 0.0382	28.953 ± 7.510
[0.35, 0.40)	[4.80, 5.10)	12 ± 3.464	2 ± 1.414	0.5159 ± 0.0694	19.383 ± 7.707
[0.35, 0.40)	[5.10, 5.40)	4 ± 2.000	1 ± 1.000	0.3753 ± 0.1570	7.994 ± 6.833
[0.35, 0.40)	[5.40, 5.70)	9 ± 3.000	0 ± 0.000	0.5552 ± 0.0419	16.210 ± 5.540
[0.35, 0.40)	[5.70, 6.00)	2 ± 1.414	0 ± 0.000	0.6177 ± 0.0934	3.238 ± 2.341
[0.35, 0.40)	[6.00, 6.30)	2 ± 1.414	0 ± 0.000	0.5861 ± 0.0861	3.412 ± 2.464
[0.35, 0.40)	[6.30, 6.60)	4 ± 2.000	0 ± 0.000	0.5202 ± 0.0874	7.689 ± 4.056
[0.40, 0.45)	[4.20, 4.50)	37 ± 6.083	6 ± 2.449	0.5358 ± 0.0349	57.861 ± 12.806
[0.40, 0.45)	[4.50, 4.80)	19 ± 4.359	2 ± 1.414	0.4980 ± 0.0530	34.137 ± 9.892
[0.40, 0.45)	[4.80, 5.10)	14 ± 3.742	0 ± 0.000	0.5157 ± 0.0494	27.149 ± 7.708
[0.40, 0.45)	[5.10, 5.40)	9 ± 3.000	1 ± 1.000	0.5325 ± 0.0912	15.024 ± 6.472
[0.40, 0.45)	[5.40, 5.70)	4 ± 2.000	0 ± 0.000	0.4579 ± 0.0820	8.736 ± 4.640
[0.40, 0.45)	[5.70, 6.00)	4 ± 2.000	1 ± 1.000	0.4610 ± 0.1630	6.507 ± 5.368
[0.40, 0.45)	[6.00, 6.30)	2 ± 1.414	0 ± 0.000	0.5872 ± 0.0695	3.406 ± 2.442
[0.40, 0.45)	[6.90, 7.50)	1 ± 1.000	0 ± 0.000	0.4984 ± 0.0992	2.006 ± 2.046
[0.40, 0.45)	[7.50, 8.70)	1 ± 1.000	0 ± 0.000	0.5270 ± 0.0951	1.897 ± 1.928
[0.45, 0.50)	[4.20, 4.50)	34 ± 5.831	2 ± 1.414	0.5362 ± 0.0353	59.677 ± 11.860
[0.45, 0.50)	[4.50, 4.80)	17 ± 4.123	0 ± 0.000	0.5342 ± 0.0361	31.825 ± 8.012
[0.45, 0.50)	[4.80, 5.10)	5 ± 2.236	0 ± 0.000	0.5356 ± 0.0541	9.336 ± 4.280
[0.45, 0.50)	[5.10, 5.40)	8 ± 2.828	0 ± 0.000	0.4936 ± 0.0650	16.208 ± 6.115
[0.45, 0.50)	[5.40, 5.70)	4 ± 2.000	0 ± 0.000	0.4989 ± 0.1445	8.018 ± 4.634
[0.45, 0.50)	[5.70, 6.00)	2 ± 1.414	0 ± 0.000	0.4955 ± 0.1031	4.037 ± 2.975
[0.45, 0.50)	[6.00, 6.30)	1 ± 1.000	0 ± 0.000	0.5952 ± 0.1150	1.680 ± 1.711
[0.50, 0.55)	[4.20, 4.50)	16 ± 4.000	3 ± 1.732	0.5562 ± 0.0695	23.373 ± 8.364
[0.50, 0.55)	[4.50, 4.80)	20 ± 4.472	0 ± 0.000	0.5218 ± 0.0354	38.326 ± 8.956
[0.50, 0.55)	[4.80, 5.10)	7 ± 2.646	1 ± 1.000	0.5807 ± 0.0554	10.332 ± 4.969
[0.50, 0.55)	[5.10, 5.40)	6 ± 2.449	0 ± 0.000	0.5210 ± 0.0771	11.515 ± 5.001
[0.50, 0.55)	[5.40, 5.70)	8 ± 2.828	0 ± 0.000	0.5241 ± 0.0602	15.265 ± 5.675
[0.50, 0.55)	[5.70, 6.00)	3 ± 1.732	0 ± 0.000	0.5340 ± 0.0861	5.618 ± 3.368
[0.50, 0.55)	[6.00, 6.30)	2 ± 1.414	0 ± 0.000	0.6153 ± 0.0908	3.251 ± 2.348
[0.50, 0.55)	[6.30, 6.60)	1 ± 1.000	0 ± 0.000	0.6484 ± 0.0893	1.542 ± 1.557
[0.55, 0.60)	[4.20, 4.50)	17 ± 4.123	4 ± 2.000	0.5798 ± 0.0580	22.421 ± 8.216
[0.55, 0.60)	[4.50, 4.80)	6 ± 2.449	1 ± 1.000	0.5250 ± 0.0870	9.524 ± 5.281
[0.55, 0.60)	[4.80, 5.10)	5 ± 2.236	0 ± 0.000	0.5208 ± 0.0425	9.600 ± 4.364
[0.55, 0.60)	[5.10, 5.40)	2 ± 1.414	0 ± 0.000	0.4518 ± 0.1725	4.426 ± 3.557
[0.55, 0.60)	[5.40, 5.70)	3 ± 1.732	0 ± 0.000	0.5334 ± 0.1243	5.624 ± 3.502
[0.55, 0.60)	[5.70, 6.00)	3 ± 1.732	0 ± 0.000	0.5032 ± 0.1194	5.962 ± 3.721
[0.55, 0.60)	[6.00, 6.30)	4 ± 2.000	0 ± 0.000	0.6467 ± 0.0421	6.185 ± 3.119
[0.55, 0.60)	[6.30, 6.60)	1 ± 1.000	0 ± 0.000	0.4926 ± 0.0795	2.030 ± 2.056
[0.60, 0.65)	[4.20, 4.50)	9 ± 3.000	2 ± 1.414	0.5587 ± 0.0641	12.528 ± 6.108

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Table 6 – continued from previous page

xF bin	**Mass (GeV)**	$Y_{total} \pm \delta Y_{total}$	$Y_{mix} \pm \delta Y_{mix}$	$\langle \epsilon_{signal} \rangle$	$Y_{corr} \pm \delta Y_{corr}$
[0.60, 0.65)	[4.50, 4.80)	9 ± 3.000	1 ± 1.000	0.5542 ± 0.0625	14.435 ± 5.934
[0.60, 0.65)	[4.80, 5.10)	4 ± 2.000	0 ± 0.000	0.5027 ± 0.1198	7.958 ± 4.408
[0.60, 0.65)	[5.10, 5.40)	4 ± 2.000	0 ± 0.000	0.5708 ± 0.0556	7.008 ± 3.570
[0.60, 0.65)	[5.40, 5.70)	2 ± 1.414	0 ± 0.000	0.4972 ± 0.1269	4.022 ± 3.024
[0.60, 0.65)	[5.70, 6.00)	3 ± 1.732	0 ± 0.000	0.4923 ± 0.1891	6.094 ± 4.226
[0.60, 0.65)	[6.00, 6.30)	2 ± 1.414	0 ± 0.000	0.4916 ± 0.1019	4.068 ± 2.998
[0.60, 0.65)	[6.60, 6.90)	1 ± 1.000	0 ± 0.000	0.4518 ± 0.2414	2.213 ± 2.509
[0.65, 0.70)	[4.20, 4.50)	4 ± 2.000	1 ± 1.000	0.5343 ± 0.0987	5.614 ± 4.311
[0.65, 0.70)	[4.50, 4.80)	5 ± 2.236	0 ± 0.000	0.5147 ± 0.1116	9.715 ± 4.829
[0.65, 0.70)	[4.80, 5.10)	4 ± 2.000	0 ± 0.000	0.4721 ± 0.0841	8.472 ± 4.497
[0.65, 0.70)	[5.10, 5.40)	5 ± 2.236	0 ± 0.000	0.4407 ± 0.1286	11.347 ± 6.059
[0.65, 0.70)	[6.00, 6.30)	1 ± 1.000	0 ± 0.000	0.6329 ± 0.0875	1.580 ± 1.595
[0.65, 0.70)	[6.30, 6.60)	1 ± 1.000	0 ± 0.000	0.5880 ± 0.0914	1.701 ± 1.721
[0.70, 0.75)	[4.20, 4.50)	5 ± 2.236	0 ± 0.000	0.5515 ± 0.0499	9.066 ± 4.137
[0.70, 0.75)	[4.50, 4.80)	3 ± 1.732	0 ± 0.000	0.5685 ± 0.1190	5.277 ± 3.241
[0.70, 0.75)	[4.80, 5.10)	2 ± 1.414	0 ± 0.000	0.4200 ± 0.1850	4.762 ± 3.967
[0.70, 0.75)	[5.10, 5.40)	1 ± 1.000	0 ± 0.000	0.4707 ± 0.1359	2.124 ± 2.211
[0.70, 0.75)	[5.40, 5.70)	1 ± 1.000	0 ± 0.000	0.6003 ± 0.0650	1.666 ± 1.676
[0.70, 0.75)	[6.60, 6.90)	1 ± 1.000	0 ± 0.000	0.5646 ± 0.0970	1.771 ± 1.797
[0.75, 0.80)	[4.20, 4.50)	2 ± 1.414	2 ± 1.414	0.0000 ± 0.0000	0.000 ± 0.000
[0.75, 0.80)	[4.50, 4.80)	5 ± 2.236	0 ± 0.000	0.5600 ± 0.0722	8.928 ± 4.156
[0.75, 0.80)	[4.80, 5.10)	2 ± 1.414	0 ± 0.000	0.7115 ± 0.0448	2.811 ± 1.996
[0.75, 0.80)	[5.40, 5.70)	1 ± 1.000	0 ± 0.000	0.6582 ± 0.0883	1.519 ± 1.533

13 Appendix: Table of Cross-Section Values

Table 7: Final LH2 Cross Section Table

xF bin	**xF centroid**	**Mass bin (GeV)**	**Mass centroid (GeV)**	$\sigma^{\pm stat.error}_{\pm syst.error}$ **(nb GeV^2)**
[0.00, 0.05)	0.03	[4.20, 4.50)	4.35	$1.5864^{\pm 1.5864}_{\pm 2.3557}$
[0.00, 0.05)	0.03	[4.50, 4.80)	4.65	$1.0362^{\pm 0.4096}_{\pm 0.3796}$
[0.00, 0.05)	0.03	[4.80, 5.10)	4.95	$0.4373^{\pm 0.3344}_{\pm 0.3292}$
[0.00, 0.05)	0.03	[5.10, 5.40)	5.25	$0.6064^{\pm 0.2077}_{\pm 0.1195}$
[0.00, 0.05)	0.03	[5.40, 5.70)	5.55	$0.2272^{\pm 0.1298}_{\pm 0.0666}$
[0.00, 0.05)	0.03	[5.70, 6.00)	5.85	$0.1644^{\pm 0.0793}_{\pm 0.0484}$
[0.00, 0.05)	0.03	[6.00, 6.30)	6.15	$0.1429^{\pm 0.0516}_{\pm 0.0270}$
[0.00, 0.05)	0.03	[6.30, 6.60)	6.45	$0.1104^{\pm 0.0285}_{\pm 0.0122}$
[0.00, 0.05)	0.03	[6.60, 6.90)	6.75	$0.0999^{\pm 0.0288}_{\pm 0.0110}$
[0.00, 0.05)	0.03	[6.90, 7.50)	7.20	$0.0616^{\pm 0.0205}_{\pm 0.0110}$
[0.00, 0.05)	0.03	[7.50, 8.70)	8.10	$0.0078^{\pm 0.0078}_{\pm 0.0025}$
[0.05, 0.10)	0.08	[4.20, 4.50)	4.35	$1.0710^{\pm 1.8550}_{\pm 1.5598}$
[0.05, 0.10)	0.08	[4.50, 4.80)	4.65	$0.8549^{\pm 0.5258}_{\pm 0.2984}$
[0.05, 0.10)	0.08	[4.80, 5.10)	4.95	$1.0045^{\pm 0.2390}_{\pm 0.1565}$
[0.05, 0.10)	0.08	[5.10, 5.40)	5.25	$0.4259^{\pm 0.1330}_{\pm 0.0851}$
[0.05, 0.10)	0.08	[5.40, 5.70)	5.55	$0.2335^{\pm 0.0647}_{\pm 0.0291}$
[0.05, 0.10)	0.08	[5.70, 6.00)	5.85	$0.1555^{\pm 0.0628}_{\pm 0.0275}$
[0.05, 0.10)	0.08	[6.00, 6.30)	6.15	$0.1491^{\pm 0.0504}_{\pm 0.0292}$
[0.05, 0.10)	0.08	[6.30, 6.60)	6.45	$0.1230^{\pm 0.0334}_{\pm 0.0153}$
[0.05, 0.10)	0.08	[6.60, 6.90)	6.75	$0.0212^{\pm 0.0187}_{\pm 0.0141}$
[0.05, 0.10)	0.08	[6.90, 7.50)	7.20	$0.0335^{\pm 0.0127}_{\pm 0.0051}$
[0.05, 0.10)	0.08	[7.50, 8.70)	8.10	$0.0311^{\pm 0.0127}_{\pm 0.0093}$
[0.10, 0.15)	0.12	[4.20, 4.50)	4.35	$1.5208^{\pm 0.4234}_{\pm 0.4561}$

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Table 7 – continued from previous page

xF bin	xF centroid	Mass bin (GeV)	Mass centroid (GeV)	$\sigma^{\pm stat.error}_{\pm syst.error}$ (nb GeV^2)
[0.10, 0.15)	0.12	[4.50, 4.80)	4.65	$1.2098^{\pm 0.3301}_{\pm 0.2190}$
[0.10, 0.15)	0.12	[4.80, 5.10)	4.95	$0.5154^{\pm 0.1636}_{\pm 0.0979}$
[0.10, 0.15)	0.12	[5.10, 5.40)	5.25	$0.5737^{\pm 0.0970}_{\pm 0.0618}$
[0.10, 0.15)	0.12	[5.40, 5.70)	5.55	$0.2304^{\pm 0.0625}_{\pm 0.0246}$
[0.10, 0.15)	0.12	[5.70, 6.00)	5.85	$0.1935^{\pm 0.0527}_{\pm 0.0185}$
[0.10, 0.15)	0.12	[6.00, 6.30)	6.15	$0.2023^{\pm 0.0315}_{\pm 0.0158}$
[0.10, 0.15)	0.12	[6.30, 6.60)	6.45	$0.0947^{\pm 0.0312}_{\pm 0.0111}$
[0.10, 0.15)	0.12	[6.60, 6.90)	6.75	$0.0488^{\pm 0.0154}_{\pm 0.0056}$
[0.10, 0.15)	0.12	[6.90, 7.50)	7.20	$0.0389^{\pm 0.0135}_{\pm 0.0075}$
[0.10, 0.15)	0.12	[7.50, 8.70)	8.10	$0.0257^{\pm 0.0097}_{\pm 0.0020}$
[0.15, 0.20)	0.17	[4.20, 4.50)	4.35	$1.4213^{\pm 0.2627}_{\pm 0.2657}$
[0.15, 0.20)	0.17	[4.50, 4.80)	4.65	$0.9905^{\pm 0.1688}_{\pm 0.1176}$
[0.15, 0.20)	0.17	[4.80, 5.10)	4.95	$0.7126^{\pm 0.1208}_{\pm 0.0889}$
[0.15, 0.20)	0.17	[5.10, 5.40)	5.25	$0.6052^{\pm 0.0847}_{\pm 0.0441}$
[0.15, 0.20)	0.17	[5.40, 5.70)	5.55	$0.3500^{\pm 0.0466}_{\pm 0.0238}$
[0.15, 0.20)	0.17	[5.70, 6.00)	5.85	$0.3121^{\pm 0.0396}_{\pm 0.0198}$
[0.15, 0.20)	0.17	[6.00, 6.30)	6.15	$0.1690^{\pm 0.0418}_{\pm 0.0169}$
[0.15, 0.20)	0.17	[6.30, 6.60)	6.45	$0.0912^{\pm 0.0261}_{\pm 0.0073}$
[0.15, 0.20)	0.17	[6.60, 6.90)	6.75	$0.0621^{\pm 0.0155}_{\pm 0.0051}$
[0.15, 0.20)	0.17	[6.90, 7.50)	7.20	$0.0258^{\pm 0.0175}_{\pm 0.0060}$
[0.15, 0.20)	0.17	[7.50, 8.70)	8.10	$0.0086^{\pm 0.0050}_{\pm 0.0019}$
[0.20, 0.25)	0.23	[4.20, 4.50)	4.35	$1.0016^{\pm 0.2234}_{\pm 0.1458}$
[0.20, 0.25)	0.23	[4.50, 4.80)	4.65	$0.9585^{\pm 0.1727}_{\pm 0.1160}$
[0.20, 0.25)	0.23	[4.80, 5.10)	4.95	$0.7832^{\pm 0.0591}_{\pm 0.0459}$
[0.20, 0.25)	0.23	[5.10, 5.40)	5.25	$0.4255^{\pm 0.0605}_{\pm 0.0318}$

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Table 7 – continued from previous page

xF bin	xF centroid	Mass bin (GeV)	Mass centroid (GeV)	$\sigma^{\pm stat.error}_{\pm syst.error}$ (nb GeV^2)
[0.20, 0.25)	0.23	[5.40, 5.70)	5.55	$0.3360^{\pm 0.0414}_{\pm 0.0219}$
[0.20, 0.25)	0.23	[5.70, 6.00)	5.85	$0.2434^{\pm 0.0384}_{\pm 0.0224}$
[0.20, 0.25)	0.23	[6.00, 6.30)	6.15	$0.1015^{\pm 0.0345}_{\pm 0.0150}$
[0.20, 0.25)	0.23	[6.30, 6.60)	6.45	$0.1309^{\pm 0.0199}_{\pm 0.0084}$
[0.20, 0.25)	0.23	[6.60, 6.90)	6.75	$0.0536^{\pm 0.0215}_{\pm 0.0053}$
[0.20, 0.25)	0.23	[6.90, 7.50)	7.20	$0.0104^{\pm 0.0132}_{\pm 0.0035}$
[0.20, 0.25)	0.23	[7.50, 8.70)	8.10	$0.0126^{\pm 0.0051}_{\pm 0.0012}$
[0.25, 0.30)	0.28	[4.20, 4.50)	4.35	$1.0498^{\pm 0.1543}_{\pm 0.1027}$
[0.25, 0.30)	0.28	[4.50, 4.80)	4.65	$1.0690^{\pm 0.1087}_{\pm 0.0771}$
[0.25, 0.30)	0.28	[4.80, 5.10)	4.95	$0.6585^{\pm 0.0600}_{\pm 0.0376}$
[0.25, 0.30)	0.28	[5.10, 5.40)	5.25	$0.4290^{\pm 0.0497}_{\pm 0.0282}$
[0.25, 0.30)	0.28	[5.40, 5.70)	5.55	$0.3491^{\pm 0.0439}_{\pm 0.0209}$
[0.25, 0.30)	0.28	[5.70, 6.00)	5.85	$0.1816^{\pm 0.0221}_{\pm 0.0101}$
[0.25, 0.30)	0.28	[6.00, 6.30)	6.15	$0.1470^{\pm 0.0190}_{\pm 0.0075}$
[0.25, 0.30)	0.28	[6.30, 6.60)	6.45	$0.0565^{\pm 0.0237}_{\pm 0.0096}$
[0.25, 0.30)	0.28	[6.60, 6.90)	6.75	$0.0584^{\pm 0.0188}_{\pm 0.0052}$
[0.25, 0.30)	0.28	[6.90, 7.50)	7.20	$0.0424^{\pm 0.0092}_{\pm 0.0039}$
[0.25, 0.30)	0.28	[7.50, 8.70)	8.10	$0.0047^{\pm 0.0034}_{\pm 0.0017}$
[0.30, 0.35)	0.32	[4.20, 4.50)	4.35	$1.2000^{\pm 0.1245}_{\pm 0.0915}$
[0.30, 0.35)	0.32	[4.50, 4.80)	4.65	$0.8546^{\pm 0.0848}_{\pm 0.0573}$
[0.30, 0.35)	0.32	[4.80, 5.10)	4.95	$0.6422^{\pm 0.0491}_{\pm 0.0326}$
[0.30, 0.35)	0.32	[5.10, 5.40)	5.25	$0.4816^{\pm 0.0427}_{\pm 0.0227}$
[0.30, 0.35)	0.32	[5.40, 5.70)	5.55	$0.2824^{\pm 0.0355}_{\pm 0.0160}$
[0.30, 0.35)	0.32	[5.70, 6.00)	5.85	$0.1618^{\pm 0.0267}_{\pm 0.0116}$
[0.30, 0.35)	0.32	[6.00, 6.30)	6.15	$0.1290^{\pm 0.0175}_{\pm 0.0075}$

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Table 7 – continued from previous page

xF bin	xF centroid	Mass bin (GeV)	Mass centroid (GeV)	$\sigma^{\pm stat.error}_{\pm syst.error}$ (nb GeV^2)
[0.30, 0.35)	0.32	[6.30, 6.60)	6.45	$0.0824^{\pm 0.0246}_{\pm 0.0154}$
[0.30, 0.35)	0.32	[6.60, 6.90)	6.75	$0.0709^{\pm 0.0142}_{\pm 0.0056}$
[0.30, 0.35)	0.32	[6.90, 7.50)	7.20	$0.0311^{\pm 0.0071}_{\pm 0.0025}$
[0.30, 0.35)	0.32	[7.50, 8.70)	8.10	$0.0156^{\pm 0.0052}_{\pm 0.0012}$
[0.35, 0.40)	0.38	[4.20, 4.50)	4.35	$0.9306^{\pm 0.0872}_{\pm 0.0626}$
[0.35, 0.40)	0.38	[4.50, 4.80)	4.65	$0.8491^{\pm 0.0569}_{\pm 0.0426}$
[0.35, 0.40)	0.38	[4.80, 5.10)	4.95	$0.6130^{\pm 0.0484}_{\pm 0.0294}$
[0.35, 0.40)	0.38	[5.10, 5.40)	5.25	$0.4210^{\pm 0.0362}_{\pm 0.0215}$
[0.35, 0.40)	0.38	[5.40, 5.70)	5.55	$0.1813^{\pm 0.0296}_{\pm 0.0104}$
[0.35, 0.40)	0.38	[5.70, 6.00)	5.85	$0.1842^{\pm 0.0216}_{\pm 0.0089}$
[0.35, 0.40)	0.38	[6.00, 6.30)	6.15	$0.1174^{\pm 0.0196}_{\pm 0.0070}$
[0.35, 0.40)	0.38	[6.30, 6.60)	6.45	$0.0480^{\pm 0.0233}_{\pm 0.0077}$
[0.35, 0.40)	0.38	[6.60, 6.90)	6.75	$0.0472^{\pm 0.0105}_{\pm 0.0033}$
[0.35, 0.40)	0.38	[6.90, 7.50)	7.20	$0.0285^{\pm 0.0065}_{\pm 0.0021}$
[0.35, 0.40)	0.38	[7.50, 8.70)	8.10	$0.0120^{\pm 0.0042}_{\pm 0.0014}$
[0.40, 0.45)	0.43	[4.20, 4.50)	4.35	$0.8713^{\pm 0.0755}_{\pm 0.0514}$
[0.40, 0.45)	0.43	[4.50, 4.80)	4.65	$0.6169^{\pm 0.0503}_{\pm 0.0313}$
[0.40, 0.45)	0.43	[4.80, 5.10)	4.95	$0.4585^{\pm 0.0376}_{\pm 0.0208}$
[0.40, 0.45)	0.43	[5.10, 5.40)	5.25	$0.3207^{\pm 0.0327}_{\pm 0.0166}$
[0.40, 0.45)	0.43	[5.40, 5.70)	5.55	$0.2464^{\pm 0.0263}_{\pm 0.0122}$
[0.40, 0.45)	0.43	[5.70, 6.00)	5.85	$0.1289^{\pm 0.0247}_{\pm 0.0116}$
[0.40, 0.45)	0.43	[6.00, 6.30)	6.15	$0.1283^{\pm 0.0188}_{\pm 0.0063}$
[0.40, 0.45)	0.43	[6.30, 6.60)	6.45	$0.0986^{\pm 0.0144}_{\pm 0.0057}$
[0.40, 0.45)	0.43	[6.60, 6.90)	6.75	$0.0513^{\pm 0.0105}_{\pm 0.0038}$
[0.40, 0.45)	0.43	[6.90, 7.50)	7.20	$0.0225^{\pm 0.0098}_{\pm 0.0034}$

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Table 7 – continued from previous page

xF bin	xF centroid	Mass bin (GeV)	Mass centroid (GeV)	$\sigma^{\pm stat.error}_{\pm syst.error}$ (nb GeV^2)
[0.40, 0.45)	0.43	[7.50, 8.70)	8.10	$0.0046^{\pm 0.0070}_{\pm 0.0015}$
[0.45, 0.50)	0.47	[4.20, 4.50)	4.35	$0.6911^{\pm 0.0554}_{\pm 0.0375}$
[0.45, 0.50)	0.47	[4.50, 4.80)	4.65	$0.5839^{\pm 0.0418}_{\pm 0.0267}$
[0.45, 0.50)	0.47	[4.80, 5.10)	4.95	$0.4492^{\pm 0.0298}_{\pm 0.0184}$
[0.45, 0.50)	0.47	[5.10, 5.40)	5.25	$0.2438^{\pm 0.0274}_{\pm 0.0126}$
[0.45, 0.50)	0.47	[5.40, 5.70)	5.55	$0.1934^{\pm 0.0235}_{\pm 0.0127}$
[0.45, 0.50)	0.47	[5.70, 6.00)	5.85	$0.1306^{\pm 0.0181}_{\pm 0.0071}$
[0.45, 0.50)	0.47	[6.00, 6.30)	6.15	$0.0924^{\pm 0.0150}_{\pm 0.0052}$
[0.45, 0.50)	0.47	[6.30, 6.60)	6.45	$0.0919^{\pm 0.0131}_{\pm 0.0051}$
[0.45, 0.50)	0.47	[6.60, 6.90)	6.75	$0.0359^{\pm 0.0095}_{\pm 0.0040}$
[0.45, 0.50)	0.47	[6.90, 7.50)	7.20	$0.0357^{\pm 0.0069}_{\pm 0.0024}$
[0.45, 0.50)	0.47	[7.50, 8.70)	8.10	$0.0101^{\pm 0.0036}_{\pm 0.0015}$
[0.50, 0.55)	0.53	[4.20, 4.50)	4.35	$0.7256^{\pm 0.0433}_{\pm 0.0356}$
[0.50, 0.55)	0.53	[4.50, 4.80)	4.65	$0.3184^{\pm 0.0325}_{\pm 0.0156}$
[0.50, 0.55)	0.53	[4.80, 5.10)	4.95	$0.3026^{\pm 0.0249}_{\pm 0.0129}$
[0.50, 0.55)	0.53	[5.10, 5.40)	5.25	$0.2265^{\pm 0.0238}_{\pm 0.0112}$
[0.50, 0.55)	0.53	[5.40, 5.70)	5.55	$0.1588^{\pm 0.0244}_{\pm 0.0095}$
[0.50, 0.55)	0.53	[5.70, 6.00)	5.85	$0.0913^{\pm 0.0165}_{\pm 0.0059}$
[0.50, 0.55)	0.53	[6.00, 6.30)	6.15	$0.0571^{\pm 0.0140}_{\pm 0.0050}$
[0.50, 0.55)	0.53	[6.30, 6.60)	6.45	$0.0645^{\pm 0.0135}_{\pm 0.0040}$
[0.50, 0.55)	0.53	[6.60, 6.90)	6.75	$0.0338^{\pm 0.0084}_{\pm 0.0029}$
[0.50, 0.55)	0.53	[6.90, 7.50)	7.20	$0.0173^{\pm 0.0046}_{\pm 0.0015}$
[0.50, 0.55)	0.53	[7.50, 8.70)	8.10	$0.0115^{\pm 0.0036}_{\pm 0.0011}$
[0.55, 0.60)	0.57	[4.20, 4.50)	4.35	$0.4597^{\pm 0.0346}_{\pm 0.0225}$
[0.55, 0.60)	0.57	[4.50, 4.80)	4.65	$0.3805^{\pm 0.0254}_{\pm 0.0157}$

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Table 7 – continued from previous page

xF bin	xF centroid	Mass bin (GeV)	Mass centroid (GeV)	$\sigma^{\pm stat.error}_{\pm syst.error}$ (nb GeV^2)
[0.55, 0.60)	0.57	[4.80, 5.10)	4.95	$0.2550^{\pm 0.0223}_{\pm 0.0111}$
[0.55, 0.60)	0.57	[5.10, 5.40)	5.25	$0.1594^{\pm 0.0167}_{\pm 0.0081}$
[0.55, 0.60)	0.57	[5.40, 5.70)	5.55	$0.1010^{\pm 0.0164}_{\pm 0.0073}$
[0.55, 0.60)	0.57	[5.70, 6.00)	5.85	$0.0609^{\pm 0.0149}_{\pm 0.0059}$
[0.55, 0.60)	0.57	[6.00, 6.30)	6.15	$0.0392^{\pm 0.0141}_{\pm 0.0034}$
[0.55, 0.60)	0.57	[6.30, 6.60)	6.45	$0.0277^{\pm 0.0111}_{\pm 0.0030}$
[0.55, 0.60)	0.57	[6.60, 6.90)	6.75	$0.0295^{\pm 0.0074}_{\pm 0.0027}$
[0.55, 0.60)	0.57	[6.90, 7.50)	7.20	$0.0169^{\pm 0.0045}_{\pm 0.0015}$
[0.55, 0.60)	0.57	[7.50, 8.70)	8.10	$0.0056^{\pm 0.0025}_{\pm 0.0014}$
[0.60, 0.65)	0.62	[4.20, 4.50)	4.35	$0.3594^{\pm 0.0283}_{\pm 0.0182}$
[0.60, 0.65)	0.62	[4.50, 4.80)	4.65	$0.2099^{\pm 0.0216}_{\pm 0.0103}$
[0.60, 0.65)	0.62	[4.80, 5.10)	4.95	$0.1467^{\pm 0.0173}_{\pm 0.0083}$
[0.60, 0.65)	0.62	[5.10, 5.40)	5.25	$0.1007^{\pm 0.0153}_{\pm 0.0052}$
[0.60, 0.65)	0.62	[5.40, 5.70)	5.55	$0.0629^{\pm 0.0123}_{\pm 0.0045}$
[0.60, 0.65)	0.62	[5.70, 6.00)	5.85	$0.0494^{\pm 0.0147}_{\pm 0.0085}$
[0.60, 0.65)	0.62	[6.00, 6.30)	6.15	$0.0405^{\pm 0.0138}_{\pm 0.0043}$
[0.60, 0.65)	0.62	[6.30, 6.60)	6.45	$0.0299^{\pm 0.0068}_{\pm 0.0022}$
[0.60, 0.65)	0.62	[6.60, 6.90)	6.75	$0.0113^{\pm 0.0120}_{\pm 0.0057}$
[0.60, 0.65)	0.62	[6.90, 7.50)	7.20	$0.0103^{\pm 0.0038}_{\pm 0.0017}$
[0.60, 0.65)	0.62	[7.50, 8.70)	8.10	$0.0037^{\pm 0.0021}_{\pm 0.0003}$
[0.65, 0.70)	0.68	[4.20, 4.50)	4.35	$0.2090^{\pm 0.0203}_{\pm 0.0118}$
[0.65, 0.70)	0.68	[4.50, 4.80)	4.65	$0.1468^{\pm 0.0172}_{\pm 0.0084}$
[0.65, 0.70)	0.68	[4.80, 5.10)	4.95	$0.0909^{\pm 0.0151}_{\pm 0.0057}$
[0.65, 0.70)	0.68	[5.10, 5.40)	5.25	$0.0711^{\pm 0.0174}_{\pm 0.0096}$
[0.65, 0.70)	0.68	[5.40, 5.70)	5.55	$0.0671^{\pm 0.0093}_{\pm 0.0037}$

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Table 7 – continued from previous page

xF bin	xF centroid	Mass bin (GeV)	Mass centroid (GeV)	$\sigma^{\pm stat.error}_{\pm syst.error}$ (nb GeV^2)
[0.65, 0.70)	0.68	[5.70, 6.00)	5.85	$0.0412^{\pm 0.0078}_{\pm 0.0036}$
[0.65, 0.70)	0.68	[6.00, 6.30)	6.15	$0.0273^{\pm 0.0090}_{\pm 0.0028}$
[0.65, 0.70)	0.68	[6.30, 6.60)	6.45	$0.0093^{\pm 0.0083}_{\pm 0.0024}$
[0.65, 0.70)	0.68	[6.60, 6.90)	6.75	$0.0164^{\pm 0.0055}_{\pm 0.0019}$
[0.65, 0.70)	0.68	[6.90, 7.50)	7.20	$0.0163^{\pm 0.0042}_{\pm 0.0013}$
[0.65, 0.70)	0.68	[7.50, 8.70)	8.10	$0.0050^{\pm 0.0022}_{\pm 0.0008}$
[0.70, 0.75)	0.72	[4.20, 4.50)	4.35	$0.1354^{\pm 0.0176}_{\pm 0.0082}$
[0.70, 0.75)	0.72	[4.50, 4.80)	4.65	$0.1212^{\pm 0.0144}_{\pm 0.0071}$
[0.70, 0.75)	0.72	[4.80, 5.10)	4.95	$0.0751^{\pm 0.0126}_{\pm 0.0063}$
[0.70, 0.75)	0.72	[5.10, 5.40)	5.25	$0.0424^{\pm 0.0097}_{\pm 0.0038}$
[0.70, 0.75)	0.72	[5.40, 5.70)	5.55	$0.0245^{\pm 0.0074}_{\pm 0.0021}$
[0.70, 0.75)	0.72	[5.70, 6.00)	5.85	$0.0222^{\pm 0.0054}_{\pm 0.0021}$
[0.70, 0.75)	0.72	[6.00, 6.30)	6.15	$0.0210^{\pm 0.0054}_{\pm 0.0019}$
[0.70, 0.75)	0.72	[6.30, 6.60)	6.45	$0.0171^{\pm 0.0051}_{\pm 0.0019}$
[0.70, 0.75)	0.72	[6.60, 6.90)	6.75	$0.0011^{\pm 0.0085}_{\pm 0.0024}$
[0.70, 0.75)	0.72	[6.90, 7.50)	7.20	$0.0040^{\pm 0.0023}_{\pm 0.0012}$
[0.70, 0.75)	0.72	[7.50, 8.70)	8.10	$0.0024^{\pm 0.0017}_{\pm 0.0013}$
[0.75, 0.80)	0.78	[4.20, 4.50)	4.35	$0.0975^{\pm 0.0112}_{\pm 0.0064}$
[0.75, 0.80)	0.78	[4.50, 4.80)	4.65	$0.0199^{\pm 0.0117}_{\pm 0.0035}$
[0.75, 0.80)	0.78	[4.80, 5.10)	4.95	$0.0286^{\pm 0.0079}_{\pm 0.0027}$
[0.75, 0.80)	0.78	[5.10, 5.40)	5.25	$0.0243^{\pm 0.0052}_{\pm 0.0020}$
[0.75, 0.80)	0.78	[5.40, 5.70)	5.55	$0.0162^{\pm 0.0064}_{\pm 0.0015}$
[0.75, 0.80)	0.78	[5.70, 6.00)	5.85	$0.0201^{\pm 0.0052}_{\pm 0.0020}$
[0.75, 0.80)	0.78	[6.00, 6.30)	6.15	$0.0175^{\pm 0.0048}_{\pm 0.0012}$
[0.75, 0.80)	0.78	[6.30, 6.60)	6.45	$0.0083^{\pm 0.0042}_{\pm 0.0039}$

Continued on next page

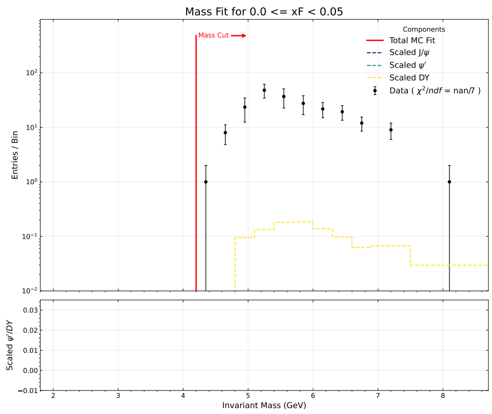
Table 7 – continued from previous page

xF bin	xF centroid	Mass bin (GeV)	Mass centroid (GeV)	$\sigma^{\pm stat.error}_{\pm syst.error}$ (nb GeV^2)
[0.75, 0.80)	0.78	[6.60, 6.90)	6.75	$0.0033^{\pm 0.0023}_{\pm 0.0003}$
[0.75, 0.80)	0.78	[6.90, 7.50)	7.20	$0.0009^{\pm 0.0009}_{\pm 0.0001}$
[0.75, 0.80)	0.78	[7.50, 8.70)	8.10	$0.0020^{\pm 0.0014}_{\pm 0.0003}$

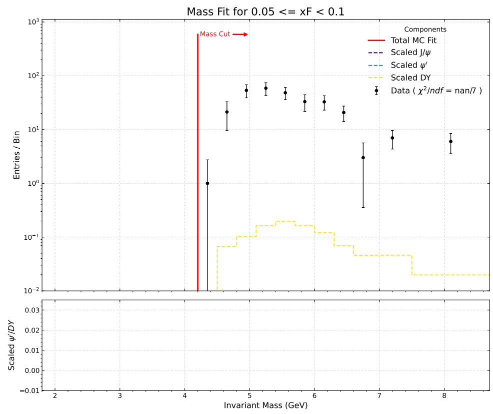
14 Appendix: ψ' Contamination Study

To quantify the systematic uncertainty arising from the ψ' resonance contamination within our signal region, we evaluated the background contribution across the invariant mass spectrum. Specifically, for each defined mass bin, this systematic uncertainty was determined by evaluating the ratio of the extracted ψ' yield to the Drell-Yan (DY) yield ($Y_{\psi'}/Y_{DY}$). The statistical and fit uncertainties from the individual yield extractions were propagated to calculate the total propagated uncertainty for each mass bin, which is visually represented in the $Y_{\psi'}/Y_{DY}$ ratio plot. The following figures illustrate the Monte Carlo (MC) fit plots utilized to model these shapes and extract the respective yields across the mass bins. The study was explained in DocDB 11454[20].

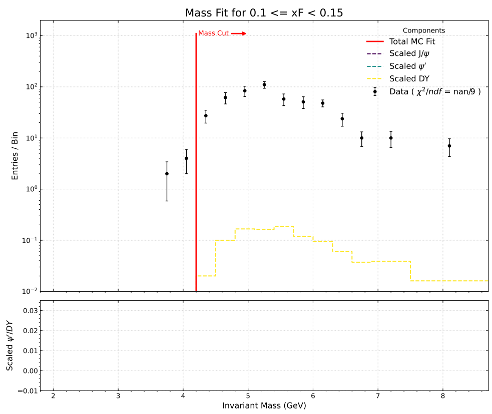
Figure 72: Plots of ψ' contamination generated by using MC with the ratio $Y_{\psi'}/Y_{DY}$



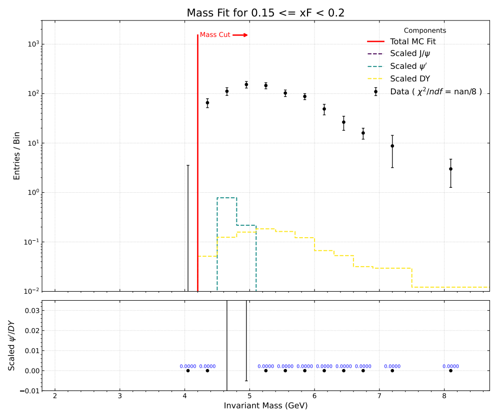
(a) $0.0 \leq x_F < 0.05$



(b) $0.05 \leq x_F < 0.1$



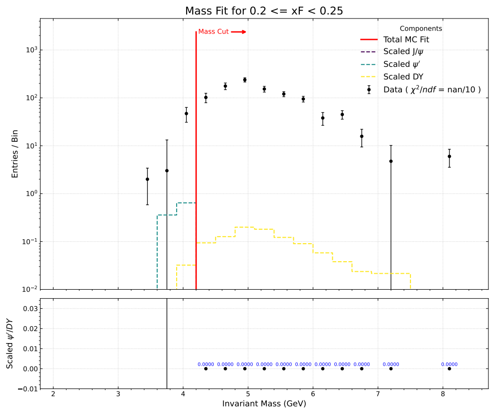
(c) $0.1 \leq x_F < 0.15$



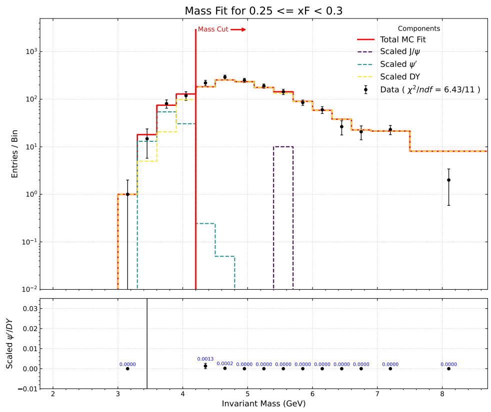
(d) $0.15 \leq x_F < 0.2$

[20] https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11454

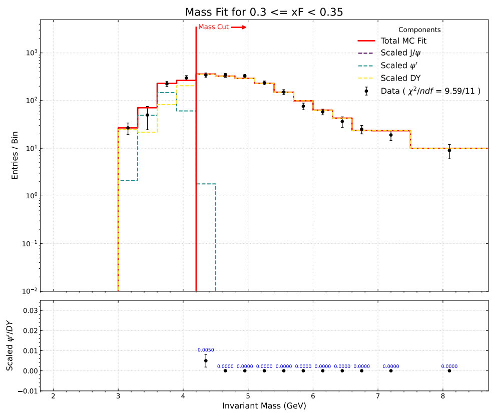
Figure 73: Plots of ψ' contamination generated by using MC with the ratio $Y_{\psi'}/Y_{DY}$



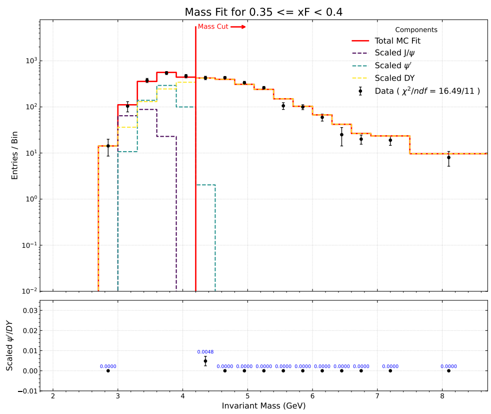
(a) $0.2 \leq x_F < 0.25$



(b) $0.25 \leq x_F < 0.3$

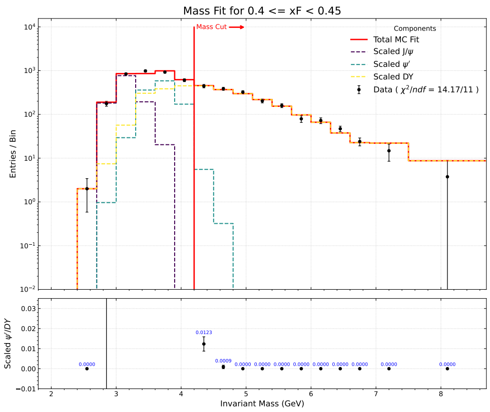


(c) $0.3 \leq x_F < 0.35$

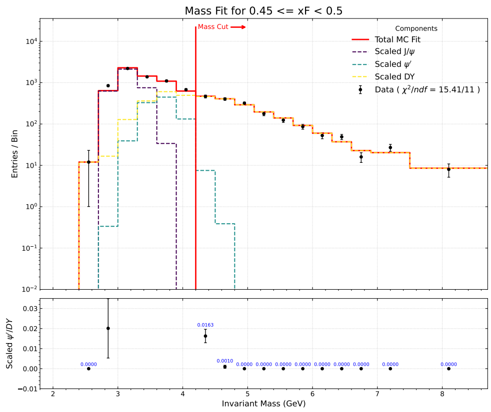


(d) $0.35 \leq x_F < 0.4$

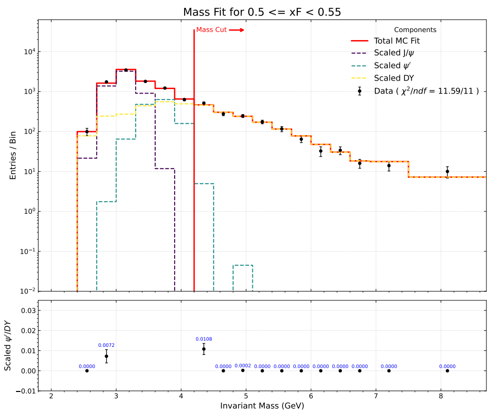
Figure 74: Plots of ψ' contamination generated by using MC with the ratio $Y_{\psi'}/Y_{DY}$



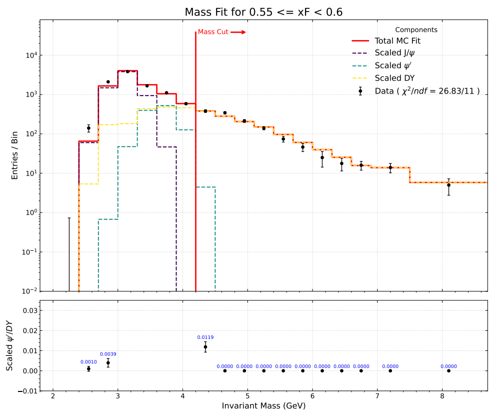
(a) $0.4 \leq x_F < 0.45$



(b) $0.45 \leq x_F < 0.5$

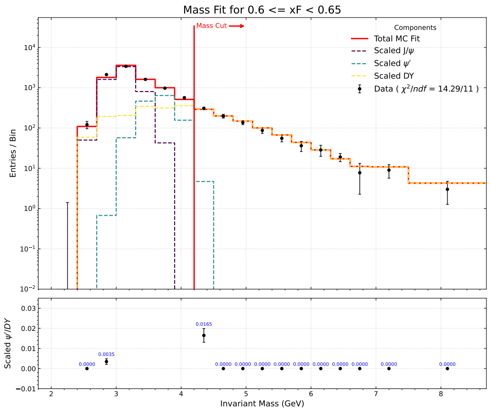


(c) $0.5 \leq x_F < 0.55$

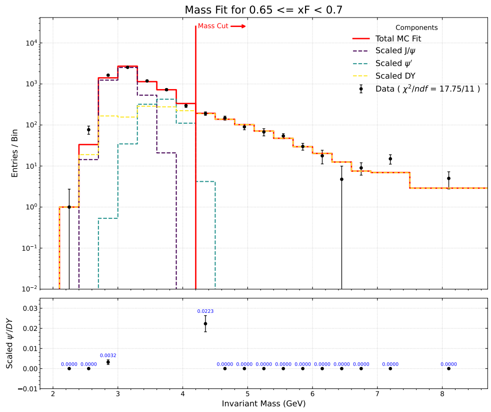


(d) $0.55 \leq x_F < 0.6$

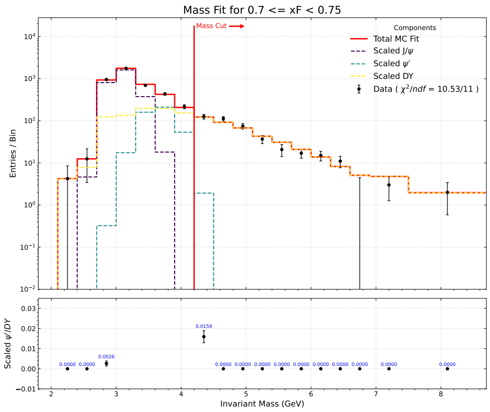
Figure 75: Plots of ψ' contamination generated by using MC with the ratio $Y_{\psi'}/Y_{DY}$



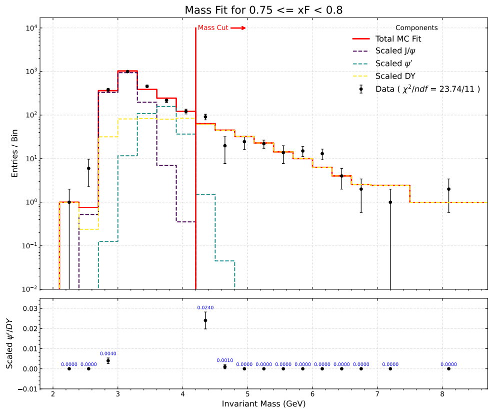
(a) $0.6 \leq x_F < 0.65$



(b) $0.65 \leq x_F < 0.7$



(c) $0.7 \leq x_F < 0.75$



(d) $0.75 \leq x_F < 0.8$

Table 8: ψ' Contamination Table for LH2

xF bin	**Mass bin** (GeV)	**ψ' contamination** ($Y_{\psi'}/Y_{DY}$)	$\sigma \pm \delta\sigma^{stat.} \pm \delta\sigma^{syst.}$ (nb GeV^2)	$\delta\sigma_{\psi'}^{\mathrm{syst.}} = \sigma \times (Y_{\psi'}/Y_{DY})$ (nb GeV^2)
[0.25, 0.30)	[4.20, 4.50)	0.0013 ± 0.0013	1.0617 ± 0.1561 ± 0.1039	0.0014
[0.25, 0.30)	[4.50, 4.80)	0.0002 ± 0.0002	1.0676 ± 0.1086 ± 0.0770	0.0002
[0.30, 0.35)	[4.20, 4.50)	0.0050 ± 0.0032	1.2002 ± 0.1245 ± 0.0915	0.0060
[0.35, 0.40)	[4.20, 4.50)	0.0048 ± 0.0024	0.9264 ± 0.0868 ± 0.0623	0.0045
[0.40, 0.45)	[4.20, 4.50)	0.0123 ± 0.0036	0.8710 ± 0.0755 ± 0.0514	0.0107
[0.40, 0.45)	[4.50, 4.80)	0.0009 ± 0.0009	0.6147 ± 0.0501 ± 0.0312	0.0005
[0.45, 0.50)	[4.20, 4.50)	0.0163 ± 0.0034	0.6885 ± 0.0552 ± 0.0374	0.0112
[0.45, 0.50)	[4.50, 4.80)	0.0010 ± 0.0008	0.5805 ± 0.0415 ± 0.0266	0.0006
[0.50, 0.55)	[4.20, 4.50)	0.0108 ± 0.0028	0.7247 ± 0.0433 ± 0.0356	0.0078
[0.50, 0.55)	[4.80, 5.10)	0.0002 ± 0.0002	0.3013 ± 0.0248 ± 0.0129	0.0001
[0.55, 0.60)	[4.20, 4.50)	0.0119 ± 0.0026	0.4574 ± 0.0344 ± 0.0224	0.0054
[0.60, 0.65)	[4.20, 4.50)	0.0165 ± 0.0034	0.3594 ± 0.0283 ± 0.0182	0.0059
[0.65, 0.70)	[4.20, 4.50)	0.0223 ± 0.0040	0.2059 ± 0.0200 ± 0.0116	0.0046
[0.70, 0.75)	[4.20, 4.50)	0.0159 ± 0.0030	0.1345 ± 0.0175 ± 0.0082	0.0021
[0.75, 0.80)	[4.20, 4.50)	0.0240 ± 0.0042	0.0972 ± 0.0111 ± 0.0063	0.0023
[0.75, 0.80)	[4.50, 4.80)	0.0010 ± 0.0010	0.0197 ± 0.0116 ± 0.0034	0.0000

15 Appendix: Determination of Mass Bin Centroids

In order to accurately plot the double-differential cross-sections, the horizontal placement of the data points must reflect the true physical average of the invariant mass within each finite mass bin, rather than the simple geometric center. Because the dimuon mass distribution falls exponentially, the true centroid is typically skewed toward the lower edge of the bin.

To account for combinatoric backgrounds, efficiency corrections, and flask contributions, the centroid calculation must be weighted using the fully corrected equations for the yields. For the $\mathrm{LH_2}$ target, the corrected yield is given by:

$$Y^{\mathrm{LH_2}}_{\mathrm{corrected}} = \frac{Y^{\mathrm{LH_2}}_{\mathrm{total}} - Y^{\mathrm{LH_2}}_{\mathrm{mixed}}}{\langle \epsilon^{\mathrm{LH_2}}_{\mathrm{signal}} \rangle} - \frac{I_{\mathrm{LH_2}}}{I_{\mathrm{flask}}} \left(\frac{Y^{\mathrm{flask}}_{\mathrm{total}} - Y^{\mathrm{flask}}_{\mathrm{mixed}}}{\langle \epsilon^{\mathrm{flask}}_{\mathrm{signal}} \rangle} \right) \tag{28}$$

For the $\mathrm{LD_2}$ target, the corrected yield accounts for both the empty flask and the hydrogen contribution:

$$\begin{aligned} Y^{\mathrm{LD_2}}_{\mathrm{corrected}} = & \frac{Y^{\mathrm{LD_2}}_{\mathrm{total}} - Y^{\mathrm{LD_2}}_{\mathrm{mixed}}}{\langle \epsilon^{\mathrm{LD_2}}_{\mathrm{signal}} \rangle} - \frac{I_{\mathrm{LD_2}}}{I_{\mathrm{flask}}} \left(\frac{Y^{\mathrm{flask}}_{\mathrm{total}} - Y^{\mathrm{flask}}_{\mathrm{mixed}}}{\langle \epsilon^{\mathrm{flask}}_{\mathrm{signal}} \rangle} \right) \\ & - \frac{T_{\mathrm{HD}}}{T_{\mathrm{HH}}} \frac{I_{\mathrm{LD_2}}}{I_{\mathrm{LH_2}}} \left[\frac{Y^{\mathrm{LH_2}}_{\mathrm{total}} - Y^{\mathrm{LH_2}}_{\mathrm{mixed}}}{\langle \epsilon^{\mathrm{LH_2}}_{\mathrm{signal}} \rangle} - \frac{I_{\mathrm{LH_2}}}{I_{\mathrm{flask}}} \left(\frac{Y^{\mathrm{flask}}_{\mathrm{total}} - Y^{\mathrm{flask}}_{\mathrm{mixed}}}{\langle \epsilon^{\mathrm{flask}}_{\mathrm{signal}} \rangle} \right) \right] \end{aligned} \tag{29}$$

The mass bin centroid for the $\mathrm{LH_2}$ target is then calculated as the efficiency-corrected, background-subtracted mass sum divided by the corrected yield:

$$\langle M \rangle_{\mathrm{LH_2}} = \frac{1}{Y^{\mathrm{LH_2}}_{\mathrm{corrected}}} \left[\frac{\sum M^{\mathrm{LH_2}}_{\mathrm{total}} - \sum M^{\mathrm{LH_2}}_{\mathrm{mixed}}}{\langle \epsilon^{\mathrm{LH_2}}_{\mathrm{signal}} \rangle} - \frac{I_{\mathrm{LH_2}}}{I_{\mathrm{flask}}} \left(\frac{\sum M^{\mathrm{flask}}_{\mathrm{total}} - \sum M^{\mathrm{flask}}_{\mathrm{mixed}}}{\langle \epsilon^{\mathrm{flask}}_{\mathrm{signal}} \rangle} \right) \right] \tag{30}$$

Similarly, we calculate the mass bin centroid for the $\mathrm{LD_2}$ target using the fully corrected yield and mass sums:

$$\begin{aligned} \langle M \rangle_{\mathrm{LD_2}} = \frac{1}{Y^{\mathrm{LD_2}}_{\mathrm{corrected}}} \Bigg\{ & \frac{\sum M^{\mathrm{LD_2}}_{\mathrm{total}} - \sum M^{\mathrm{LD_2}}_{\mathrm{mixed}}}{\langle \epsilon^{\mathrm{LD_2}}_{\mathrm{signal}} \rangle} - \frac{I_{\mathrm{LD_2}}}{I_{\mathrm{flask}}} \left(\frac{\sum M^{\mathrm{flask}}_{\mathrm{total}} - \sum M^{\mathrm{flask}}_{\mathrm{mixed}}}{\langle \epsilon^{\mathrm{flask}}_{\mathrm{signal}} \rangle} \right) \\ & - \frac{T_{\mathrm{HD}}}{T_{\mathrm{HH}}} \frac{I_{\mathrm{LD_2}}}{I_{\mathrm{LH_2}}} \left[\frac{\sum M^{\mathrm{LH_2}}_{\mathrm{total}} - \sum M^{\mathrm{LH_2}}_{\mathrm{mixed}}}{\langle \epsilon^{\mathrm{LH_2}}_{\mathrm{signal}} \rangle} - \frac{I_{\mathrm{LH_2}}}{I_{\mathrm{flask}}} \left(\frac{\sum M^{\mathrm{flask}}_{\mathrm{total}} - \sum M^{\mathrm{flask}}_{\mathrm{mixed}}}{\langle \epsilon^{\mathrm{flask}}_{\mathrm{signal}} \rangle} \right) \right] \Bigg\} \end{aligned} \tag{31}$$

Here, we plot dimuon yield distributions for $\mathrm{LH_2}$ cross-section calculations for the following components:

- $\mathrm{LH_2}$ total (red)
- $\mathrm{LH_2}$ mixed (blue)
- Flask total (green)
- Flask mixed (orange)

Then, the table of each dimuon component, efficiency corrections, and resulting mass bin centroids are presented for each x_F bin. Similarly, for $\mathrm{LD_2}$ cross-section calculations, we plot the following components:

- LD_2 total (red)
- LD_2 mixed (blue)
- Flask total (green)
- Flask mixed (orange)

These represent the total, mixed background, and flask components used in these calculations. Alongside the distributions are the precise kinematic values, raw counts, and resulting mass bin centroids for each x_F bin. Finally, the table of each dimuon component, efficiency corrections, and resulting mass bin centroids are presented for each x_F bin for the LD_2 target as well. This detailed breakdown allows for a clear understanding of how the final cross-section points are derived from the raw data and corrections.

15.1 Mass bin centroid calculations in $x_F \in [0.00, 0.05)$

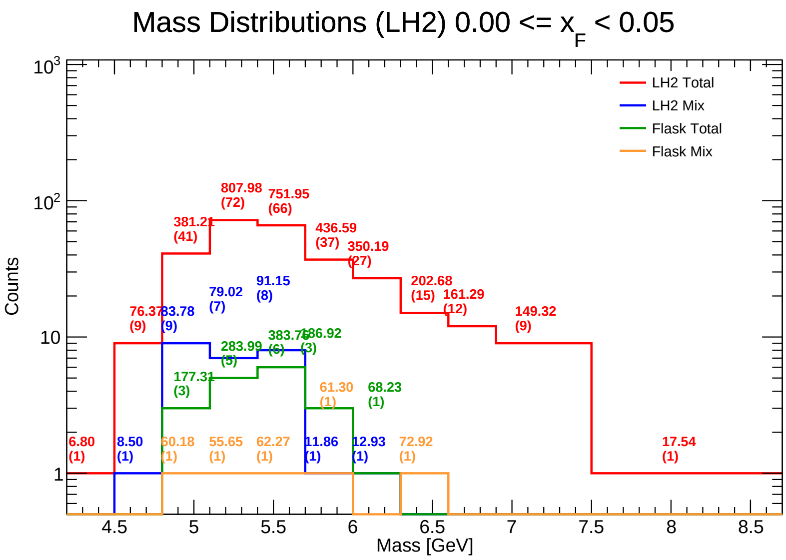


Figure 76: Invariant Mass distributions for LH_2 in the range $x_F \in [0.00, 0.05)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 9: LH_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.00, 0.05)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LH_2)	$N_{\mathrm{tot}}^{\mathrm{LH_2}}$	$N_{\mathrm{mix}}^{\mathrm{LH_2}}$	$N_{\mathrm{tot}}^{\mathrm{fl}}$	$N_{\mathrm{mix}}^{\mathrm{fl}}$	$\langle \epsilon_{\mathrm{sig.}}^{\mathrm{LH_2}} \rangle$	$\langle \epsilon_{\mathrm{sig.}}^{\mathrm{LD_2}} \rangle$	$\langle \epsilon_{\mathrm{sig.}}^{\mathrm{fl}} \rangle$	$\sum M_{\mathrm{corr}}^{\mathrm{LH_2}}$ (Num.)	$Y_{\mathrm{corr}}^{\mathrm{LH_2}}$ (Denom.)
[4.20, 4.50)	4.35	4.416	1	0	0	0	0.6494	0.5373	0.0000	6.7998	1.5398
[4.50, 4.80)	4.65	4.740	9	1	0	0	0.5586	0.5757	0.0000	67.8769	14.3209
[4.80, 5.10)	4.95	5.033	41	9	3	1	0.5349	0.5310	0.3663	180.2936	35.8247
[5.10, 5.40)	5.25	5.250	72	7	5	1	0.4684	0.4850	0.4049	500.6185	95.3613
[5.40, 5.70)	5.55	5.499	66	8	6	1	0.4860	0.4636	0.3813	339.3160	61.7094
[5.70, 6.00)	5.85	5.826	37	1	3	1	0.4947	0.5223	0.4103	299.1022	51.3405
[6.00, 6.30)	6.15	6.180	27	1	1	0	0.4757	0.4706	0.3954	269.0300	43.5350
[6.30, 6.60)	6.45	6.433	15	0	0	1	0.4754	0.4572	0.3895	275.5921	42.8396
[6.60, 6.90)	6.75	6.749	12	0	0	0	0.5021	0.5216	0.0000	161.2926	23.8975
[6.90, 7.50)	7.20	7.171	9	0	0	0	0.4322	0.5026	0.0000	149.3222	20.8238
[7.50, 8.70)	8.10	7.914	1	0	0	0	0.4512	0.0000	0.0000	17.5383	2.2161

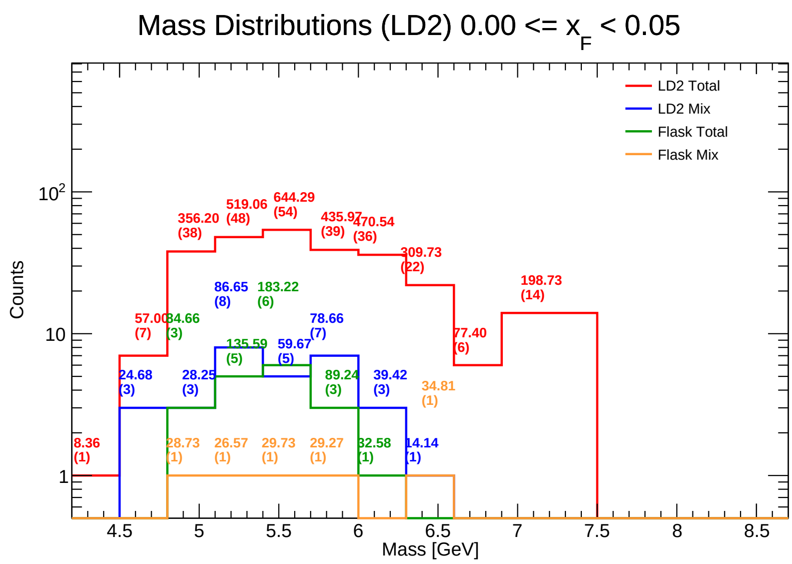


Figure 77: Invariant Mass distributions for LD_2 in the range $x_F \in [0.00, 0.05)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 10: LD_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.00, 0.05)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LD2)	$N_{\rm tot}^{\rm LD2}$	$N_{\rm mix}^{\rm LD2}$	$N_{\rm tot}^{\rm LH2}$	$N_{\rm mix}^{\rm LH2}$	$N_{\rm tot}^{\rm fl}$	$N_{\rm mix}^{\rm fl}$	$\langle \epsilon_{\rm sig.}^{\rm LH2} \rangle$	$\langle \epsilon_{\rm sig.}^{\rm LD2} \rangle$	$\langle \epsilon_{\rm sig.}^{\rm fl} \rangle$	$\sum M_{\rm corr}^{\rm LD2}$ (Num.)	$Y_{\rm corr}^{\rm LD2}$ (Denom.)
[4.20, 4.50)	4.35	4.490	1	0	1	0	0	0	0.6494	0.5373	0.0000	8.2585	1.8392
[4.50, 4.80)	4.65	4.649	7	3	9	1	0	0	0.5586	0.5757	0.0000	31.3423	6.7416
[4.80, 5.10)	4.95	4.995	38	3	41	9	3	1	0.5349	0.5310	0.3663	269.4289	53.9374
[5.10, 5.40)	5.25	5.237	48	8	72	7	5	1	0.4684	0.4850	0.4049	316.1812	60.3762
[5.40, 5.70)	5.55	5.514	54	5	66	8	6	1	0.4860	0.4636	0.3813	426.2461	77.2961
[5.70, 6.00)	5.85	5.826	39	7	37	1	3	1	0.4947	0.5223	0.4103	293.0269	50.2939
[6.00, 6.30)	6.15	6.149	36	3	27	1	1	0	0.4757	0.4706	0.3954	394.6746	64.1881
[6.30, 6.60)	6.45	6.438	22	1	15	0	0	1	0.4754	0.4572	0.3895	326.4352	50.7049
[6.60, 6.90)	6.75	6.728	6	0	12	0	0	0	0.5021	0.5216	0.0000	75.0796	11.1588
[6.90, 7.50)	7.20	7.134	14	0	9	0	0	0	0.4322	0.5026	0.0000	196.5794	27.5545
[7.50, 8.70)	8.10	7.914	0	0	1	0	0	0	0.4512	0.0000	0.0000	-0.2524	-0.0319

15.2 Mass bin centroid calculations in $x_F \in [0.05, 0.10)$

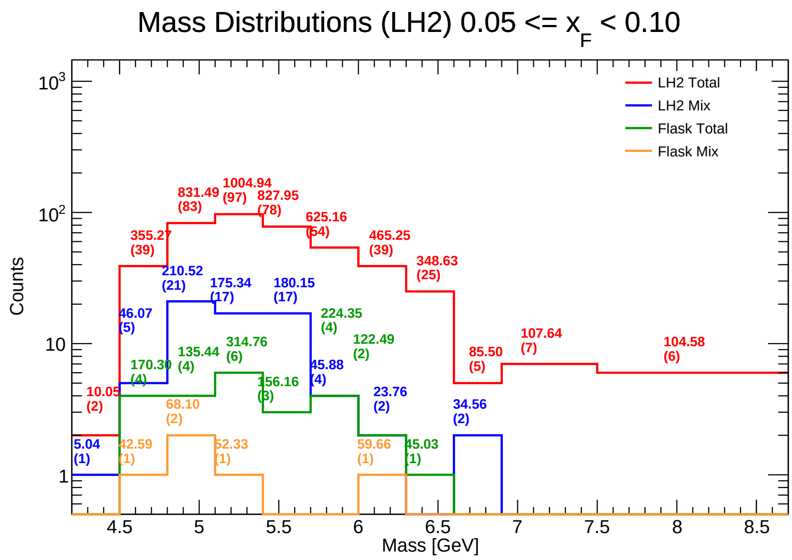


Figure 78: Invariant Mass distributions for LH_2 in the range $x_F \in [0.05, 0.10)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 11: LH_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.05, 0.10)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LH_2)	$N_{\text{tot}}^{\text{LH}_2}$	$N_{\text{mix}}^{\text{LH}_2}$	$N_{\text{tot}}^{\text{fl}}$	$N_{\text{mix}}^{\text{fl}}$	$\langle \epsilon_{\text{sig.}}^{\text{LH}_2} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{LD}_2} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{fl}} \rangle$	$\sum M_{\text{corr}}^{\text{LH}_2}$ (Num.)	$Y_{\text{corr}}^{\text{LH}_2}$ (Denom.)
[4.20, 4.50)	4.35	4.370	2	1	0	0	0.8712	0.3502	0.0000	5.0156	1.1479
[4.50, 4.80)	4.65	4.625	39	5	4	1	0.5110	0.5031	0.4830	181.4894	39.2381
[4.80, 5.10)	4.95	4.942	83	21	4	2	0.4931	0.4643	0.6403	553.6282	112.0150
[5.10, 5.40)	5.25	5.250	97	17	6	1	0.5069	0.4868	0.4415	567.1652	108.0304
[5.40, 5.70)	5.55	5.524	78	17	3	0	0.5219	0.4924	0.4729	491.6348	88.9959
[5.70, 6.00)	5.85	5.867	54	4	4	0	0.5055	0.4750	0.4576	354.9319	60.5006
[6.00, 6.30)	6.15	6.080	39	2	2	1	0.5134	0.4800	0.4489	378.6591	62.2802
[6.30, 6.60)	6.45	6.415	25	0	1	0	0.4610	0.4526	0.6373	303.5984	47.3295
[6.60, 6.90)	6.75	6.725	5	2	0	0	0.3960	0.4969	0.0000	50.9435	7.5757
[6.90, 7.50)	7.20	7.125	7	0	0	0	0.4634	0.5029	0.0000	107.6402	15.1065
[7.50, 8.70)	8.10	7.731	6	0	0	0	0.4436	0.5684	0.0000	104.5820	13.5269

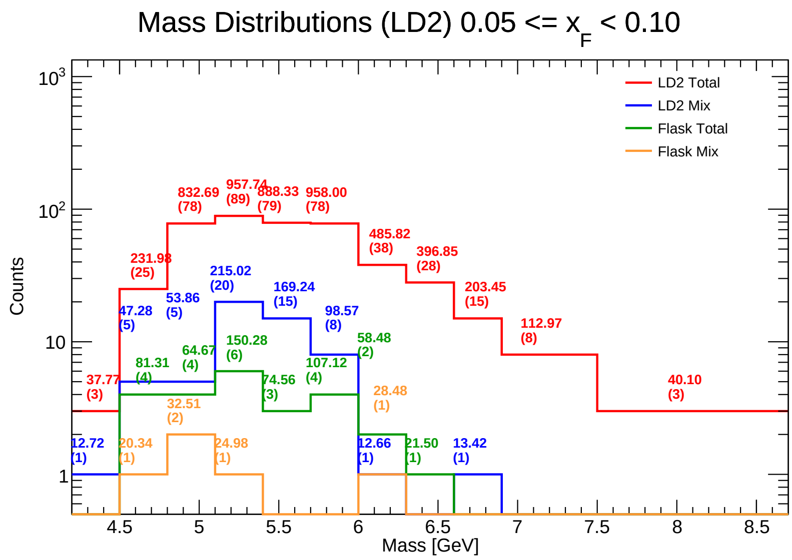


Figure 79: Invariant Mass distributions for LD_2 in the range $x_F \in [0.05, 0.10)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 12: LD_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.05, 0.10)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LD2)	$N_{\text{tot}}^{\text{LD2}}$	$N_{\text{mix}}^{\text{LD2}}$	$N_{\text{tot}}^{\text{LH2}}$	$N_{\text{mix}}^{\text{LH2}}$	$N_{\text{tot}}^{\text{fl}}$	$N_{\text{mix}}^{\text{fl}}$	$\langle \epsilon_{\text{sig.}}^{\text{LH2}} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{LD2}} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{fl}} \rangle$	$\sum M_{\text{corr}}^{\text{LD2}}$ (Num.)	$Y_{\text{corr}}^{\text{LD2}}$ (Denom.)
[4.20, 4.50)	4.35	4.386	3	1	2	1	0	0	0.8712	0.3502	0.0000	24.9751	5.6949
[4.50, 4.80)	4.65	4.630	25	5	39	5	4	1	0.5110	0.5031	0.4830	121.1075	26.1544
[4.80, 5.10)	4.95	4.956	78	5	83	21	4	2	0.4931	0.4643	0.6403	738.7109	149.0451
[5.10, 5.40)	5.25	5.234	89	20	97	17	6	1	0.5069	0.4868	0.4415	609.2577	116.4136
[5.40, 5.70)	5.55	5.525	79	15	78	17	3	0	0.5219	0.4924	0.4729	637.4566	115.3832
[5.70, 6.00)	5.85	5.830	78	8	54	4	4	0	0.5055	0.4750	0.4576	747.2044	128.1574
[6.00, 6.30)	6.15	6.121	38	1	39	2	2	1	0.5134	0.4800	0.4489	437.7090	71.5117
[6.30, 6.60)	6.45	6.408	28	0	25	0	1	0	0.4610	0.4526	0.6373	370.9822	57.8947
[6.60, 6.90)	6.75	6.745	15	1	5	2	0	0	0.3960	0.4969	0.0000	189.2935	28.0663
[6.90, 7.50)	7.20	7.101	8	0	7	0	0	0	0.4634	0.5029	0.0000	111.4172	15.6908
[7.50, 8.70)	8.10	7.594	3	0	6	0	0	0	0.4436	0.5684	0.0000	38.5989	5.0830

15.3 Mass bin centroid calculations in $x_F \in [0.10, 0.15)$

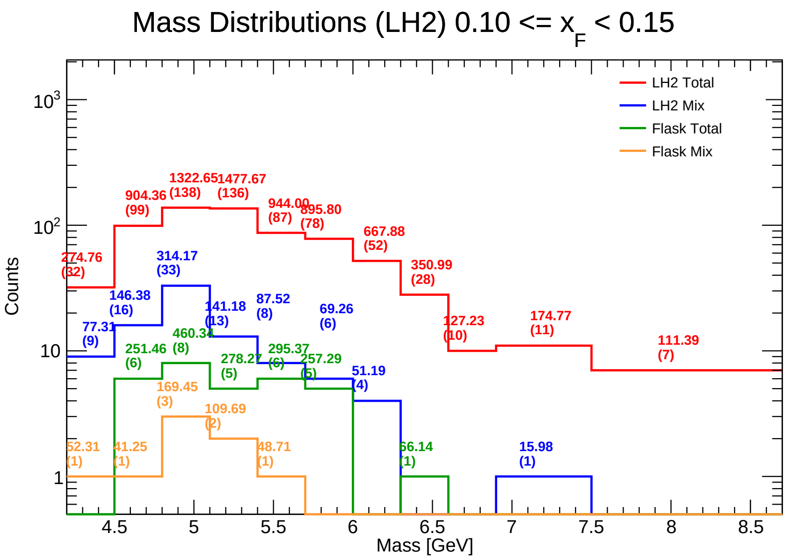


Figure 80: Invariant Mass distributions for LH_2 in the range $x_F \in [0.10, 0.15)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 13: LH_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.10, 0.15)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LH_2)	$N_{\text{tot}}^{\text{LH}_2}$	$N_{\text{mix}}^{\text{LH}_2}$	$N_{\text{tot}}^{\text{fl}}$	$N_{\text{mix}}^{\text{fl}}$	$\langle \epsilon_{\text{sig.}}^{\text{LH}_2} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{LD}_2} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{fl}} \rangle$	$\sum M_{\text{corr}}^{\text{LH}_2}$ (Num.)	$Y_{\text{corr}}^{\text{LH}_2}$ (Denom.)
[4.20, 4.50)	4.35	4.374	32	9	0	1	0.5097	0.4898	0.3670	249.7608	57.0975
[4.50, 4.80)	4.65	4.686	99	16	6	1	0.5114	0.5006	0.4839	547.7712	116.9004
[4.80, 5.10)	4.95	4.949	138	33	8	3	0.5173	0.4962	0.3791	717.5914	144.9975
[5.10, 5.40)	5.25	5.229	136	13	5	2	0.4822	0.4829	0.4153	1167.9167	223.3499
[5.40, 5.70)	5.55	5.499	87	8	6	1	0.5104	0.4936	0.5008	609.8187	110.8903
[5.70, 6.00)	5.85	5.815	78	6	5	0	0.5077	0.4932	0.5004	569.2487	97.8856
[6.00, 6.30)	6.15	6.139	52	4	0	0	0.4778	0.5080	0.0000	616.6852	100.4597
[6.30, 6.60)	6.45	6.441	28	0	1	0	0.5134	0.5011	0.4263	284.8498	44.2233
[6.60, 6.90)	6.75	6.746	10	0	0	0	0.5302	0.5025	0.0000	127.2270	18.8606
[6.90, 7.50)	7.20	7.114	11	1	0	0	0.4480	0.5246	0.0000	158.7891	22.3205
[7.50, 8.70)	8.10	7.838	7	0	0	0	0.4926	0.4386	0.0000	111.3862	14.2112

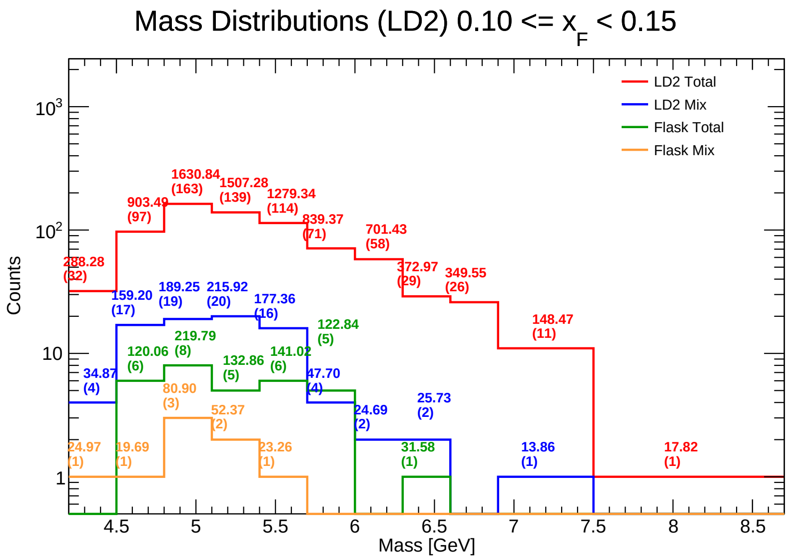


Figure 81: Invariant Mass distributions for LD_2 in the range $x_F \in [0.10, 0.15)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 14: LD_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.10, 0.15)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LD2)	$N_{\mathrm{tot}}^{\mathrm{LD2}}$	$N_{\mathrm{mix}}^{\mathrm{LD2}}$	$N_{\mathrm{tot}}^{\mathrm{LH2}}$	$N_{\mathrm{mix}}^{\mathrm{LH2}}$	$N_{\mathrm{tot}}^{\mathrm{fl}}$	$N_{\mathrm{mix}}^{\mathrm{fl}}$	$\langle \epsilon_{\mathrm{sig.}}^{\mathrm{LH2}} \rangle$	$\langle \epsilon_{\mathrm{sig.}}^{\mathrm{LD2}} \rangle$	$\langle \epsilon_{\mathrm{sig.}}^{\mathrm{fl}} \rangle$	$\sum M_{\mathrm{corr}}^{\mathrm{LD2}}$ (Num.)	$Y_{\mathrm{corr}}^{\mathrm{LD2}}$ (Denom.)
[4.20, 4.50)	4.35	4.428	32	4	32	9	0	1	0.5097	0.4898	0.3670	274.7899	62.0616
[4.50, 4.80)	4.65	4.661	97	17	99	16	6	1	0.5114	0.5006	0.4839	636.0483	136.4495
[4.80, 5.10)	4.95	4.963	163	19	138	33	8	3	0.5173	0.4962	0.3791	1292.3793	260.4249
[5.10, 5.40)	5.25	5.236	139	20	136	13	5	2	0.4822	0.4829	0.4153	1194.0605	228.0422
[5.40, 5.70)	5.55	5.542	114	16	87	8	6	1	0.5104	0.4936	0.5008	975.4395	176.0041
[5.70, 6.00)	5.85	5.822	71	4	78	6	5	0	0.5077	0.4932	0.5004	660.6345	113.4654
[6.00, 6.30)	6.15	6.139	58	2	52	4	0	0	0.4778	0.5080	0.0000	667.8679	108.7901
[6.30, 6.60)	6.45	6.448	29	2	28	0	1	0	0.5134	0.5011	0.4263	311.5571	48.3221
[6.60, 6.90)	6.75	6.755	26	0	10	0	0	0	0.5302	0.5025	0.0000	347.7162	51.4723
[6.90, 7.50)	7.20	7.062	11	1	11	1	0	0	0.4480	0.5246	0.0000	132.3330	18.7401
[7.50, 8.70)	8.10	7.813	1	0	7	0	0	0	0.4926	0.4386	0.0000	16.2159	2.0755

15.4 Mass bin centroid calculations in $x_F \in [0.15, 0.20)$

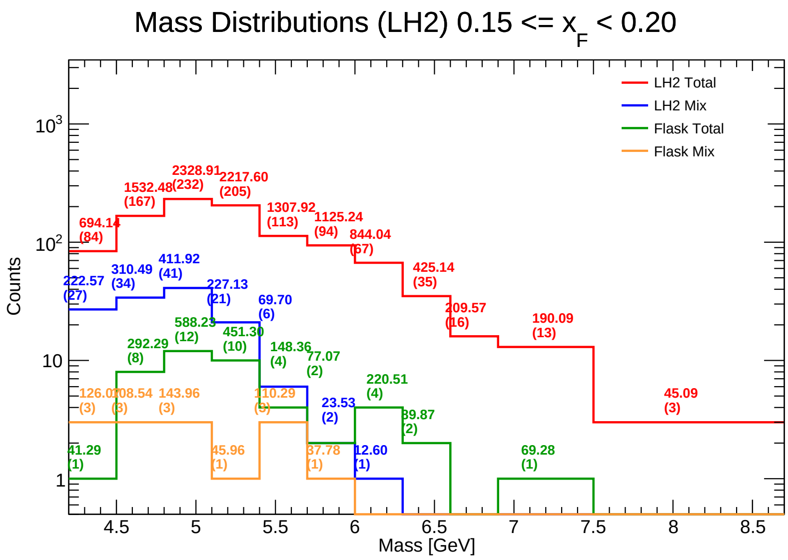


Figure 82: Invariant Mass distributions for LH_2 in the range $x_F \in [0.15, 0.20)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 15: LH_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.15, 0.20)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LH_2)	$N_{\text{tot}}^{\text{LH}_2}$	$N_{\text{mix}}^{\text{LH}_2}$	$N_{\text{tot}}^{\text{fl}}$	$N_{\text{mix}}^{\text{fl}}$	$\langle \epsilon_{\text{sig.}}^{\text{LH}_2} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{LD}_2} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{fl}} \rangle$	$\sum M_{\text{corr}}^{\text{LH}_2}$ (Num.)	$Y_{\text{corr}}^{\text{LH}_2}$ (Denom.)
[4.20, 4.50)	4.35	4.373	84	27	1	3	0.5286	0.5172	0.4530	556.3495	127.2349
[4.50, 4.80)	4.65	4.658	167	34	8	3	0.5081	0.4913	0.5656	1038.2458	222.9144
[4.80, 5.10)	4.95	4.917	232	41	12	3	0.4922	0.4962	0.4467	1472.7202	299.4881
[5.10, 5.40)	5.25	5.251	205	21	10	1	0.4846	0.4951	0.5086	1585.1301	301.8812
[5.40, 5.70)	5.55	5.522	113	6	4	3	0.4776	0.5415	0.6553	1200.1446	217.3331
[5.70, 6.00)	5.85	5.835	94	2	2	1	0.4877	0.4980	0.6695	1062.4184	182.0682
[6.00, 6.30)	6.15	6.138	67	1	4	0	0.4876	0.5231	0.4907	610.9264	99.5338
[6.30, 6.60)	6.45	6.467	35	0	2	0	0.5322	0.4876	0.6315	335.2777	51.8473
[6.60, 6.90)	6.75	6.748	16	0	0	0	0.5152	0.4854	0.0000	209.5684	31.0548
[6.90, 7.50)	7.20	6.912	13	0	1	0	0.4856	0.4978	0.4731	120.8094	17.4787
[7.50, 8.70)	8.10	7.632	3	0	0	0	0.5078	0.4635	0.0000	45.0881	5.9077

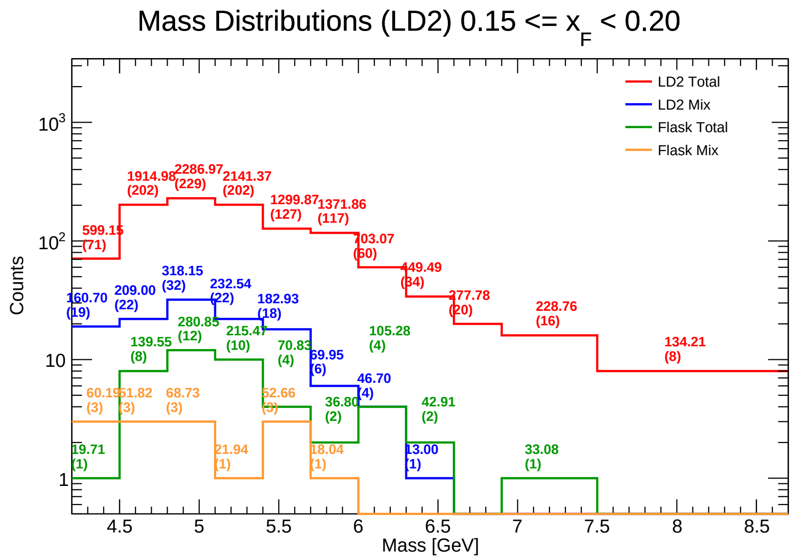


Figure 83: Invariant Mass distributions for LD_2 in the range $x_F \in [0.15, 0.20)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 16: LD_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.15, 0.20)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LD2)	$N_{\text{tot}}^{\text{LD2}}$	$N_{\text{mix}}^{\text{LD2}}$	$N_{\text{tot}}^{\text{LH2}}$	$N_{\text{mix}}^{\text{LH2}}$	$N_{\text{tot}}^{\text{fl}}$	$N_{\text{mix}}^{\text{fl}}$	$\langle \epsilon_{\text{sig.}}^{\text{LH2}} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{LD2}} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{fl}} \rangle$	$\sum M_{\text{corr}}^{\text{LD2}}$ (Num.)	$Y_{\text{corr}}^{\text{LD2}}$ (Denom.)
[4.20, 4.50)	4.35	4.361	71	19	84	27	1	3	0.5286	0.5172	0.4530	470.9254	107.9859
[4.50, 4.80)	4.65	4.652	202	22	167	34	8	3	0.5081	0.4913	0.5656	1603.3125	344.6176
[4.80, 5.10)	4.95	4.953	229	32	232	41	12	3	0.4922	0.4962	0.4467	1735.5105	350.4244
[5.10, 5.40)	5.25	5.255	202	22	205	21	10	1	0.4846	0.4951	0.5086	1692.4898	322.0962
[5.40, 5.70)	5.55	5.548	127	18	113	6	4	3	0.4776	0.5415	0.6553	1081.4909	194.9510
[5.70, 6.00)	5.85	5.839	117	6	94	2	2	1	0.4877	0.4980	0.6695	1267.8607	217.1474
[6.00, 6.30)	6.15	6.127	60	4	67	1	4	0	0.4876	0.5231	0.4907	542.2939	88.5115
[6.30, 6.60)	6.45	6.449	34	1	35	0	2	0	0.5322	0.4876	0.6315	388.7509	60.2847
[6.60, 6.90)	6.75	6.742	20	0	16	0	0	0	0.5152	0.4854	0.0000	274.7661	40.7526
[6.90, 7.50)	7.20	7.064	16	0	13	0	1	0	0.4856	0.4978	0.4731	193.9430	27.4548
[7.50, 8.70)	8.10	7.777	8	0	3	0	0	0	0.5078	0.4635	0.0000	133.5582	17.1744

15.5 Mass bin centroid calculations in $x_F \in [0.20, 0.25)$

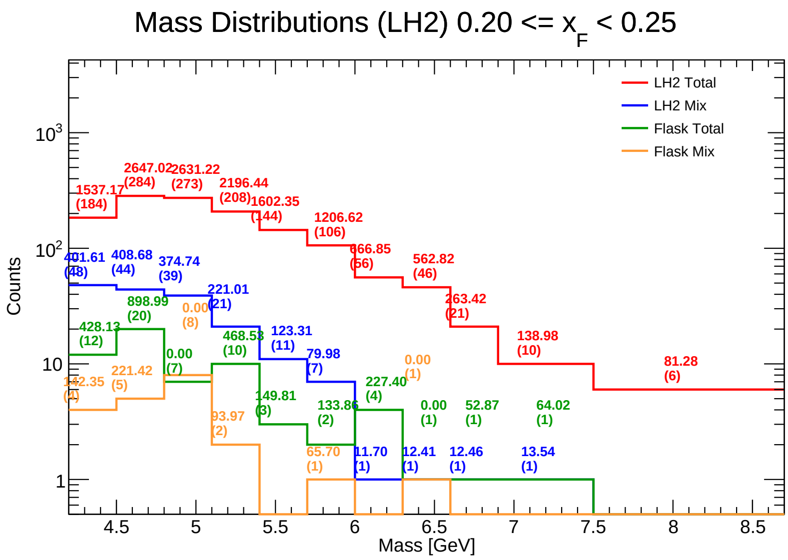


Figure 84: Invariant Mass distributions for LH_2 in the range $x_F \in [0.20, 0.25)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 17: LH_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.20, 0.25)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LH_2)	$N_{\text{tot}}^{\text{LH}_2}$	$N_{\text{mix}}^{\text{LH}_2}$	$N_{\text{tot}}^{\text{fl}}$	$N_{\text{mix}}^{\text{fl}}$	$\langle \epsilon_{\text{sig.}}^{\text{LH}_2} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{LD}_2} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{fl}} \rangle$	$\sum M_{\text{corr}}^{\text{LH}_2}$ (Num.)	$Y_{\text{corr}}^{\text{LH}_2}$ (Denom.)
[4.20, 4.50)	4.35	4.348	184	48	12	4	0.5225	0.5148	0.5424	849.7774	195.4540
[4.50, 4.80)	4.65	4.653	284	44	20	5	0.4995	0.5055	0.4545	1560.7659	335.4556
[4.80, 5.10)	4.95	4.956	273	39	7	8	0.5139	0.4887	-0.4617	2256.4780	455.3256
[5.10, 5.40)	5.25	5.237	208	21	10	2	0.4960	0.4944	0.4927	1600.8694	305.6673
[5.40, 5.70)	5.55	5.539	144	11	3	0	0.4977	0.5047	0.4842	1329.2264	239.9713
[5.70, 6.00)	5.85	5.832	106	7	2	1	0.5132	0.4811	0.3850	1058.4766	181.4801
[6.00, 6.30)	6.15	6.157	56	1	4	0	0.5163	0.5179	0.4745	427.7471	69.4688
[6.30, 6.60)	6.45	6.438	46	1	1	1	0.5264	0.5080	0.0000	550.4098	85.4929
[6.60, 6.90)	6.75	6.760	21	1	1	0	0.5377	0.5693	0.5570	198.0963	29.3030
[6.90, 7.50)	7.20	7.133	10	1	1	0	0.5145	0.4863	0.4949	61.4268	8.6110
[7.50, 8.70)	8.10	7.634	6	0	0	0	0.5635	0.5460	0.0000	81.2758	10.6471

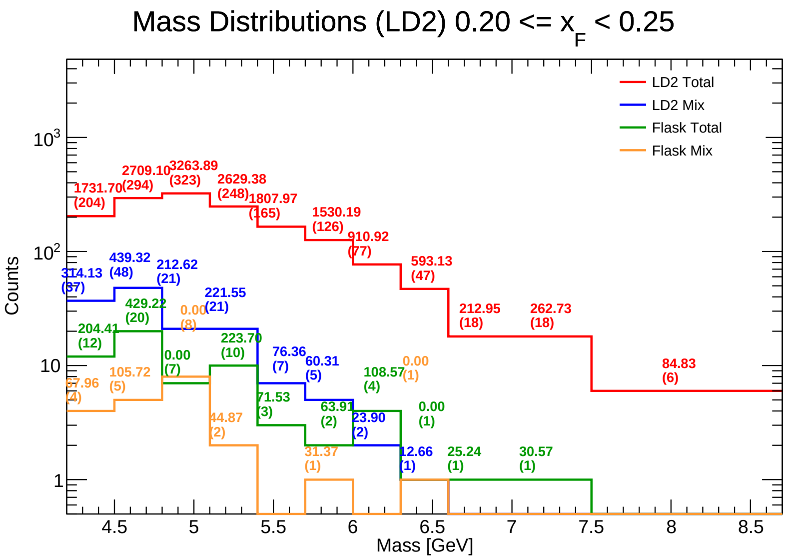


Figure 85: Invariant Mass distributions for LD_2 in the range $x_F \in [0.20, 0.25)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 18: LD_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.20, 0.25)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LD2)	$N_{\rm tot}^{\rm LD2}$	$N_{\rm mix}^{\rm LD2}$	$N_{\rm tot}^{\rm LH2}$	$N_{\rm mix}^{\rm LH2}$	$N_{\rm tot}^{\rm fl}$	$N_{\rm mix}^{\rm fl}$	$\langle \epsilon_{\rm sig.}^{\rm LH2} \rangle$	$\langle \epsilon_{\rm sig.}^{\rm LD2} \rangle$	$\langle \epsilon_{\rm sig.}^{\rm fl} \rangle$	$\sum M_{\rm corr}^{\rm LD2}$ (Num.)	$Y_{\rm corr}^{\rm LD2}$ (Denom.)
[4.20, 4.50)	4.35	4.366	204	37	184	48	12	4	0.5225	0.5148	0.5424	1268.8937	290.6383
[4.50, 4.80)	4.65	4.663	294	48	284	44	20	5	0.4995	0.5055	0.4545	1923.8186	412.5364
[4.80, 5.10)	4.95	4.937	323	21	273	39	7	8	0.5139	0.4887	-0.4617	3018.7986	611.4060
[5.10, 5.40)	5.25	5.244	248	21	208	21	10	2	0.4960	0.4944	0.4927	2205.9561	420.6262
[5.40, 5.70)	5.55	5.533	165	7	144	11	3	0	0.4977	0.5047	0.4842	1640.9515	296.5898
[5.70, 6.00)	5.85	5.841	126	5	106	7	2	1	0.5132	0.4811	0.3850	1422.1124	243.4704
[6.00, 6.30)	6.15	6.123	77	2	56	1	4	0	0.5163	0.5179	0.4745	772.2864	126.1341
[6.30, 6.60)	6.45	6.410	47	1	46	1	1	1	0.5264	0.5080	0.0000	572.5510	89.3178
[6.60, 6.90)	6.75	6.740	18	0	21	1	1	0	0.5377	0.5693	0.5570	184.8543	27.4285
[6.90, 7.50)	7.20	7.084	18	0	10	1	1	0	0.5145	0.4863	0.4949	231.2807	32.6502
[7.50, 8.70)	8.10	7.720	6	0	6	0	0	0	0.5635	0.5460	0.0000	83.6588	10.8368

15.6 Mass bin centroid calculations in $x_F \in [0.25, 0.30)$

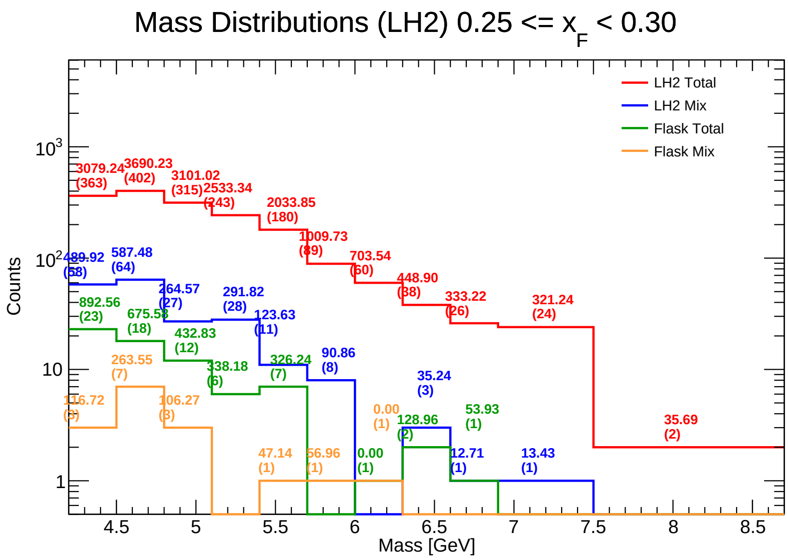


Figure 86: Invariant Mass distributions for LH_2 in the range $x_F \in [0.25, 0.30)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 19: LH_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.25, 0.30)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LH_2)	$N_{\mathrm{tot}}^{\mathrm{LH_2}}$	$N_{\mathrm{mix}}^{\mathrm{LH_2}}$	$N_{\mathrm{tot}}^{\mathrm{fl}}$	$N_{\mathrm{mix}}^{\mathrm{fl}}$	$\langle \epsilon_{\mathrm{sig.}}^{\mathrm{LH_2}} \rangle$	$\langle \epsilon_{\mathrm{sig.}}^{\mathrm{LD_2}} \rangle$	$\langle \epsilon_{\mathrm{sig.}}^{\mathrm{fl}} \rangle$	$\sum M_{\mathrm{corr}}^{\mathrm{LH_2}}$ (Num.)	$Y_{\mathrm{corr}}^{\mathrm{LH_2}}$ (Denom.)
[4.20, 4.50)	4.35	4.392	363	58	23	3	0.5147	0.5157	0.4894	1813.4698	412.9125
[4.50, 4.80)	4.65	4.652	402	64	18	7	0.5063	0.5026	0.5422	2690.7315	578.3617
[4.80, 5.10)	4.95	4.949	315	27	12	3	0.5032	0.5037	0.6070	2509.8875	507.2003
[5.10, 5.40)	5.25	5.244	243	28	6	0	0.5028	0.5012	0.4081	1903.3407	362.9712
[5.40, 5.70)	5.55	5.555	180	11	7	1	0.4907	0.4937	0.5195	1631.1220	293.6527
[5.70, 6.00)	5.85	5.839	89	8	0	1	0.5153	0.5043	0.4428	975.8315	167.1199
[6.00, 6.30)	6.15	6.144	60	0	1	1	0.5239	0.4989	0.0000	703.5422	114.5149
[6.30, 6.60)	6.45	6.473	38	3	2	0	0.5455	0.5098	0.4356	284.7006	43.9800
[6.60, 6.90)	6.75	6.755	26	1	1	0	0.5265	0.5124	0.5485	266.5882	39.4682
[6.90, 7.50)	7.20	7.107	24	1	0	0	0.5310	0.5156	0.0000	307.8096	43.3117
[7.50, 8.70)	8.10	7.598	2	0	0	0	0.4258	0.4489	0.0000	35.6911	4.6973

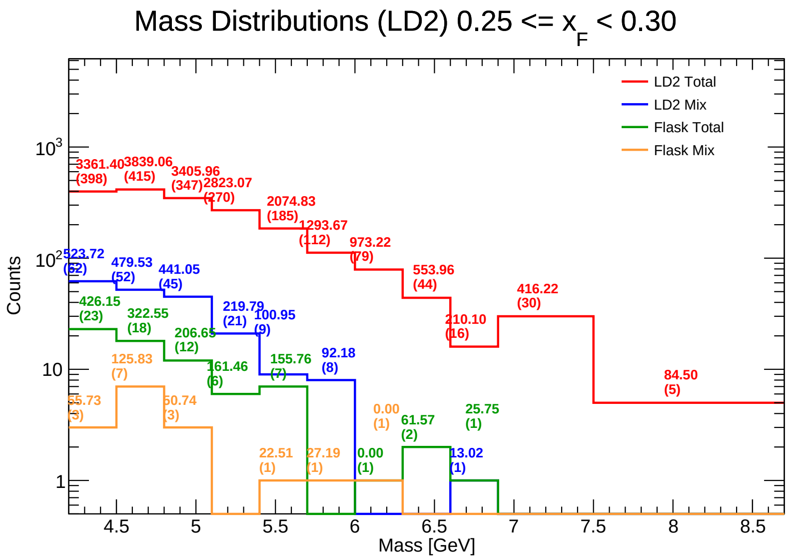


Figure 87: Invariant Mass distributions for LD_2 in the range $x_F \in [0.25, 0.30)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 20: LD_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.25, 0.30)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LD2)	$N_{\text{tot}}^{\text{LD2}}$	$N_{\text{mix}}^{\text{LD2}}$	$N_{\text{tot}}^{\text{LH2}}$	$N_{\text{mix}}^{\text{LH2}}$	$N_{\text{tot}}^{\text{fl}}$	$N_{\text{mix}}^{\text{fl}}$	$\langle \epsilon_{\text{sig.}}^{\text{LH2}} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{LD2}} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{fl}} \rangle$	$\sum M_{\text{corr}}^{\text{LD2}}$ (Num.)	$Y_{\text{corr}}^{\text{LD2}}$ (Denom.)
[4.20, 4.50)	4.35	4.361	398	62	363	58	23	3	0.5147	0.5157	0.4894	2441.1578	559.8268
[4.50, 4.80)	4.65	4.653	415	52	402	64	18	7	0.5063	0.5026	0.5422	3124.0930	671.3725
[4.80, 5.10)	4.95	4.942	347	45	315	27	12	3	0.5032	0.5037	0.6070	2772.8869	561.1319
[5.10, 5.40)	5.25	5.241	270	21	243	28	6	0	0.5028	0.5012	0.4081	2414.4346	460.7259
[5.40, 5.70)	5.55	5.539	185	9	180	11	7	1	0.4907	0.4937	0.5195	1817.1494	328.0498
[5.70, 6.00)	5.85	5.824	112	8	89	8	0	1	0.5153	0.5043	0.4428	1214.6447	208.5595
[6.00, 6.30)	6.15	6.146	79	0	60	0	1	1	0.5239	0.4989	0.0000	963.0914	156.7021
[6.30, 6.60)	6.45	6.422	44	0	38	3	2	0	0.5455	0.5098	0.4356	488.2934	76.0358
[6.60, 6.90)	6.75	6.732	16	1	26	1	1	0	0.5265	0.5124	0.5485	167.4945	24.8815
[6.90, 7.50)	7.20	7.155	30	0	24	1	0	0	0.5310	0.5156	0.0000	411.7907	57.5562
[7.50, 8.70)	8.10	7.587	5	0	2	0	0	0	0.4258	0.4489	0.0000	83.9894	11.0699

715 15.7 Mass bin centroid calculations in $x_F \in [0.30, 0.35)$

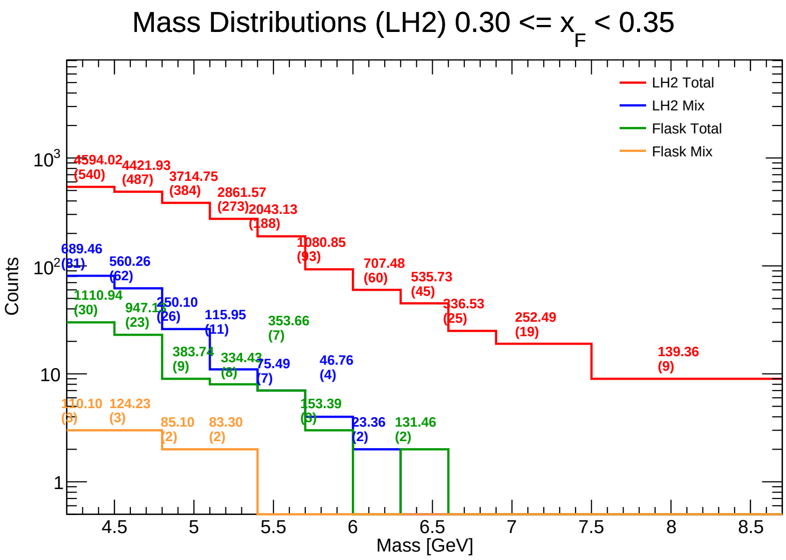


Figure 88: Invariant Mass distributions for LH_2 in the range $x_F \in [0.30, 0.35)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 21: LH_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.30, 0.35)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LH_2)	$N_{\text{tot}}^{LH_2}$	$N_{\text{mix}}^{LH_2}$	$N_{\text{tot}}^{\text{fl}}$	$N_{\text{mix}}^{\text{fl}}$	$\langle \epsilon_{\text{sig.}}^{LH_2} \rangle$	$\langle \epsilon_{\text{sig.}}^{LD_2} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{fl}} \rangle$	$\sum M_{\text{corr}}^{LH_2}$ (Num.)	$Y_{\text{corr}}^{LH_2}$ (Denom.)
[4.20, 4.50)	4.35	4.355	540	81	30	3	0.5114	0.4917	0.5141	2903.7186	666.7770
[4.50, 4.80)	4.65	4.666	487	62	23	3	0.5119	0.4937	0.4912	3038.7688	651.3263
[4.80, 5.10)	4.95	4.947	384	26	9	2	0.5109	0.4923	0.5068	3166.0168	639.9740
[5.10, 5.40)	5.25	5.249	273	11	8	2	0.5007	0.4902	0.5486	2494.4808	475.2328
[5.40, 5.70)	5.55	5.542	188	7	7	0	0.5094	0.5098	0.4799	1613.9740	291.2323
[5.70, 6.00)	5.85	5.844	93	4	3	0	0.5021	0.5055	0.4970	880.7042	150.7089
[6.00, 6.30)	6.15	6.133	60	2	0	0	0.5199	0.5260	0.0000	684.1149	111.5505
[6.30, 6.60)	6.45	6.376	45	0	2	0	0.5385	0.4886	0.4358	404.2769	63.4017
[6.60, 6.90)	6.75	6.741	25	0	0	0	0.5008	0.5025	0.0000	336.5274	49.9209
[6.90, 7.50)	7.20	7.045	19	0	0	0	0.5302	0.5058	0.0000	252.4916	35.8386
[7.50, 8.70)	8.10	7.919	9	0	0	0	0.5114	0.5042	0.0000	139.3574	17.5975

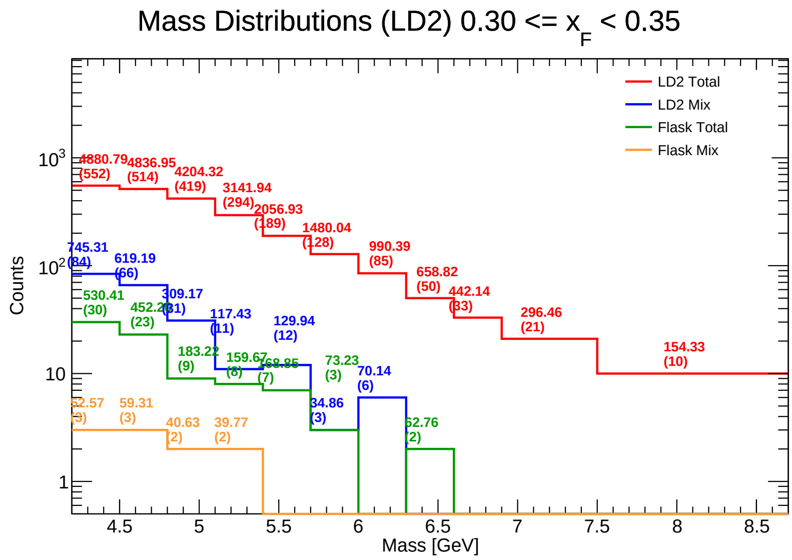


Figure 89: Invariant Mass distributions for LD_2 in the range $x_F \in [0.30, 0.35)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 22: LD_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.30, 0.35)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LD2)	$N_{\mathrm{tot}}^{\mathrm{LD2}}$	$N_{\mathrm{mix}}^{\mathrm{LD2}}$	$N_{\mathrm{tot}}^{\mathrm{LH2}}$	$N_{\mathrm{mix}}^{\mathrm{LH2}}$	$N_{\mathrm{tot}}^{\mathrm{fl}}$	$N_{\mathrm{mix}}^{\mathrm{fl}}$	$\langle \epsilon_{\mathrm{sig.}}^{\mathrm{LH2}} \rangle$	$\langle \epsilon_{\mathrm{sig.}}^{\mathrm{LD2}} \rangle$	$\langle \epsilon_{\mathrm{sig.}}^{\mathrm{fl}} \rangle$	$\sum M_{\mathrm{corr}}^{\mathrm{LD2}}$ (Num.)	$Y_{\mathrm{corr}}^{\mathrm{LD2}}$ (Denom.)
[4.20, 4.50)	4.35	4.346	552	84	540	81	30	3	0.5114	0.4917	0.5141	3615.8430	831.9812
[4.50, 4.80)	4.65	4.653	514	66	487	62	23	3	0.5119	0.4937	0.4912	3781.1345	812.5723
[4.80, 5.10)	4.95	4.943	419	31	384	26	9	2	0.5109	0.4923	0.5068	3707.0056	749.9868
[5.10, 5.40)	5.25	5.240	294	11	273	11	8	2	0.5007	0.4902	0.5486	2868.7115	547.4693
[5.40, 5.70)	5.55	5.554	189	12	188	7	7	0	0.5094	0.5098	0.4799	1734.9115	312.3921
[5.70, 6.00)	5.85	5.848	128	3	93	4	3	0	0.5021	0.5055	0.4970	1359.2760	232.4282
[6.00, 6.30)	6.15	6.127	85	6	60	2	0	0	0.5199	0.5260	0.0000	910.4013	148.5880
[6.30, 6.60)	6.45	6.431	50	0	45	0	2	0	0.5385	0.4886	0.4358	590.2432	91.7817
[6.60, 6.90)	6.75	6.732	33	0	25	0	0	0	0.5008	0.5025	0.0000	437.2951	64.9546
[6.90, 7.50)	7.20	7.142	21	0	19	0	0	0	0.5302	0.5058	0.0000	292.8265	41.0018
[7.50, 8.70)	8.10	7.780	10	0	9	0	0	0	0.5114	0.5042	0.0000	152.3259	19.5804

716 15.8 Mass bin centroid calculations in $x_F \in [0.35, 0.40)$

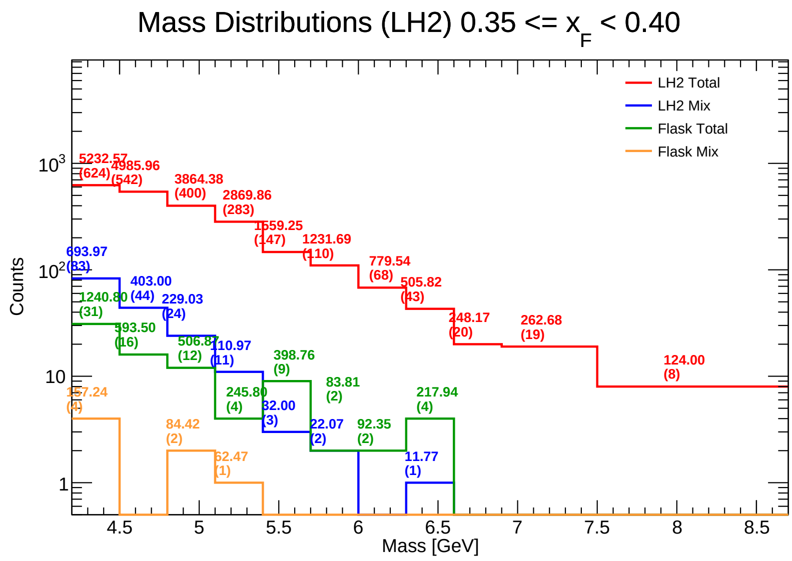


Figure 90: Invariant Mass distributions for LH_2 in the range $x_F \in [0.35, 0.40)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 23: LH_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.35, 0.40)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LH_2)	$N_{\rm tot}^{\rm LH_2}$	$N_{\rm mix}^{\rm LH_2}$	$N_{\rm tot}^{\rm fl}$	$N_{\rm mix}^{\rm fl}$	$\langle \epsilon_{\rm sig.}^{\rm LH_2} \rangle$	$\langle \epsilon_{\rm sig.}^{\rm LD_2} \rangle$	$\langle \epsilon_{\rm sig.}^{\rm fl} \rangle$	$\sum M_{\rm corr}^{\rm LH_2}$ (Num.)	$Y_{\rm corr}^{\rm LH_2}$ (Denom.)
[4.20, 4.50)	4.35	4.336	624	83	31	4	0.5180	0.5083	0.4793	3455.0394	796.8315
[4.50, 4.80)	4.65	4.640	542	44	16	0	0.5045	0.4974	0.5526	3989.4672	859.8243
[4.80, 5.10)	4.95	4.944	400	24	12	2	0.5115	0.4967	0.5159	3212.9033	649.9225
[5.10, 5.40)	5.25	5.239	283	11	4	1	0.5164	0.4993	0.3753	2575.5627	491.5673
[5.40, 5.70)	5.55	5.517	147	3	9	0	0.5221	0.5140	0.5552	1128.4843	204.5489
[5.70, 6.00)	5.85	5.835	110	2	2	0	0.5213	0.4985	0.6177	1125.8140	192.9485
[6.00, 6.30)	6.15	6.125	68	0	2	0	0.5347	0.5167	0.5861	687.1916	112.1869
[6.30, 6.60)	6.45	6.446	43	1	4	0	0.5481	0.5003	0.5202	276.1103	42.8332
[6.60, 6.90)	6.75	6.727	20	0	0	0	0.5421	0.5070	0.0000	248.1659	36.8926
[6.90, 7.50)	7.20	7.175	19	0	0	0	0.5190	0.4941	0.0000	262.6775	36.6110
[7.50, 8.70)	8.10	7.764	8	0	0	0	0.5009	0.5038	0.0000	124.0016	15.9705

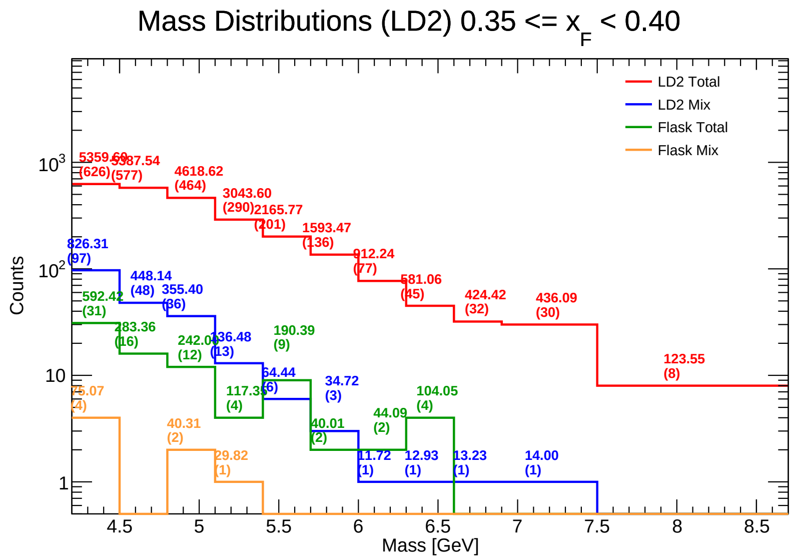


Figure 91: Invariant Mass distributions for LD_2 in the range $x_F \in [0.35, 0.40)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 24: LD_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.35, 0.40)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LD2)	$N^{\text{LD2}}_{\text{tot}}$	$N^{\text{LD2}}_{\text{mix}}$	$N^{\text{LH2}}_{\text{tot}}$	$N^{\text{LH2}}_{\text{mix}}$	$N^{\text{fl}}_{\text{tot}}$	$N^{\text{fl}}_{\text{mix}}$	$\langle \epsilon^{\text{LH2}}_{\text{sig.}} \rangle$	$\langle \epsilon^{\text{LD2}}_{\text{sig.}} \rangle$	$\langle \epsilon^{\text{fl}}_{\text{sig.}} \rangle$	$\sum M^{\text{LD2}}_{\text{corr}}$ (Num.)	$Y^{\text{LD2}}_{\text{corr}}$ (Denom.)
[4.20, 4.50)	4.35	4.353	626	97	624	83	31	4	0.5180	0.5083	0.4793	3966.2293	911.0519
[4.50, 4.80)	4.65	4.643	577	48	542	44	16	0	0.5045	0.4974	0.5526	4598.6303	990.4119
[4.80, 5.10)	4.95	4.947	464	36	400	24	12	2	0.5115	0.4967	0.5159	4015.2907	811.7330
[5.10, 5.40)	5.25	5.241	290	13	283	11	4	1	0.5164	0.4993	0.3753	2782.5322	530.8908
[5.40, 5.70)	5.55	5.533	201	6	147	3	9	0	0.5221	0.5140	0.5552	1894.7067	342.4499
[5.70, 6.00)	5.85	5.842	136	3	110	2	2	0	0.5213	0.4985	0.6177	1502.5416	257.2152
[6.00, 6.30)	6.15	6.120	77	1	68	0	2	0	0.5347	0.5167	0.5861	846.5433	138.3250
[6.30, 6.60)	6.45	6.462	45	1	43	1	4	0	0.5481	0.5003	0.5202	460.0987	71.1969
[6.60, 6.90)	6.75	6.726	32	1	20	0	0	0	0.5421	0.5070	0.0000	407.6201	60.6077
[6.90, 7.50)	7.20	7.191	30	1	19	0	0	0	0.5190	0.4941	0.0000	418.3077	58.1689
[7.50, 8.70)	8.10	7.782	8	0	8	0	0	0	0.5009	0.5038	0.0000	121.7674	15.6483

717 15.9 Mass bin centroid calculations in $x_F \in [0.40, 0.45)$

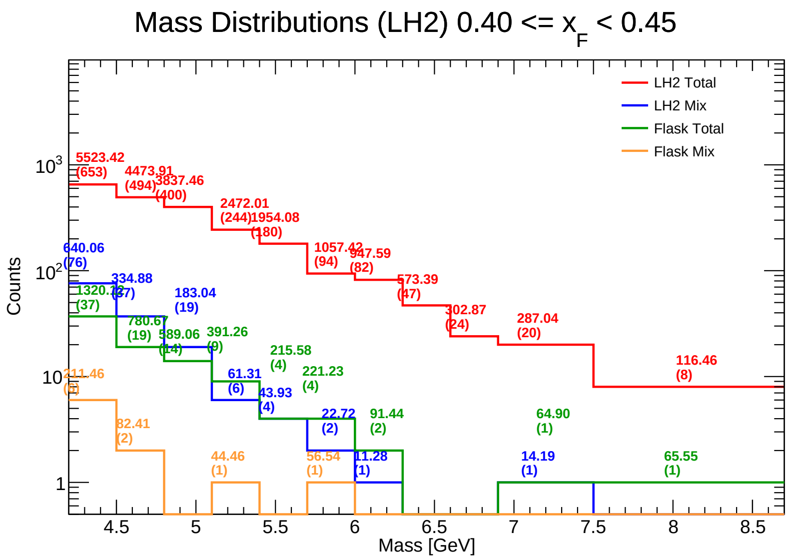


Figure 92: Invariant Mass distributions for LH_2 in the range $x_F \in [0.40, 0.45)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 25: LH_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.40, 0.45)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LH_2)	$N_{\rm tot}^{\rm LH_2}$	$N_{\rm mix}^{\rm LH_2}$	$N_{\rm tot}^{\rm fl}$	$N_{\rm mix}^{\rm fl}$	$\langle \epsilon_{\rm sig.}^{\rm LH_2} \rangle$	$\langle \epsilon_{\rm sig.}^{\rm LD_2} \rangle$	$\langle \epsilon_{\rm sig.}^{\rm fl} \rangle$	$\sum M_{\rm corr}^{\rm LH_2}$ (Num.)	$Y_{\rm corr}^{\rm LH_2}$ (Denom.)
[4.20, 4.50)	4.35	4.350	653	76	37	6	0.5142	0.5034	0.5358	3774.7043	867.7781
[4.50, 4.80)	4.65	4.643	494	37	19	2	0.5128	0.5003	0.4980	3440.7783	741.0819
[4.80, 5.10)	4.95	4.934	400	19	14	0	0.5144	0.4979	0.5157	3065.3669	621.2994
[5.10, 5.40)	5.25	5.231	244	6	9	1	0.5167	0.5083	0.5325	2063.8964	394.5657
[5.40, 5.70)	5.55	5.514	180	4	4	0	0.5090	0.4949	0.4579	1694.5716	307.3449
[5.70, 6.00)	5.85	5.856	94	2	4	1	0.5193	0.5050	0.4610	870.0097	148.5659
[6.00, 6.30)	6.15	6.175	82	1	2	0	0.5336	0.5240	0.5872	844.8677	136.8225
[6.30, 6.60)	6.45	6.422	47	0	0	0	0.5264	0.5222	0.0000	573.3919	89.2836
[6.60, 6.90)	6.75	6.751	24	0	0	0	0.5349	0.4834	0.0000	302.8656	44.8649
[6.90, 7.50)	7.20	7.100	20	1	1	0	0.4986	0.5283	0.4984	207.9480	29.2882
[7.50, 8.70)	8.10	7.820	8	0	1	0	0.5387	0.5157	0.5270	50.9102	6.5107

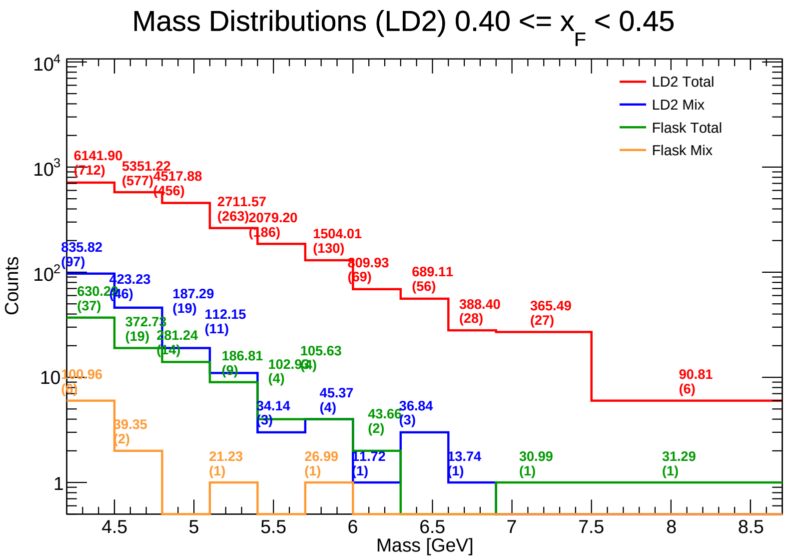


Figure 93: Invariant Mass distributions for LD_2 in the range $x_F \in [0.40, 0.45)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 26: LD_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.40, 0.45)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LD2)	$N_{\text{tot}}^{\text{LD2}}$	$N_{\text{mix}}^{\text{LD2}}$	$N_{\text{tot}}^{\text{LH2}}$	$N_{\text{mix}}^{\text{LH2}}$	$N_{\text{tot}}^{\text{fl}}$	$N_{\text{mix}}^{\text{fl}}$	$\langle \epsilon_{\text{sig.}}^{\text{LH2}} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{LD2}} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{fl}} \rangle$	$\sum M_{\text{corr}}^{\text{LD2}}$ (Num.)	$Y_{\text{corr}}^{\text{LD2}}$ (Denom.)
[4.20, 4.50)	4.35	4.341	712	97	653	76	37	6	0.5142	0.5034	0.5358	4722.4401	1087.7501
[4.50, 4.80)	4.65	4.642	577	46	494	37	19	2	0.5128	0.5003	0.4980	4545.0959	979.1113
[4.80, 5.10)	4.95	4.934	456	19	400	19	14	0	0.5144	0.4979	0.5157	4005.2284	811.7660
[5.10, 5.40)	5.25	5.243	263	11	244	6	9	1	0.5167	0.5083	0.5325	2404.1459	458.5667
[5.40, 5.70)	5.55	5.527	186	3	180	4	4	0	0.5090	0.4949	0.4579	1917.7415	346.9796
[5.70, 6.00)	5.85	5.851	130	4	94	2	4	1	0.5193	0.5050	0.4610	1367.4861	233.7068
[6.00, 6.30)	6.15	6.153	69	1	82	1	2	0	0.5336	0.5240	0.5872	742.3969	120.6594
[6.30, 6.60)	6.45	6.426	56	3	47	0	0	0	0.5264	0.5222	0.0000	644.0226	100.2172
[6.60, 6.90)	6.75	6.708	28	1	24	0	0	0	0.5349	0.4834	0.0000	370.3046	55.2039
[6.90, 7.50)	7.20	7.134	27	0	20	1	1	0	0.4986	0.5283	0.4984	331.5057	46.4713
[7.50, 8.70)	8.10	7.776	6	0	8	0	1	0	0.5387	0.5157	0.5270	58.7789	7.5586

718 15.10 Mass bin centroid calculations in $x_F \in [0.45, 0.50)$

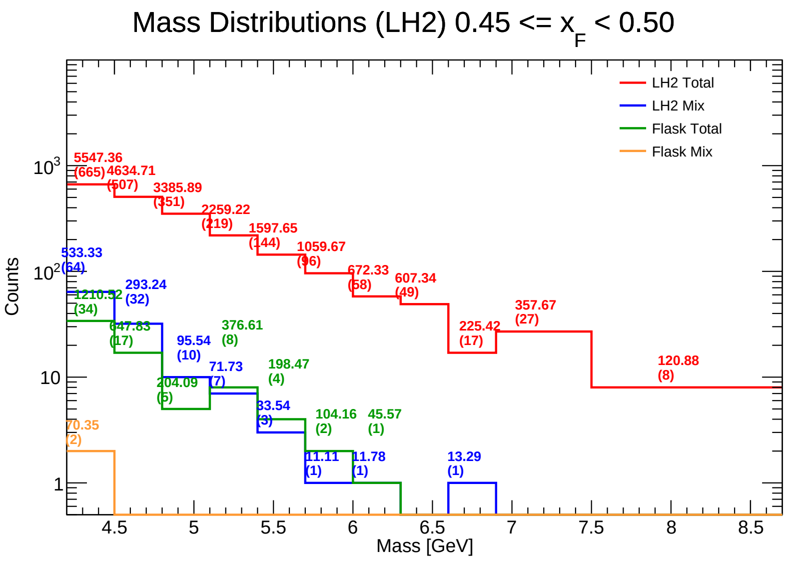


Figure 94: Invariant Mass distributions for LH_2 in the range $x_F \in [0.45, 0.50)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 27: LH_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.45, 0.50)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LH_2)	$N_{\text{tot}}^{\text{LH}_2}$	$N_{\text{mix}}^{\text{LH}_2}$	$N_{\text{tot}}^{\text{fl}}$	$N_{\text{mix}}^{\text{fl}}$	$\langle \epsilon_{\text{sig.}}^{\text{LH}_2} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{LD}_2} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{fl}} \rangle$	$\sum M_{\text{corr}}^{\text{LH}_2}$ (Num.)	$Y_{\text{corr}}^{\text{LH}_2}$ (Denom.)
[4.20, 4.50)	4.35	4.345	665	64	34	2	0.5208	0.4965	0.5362	3873.8623	891.6476
[4.50, 4.80)	4.65	4.642	507	32	17	0	0.5077	0.4946	0.5342	3693.6419	795.7268
[4.80, 5.10)	4.95	4.941	351	10	5	0	0.5123	0.5022	0.5356	3086.2595	624.6048
[5.10, 5.40)	5.25	5.238	219	7	8	0	0.5084	0.4926	0.4936	1810.8808	345.7319
[5.40, 5.70)	5.55	5.523	144	3	4	0	0.4991	0.5129	0.4989	1365.6411	247.2480
[5.70, 6.00)	5.85	5.841	96	1	2	0	0.5294	0.5013	0.4955	944.4068	161.6899
[6.00, 6.30)	6.15	6.135	58	1	1	0	0.5296	0.4931	0.5952	614.9792	100.2385
[6.30, 6.60)	6.45	6.437	49	0	0	0	0.5194	0.4905	0.0000	607.3368	94.3451
[6.60, 6.90)	6.75	6.741	17	1	0	0	0.5085	0.4813	0.0000	212.1303	31.4677
[6.90, 7.50)	7.20	7.198	27	0	0	0	0.5433	0.5015	0.0000	357.6699	49.6922
[7.50, 8.70)	8.10	7.828	8	0	0	0	0.5181	0.4924	0.0000	120.8757	15.4422

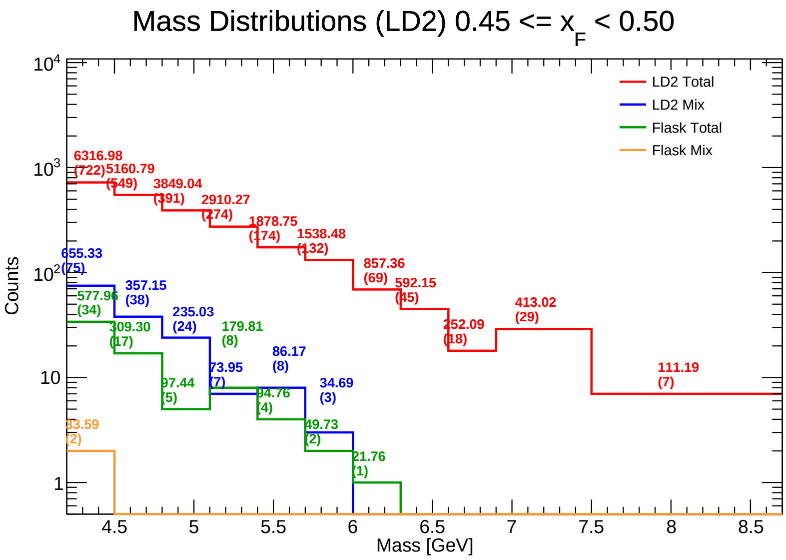


Figure 95: Invariant Mass distributions for LD_2 in the range $x_F \in [0.45, 0.50)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 28: LD_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.45, 0.50)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LD2)	$N_{\text{tot}}^{\text{LD2}}$	$N_{\text{mix}}^{\text{LD2}}$	$N_{\text{tot}}^{\text{LH2}}$	$N_{\text{mix}}^{\text{LH2}}$	$N_{\text{tot}}^{\text{fl}}$	$N_{\text{mix}}^{\text{fl}}$	$\langle \epsilon_{\text{sig.}}^{\text{LH2}} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{LD2}} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{fl}} \rangle$	$\sum M_{\text{corr}}^{\text{LD2}}$ (Num.)	$Y_{\text{corr}}^{\text{LD2}}$ (Denom.)
[4.20, 4.50)	4.35	4.345	722	75	665	64	34	2	0.5208	0.4965	0.5362	5061.5456	1165.0141
[4.50, 4.80)	4.65	4.651	549	38	507	32	17	0	0.5077	0.4946	0.5342	4441.1827	954.8695
[4.80, 5.10)	4.95	4.945	391	24	351	10	5	0	0.5123	0.5022	0.5356	3472.1561	702.2115
[5.10, 5.40)	5.25	5.230	274	7	219	7	8	0	0.5084	0.4926	0.4936	2630.4452	502.9811
[5.40, 5.70)	5.55	5.534	174	8	144	3	4	0	0.4991	0.5129	0.4989	1678.1735	303.2372
[5.70, 6.00)	5.85	5.843	132	3	96	1	2	0	0.5294	0.5013	0.4955	1440.4735	246.5342
[6.00, 6.30)	6.15	6.126	69	0	58	1	1	0	0.5296	0.4931	0.5952	826.7524	134.9659
[6.30, 6.60)	6.45	6.454	45	0	49	0	0	0	0.5194	0.4905	0.0000	583.4090	90.3928
[6.60, 6.90)	6.75	6.740	18	0	17	1	0	0	0.5085	0.4813	0.0000	249.0349	36.9471
[6.90, 7.50)	7.20	7.142	29	0	27	0	0	0	0.5433	0.5015	0.0000	407.8748	57.1108
[7.50, 8.70)	8.10	7.821	7	0	8	0	0	0	0.5181	0.4924	0.0000	109.4477	13.9942

15.11 Mass bin centroid calculations in $x_F \in [0.50, 0.55)$

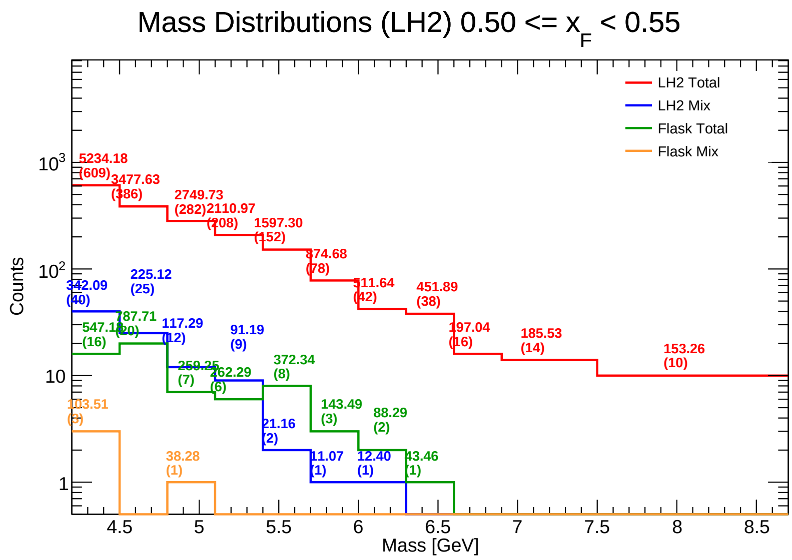


Figure 96: Invariant Mass distributions for LH_2 in the range $x_F \in [0.50, 0.55)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 29: LH_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.50, 0.55)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LH_2)	$N_{\text{tot}}^{\text{LH}_2}$	$N_{\text{mix}}^{\text{LH}_2}$	$N_{\text{tot}}^{\text{fl}}$	$N_{\text{mix}}^{\text{fl}}$	$\langle \epsilon_{\text{sig.}}^{\text{LH}_2} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{LD}_2} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{fl}} \rangle$	$\sum M_{\text{corr}}^{\text{LH}_2}$ (Num.)	$Y_{\text{corr}}^{\text{LH}_2}$ (Denom.)
[4.20, 4.50)	4.35	4.353	609	40	16	3	0.5059	0.5016	0.5562	4448.4213	1021.9635
[4.50, 4.80)	4.65	4.621	386	25	20	0	0.5143	0.4966	0.5218	2464.7999	533.3947
[4.80, 5.10)	4.95	4.950	282	12	7	1	0.5069	0.4972	0.5807	2411.4753	487.1827
[5.10, 5.40)	5.25	5.246	208	9	6	0	0.5160	0.4966	0.5210	1757.4828	335.0367
[5.40, 5.70)	5.55	5.534	152	2	8	0	0.5270	0.5090	0.5241	1203.7964	217.5327
[5.70, 6.00)	5.85	5.843	78	1	3	0	0.5205	0.5089	0.5340	720.1191	123.2478
[6.00, 6.30)	6.15	6.124	42	1	2	0	0.5037	0.5153	0.6153	410.9590	67.1047
[6.30, 6.60)	6.45	6.417	38	0	1	0	0.5396	0.5031	0.6484	408.4294	63.6487
[6.60, 6.90)	6.75	6.690	16	0	0	0	0.5432	0.5135	0.0000	197.0444	29.4524
[6.90, 7.50)	7.20	7.135	14	0	0	0	0.5384	0.4827	0.0000	185.5301	26.0023
[7.50, 8.70)	8.10	7.861	10	0	0	0	0.5129	0.5207	0.0000	153.2571	19.4954

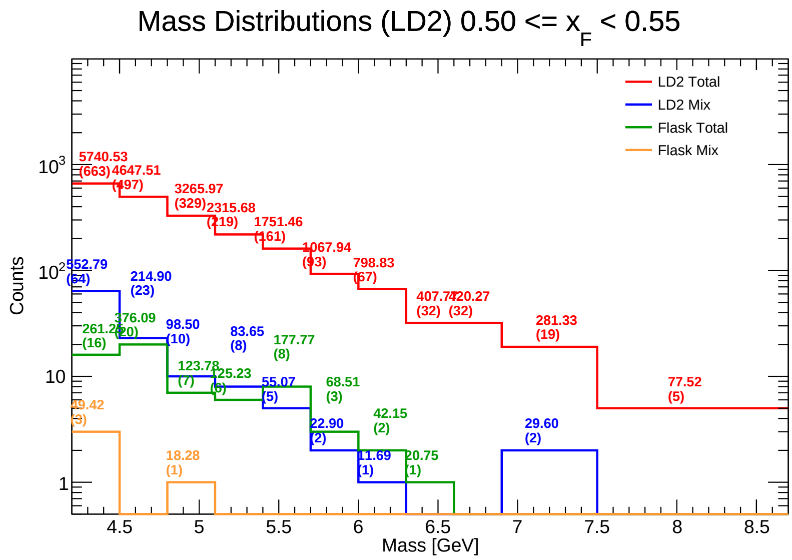


Figure 97: Invariant Mass distributions for LD_2 in the range $x_F \in [0.50, 0.55)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 30: LD_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.50, 0.55)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LD2)	$N_{\text{tot}}^{\text{LD2}}$	$N_{\text{mix}}^{\text{LD2}}$	$N_{\text{tot}}^{\text{LH2}}$	$N_{\text{mix}}^{\text{LH2}}$	$N_{\text{tot}}^{\text{fl}}$	$N_{\text{mix}}^{\text{fl}}$	$\langle \epsilon_{\text{sig.}}^{\text{LH2}} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{LD2}} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{fl}} \rangle$	$\sum M_{\text{corr}}^{\text{LD2}}$ (Num.)	$Y_{\text{corr}}^{\text{LD2}}$ (Denom.)
[4.20, 4.50)	4.35	4.346	663	64	609	40	16	3	0.5059	0.5016	0.5562	4911.9052	1130.3203
[4.50, 4.80)	4.65	4.642	497	23	386	25	20	0	0.5143	0.4966	0.5218	4021.0474	866.3062
[4.80, 5.10)	4.95	4.940	329	10	282	12	7	1	0.5069	0.4972	0.5807	3027.2644	612.8650
[5.10, 5.40)	5.25	5.257	219	8	208	9	6	0	0.5160	0.4966	0.5210	2081.5029	395.9439
[5.40, 5.70)	5.55	5.534	161	5	152	2	8	0	0.5270	0.5090	0.5241	1501.2938	271.3053
[5.70, 6.00)	5.85	5.847	93	2	78	1	3	0	0.5205	0.5089	0.5340	966.1701	165.2398
[6.00, 6.30)	6.15	6.144	67	1	42	1	2	0	0.5037	0.5153	0.6153	739.0735	120.2880
[6.30, 6.60)	6.45	6.410	32	0	38	0	1	0	0.5396	0.5031	0.6484	381.1390	59.4577
[6.60, 6.90)	6.75	6.745	32	0	16	0	0	0	0.5432	0.5135	0.0000	417.4386	61.8928
[6.90, 7.50)	7.20	7.148	19	2	14	0	0	0	0.5384	0.4827	0.0000	249.0644	34.8456
[7.50, 8.70)	8.10	8.079	5	0	10	0	0	0	0.5129	0.5207	0.0000	75.3126	9.3219

720 15.12 Mass bin centroid calculations in $x_F \in [0.55, 0.60)$

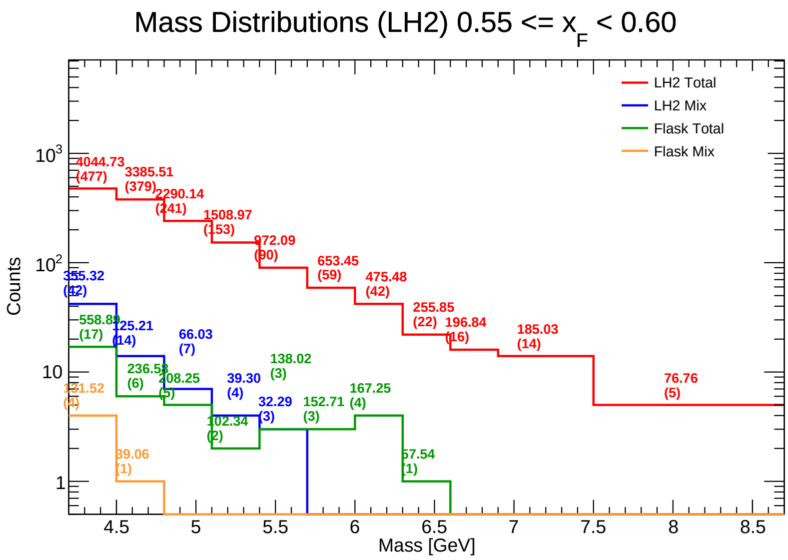


Figure 98: Invariant Mass distributions for LH_2 in the range $x_F \in [0.55, 0.60)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 31: LH_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.55, 0.60)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LH_2)	$N_{\text{tot}}^{\text{LH}_2}$	$N_{\text{mix}}^{\text{LH}_2}$	$N_{\text{tot}}^{\text{fl}}$	$N_{\text{mix}}^{\text{fl}}$	$\langle \epsilon_{\text{sig.}}^{\text{LH}_2} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{LD}_2} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{fl}} \rangle$	$\sum M_{\text{corr}}^{\text{LH}_2}$ (Num.)	$Y_{\text{corr}}^{\text{LH}_2}$ (Denom.)
[4.20, 4.50)	4.35	4.345	477	42	17	4	0.5122	0.4986	0.5798	3262.0459	750.7943
[4.50, 4.80)	4.65	4.635	379	14	6	1	0.5194	0.5029	0.5250	3062.7927	660.8116
[4.80, 5.10)	4.95	4.951	241	7	5	0	0.5207	0.5058	0.5208	2015.8644	407.1656
[5.10, 5.40)	5.25	5.247	153	4	2	0	0.5320	0.4999	0.4518	1367.3328	260.5952
[5.40, 5.70)	5.55	5.525	90	3	3	0	0.5122	0.4975	0.5334	801.7830	145.1304
[5.70, 6.00)	5.85	5.830	59	0	3	0	0.5263	0.5356	0.5032	500.7378	85.8927
[6.00, 6.30)	6.15	6.122	42	0	4	0	0.5417	0.5135	0.6467	308.2363	50.3521
[6.30, 6.60)	6.45	6.419	22	0	1	0	0.5525	0.4884	0.4926	198.3128	30.8924
[6.60, 6.90)	6.75	6.697	16	0	0	0	0.5444	0.5190	0.0000	196.8407	29.3927
[6.90, 7.50)	7.20	7.185	14	0	0	0	0.5437	0.5198	0.0000	185.0287	25.7514
[7.50, 8.70)	8.10	7.799	5	0	0	0	0.5080	0.5289	0.0000	76.7604	9.8428

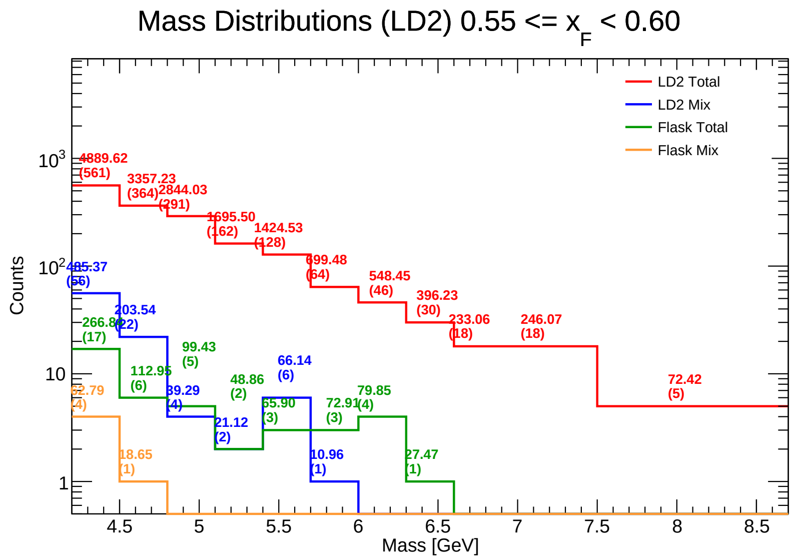


Figure 99: Invariant Mass distributions for LD_2 in the range $x_F \in [0.55, 0.60)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 32: LD_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.55, 0.60)$.

Mass Bin (GeV)	Geo. $\langle M\rangle$	$\langle M\rangle$ (LD2)	$N_{\rm tot}^{\rm LD2}$	$N_{\rm mix}^{\rm LD2}$	$N_{\rm tot}^{\rm LH2}$	$N_{\rm mix}^{\rm LH2}$	$N_{\rm tot}^{\rm fl}$	$N_{\rm mix}^{\rm fl}$	$\langle\epsilon_{\rm sig.}^{\rm LH2}\rangle$	$\langle\epsilon_{\rm sig.}^{\rm LD2}\rangle$	$\langle\epsilon_{\rm sig.}^{\rm fl}\rangle$	$\sum M_{\rm corr}^{\rm LD2}$ (Num.)	$Y_{\rm corr}^{\rm LD2}$ (Denom.)
[4.20, 4.50)	4.35	4.349	561	56	477	42	17	4	0.5122	0.4986	0.5798	4153.2590	955.0744
[4.50, 4.80)	4.65	4.635	364	22	379	14	6	1	0.5194	0.5029	0.5250	3015.3126	650.5229
[4.80, 5.10)	4.95	4.944	291	4	241	7	5	0	0.5207	0.5058	0.5208	2676.3040	541.3583
[5.10, 5.40)	5.25	5.231	162	2	153	4	2	0	0.5320	0.4999	0.4518	1605.8395	306.9928
[5.40, 5.70)	5.55	5.537	128	6	90	3	3	0	0.5122	0.4975	0.5334	1280.9565	231.3347
[5.70, 6.00)	5.85	5.857	64	1	59	0	3	0	0.5263	0.5356	0.5032	608.4089	103.8743
[6.00, 6.30)	6.15	6.117	46	0	42	0	4	0	0.5417	0.5135	0.6467	464.1622	75.8760
[6.30, 6.60)	6.45	6.451	30	0	22	0	1	0	0.5525	0.4884	0.4926	365.9023	56.7161
[6.60, 6.90)	6.75	6.720	18	0	16	0	0	0	0.5444	0.5190	0.0000	230.2276	34.2582
[6.90, 7.50)	7.20	7.104	18	0	14	0	0	0	0.5437	0.5198	0.0000	243.4042	34.2612
[7.50, 8.70)	8.10	7.658	5	0	5	0	0	0	0.5080	0.5289	0.0000	71.3107	9.3116

15.13 Mass bin centroid calculations in $x_F \in [0.60, 0.65)$

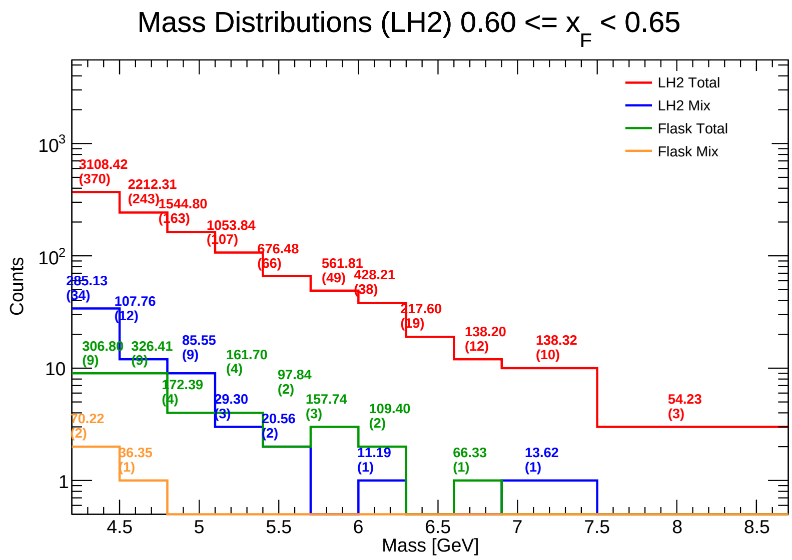


Figure 100: Invariant Mass distributions for LH_2 in the range $x_F \in [0.60, 0.65)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 33: LH_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.60, 0.65)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LH_2)	$N_{\text{tot}}^{\text{LH}_2}$	$N_{\text{mix}}^{\text{LH}_2}$	$N_{\text{tot}}^{\text{fl}}$	$N_{\text{mix}}^{\text{fl}}$	$\langle \epsilon_{\text{sig.}}^{\text{LH}_2} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{LD}_2} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{fl}} \rangle$	$\sum M_{\text{corr}}^{\text{LH}_2}$ (Num.)	$Y_{\text{corr}}^{\text{LH}_2}$ (Denom.)
[4.20, 4.50)	4.35	4.356	370	34	9	2	0.5178	0.4932	0.5587	2586.7099	593.8374
[4.50, 4.80)	4.65	4.664	243	12	9	1	0.5105	0.4915	0.5542	1814.4824	389.0589
[4.80, 5.10)	4.95	4.944	163	9	4	0	0.5216	0.4932	0.5027	1286.8584	260.2670
[5.10, 5.40)	5.25	5.240	107	3	4	0	0.5321	0.5152	0.5708	862.8458	164.6604
[5.40, 5.70)	5.55	5.547	66	2	2	0	0.5410	0.4870	0.4972	558.0861	100.6093
[5.70, 6.00)	5.85	5.817	49	0	3	0	0.5091	0.5286	0.4923	404.0685	69.4596
[6.00, 6.30)	6.15	6.147	38	1	2	0	0.5447	0.4886	0.4916	307.6228	50.0404
[6.30, 6.60)	6.45	6.406	19	0	0	0	0.5594	0.4924	0.0000	217.5984	33.9670
[6.60, 6.90)	6.75	6.685	12	0	1	0	0.5859	0.5400	0.4518	71.8776	10.7519
[6.90, 7.50)	7.20	7.225	10	1	0	0	0.5215	0.4865	0.0000	124.6918	17.2574
[7.50, 8.70)	8.10	8.039	3	0	0	0	0.4447	0.4827	0.0000	54.2329	6.7460

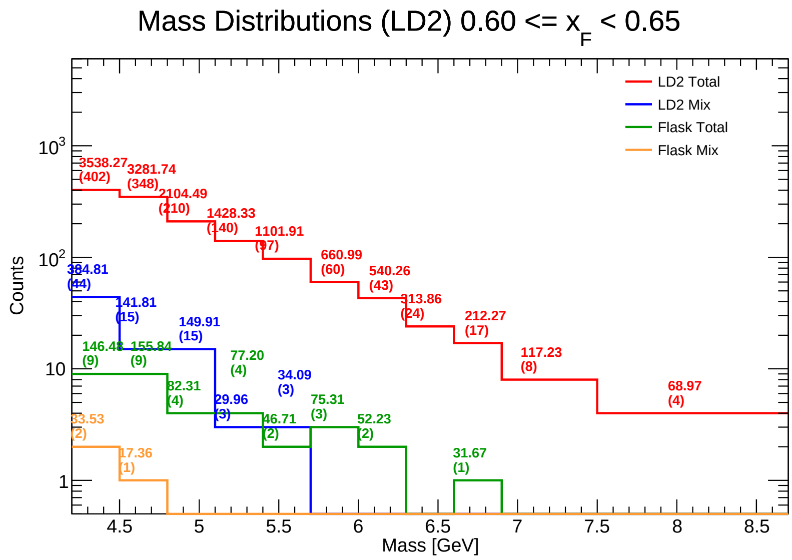


Figure 101: Invariant Mass distributions for LD_2 in the range $x_F \in [0.60, 0.65)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 34: LD_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.60, 0.65)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LD2)	$N_{\text{tot}}^{\text{LD2}}$	$N_{\text{mix}}^{\text{LD2}}$	$N_{\text{tot}}^{\text{LH2}}$	$N_{\text{mix}}^{\text{LH2}}$	$N_{\text{tot}}^{\text{fl}}$	$N_{\text{mix}}^{\text{fl}}$	$\langle \epsilon_{\text{sig.}}^{\text{LH2}} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{LD2}} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{fl}} \rangle$	$\sum M_{\text{corr}}^{\text{LD2}}$ (Num.)	$Y_{\text{corr}}^{\text{LD2}}$ (Denom.)
[4.20, 4.50)	4.35	4.346	402	44	370	34	9	2	0.5178	0.4932	0.5587	3003.2821	691.0433
[4.50, 4.80)	4.65	4.637	348	15	243	12	9	1	0.5105	0.4915	0.5542	2975.3272	641.6334
[4.80, 5.10)	4.95	4.944	210	15	163	9	4	0	0.5216	0.4932	0.5027	1853.7536	374.9246
[5.10, 5.40)	5.25	5.259	140	3	107	3	4	0	0.5321	0.5152	0.5708	1308.7492	248.8414
[5.40, 5.70)	5.55	5.532	97	3	66	2	2	0	0.5410	0.4870	0.4972	1013.0783	183.1413
[5.70, 6.00)	5.85	5.814	60	0	49	0	3	0	0.5091	0.5286	0.4923	579.8578	99.7293
[6.00, 6.30)	6.15	6.141	43	0	38	1	2	0	0.5447	0.4886	0.4916	483.5994	78.7520
[6.30, 6.60)	6.45	6.440	24	0	19	0	0	0	0.5594	0.4924	0.0000	310.7289	48.2535
[6.60, 6.90)	6.75	6.729	17	0	12	0	1	0	0.5859	0.5400	0.4518	179.5672	26.6842
[6.90, 7.50)	7.20	7.128	8	0	10	1	0	0	0.5215	0.4865	0.0000	115.4384	16.1951
[7.50, 8.70)	8.10	8.326	4	0	3	0	0	0	0.4447	0.4827	0.0000	68.1898	8.1895

15.14 Mass bin centroid calculations in $x_F \in [0.65, 0.70)$

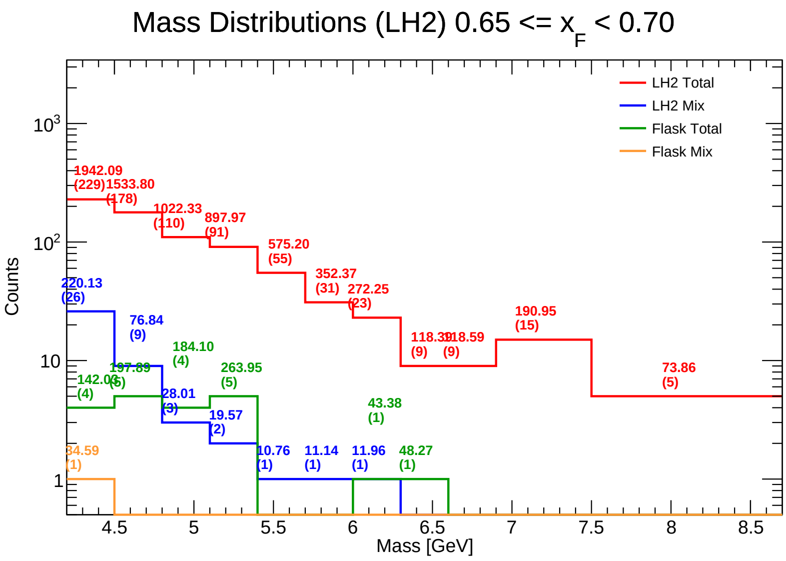


Figure 102: Invariant Mass distributions for LH_2 in the range $x_F \in [0.65, 0.70)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 35: LH_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.65, 0.70)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LH_2)	$N_{tot}^{LH_2}$	$N_{mix}^{LH_2}$	N_{tot}^{fl}	N_{mix}^{fl}	$\langle \epsilon_{sig.}^{LH_2} \rangle$	$\langle \epsilon_{sig.}^{LD_2} \rangle$	$\langle \epsilon_{sig.}^{fl} \rangle$	$\sum M_{corr}^{LH_2}$ (Num.)	$Y_{corr}^{LH_2}$ (Denom.)
[4.20, 4.50)	4.35	4.328	229	26	4	1	0.5104	0.5060	0.5343	1614.5213	373.0327
[4.50, 4.80)	4.65	4.650	178	9	5	0	0.5391	0.5126	0.5147	1259.0746	270.7620
[4.80, 5.10)	4.95	4.921	110	3	4	0	0.5300	0.5031	0.4721	810.2185	164.6342
[5.10, 5.40)	5.25	5.213	91	2	5	0	0.5306	0.5154	0.4407	614.4418	117.8617
[5.40, 5.70)	5.55	5.535	55	1	0	0	0.5295	0.4890	0.0000	564.4366	101.9790
[5.70, 6.00)	5.85	5.840	31	1	0	0	0.5134	0.5113	0.0000	341.2306	58.4309
[6.00, 6.30)	6.15	6.130	23	1	1	0	0.5197	0.5625	0.6329	216.9123	35.3850
[6.30, 6.60)	6.45	6.442	9	0	1	0	0.4902	0.5597	0.5880	70.1225	10.8851
[6.60, 6.90)	6.75	6.734	9	0	0	0	0.5111	0.5134	0.0000	118.5885	17.6100
[6.90, 7.50)	7.20	7.164	15	0	0	0	0.5628	0.5462	0.0000	190.9453	26.6547
[7.50, 8.70)	8.10	7.654	5	0	0	0	0.5181	0.5731	0.0000	73.8638	9.6509

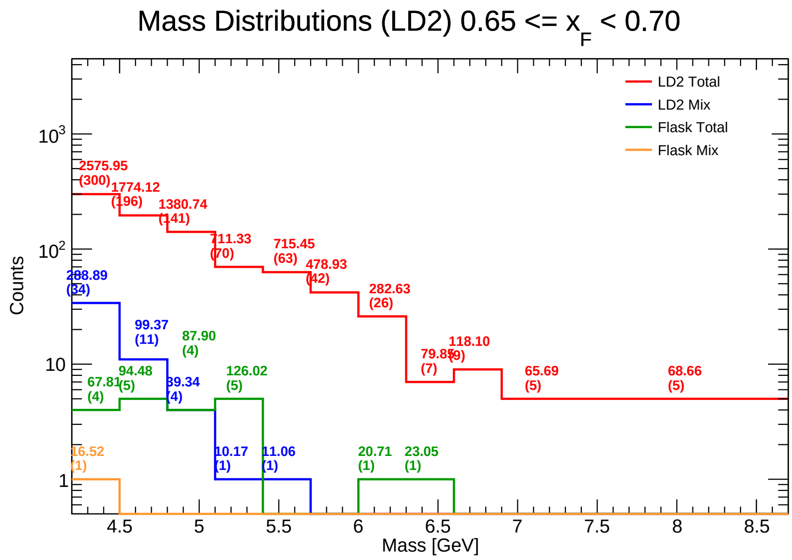


Figure 103: Invariant Mass distributions for LD_2 in the range $x_F \in [0.65, 0.70)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 36: LD_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.65, 0.70)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LD2)	$N_{\text{tot}}^{\text{LD2}}$	$N_{\text{mix}}^{\text{LD2}}$	$N_{\text{tot}}^{\text{LH2}}$	$N_{\text{mix}}^{\text{LH2}}$	$N_{\text{tot}}^{\text{fl}}$	$N_{\text{mix}}^{\text{fl}}$	$\langle \epsilon_{\text{sig.}}^{\text{LH2}} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{LD2}} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{fl}} \rangle$	$\sum M_{\text{corr}}^{\text{LD2}}$ (Num.)	$Y_{\text{corr}}^{\text{LD2}}$ (Denom.)
[4.20, 4.50)	4.35	4.351	300	34	229	26	4	1	0.5104	0.5060	0.5343	2212.5222	508.5613
[4.50, 4.80)	4.65	4.641	196	11	178	9	5	0	0.5391	0.5126	0.5147	1562.1423	336.6246
[4.80, 5.10)	4.95	4.925	141	4	110	3	4	0	0.5300	0.5031	0.4721	1241.8401	252.1463
[5.10, 5.40)	5.25	5.226	70	1	91	2	5	0	0.5306	0.5154	0.4407	566.3023	108.3621
[5.40, 5.70)	5.55	5.556	63	1	55	1	0	0	0.5295	0.4890	0.0000	696.2754	125.3288
[5.70, 6.00)	5.85	5.831	42	0	31	1	0	0	0.5134	0.5113	0.0000	474.0176	81.2966
[6.00, 6.30)	6.15	6.104	26	0	23	1	1	0	0.5197	0.5625	0.6329	258.7918	42.4004
[6.30, 6.60)	6.45	6.354	7	0	9	0	1	0	0.4902	0.5597	0.5880	55.7963	8.7812
[6.60, 6.90)	6.75	6.737	9	0	9	0	0	0	0.5111	0.5134	0.0000	116.3977	17.2774
[6.90, 7.50)	7.20	7.177	5	0	15	0	0	0	0.5628	0.5462	0.0000	62.9414	8.7699
[7.50, 8.70)	8.10	7.873	5	0	5	0	0	0	0.5181	0.5731	0.0000	67.6008	8.5864

15.15 Mass bin centroid calculations in $x_F \in [0.70, 0.75)$

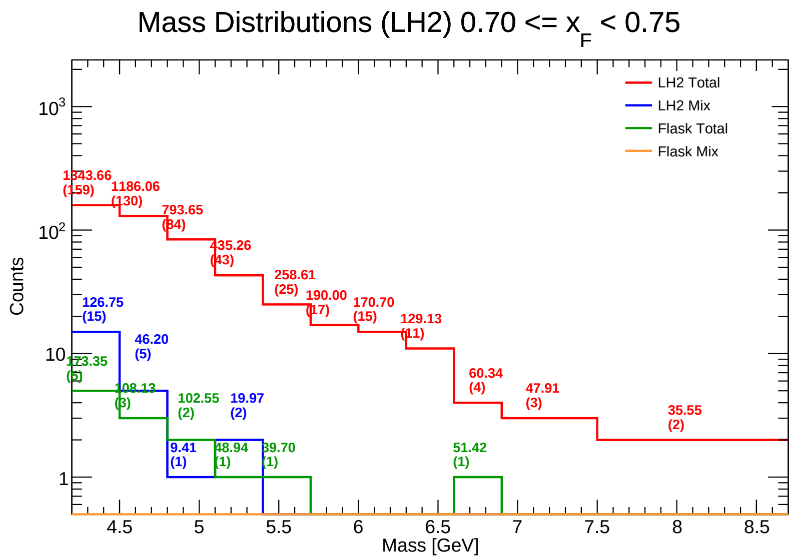


Figure 104: Invariant Mass distributions for LH_2 in the range $x_F \in [0.70, 0.75)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 37: LH_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.70, 0.75)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LH_2)	$N_{\text{tot}}^{\text{LH}_2}$	$N_{\text{mix}}^{\text{LH}_2}$	$N_{\text{tot}}^{\text{fl}}$	$N_{\text{mix}}^{\text{fl}}$	$\langle \epsilon_{\text{sig.}}^{\text{LH}_2} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{LD}_2} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{fl}} \rangle$	$\sum M_{\text{corr}}^{\text{LH}_2}$ (Num.)	$Y_{\text{corr}}^{\text{LH}_2}$ (Denom.)
[4.20, 4.50)	4.35	4.338	159	15	5	0	0.5135	0.5005	0.5515	1043.5578	240.5730
[4.50, 4.80)	4.65	4.634	130	5	3	0	0.5084	0.5199	0.5685	1031.7333	222.6551
[4.80, 5.10)	4.95	4.948	84	1	2	0	0.5230	0.4972	0.4200	681.6890	137.7682
[5.10, 5.40)	5.25	5.259	43	2	1	0	0.5190	0.5329	0.4707	366.3536	69.6586
[5.40, 5.70)	5.55	5.581	25	0	1	0	0.5371	0.4992	0.6003	218.9100	39.2232
[5.70, 6.00)	5.85	5.829	17	0	0	0	0.5215	0.4814	0.0000	189.9993	32.5959
[6.00, 6.30)	6.15	6.128	15	0	0	0	0.5385	0.4348	0.0000	170.7012	27.8546
[6.30, 6.60)	6.45	6.475	11	0	0	0	0.5516	0.4718	0.0000	129.1253	19.9419
[6.60, 6.90)	6.75	7.743	4	0	1	0	0.4475	0.5303	0.5646	8.9237	1.1525
[6.90, 7.50)	7.20	7.168	3	0	0	0	0.4489	0.4396	0.0000	47.9088	6.6837
[7.50, 8.70)	8.10	8.070	2	0	0	0	0.4539	0.5278	0.0000	35.5542	4.4058

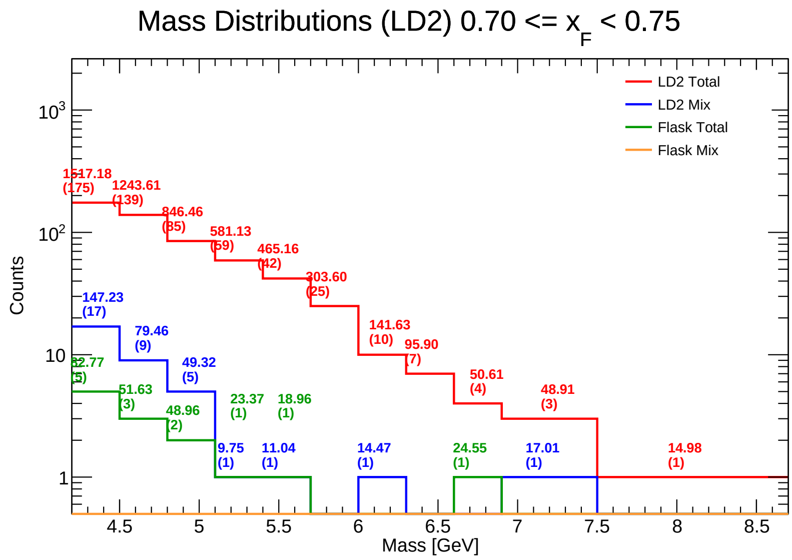


Figure 105: Invariant Mass distributions for LD_2 in the range $x_F \in [0.70, 0.75)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 38: LD_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.70, 0.75)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LD2)	$N_{\text{tot}}^{\text{LD2}}$	$N_{\text{mix}}^{\text{LD2}}$	$N_{\text{tot}}^{\text{LH2}}$	$N_{\text{mix}}^{\text{LH2}}$	$N_{\text{tot}}^{\text{fl}}$	$N_{\text{mix}}^{\text{fl}}$	$\langle \epsilon_{\text{sig.}}^{\text{LH2}} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{LD2}} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{fl}} \rangle$	$\sum M_{\text{corr}}^{\text{LD2}}$ (Num.)	$Y_{\text{corr}}^{\text{LD2}}$ (Denom.)
[4.20, 4.50)	4.35	4.339	175	17	159	15	5	0	0.5135	0.5005	0.5515	1272.1660	293.1846
[4.50, 4.80)	4.65	4.656	139	9	130	5	3	0	0.5084	0.5199	0.5685	1097.6753	235.7674
[4.80, 5.10)	4.95	4.958	85	5	84	1	2	0	0.5230	0.4972	0.4200	738.3730	148.9338
[5.10, 5.40)	5.25	5.250	59	1	43	2	1	0	0.5190	0.5329	0.4707	542.7367	103.3736
[5.40, 5.70)	5.55	5.533	42	1	25	0	1	0	0.5371	0.4992	0.6003	432.0106	78.0721
[5.70, 6.00)	5.85	5.846	25	0	17	0	0	0	0.5215	0.4814	0.0000	300.8615	51.4601
[6.00, 6.30)	6.15	6.143	10	1	15	0	0	0	0.5385	0.4348	0.0000	124.7041	20.2999
[6.30, 6.60)	6.45	6.464	7	0	11	0	0	0	0.5516	0.4718	0.0000	94.0442	14.5498
[6.60, 6.90)	6.75	6.809	4	0	4	0	1	0	0.4475	0.5303	0.5646	25.9348	3.8088
[6.90, 7.50)	7.20	7.010	3	1	3	0	0	0	0.4489	0.4396	0.0000	31.2165	4.4532
[7.50, 8.70)	8.10	7.903	1	0	2	0	0	0	0.4539	0.5278	0.0000	14.4727	1.8314

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15.16 Mass bin centroid calculations in $x_F \in [0.75, 0.80)$

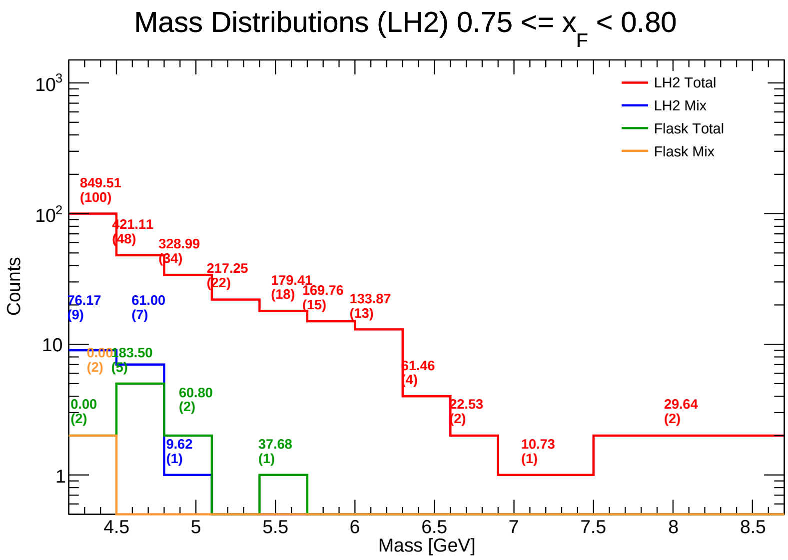

Figure 106: Invariant Mass distributions for LH_2 in the range $x_F \in [0.75, 0.80)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 39: LH_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.75, 0.80)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LH_2)	$N_{\rm tot}^{\rm LH_2}$	$N_{\rm mix}^{\rm LH_2}$	$N_{\rm tot}^{\rm fl}$	$N_{\rm mix}^{\rm fl}$	$\langle \epsilon_{\rm sig.}^{\rm LH_2} \rangle$	$\langle \epsilon_{\rm sig.}^{\rm LD_2} \rangle$	$\langle \epsilon_{\rm sig.}^{\rm fl} \rangle$	$\sum M_{\rm corr}^{\rm LH_2}$ (Num.)	$Y_{\rm corr}^{\rm LH_2}$ (Denom.)
[4.20, 4.50)	4.35	4.346	100	9	2	2	0.5115	0.5097	0.0000	773.3347	177.9246
[4.50, 4.80)	4.65	4.609	48	7	5	0	0.5286	0.5246	0.5600	176.6078	38.3167
[4.80, 5.10)	4.95	4.955	34	1	2	0	0.5113	0.5142	0.7115	258.5722	52.1815
[5.10, 5.40)	5.25	5.257	22	0	0	0	0.5323	0.4824	0.0000	217.2482	41.3278
[5.40, 5.70)	5.55	5.457	18	0	1	0	0.5513	0.4862	0.6582	141.7319	25.9716
[5.70, 6.00)	5.85	5.823	15	0	0	0	0.5146	0.4888	0.0000	169.7575	29.1515
[6.00, 6.30)	6.15	6.130	13	0	0	0	0.5953	0.5113	0.0000	133.8715	21.8384
[6.30, 6.60)	6.45	6.448	4	0	0	0	0.4197	0.5162	0.0000	61.4569	9.5313
[6.60, 6.90)	6.75	6.749	2	0	0	0	0.5992	0.6439	0.0000	22.5261	3.3378
[6.90, 7.50)	7.20	6.988	1	0	0	0	0.6510	0.5786	0.0000	10.7348	1.5361
[7.50, 8.70)	8.10	8.004	2	0	0	0	0.5401	0.5301	0.0000	29.6385	3.7027

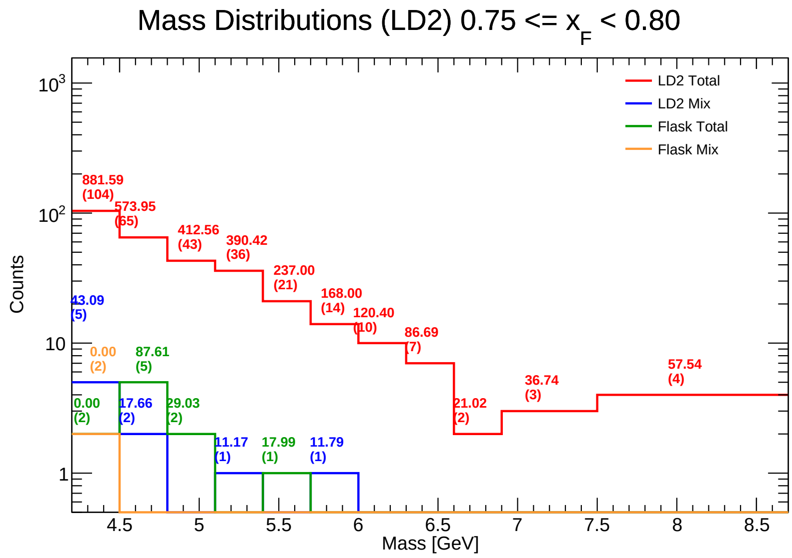

Figure 107: Invariant Mass distributions for LD_2 in the range $x_F \in [0.75, 0.80)$. Components shown include target total (red), target mix (blue), flask total (green), and flask mix (orange). The values listed above each bin represent the efficiency-weighted corrected mass $\sum M/\epsilon$ and the raw counts (N).

Table 40: LD_2 raw counts, efficiency corrections, yields, and final data-driven centroids for $x_F \in [0.75, 0.80)$.

Mass Bin (GeV)	Geo. $\langle M \rangle$	$\langle M \rangle$ (LD2)	$N_{\text{tot}}^{\text{LD2}}$	$N_{\text{mix}}^{\text{LD2}}$	$N_{\text{tot}}^{\text{LH2}}$	$N_{\text{mix}}^{\text{LH2}}$	$N_{\text{tot}}^{\text{fl}}$	$N_{\text{mix}}^{\text{fl}}$	$\langle \epsilon_{\text{sig.}}^{\text{LH2}} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{LD2}} \rangle$	$\langle \epsilon_{\text{sig.}}^{\text{fl}} \rangle$	$\sum M_{\text{corr}}^{\text{LD2}}$ (Num.)	$Y_{\text{corr}}^{\text{LD2}}$ (Denom.)
[4.20, 4.50)	4.35	4.317	104	5	100	9	2	2	0.5115	0.5097	0.0000	827.3672	191.6632
[4.50, 4.80)	4.65	4.624	65	2	48	7	5	0	0.5286	0.5246	0.5600	466.1328	100.8035
[4.80, 5.10)	4.95	4.934	43	0	34	1	2	0	0.5113	0.5142	0.7115	379.8061	76.9715
[5.10, 5.40)	5.25	5.227	36	1	22	0	0	0	0.5323	0.4824	0.0000	376.1295	71.9579
[5.40, 5.70)	5.55	5.475	21	0	18	0	1	0	0.5513	0.4862	0.6582	216.9670	39.6300
[5.70, 6.00)	5.85	5.874	14	1	15	0	0	0	0.5146	0.4888	0.0000	153.7675	26.1761
[6.00, 6.30)	6.15	6.157	10	0	13	0	0	0	0.5953	0.5113	0.0000	118.4778	19.2434
[6.30, 6.60)	6.45	6.392	7	0	4	0	0	0	0.4197	0.5162	0.0000	85.8021	13.4225
[6.60, 6.90)	6.75	6.766	2	0	2	0	0	0	0.5992	0.6439	0.0000	20.6920	3.0582
[6.90, 7.50)	7.20	7.087	3	0	1	0	0	0	0.6510	0.5786	0.0000	36.5846	5.1625
[7.50, 8.70)	8.10	7.622	4	0	2	0	0	0	0.5401	0.5301	0.0000	57.1093	7.4930

References

[1] S. D. Drell and T. M. Yan, Phys. Rev. Lett. 25, 316 (1970).

[2] DocDB 2111, https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=2111

[3] DocDB 8588, https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=8588

[4] DocDB 10809, https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=10809

[5] DocDB 11227, https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11227

[6] DocDB 11322, https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11322

[7] DocDB 11383, https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11383

[8] DocDB 11427, https://seaquest-docdb.fnal.gov/cgi-bin/sso/RetrieveFile?docid=11427&filename=reco_eff_kinematics.pdf&version=1

[9] DocDB 11431, https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11431

[10] DocDB 11445, https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11445

[11] DocDB 11448-v2, https://seaquest-docdb.fnal.gov/cgi-bin/sso/RetrieveFile?docid=11448&filename=eff_mix_corr.pdf&version=2

[12] DocDB 11467-v4, https://seaquest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11467